



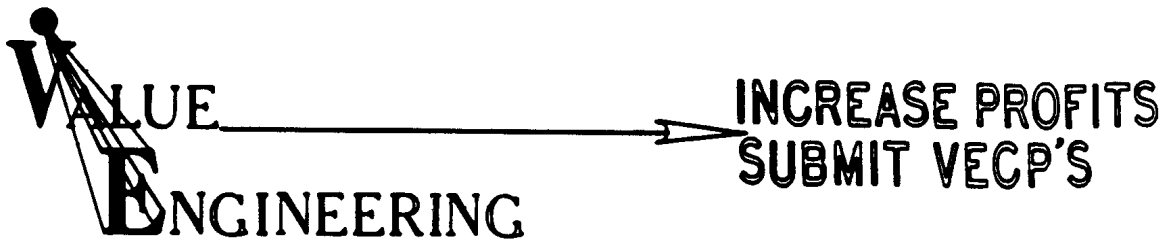
US Army Corps
of Engineers®

Jacksonville District

U.S. NAVAL STATION MAYPORT MAINTENANCE DREDGING

Duval County, Florida

Construction Solicitation and Specifications
Volume 2 of 2



See Volume 1 of 2 for issue date

PROJECT TABLE OF CONTENTS

DIVISION 00 - PROCUREMENT AND CONTRACTING REQUIREMENTS

00 31 21 CONTROL MONUMENT DESCRIPTIONS
00 33 50 WEATHER, WATER STAGE AND TIDE DATA

DIVISION 01 - GENERAL REQUIREMENTS

01 11 00 SUMMARY OF WORK
01 20 00 PRICE AND PAYMENT PROCEDURES
01 30 00 ADMINISTRATIVE REQUIREMENTS
01 32 01.10 35 PROJECT SCHEDULE (Alternate)
01 33 00 SUBMITTAL PROCEDURES
01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS
01 45 00.15 10 RESIDENT MANAGEMENT SYSTEM CONTRACTOR MODE (RMS CM)
01 45 05 DREDGING/BEACH FILL PLACEMENT - QUALITY CONTROL
01 50 00.00 35 TEMPORARY CONSTRUCTION FACILITIES AND CONTROLS
01 57 20.00 35 ENVIRONMENTAL PROTECTION
01 57 25.00 35 TURBIDITY MONITORING

DIVISION 35 - WATERWAY AND MARINE CONSTRUCTION

35 20 23.00 35 DREDGING
35 20 23.13 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM - SCOW ULLAGE
PROFILE
35 20 23.23 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM - HOPPER
DREDGE
35 20 23.33 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM - PIPELINE
HYDRAULIC DREDGE

-- End of Project Table of Contents --

SECTION 00 31 21 CONTROL MONUMENT DESCRIPTION

DESIGNATION - MYPT-SBB-1

SETTING - BRASS USACE DISK SET IN THE CONCRETE SEAWALL

AT THE SMALL BOAT BASIN. MARK IS SET IN THE CONCRETE SEAWALL. MARKER WAS SET DURING USACE SURVEY 22-005.

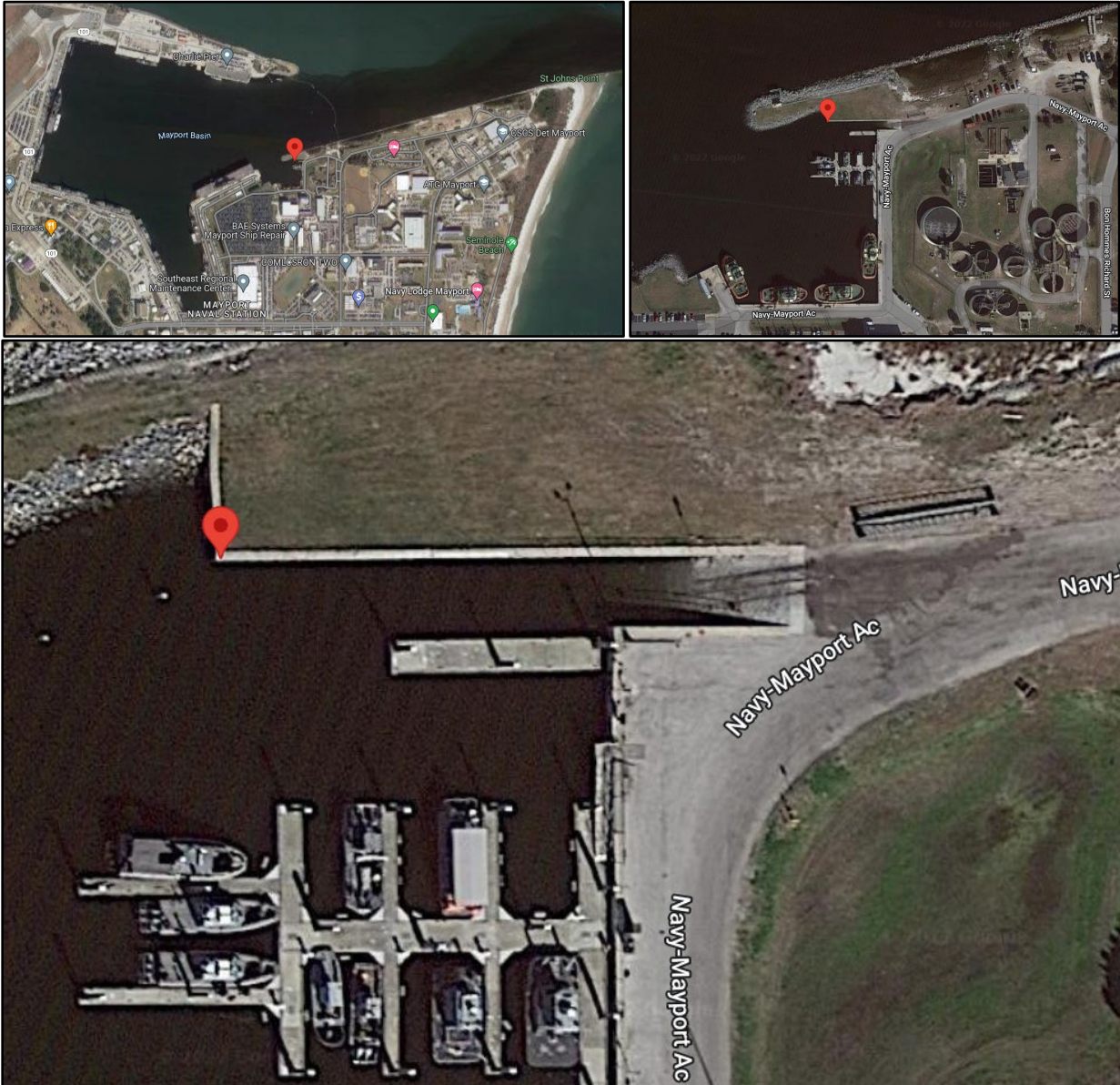


SECTION 00 31 21 CONTROL MONUMENT DESCRIPTION

DESIGNATION - MYPT-SBB-2

SETTING - BRASS USACE DISK SET IN THE CONCRETE SEAWALL

AT THE SMALL BOAT BASIN. MARK IS SET IN THE CONCRETE SEAWALL. MARKER WAS SET DURING USACE SURVEY 22-005.



SECTION 00 31 21 CONTROL MONUMENT DESCRIPTION

DESIGNATION - 872 0214 A

MARKER: FLANGE-ENCASED STAINLESS STEEL ROD W/O SLEEVE
STAMPING: 0214 A 1995
MARK LOGO: NOS
PROJECTION: FLUSH
MAGNETIC: NO MAGNETIC MATERIAL
STABILITY: PROBABLY HOLD POSITION/ELEVATION WELL
SATELLITE: THE SITE LOCATION WAS REPORTED AS SUITABLE FOR SATELLITE
OBSERVATIONS - May 23, 2007

STATION DESCRIPTION

RECOVERY NOTE BY FL DEPT OF ENV PRO 2007 (JLM)
THE MARK IS IN MAYPORT, ON THE MAYPORT NAVAL BASIN, AT THE END OF BALTIMORE STREET. TO REACH THE MARK FROM THE OVERPASS OF ATLANTIC BLVD (STATE ROAD 10) AND MAYPORT ROAD (STATE ROAD A1A) ON THE SOUTH SIDE OF MAYPORT, GO NORTH ON MAYPORT ROAD (STATE ROAD A1A) FOR 2.25 MI TO THE JUNCTION OF COUNTY ROAD 101 (MAYPORT ROAD-MAIN STREET), CONTINUE NORTH ON MAYPORT ROAD (COUNTY ROAD 101) FOR 1.8 MI TO THE ENTRANCE OF THE MAYPORT NAVAL BASIN STATION, PASSING THROUGH THE CHECK-IN STATION CONTINUE NORTH ON MAIN STREET FOR 0.1 MI TO THE JUNCTION OF MOALE AVENUE, TURN RIGHT ON MOALE AVENUE AND GO EAST FOR 0.65 MI TO THE JUNCTION OF BALTIMORE STREET, TURN LEFT ON BALTIMORE STREET AND GO NORTH FOR 0.9 MI TO THE END OF THE PAVEMENT AND THE MARK ON THE RIGHT, A STAINLESS-STEEL ROD DRIVEN TO REFUSAL AT A DEPTH OF 39.6 FT WITH A NOS LOGO CAP FLUSH WITH THE GROUND AND LEVEL WITH BALTIMORE STREET, THE DATUM POINT IS RECESSED 0.5 FT BELOW THE LEVEL OF THE NOS LOGO CAP.

LOCATED 77.5 FT SOUTHWEST OF THE CENTER AND END OF BALTIMORE STREET, 23.8 FT SOUTH-SOUTHEAST OF THE CENTERLINE OF BALTIMORE STREET, 1.8 FT NORTHEAST OF THE SOUTHEAST CORNER OF A 10.0 FT SQUARE CONCRETE PAD THAT IS AROUND A DRAIN AND 1.3 FT NORTHEAST OF A 3-INCH PVC WITNESS POST.

NOTE: ACCESS TO THE DATUM POINT IS HAD THROUGH A 5-INCH NOS LOGO CAP.



SECTION 00 31 21 CONTROL MONUMENT DESCRIPTION

DESIGNATION - 872 0214 C

MARKER: VERTICAL CONTROL DISK
SETTING: SET IN THE FOOTINGS OF SMALL/MEDIUM STRUCTURE/FOUNDATION
STAMPING: 0214 C 1995
MARK LOGO: NOS
MAGNETIC: NO MAGNETIC MATERIAL
STABILITY: MARK OF QUESTIONABLE OR UNKNOWN STABILITY
SATELLITE: THE SITE LOCATION WAS REPORTED AS SUITABLE FOR SATELLITE
OBSERVATIONS - May 23, 2007

STATION DESCRIPTION

DESCRIBED BY FL DEPT OF ENV PRO 2007 (JLM)
THE MARK IS AT MAYPORT NAVAL STATION NEAR SAINT JOHNS POINT.

TO REACH THE MARK FROM THE OVERPASS OF ATLANTIC BOULEVARD (STATE ROAD 10) AND MAYPORT ROAD (U.S. HIGHWAY A1A) ON THE SOUTH SIDE OF MAYPORT, GO NORTH ON MAYPORT ROAD (U.S. HIGHWAY A1A) FOR 2.25 MI TO THE JUNCTION OF MAYPORT ROAD (U.S. HIGHWAY A1A) AND MIAN STREET (COUNTY ROAD 101), CONTINUE NORTH ON MAIN STREET (COUNTY ROAD 101) FOR 1.8 MI TO THE ENTRANCE OF MAYPORT NAVAL STATION, PASSING THROUGH THE CHECK IN STATION, CONTINUE NORTH ON MAIN STREET FOR 0.1 MI TO THE JUNCTION OF MAIN STREET AND MOALE AVENUE, TURN RIGHT ON MOALE AVENUE AND GO EAST FOR 0.65 MI TO THE JUNCTION OF MOALE AVENUE AND BALTIMORE STREET, TURN LEFT ON BALTIMORE STREET AND GO NORTH FOR 0.9 MI TO THE END OF PAVEMENT AND THE MARK ON THE LEFT, SET IN THE NORTHWEST CORNER OF A SMALL CONCRETE FOUNDATION WITH A HIGH VOLTAGE TRANSFORMER BOX NUMBER 1410 ATTACHED, SOUTH OF A SMALL RED BRICK BUILDING NUMBER 1375.

LOCATED ABOUT 275.0 FT NORTH OF THE CENTERLINE OF BALTIMORE STREET, 76.2 FT NORTH OF THE APPROXIMATE CENTER OF A DIRT ROAD AROUND A PARKING LOT, 32.4 FT SOUTHEAST OF THE SOUTHWEST CORNER OF BUILDING NUMBER 1375 AND 16.0 FT EAST OF A CHAIN LINK FENCELINE.



Designation: USACE MAYPORT ARP

Stamping: N/A

Stability: May hold, commonly subject to ground movement.

Setting: Large structures with deep foundations.

Description: Station is located on US Naval Station Mayport and is operated by the USACE Jacksonville District Hydrographic Survey Section. Station is located approximately 50 feet above ground level on a metal tower structure. Tower is adjacent to building 351 & 1388, Training Support Detachment Mayport. Reference point is located approximately 8 feet south of the northeast corner of the top level of the tower and 15 feet north of the southeast corner of the tower.



Designation: 872 0218 A**Stamping:** 0218 A 1995**Setting:** Stainless steel rod without sleeve (10FT+ or 3.048M+)

Description: The primary bench mark is a flange encased rod, located near the entrance to the Saint Johns Bar Pilot Association, 140.97 m (462.5 ft) north of the approximate center of Ocean Street (State Highway A1A), 66.60 m (218.5 ft) west of the center of the Pilot Association Dock, 66.45 m (218.0 ft) SW of the Pilot Association Building, 37.37 m (122.6 ft) west of the NW corner of the Pilot Association Club, 19.20 m (63.0 ft) NNW of a power pole with two lights on the top and 0.30 m (1.0 ft) SW of the NW



Designation: 872 0218 C**Stamping:** 0218 C 1995**Setting:** Unspecified setting

Description: To reach the mark from the overpass of Atlantic Boulevard (State Highway 10) and Mayport Road (State Highway A1A) on the South Side of Mayport, go North on Mayport Road (State Highway A1A) for 2.25 MI (3.6 KM) to the Junction of State Highway 101 (Mayport Road-Main Street) and State Highway A1A North (State Road) on the left, Turn left on the State Highway A1A North (State Road-Ocean Street) and go Northwest for 3.6 MI (5.8 KM) to the entrance to the Mayport City Park and the mark on the left, set flush in the base of a 12.0 FT (3.7 M) diameter concrete foundation for a Wooden Umbrella Manatees sign, 0.4 FT (0.1 M) above the level of the ground and 0.4 FT (0.1 M) above the level of the entrance road.



SECTION 00 33 50 WEATHER, WATER STAGE AND TIDE DATA

1. The following water stage, wind, wave, and weather data are provided for inclusion in the subject project's P&S. The water level datums provided here are for information only and are not for use in conjunction with any contract-related hydrographic surveying.

a) Water Stages. Water levels in the project area are mainly affected by wind and semi-diurnal tidal fluctuations in the Atlantic Ocean. The project area is subject to storm surge from tropical systems from June through November. Surge from extratropical storms may affect the area any time of the year.

Elevations of tidal datums in the vicinity of the project area were gathered using the National Oceanic and Atmospheric Administration's (NOAA) National Ocean Service (NOS) tide stations shown in Figure 1 and Tables 1 through 2. These data are referenced to Mean Lower-Low Water (MLLW) from the tidal epoch period of 1983 to 2001 and are based on controlling tide stations near Mayport, Florida (NOAA Tide Stations 8720218 and 8720220). Additional tidal data on stations in the vicinity of the subject project area can be found at the following website: <https://tidesandcurrents.noaa.gov/map/index.html?region=Florida>.

The elevations in Table 1 are taken from NOAA's Tide Station 8720211 (Mayport Naval Sta., St Johns River, FL). These datums are based on a 12-month analysis from 01 November 2003 to 31 October 2004 from a gauge on the St Johns River, which is roughly 0.25 miles outside of the Mayport Basin. Table 2 provides datums taken from NOAA's Tide Station 8720214 (Degaussing Structure, FL). These elevations are based on data from a six-month analysis from 01 May 1998 to 31 October 1998 from a location approximately 0.5 miles east of the Mayport Basin.

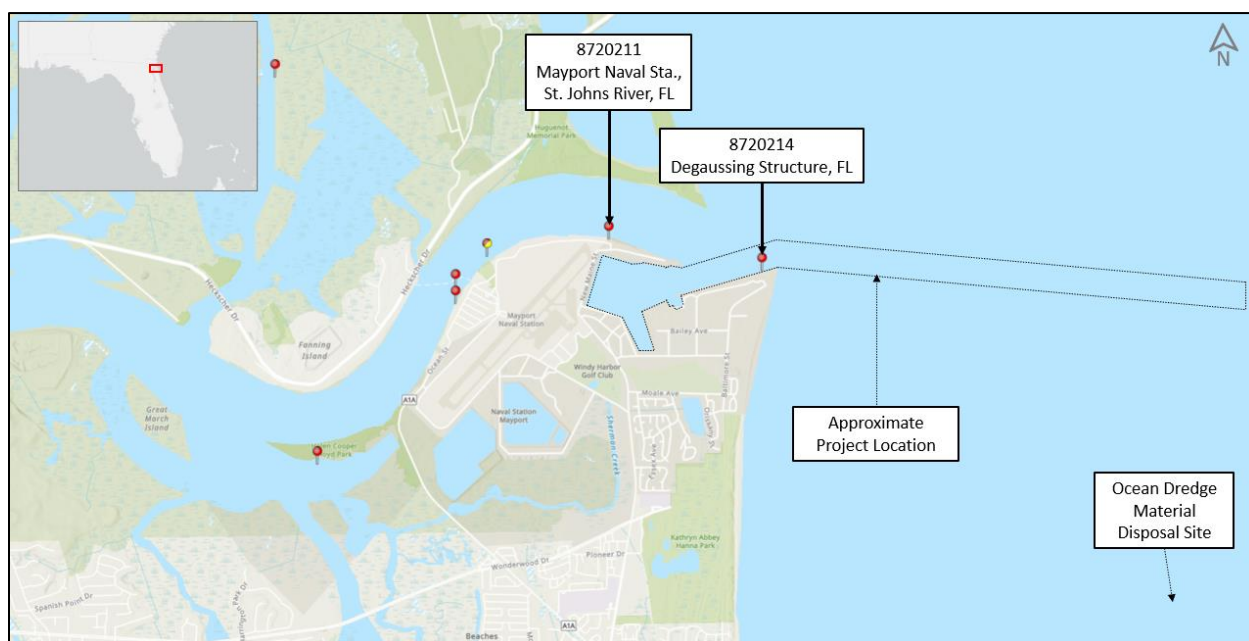


Figure 1. NOAA Tide Stations in the Vicinity of U.S. Naval Station Mayport in Duval County, Florida

SECTION 00 33 50 WEATHER, WATER STAGE AND TIDE DATA

Table 1. Datums from NOAA Station 8720211 (Mayport Naval Sta., St Johns River, FL)

Location: 30° 24.0' N, 81° 24.8' W	
Analysis Period: 01 November 2003 to 31 October 2004	
Datum	Elevation (ft-MLLW)
Highest Observed (9/27/2004)	7.81
Mean Higher High Water (MHHW)	5.11
Mean High Water (MHW)	4.82
North American Vertical Datum of 1988 (NAVD88)	3.11
Mean Sea Level (MSL)	2.54
Mean Tide Level (MTL)	2.49
Mean Low Water (MLW)	0.15
Mean Lower Low Water (MLLW)	0.00
Lowest Observed (3/08/2005)	-1.87

Table 2. Datums from NOAA Station 8720214 (Degaussing Structure, FL)

Location: 30° 23.8' N, 81° 23.7' W	
Analysis Period: 01 May 1998 to 31 October 1998	
Datum	Elevation (ft-MLLW)
Mean Higher High Water (MHHW)	5.29
Mean High Water (MHW)	4.96
North American Vertical Datum of 1988 (NAVD88)	3.18
Mean Sea Level (MSL)	2.56
Mean Tide Level (MTL)	2.56
Mean Low Water (MLW)	0.16
Mean Lower Low Water (MLLW)	0.00

Daily tidal predictions at locations along the coastline of North and South America, including locations in the vicinity of the project area, are included in the publication East Coast of North and South America Tide Tables, United States (U.S.) Department of Commerce, National Oceanic and Atmospheric Administration, National Ocean Service. In addition to daily tidal predictions, this publication provides mean tide levels, diurnal tide ranges, and some astronomical data (sunrise, sunset, moonrise, and moonset). Of note, tidal data provided in this publication are also listed at the following website: https://tidesandcurrents.noaa.gov/tide_predictions.html.

b) Wind and Wave Data. The publications in this section include wind and wave information in the vicinity of the project area. They are available for review at the websites or agencies indicated.

i. U.S. Coast Pilot Volume 4, Atlantic Coast: Cape Henry to Key West, U.S. Department of Commerce, National Oceanic and Atmospheric Administration, National Ocean Service is a publication that supplements the navigational information shown on NOAA's NOS nautical charts and provides miscellaneous meteorological data. This publication is available through NOAA or on the following

website: <https://nauticalcharts.noaa.gov/publications/coast-pilot/index.html>.

ii. U.S. Army Corps of Engineers' Coastal & Hydraulics Laboratory, Wave Information Study (WIS) Database is a wave hindcast database for various stations located along the U.S. Atlantic, Gulf of America, and Caribbean coastlines. WIS Station ST63407 is in roughly 13.0 meters of water relative to the Mean Sea Level (MSL) datum (i.e. 42.7 ft-MSL) at 30° 25.0' N, 81° 15.0' W. This station lies approximately 9.6 miles east of Mayport. Available data includes a hindcast time series of wave heights, periods, and directions taken at 1-hour intervals for the 44-year period of 1980-2024. The following website provides links to WIS summary tables and plots, as well as other sources of wave data: <https://wisportal.erdc.dren.mil/>.

iii. National Data Buoy Center (NDBC) Website provides a wide range of real-time and archived meteorological and oceanographic data collected at offshore and nearshore stations worldwide. Data provided on this website include wind speed, wind gusts, atmospheric pressure, air temperature, sea temperature, wave height, and wave period. Gauge readings are updated hourly, and archived data is available for most stations. Data relevant to Florida projects are found at the following website: <https://www.ndbc.noaa.gov/>.

c) Weather Conditions. The project area is subject to tropical systems June through November and to windy and rainy weather any time of the year. The climate of the area is essentially tropical, as temperatures below freezing are seldom. During the wet season (May through October), rainfall events are usually short and sporadic and closely associated with convective activity. Winter months generally constitute the dry season, where rainfall is typically associated with mid-latitude, or extratropical, systems and distributed in a more spatially uniform pattern. Daily rainfall events in the dry season are occasionally heavy when mid-latitude systems impact northeast Florida. Notably, hazardous thunderstorms can occur in this area at any time of the year.

i. It is the contractor's responsibility to obtain information concerning weather conditions in the project area. The Climate Data Online website (<https://www.ncdc.noaa.gov/cdo-web/>) published by NOAA's National Centers for Environmental Information (formerly the National Climatic Data Center) in Asheville, North Carolina contains climatological and meteorological data relevant to this region. Available data include quality-controlled daily, monthly, seasonal, and yearly measurements of temperature, precipitation, wind, and degree days as well as radar data and 15- to 30-year climate normals.

NOAA's Climate Data Online website includes Monthly Normals (average values) for temperature, precipitation, and other climatological information over three periods: 1981-2010, 1991-2020, and 2006-2020. NOAA's Mayport Pilot STN, FL (ID USW00003853) is closest to the project area and lies roughly 0.5 miles north of the project site at 30° 24.0' N, 81° 25.0' W. The most applicable period for data at this site is likely 2006-2020 and is found at the following website: <https://www.ncei.noaa.gov/maps/normals/>.

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 11 00

SUMMARY OF WORK

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 WORK COVERED BY CONTRACT DOCUMENTS
 - 1.2.1 Project Description
 - 1.2.2 Location
 - 1.2.3 Safety Requirements
 - 1.2.4 Work Performed by Others
 - 1.2.4.1 Adjacent Contractors
 - 1.2.5 Contractor Access and Use of Premises for Mayport Project
 - 1.2.5.1 Subcontractors and Personnel Contacts
 - 1.2.5.2 Installation Access
 - 1.2.5.2.1 Registration for DBIDS
 - 1.2.5.2.2 DBIDS Eligibility Requirements
 - 1.2.5.2.3 DBIDS Notification Requirements
 - 1.2.5.2.4 Pier Area Access Passes
 - 1.2.5.2.5 Port Security Barrier (PSB) Access
 - 1.2.5.3 No Smoking Policy
 - 1.2.5.4 DOD Condition of Readiness
- 1.3 COMMENCEMENT, PROSECUTION AND COMPLETION OF WORK
- 1.4 LIQUIDATED DAMAGES - CONSTRUCTION
- 1.5 PHYSICAL DATA
 - 1.5.1 Physical Conditions
 - 1.5.2 Weather, Water Stage and Tide Data
 - 1.5.3 Contractor Investigation
 - 1.5.4 Maritime Traffic
 - 1.5.5 Obstruction of Channel
 - 1.5.6 Hydrographic Surveys
- 1.6 LAYOUT OF WORK
 - 1.6.1 Established Project Control
 - 1.6.2 Layout
 - 1.6.3 Survey
- 1.7 DAMAGE TO WORK
- 1.8 EXISTING WORK

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

-- End of Section Table of Contents --

SECTION 01 11 00

SUMMARY OF WORK

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

GDR Section 00 31 32, Geotechnical Data Report
for U.S. Naval Station Mayport Maintenance
Dredging Duval County, Florida

EM 385-1-1 (2024) Safety and Occupational Health
(SOH) Requirements

1.2 WORK COVERED BY CONTRACT DOCUMENTS

1.2.1 Project Description

Project work consists of maintenance dredging portions of U.S. Naval Station Mayport including Bar Cut-3, Mayport Entrance Channel, Turning Basin, Destroyer Slip, and Small Boat Basin. Excavated material will be transported to and disposed in the Jacksonville Ocean Dredged Material Disposal Site (ODMDS) Release Zone "A". Incidental work includes turbidity monitoring, environmental species monitoring, and sea turtle trawling.

1.2.2 Location

U.S. Naval Station Mayport is located at the mouth of the St. Johns River in Duval County, Florida where it enters the Atlantic Ocean and is accessible from both bodies of water. Land access is available from Florida SR A1A. The center of the Jacksonville ODMDS Release Zone "A" lies in the Atlantic Ocean approximately 4.9 nautical miles southeast of the seaward tip of the southern river jetty.

1.2.3 Safety Requirements

See Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS and EM 385-1-1.

1.2.4 Work Performed by Others

See paragraph PROJECT COORDINATION of Section 01 30 00 ADMINISTRATIVE REQUIREMENTS.

1.2.4.1 Adjacent Contractors

Active dredging near the project site may impact Contractor work. Separate contracts are underway for maintenance dredging within Jacksonville Harbor between Bar Cut-3 Station 198+00 to Cut-17 Station and

U.S. Naval Station Mayport
Duval County, Florida

Cut-55 Station 0+00 to Lower Terminal Channel Station 65+00 (one contract), and within various Jacksonville Port Authority's berths (second contract), in which each contractor is using the same ODMDS and possibly the same zone that will be used under this contract. Coordinate activities to avoid impact to the adjacent Contractors and prevent problems from the adjacent contracts.

1.2.5 Contractor Access and Use of Premises for Mayport Project

Activity Regulations

Ensure that Contractor personnel employed on the Activity become familiar with and obey Activity regulations including safety, fire, traffic and security regulations. Keep within the limits of the work and avenues of ingress and egress. To minimize traffic congestion, delivery of materials to the Base shall be outside of peak traffic hours (6:30 to 8:00 a.m. and 3:30 to 5:00 p.m.) unless otherwise approved by the Contracting Officer. Wear hard hats in designated areas. Do not enter any restricted areas unless required to do so and until cleared for such entry. Mark Contractor equipment for identification.

A partial list of key Naval Station Mayport regulations is provided in APPENDIX A after the end of this section: SOPAMYPTINST 5440.3H, 9 May 2017, MANUAL OF THE SENIOR OFFICER PRESENT AFLOAT (SOPA), MAYPORT, FLORIDA, CHAPTER 5 DESTRUCTIVE WEATHER PLAN.

1.2.5.1 Subcontractors and Personnel Contacts

Provide a list of contact personnel of the Contractor and subcontractors including addresses and telephone numbers for use in the event of an emergency. As changes occur and additional information becomes available, correct and change the information contained in previous lists.

1.2.5.2 Installation Access

Obtain access to Navy installations through participation in the Defense Biometrics Identification System (DBIDS). Requirements for Contractor employee registration, and transition for employees currently under Navy Commercial Access Control System (NCACS), are available at <https://www.cnic.navy.mil/om/dbids.html>. No fees are associated with obtaining a DBIDS credential.

Participation in the DBIDS is not mandatory, and Contractor personnel may apply for One-Day Passes at the Base Visitor Control Office to access an installation. If the Contractor chooses to not participate, the Contractor's personnel will have to obtain daily passes, be subject to daily mandatory vehicle inspection, and will have limited access to the installation. The Government will not be responsible for any cost or lost time associated with obtaining daily passes or added vehicle inspections.

1.2.5.2.1 Registration for DBIDS

Registration for DBIDS is available at <https://www.cnic.navy.mil/om/dbids.html>. Procedure includes:

- a. Present a letter or official award document (i.e. DD Form 1155 or SF 1442) from the Contracting Officer, that provides the purpose for access, to the Base Visitor Control Center representative.

- b. Present two forms of valid identification, such as a passport or Real ID Act-compliant state driver's license.
- c. Provide completed SECNAV FORM 5512/1 to the Base Visitor Control Center representative to obtain a background check. This form is available for download at <https://www.cnmc.navy.mil/om/dbids.html>.
- d. Upon successful completion of the background check, the Government will complete the DBIDS enrollment process, which includes Contractor employee photo, finger prints, Base restriction and several other assessments.
- e. Upon successful completion of the enrollment process, the Contractor employee will be issued a DBIDS credential, and will be allowed to proceed to worksite.

1.2.5.2.2 DBIDS Eligibility Requirements

Throughout the length of the contract, the Contractor employee must continue to meet background screen standards. Periodic background screenings are conducted to verify continued DBIDS participation and installation access privileges. DBIDS access privileges will be immediately suspended or revoked if at any time a Contractor employee becomes ineligible.

An adjudication process may be initiated when a background screen failure results in disqualification from participation in the DBIDS, and Contractor employee does not agree with the reason for disqualification. The Government is the final authority.

1.2.5.2.3 DBIDS Notification Requirements

- a. Immediately report instances of lost or stolen badges to the Contracting Officer.
- b. Immediately collect DBIDS credentials and notify the Contracting Officer in writing under the following circumstances:
 - (1). An employee has departed the company without having properly returned or surrendered their DBIDS credentials.
 - (2). There is a reasonable basis to conclude that an employee, or former employee, might pose a risk, compromise, or threat to the safety or security of the Installation or anyone therein.

1.2.5.2.4 Pier Area Access Passes

Access to the wharf will require all Contractor personnel to obtain pier area passes. Contractor must prepare personnel lists with supporting documentation and coordinate with the Contracting Officer for these passes. Typically passes will need to be renewed on approximately a weekly basis.

1.2.5.2.5 Port Security Barrier (PSB) Access

All maritime access to the turning basin and berthing areas will require access through the Port Security Barrier. Access through the PSB will be directly coordinated with Naval Station Mayport Port Operations. Contractor will provide Naval Station Mayport Port Operations a minimum of

two hours notice in advance of a required PSB opening. Emergency/urgent situations necessitating an opening in less than one hour will be authorized on a case by case basis. Contractor will schedule the work and conduct operations in a manner that minimize opens during hours of darkness, keeps opening time to a minimum, and only necessary to meet mission requirements.

1.2.5.3 No Smoking Policy

Smoking is prohibited within and outside of all buildings on installation, except in designated smoking areas. This applies to existing buildings, buildings under construction and buildings under renovation. Discarding tobacco materials other than into designated tobacco receptacles is considered littering and is subject to fines. The Contracting Officer will identify designated smoking areas.

1.2.5.4 DOD Condition of Readiness

DOD will set the Condition of Readiness based on the weather forecast for sustained winds 50 knots (58 mph) or greater. Contact the Contracting Officer for the current condition setting. Monitor weather conditions a minimum of twice a day and take appropriate actions according to the approved Emergency Plan in the accepted Accident Prevention Plan, [EM 385-1-1](#), Section 01 Emergency Planning and the instructions below. Unless otherwise directed by the Contracting Officer, comply with the following:

- a. Condition FOUR (Sustained winds of 58 mph or greater expected within 72 hours): Normal daily jobsite cleanup and good housekeeping practices. Collect and store in piles or containers scrap lumber, waste material, and rubbish for removal and disposal at the close of each work day. Maintain the construction site including storage areas, free of accumulation of debris. Stack form lumber in neat piles less than 3.3 feet high. Remove all debris, trash, or objects that could become missile hazards. Review requirements pertaining to "Condition THREE" and continue action as necessary to attain "Condition FOUR" readiness. Contact Contracting Officer for weather and Condition of Readiness updates and completion of required actions.
- b. Condition THREE (Sustained winds of 58 mph or greater expected within 48 hours): Maintain "Condition FOUR" requirements and commence securing operations necessary for "Condition ONE" which cannot be completed within 18 hours. Cease all routine activities which might interfere with securing operations. Commence securing and stow all gear and portable equipment. Make preparations for securing buildings. Reinforce or remove formwork and scaffolding. Secure machinery, tools, equipment, materials, or remove from the jobsite. Expend every effort to clear all missile hazards and loose equipment from general base areas. Contact Contracting Officer for weather and Condition of Readiness updates and completion of required actions. Review requirements pertaining to "Condition TWO" and continue action as necessary to attain "Condition THREE" readiness.
- c. Condition TWO (Sustained winds of 58 mph or greater expected within 24 hours): Secure the jobsite, and leave Government premises.
- d. Condition ONE (Sustained winds of 58 mph or greater expected within 12 hours): Contractor access to the jobsite and Government premises is prohibited.

1.3 COMMENCEMENT, PROSECUTION AND COMPLETION OF WORK

Read this paragraph in conjunction with the Clause COMMENCEMENT, PROSECUTION, AND COMPLETION OF WORK (FAR 52.211-10).

1.4 LIQUIDATED DAMAGES - CONSTRUCTION

Refer to the Clause LIQUIDATED DAMAGES - CONSTRUCTION (FAR 52.211-12).

1.5 PHYSICAL DATA

Read this paragraph in conjunction with the Clause PHYSICAL DATA (FAR 52.236-4).

1.5.1 Physical Conditions

The indications of physical conditions on the drawings and in the specifications are the result of site investigations by surveys and/or by core borings. When the indicated physical conditions are the result of site investigations by core borings, the core boring logs and laboratory data are in GEOTECHNICAL DATA REPORT (GDR) and the core boring locations are shown on the drawings. Also, see the GDR for instructions concerning availability of core borings for inspection.

1.5.2 Weather, Water Stage and Tide Data

See Section 00 33 50 WEATHER, WATER STAGE AND TIDE DATA.

1.5.3 Contractor Investigation

Refer to the Clause SITE INVESTIGATION AND CONDITIONS AFFECTING THE WORK (FAR 52.236-3).

1.5.4 Maritime Traffic

Marine Traffic in the project area consists of Naval, cruise ships, commercial, pleasure, and small recreational vessels of all types and sizes which can be accommodated by existing depths.

1.5.5 Obstruction of Channel

The Government will not undertake to keep the channel free from vessels or other obstructions, except to the extent of such regulations, if any, as may be prescribed by the Secretary of the Army, in accordance with the provisions of Section 7 of the River and Harbor Act approved 8 August 1917. Conduct the work in such manner as to obstruct navigation as little as possible, and in case the Contractor's plant so obstructs the channel as to make difficult or endanger the passage of any vessels, promptly move said plant on the approach of any vessel to such an extent as may be necessary to afford a practicable passage. Upon completion of the work, promptly remove the plant, including ranges, buoys, piles, and other marks placed under the Contract in navigable waters or on shore.

1.5.6 Hydrographic Surveys

Hydrographic surveys are provided on the ehydro web site:
<https://www.arcgis.com/apps/opstdashboard/index.html#/4b8f2ba307684cf597617bf1b6d2f85d>
Survey numbers are indicated on the drawings.

1.6 LAYOUT OF WORK

Read this paragraph in conjunction with the Clause LAYOUT OF WORK (FAR 52.236-17).

1.6.1 Established Project Control

The Government has established monuments along the work site(s). Each monument is represented in the contract drawings by a Designation and in most cases, an alphanumeric Permanent Identifier (PID). Information concerning the location and elevations can be found within the Survey Notes on the plan sheets. The Survey Notes for each included field survey will cite a particular survey number, and specify the control used to conduct that survey. Additional information is included in the Survey Notes that must be duplicated if an attempt to recreate the subject survey is desired.

For navigation channels, the Government has established tidal calibration sites for projects constructed and maintained to a tidal datum. Each tidal calibration site is represented in the plan sheet drawings by Station ID, and is cited in the Survey Notes. Calibrate all projects constructed and maintained to a tidal datum to the Station ID specified in the plan sheet drawings. Information regarding the location and associated benchmarks of the referenced Station ID is included in the Monument Description section of these specifications.

See Section 00 31 21 CONTROL MONUMENT DESCRIPTIONS.

1.6.2 Layout

From the monuments, control data and elevations established by the Government, complete the layout of the work and obtain all measurements that may be required for the execution of the work to the location and limit marks prescribed in the specifications or on the contract drawings, subject to such modifications as the Contracting Officer may require to meet changed conditions or as a result of necessary modifications to the contract work.

1.6.3 Survey

Provide equipment, tools, material, labor for laying out any part of the work from the monuments, control data, and elevations established by the Government. Maintain and preserve all monuments, stakes and other marks established by the Contracting Officer until authorized to remove them, and if such marks or monuments are destroyed by the Contractor or through negligence, prior to their authorized removal, they may be replaced by the Contracting Officer, at their discretion, and the expense of replacement will be deducted from any amounts due or to become due the Contractor. The Contracting Officer may require that work be suspended at any time when location and limit marks established by the Contractor are not reasonably adequate to permit checking of the work.

1.7 DAMAGE TO WORK

The responsibility for damage to any part of the permanent work will be as set forth in Clause PERMITS AND RESPONSIBILITIES. However, if any part of the permanent work performed by the Contractor is damaged by flood, earthquake, hurricane, or tornado, which damage is not due to the failure

of the Contractor to take reasonable precautions or to exercise sound engineering and construction practices in the conduct of the work, make the repairs as ordered by the Contracting Officer and full compensation for such repairs will be made at the applicable Contract unit or job prices as fixed and established in the Contract. If, in the opinion of the Contracting Officer, there are no contract unit or job prices applicable to any part of such work, an equitable adjustment pursuant to Clause CHANGES will be made as full compensation for the repairs of that part of the permanent work for which there are no applicable contract unit or job prices. Except as herein provided, repair damage to all work (including temporary construction), utilities, materials, equipment and plant at the Contractor's expense, regardless of the cause of such damage.

1.8 EXISTING WORK

In addition to FAR 52.236-9, Protection of Existing Vegetation, Structures, Equipment, Utilities, and Improvements:

- a. Remove or alter existing work in such a manner as to prevent injury or damage to any portions of the existing work which remain.
- b. Repair or replace portions of existing work which have been altered during construction operations to match existing or adjoining work, as approved by the Contracting Officer. At the completion of operations, existing work must be in a condition equal to or better than that which existed before new work started.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

-- End of Section --

SOPAMYPTINST 5440.3H
9 May 2017

CHAPTER 5
DESTRUCTIVE WEATHER PLAN

500. DESTRUCTIVE WEATHER PLAN. Northeast Florida is subject to the following types of destructive weather: hurricanes, tropical storms, gales, tornadoes, thunderstorms, freezes, ice, and snowstorms. Due to the destructive nature of these phenomena, it is essential that NAVSTA MAYPORT afloat commands become familiar with the Destructive Weather information of this plan to reduce injury to personnel and damage to equipment. In addition to this plan, each command must develop and maintain a detailed Destructive Weather Plan tailored to their specific requirements.

501. READINESS FOR GETTING UNDERWAY

a. Ships must notify CTF 80 and SOPA MAYPORT by message at least 72 hours in advance of intentions to disable critical machinery for more than 96 hours.

b. Ships under CTF 80 will submit DMR to CTF 80 and INFO USFF NORFOLK VA, COMNAVSURFLANT, COMUSNAVSO, CNSS FOURTEEN, and SERMC if unable to support a 96 hour RFS.

(1) Submit DMRs at least 72 hours in advance. CTF 80 permission is required prior to disabling machinery that would restrict ship movement on short notice.

(2) CTF 80 will respond by record message traffic if CTF 80 non-concurs with DMR.

(3) Subject requests in support of repairs to a propulsion limiting casualty must be submitted as soon as practical in addition to required CASREPS.

(4) DMRs are NOT required if a ship can maintain a 96 hour RFS, or if a ship is in a CNO Depot-level availability.

c. Commanding Officers are reminded that, in accordance with Navy Regulations, they are responsible for the safety of their ship and crew. Maintenance availabilities do not obviate that requirement.

502. DESCRIPTION OF HAZARDOUS WEATHER PHENOMENA. A list of potentially destructive weather systems with associated definition follows:

a. Tropical Cyclone. A warm-core, non-frontal synoptic scale Low pressure system originating over tropical or sub-tropical waters with organized deep convection, i.e., thunderstorm activity, and definite cyclonic surface wind circulation.

b. Hazardous Winds. Sustained winds of 35 knots or greater.

SOPAMYPTINST 5440.3H
9 May 2017

- c. Destructive Winds. Sustained winds of 50 knots or greater.
- d. Tropical Depression. A tropical cyclone with associated maximum sustained wind of 33 knots or less.
- e. Tropical Storm. A tropical cyclone with associated maximum sustained wind 34 to 63 knots.
- f. Hurricane. A tropical cyclone with associated maximum sustained wind of 64 knots or more. Hurricane Classifications (sustained winds):
 - (1) Category 1 (64-82 knots)
 - (2) Category 2 (83-95 knots)
 - (3) Category 3 (96-113 knots)
 - (4) Category 4 (114-135 knots)
 - (5) Category 5 (>135)
- g. Subtropical Cyclone. A low pressure system that develops over subtropical waters that initially has a nontropical circulation but in which some elements of tropical cyclone cloud structure are present.
- h. Wind Hazardous to Small Craft. Sustained wind between 16 and 33 knots.
- i. Gale. Sustained wind (non-tropical) between 34 and 47 knots.
- j. Storm. Sustained wind (non-tropical) of 48 knots or more.
- k. Thunderstorm. Localized storm typically of short duration that potentially can produce wind gusts less than 50 knots and hail less than 3/4" in diameter.
- l. Severe Thunderstorm. Thunderstorm that potentially can produce wind gusts of 50 knots or more and/or hail 3/4" or greater in diameter.
- m. Tornado. Violent rotating column of air associated with thunderstorms that touches the ground with wind inside the column from 100 to 250 knots. Updrafts may reach 175 knots.
- n. Freeze. Air temperature below 32 degrees F/0 degrees C.

SOPAMYPTINST 5440.3H
9 May 2017

503. LOCAL HAZARDOUS WEATHER WARNINGS AND THUNDERSTORM
CONDITIONS OF READINESS

a. Promulgation. Fleet Weather Center Norfolk will issue weather advisories and warnings based on predicted conditions. Port Operations will disseminate weather and warnings to afloat commands via VHF Channel 12 and 74. Afloat units are required to guard Channels 12 and 74.

b. Gale/Storm Warnings. Gale/Storm Warnings will be issued whenever gale/storm conditions (not associated with tropical cyclones or thunderstorms) are expected in the vicinity of Mayport within 24 hours. The warning message/notice will provide estimated time of onset and cessation of conditions. Ships will review heavy weather bills, ensure mooring configuration is appropriate, and police decks and berth for possible missile hazards. Ships will notify SOPA of any factors which may affect ship or personnel safety due to the predicted conditions.

c. Thunderstorms Conditions of Readiness

(1) Thunderstorm Conditions of Readiness are defined and established in reference (r) to prepare commands for the arrival of destructive or hazardous weather and to minimize resultant damage.

(2) CO NAVSTA MAYPORT will set hazardous weather conditions associated with thunderstorm activity as described below:

(a) Thunderstorm/Severe Thunderstorm Condition II. Hazardous wind and associated phenomena are expected within 25nm within six hours. Associated lighting/thunder, torrential rain, severe hail downbursts, and/or sudden wind shifts are possible. Additionally, tornado formation is possible in the vicinity of severe thunderstorms. Take precautions that will permit establishment of an appropriate state of readiness on short notice, such as checking lines and striking down all missile hazards.

(b) Thunderstorm/Severe Thunderstorm Condition I. Hazardous wind and associated phenomena are occurring. Associated lighting/thunder, torrential rain, severe hail downbursts, and/or sudden wind shifts are possible. Additionally, tornado formation is possible in the vicinity of severe thunderstorms. Upon notification of Thunderstorm/Severe Thunderstorm I, refueling and ammunition handling operations will cease. For ammunition handling evolutions, ships will secure hatches, strike ammunition below, and have loaded trucks moved to a safe parking area until the condition is relaxed.

504. TROPICAL CYCLONE CONDITIONS OF READINESS/SORTIE CONDITIONS

a. Discussion. Per MOA between USFF and C4F, C4F has been assigned as Sortie Commander Mayport (CTG 883.2). In most cases, SOPA MAYPORT will also serve as CTG 883.2. However, in the event that SOPA MAYPORT and CTG 883.2 are not the same command, SOPA Mayport will assist the assigned sortie commander as necessary. Send all hurricane

SOPAMYPTINST 5440.3H
9 May 2017

related reports CTG 883.2, info SOPA MAYPORT. Additionally, CNSS 14 has been designated by USFF in the same MOA as the EA for SOPA MAYPORT. The SOPA MAYPORT EA will carry out the orders of the SOPA MAYPORT for Hurricane Readiness reporting, material condition assessment of all afloat units in Mayport and execute the actions required per Sortie Condition directed.

b. Sortie Commander Mayport has been given the authority to recommend sortie and designate a sortie task unit commander, CTU 883.2.1 for NAVSTA MAYPORT. The designated sortie task unit commander will be approved by CTF-80 and will be assigned as CTU 883.2.1. Additional SAG commanders will be assigned as CTU 883.2.2 and CTU 883.2.3 as required. SAG commanders and members will be assigned at a sortie coordination meeting hosted by the hurricane sortie commander. SAG units will be identified in the Hurricane Sortie OPGEN.

c. NAVSTA MAYPORT Emergency Sortie Task Group Chain of Command

- (1) CUSFFCTF 883
- (2) CTF 80 CTG 883.0
- (3) Sortie Commander Mayport CTG 883.2
- (4) Sortie Task Unit Commander CTU 883.2.1
- (5) SAG Commander(s) CTU 883.2.X

d. Tropical Cyclone COR

(1) Definitions

(a) COR 5. Onset of destructive winds possible within 96 hours.[COR 5 is set 01 June until 30 November for the Mayport area each year.]

(b) 120 Hour Checklist. Onset of destructive winds possible within 120 hours.

(c) 96 Hour Checklist. Onset of destructive winds possible within 96 hours.

(d) COR 4. Onset of destructive winds possible within 72 hours.

(e) COR 3. Onset of destructive winds possible within 48 hours.

(f) COR 2. Onset of destructive winds possible within 24 hours.

(g) COR 1. Onset of destructive winds possible within 12 hours.

SOPAMYPTINST 5440.3H
9 May 2017

(2) Ordering of Tropical Cyclone Conditions of Readiness

(a) CO NAVSTA MAYPORT orders Tropical Cyclone Conditions of Readiness based on the onset of destructive winds associated with a tropical cyclone for NAVSTA MAYPORT shore commands. CO NAVSTA MAYPORT will inform CTG 883.2/SOPA when ordering all CORs.

e. Sortie Conditions

(1) Discussion. Sortie conditions are established to support afloat unit requirements to depart well before onset of heavy weather, and will be based on forecast weather along the potential sortie track. Sortie conditions will be set by CTG 883.2, based upon the estimated onset of destructive winds in the Mayport area, storm track relationship to evasion routes, and Mayport weather conditions. Sortie conditions will be set in concert with, but independent of, NAVSTA MAYPORT COR. Anticipated sortie time will be promulgated by message when setting Sortie Conditions.

(2) Sortie CHARLIE. Prepare to sortie within 36 to 48 hours to avoid heavy weather. Anticipated sortie commencement time will be included in the message setting Sortie C. Depending on predicted storm track, Sortie C may be set well in advance of Condition IV for shore activities.

(3) Sortie BRAVO. Sortie expected within 24 hours to avoid heavy weather.

(4) Sortie ALPHA. Commence sortie to avoid heavy weather.

f. Climatology. While no tropical cyclone should be underestimated, a study of the Mayport basin reveals the greatest threat occurs from hurricanes approaching from the east-southeast octant for the months August through October. The following factors are major considerations for effects to ships in Mayport:

(1) The geographical position of Mayport severely limits evasion routes. Early sortie may be required to avoid being caught either crossing the “T” of the approaching cyclone and steaming in the dangerous semicircle or transiting along the coast (in the preferred semicircle) on a poor sea-keeping course due to inadequate sea room.

(2) Remaining securely moored in Mayport during a heavy weather event is not feasible due to the potentially disastrous effects of winds and storm surge. Storm surge occurs when wind and pressure effects of the cyclone produce a sea height greater than mean sea level. When this “bulge” of water preceding a cyclone interacts with bottom topography and the coastline and is superimposed upon tidal fluctuations, it can produce devastating flooding and huge waves along the coast.

g. Historical Data

SOPAMYPTINST 5440.3H
9 May 2017

(1) In 1964, Hurricane Dora, with 100 knot winds, struck the coast just north of St. Augustine on 09 September. A 6-8 foot storm surge was reported in Mayport. Fortunately, this coincided with an astronomical low tide, minimizing surge damage to the wharves and surrounding area. Maximum sustained winds of 65 knots with gusts to 80 knots were recorded on station Mayport.

(2) In 2004, Hurricane Frances made landfall as a category 2 storm near Stuart, FL (~200 nm south) on 05 September. Frances' CPA with Mayport was 105 nm. Gale force winds lasted 13 hours with 50 knots sustained winds lasting 3 hours. One ship reported a maximum gust of 75 knots in the basin. USCG reported a 2-3 foot surge above high tide. Of the 14 ships in Mayport basin, four lines were parted (3 from one ship).

(3) Again in 2004, just 18 days after Hurricane Frances, hurricane Jeanne also made landfall in Stuart, FL as a category 3 storm on 26 September. Jeanne's CPA with Mayport was 75 nm. Gale force winds lasted 6 hours with sustained 54 knots lasting 1 hour. One ship reported a maximum gust of 64 knots in the basin. USCG reported a 2.5 foot surge above high tide. Of the 13 ships in Mayport basin, none reported significant damage.

(4) In 2008, Tropical Storm Fay made its third Florida landfall near Flagler Beach 21 August, with sustained winds of 30 knots gusting to 40 knots. Extreme rainfall amounts of 27.65 inches were measured near Melbourne, Florida breaking a 50-year old record for a rainfall event, although amounts in excess of 5-10 inches were common across northern Florida. Storm surge from Fay was minimal, with storm surge of 3.06 ft and storm tide of 4.71 observed at the Mayport Bar Pilot.

(5) In 2016, Hurricane Matthew had CPA of 40 miles with Mayport on 7 October as a category 2 hurricane. At 1930 on 7 Oct, maximum sustained winds of 55 knots gusting to 63 knots, Sea State: 16.2 ft and heavy rains were observed on station Mayport. Storm surge was 6-6 substantial, 11 feet, at the top of the wharf; during high tide, low level flooding at Alpha/Delta wharves caused substantial damage to (1) COMUSNAVSO building and to 45 vehicles parked in the area.

(a) Afloat. (3) ships SORTIE with CTF 883.2. (2) vessels remained at BAE Heckscher Drive facility. (8) vessels remained in the Mayport basin. During the height of the storm, (5) lines were parted from (1) DDG. (2) Sea-Tractors were sent alongside to assist in maintaining moor. Due to sea state in the basin, (2) PC's evacuated crews to a co-located DDG.

(b) Air. (9) H-60R remained hangared in HGR-1552. (25) H-60R evacuated to NAS Jacksonville, HGR-1122

(c) Personnel. On 6 October, CNRSE issued an evacuation order; a total of 2886 personnel evacuated Mayport Navy housing; 68 residents remained. MEP plan was activated

SOPAMYPTINST 5440.3H
9 May 2017

with a total of 557 support personnel remaining on station; 458 afloat personnel and 99 shore personnel.

(d) MEF COOP. Mayport did not COOP. COMUSNAVSO initiated MEF COOP to NAS Jacksonville.

(e) Damage Assessment. Number of Buildings Inspected by PWD: 93; Number of Buildings With Damage: 82; Number of Buildings With Water Damage: 31; Number of Buildings With Rising Water Damage: 14; Number of Buildings With Occupants Displaced: 1; Number of Emergency Service Calls Issued: 45; Number of Service Calls Issued: 28.

(f) Total damage costs; Facilities: \$9,177,600 and Beach: \$3,500,000.

h. Sortie and Evasion

(1) Because of the storm surge potential and lack of any high blocking terrain to provide wind breaks, none of the inland waterways or Ports in the Jacksonville area are designated safe havens during sustained hurricane force winds. There may be some berth space available for Mayport-based ships in the JAXPORT area (Talleyrand wharves) and buoys are available northwest of Blount Island for Navy yard craft and small ships. However, evasion at sea or relocation to a safe haven is the recommended course of action for all seaworthy vessels when Mayport is directly threatened with sustained destructive force winds of 50 knots or greater. Ships with large sail areas, should evade at sea when threatened by greater than 35 knot sustained winds. Storm surge from nearby tropical storms is much more common than actual strikes for Mayport. Surge dissipating effects of inland wetlands and the intracoastal make alternate, upriver berthing locations generally safe in instances where winds are not a major concern. Hurricane berths will be coordinated by SOPA (ADMIN) MAYPORT with the JAX Port Authority. Additionally, barges, floating cranes and tugs will be moved to more secure areas designated by SOPA (ADMIN) MAYPORT during preparations for hazardous weather conditions.

(2) Fleet safety is paramount and must be a key factor in a timely decision to sortie. A major hurricane (Category 3-5) on a track towards Jacksonville would devastate the fleet if caught in port.

(3) Timeliness of a sortie decision is essential. Ideally, ships should be prepared and expect to get underway at about the same time that shore stations go to COR 3. However, the forecast storm track may dictate an earlier start to the sortie. The following time considerations are critical factors of a sortie decision:

(a) A Mayport sortie may require up to 24 hours if disabled ships must be moved dead-stick to Talleyrand wharves. Normal sorties can be completed in no more than 8 hours.

SOPAMYPTINST 5440.3H
9 May 2017

(b) Due to potential storm curvature and inherent forecast error, ships may need to transit nearly 200 NM to evade the storm (13 hour transit at 15 knots). This transit will be in less than optimal sea states due to rapid swell propagation in advance of tropical cyclones. Factoring in the above delays, sortie could take more than 36 hours from first ship's departure until the last ship reaches the rendezvous point.

i. Seasonal Tropical Cyclone Conditions of Readiness. Due to the geographical location of NAVSTA MAYPORT, COR 5 will be ordered and maintained during the Atlantic Hurricane Season which runs from 01 June to 30 November each year. This enhanced level of readiness includes additional training, special reporting and frequent reviews of command Hurricane Readiness. This condition shall be consistent with ongoing routine but will include additional emphasis on hurricane preparedness. Factors which need to receive special attention include levels of fuel, lube oil and feed water on board; amount of supplies and consumables; and condition and number of fenders, lines and other mooring tackle. During this condition:

(1) Ships will provide SERMC MAYPORT, SOPA MAYPORT/CTG 883.2, ISIC, NAVSTA MAYPORT, and RSO Mayport (CNSS 14) with the current status of their Hurricane Readiness beginning 15 May via the following message format:

FROM: USS EVERREADY
TO: SOUTHEAST RMC MAYPORT FL
INFO: CTG 883.2 (AS APPROPRIATE)
SOPA MAYPORT FL
(ISIC)
COMNAVSURFRON FOURTEEN
NAVSTA MAYPORT FL

CONFIDENTIAL //N03000//

SUBJ: HURRICANE READINESS STATUS

1. FUEL PERCENTAGE ONBOARD.
2. PROJECTED HURRICANE SORTIE UNIT ASSIGNMENT (one of the following):

UNIT ONE (greater than 85 percent fuel onboard and able to get underway with 50 percent propulsion power within four hours of notification by SOPA MAYPORT).

UNIT ONE ALPHA (same as unit one but less than 85 percent fuel onboard).

UNIT TWO (able to get underway with 50 percent propulsion power within 24 hours of notification by SOPA MAYPORT).

SOPAMYPTINST 5440.3H
9 May 2017

UNIT THREE (greater than 24 hours required to get underway with 50 percent propulsion power. Include number of hours required to achieve ship's electrical and propulsion power).

UNIT FOUR (smaller ships which can proceed under own power to hurricane berths or anchorages up river. ATF, MCM, PC and similar size ships will normally be assigned to this unit).

UNIT FIVE (Ships in dry dock or otherwise restricted in ability to move from current location).

3. Special circumstances affecting readiness to sortie; e.g., degradation of watertight integrity, degradation of ships boats or life rafts, critical personnel not available to sail.

- 4. A. ABILITY TO GENERATE OWN ELECTRIC POWER:
- B. ABILITY TO MAINTAIN OWN FIREMAIN:
- C. STATUS OF SHIP'S BOATS:
- D. STATUS OF SUPPLIES AND EQUIPMENT INSIDE THE BASIN:

(Note: After initial input, changes in Hurricane Readiness status will be reported as they occur.)

(2) Foreign Ships. Visiting foreign ships' Hurricane Readiness will be the responsibility of the host ships. Host ships will assist the visiting ship with the message described above and ensure SOPA direction is fully understood, including conditions of readiness and sortie instruction.

j. Tropical Cyclone COR 4 Destructive Winds Within 72 Hours. SOPA MAYPORT/CTG 883.2, will normally convene a meeting to brief all commands on the latest data regarding the threatening storm, review individual unit readiness, assign sortie units and Units Commanders, identify unique requirements (i.e., generators, cofferdams, cranes), and provide sortie commander Mayport intentions. Upon notification, ships will commence preparation for destructive weather/sortie in accordance with the following SOPA MAYPORT instructions as well as local ships' instructions.

(1) Imperative Actions. (Required of all ships regardless of sortie decision.)

(a) SOPA MAYPORT/CTG 883.2 will provide a sortie intentions message that confirms sortie intentions, provides movement plan (if deemed necessary) and assigns sortie units and task unit commanders based on criteria previously described.

(b) Ships moored, anchored, or nested take preparatory measures for placing main engines in commission and getting underway.

SOPAMYPTINST 5440.3H
9 May 2017

(c) Commence monitoring NAVSTA MAYPORT Port Operations advisories on VHF (Bridge-to-Bridge radio telephone) Channel 12.

(d) Confirm unit readiness status to SOPA MAYPORT/CTG 883.2, CNSS 14 and SERMC via telephone.

(e) Review hurricane and heavy weather bills. Remove all unnecessary material topside and on the adjacent berth to a secure stowage area.

(f) Brief crews on storm situation and possibility the ship may be required to sortie or move to a safer mooring. Be prepared to modify or suspend liberty or to execute the recall bill.

(g) Review the Mayport Emergency Sortie and dispersal OPTASK Communications message and be prepared to institute on short notice.

(h) Upon attaining COR 4, notify SOPA MAYPORT/CTG 883.2, CNSS 14 and ISIC.

(2) Provisional Actions. (Required based on decision to sortie or as otherwise directed by SOPA/CTG 883.2 when the sortie OPGEN is promulgated):

(a) Designated flagships prepare to embark Unit Commanders. Submit guard shifts and establish communications required to support the embarked commander. Conduct communication checks with other ships in Unit.

(b) Sortie Commanders assume tactical control of units assigned and assume reporting requirements.

(c) UNIT TWO and THREE ships, assisted by SERMC, commence restoring ships machinery as necessary. SERMC will coordinate this effort. Unit Commanding Officers will notify ISIC of changes and as progress is made.

(d) UNIT FOUR ships prepare to proceed up river to available berths or anchorages as assigned by SOPA MAYPORT and NAVSTA MAYPORT Port Operations Officer.

(e) UNIT FIVE ships work with NAVSTA MAYPORT Port Operations Officer and local contractors to acquire additional mooring tackle and other material necessary to provide greatest personnel safety and protection against damage.

k. Hurricane/Tropical Cyclone III (COR 3) (Destructive Winds Within 48 Hours). If SOPA MAYPORT/CTG 883.2 determines there is a significant risk of damage to ships in Mayport due to the approaching storm, execution of the planned sortie will normally be ordered in accordance with timelines to support movement.

(1) Imperative Actions. (Required of all ships regardless of sortie decision).

SOPAMYPTINST 5440.3H
9 May 2017

(a) Unit Commanders report any changes in the status of their ships to SOPA MAYPORT (negative reports required).

(b) Execute recall bill unless otherwise directed by SOPA MAYPORT/CTG 883.2.

(c) Secure access to ship by visitors, contractors and maintenance personnel.

(d) A DAP may be established ashore to assist NAVSTA MAYPORT with preparations in basin areas, assist in possible evacuation of dependents and respond to contingencies and emergencies which may arise. The DAP will consist of at least 40 personnel, including as a minimum one officer, one CPO, two PO2s or above and five PO3s or above. Unit Commanders will provide parties from assigned ships as directed by SOPA. Personnel assigned to the DAP will report to NAVSTA MAYPORT, Building 365 (Fire Station). DAP personnel will remain ashore under the control of NAVSTA MAYPORT until conditions are relaxed. Upon sortie, all vehicles assigned to afloat units will be turned over to the DAP.

(e) Upon attaining COR 3, notify SOPA/CTG 883.2, CNSS 14 and Sortie Commander.

(f) Once the last task unit ship is underway, Sortie Task Unit Commander will make a consolidated SORTIE ATTAINMENT report to CTG 883.2, include SOPA MAYPORT and NAVSTA MAYPORT as info addressees. Include the following for each ship:

1. Narrative (include how ship is riding)
2. Loss/Damage (including effect on readiness)
3. Logistics
 - a. Fuel (percent)
 - b. Feed Water (percent)
 - c. Fresh Water (percent)
4. Remarks (include number of unauthorized absentees)

1. Hurricane/Tropical Storm Condition I (COR 1) (Destructive Winds Within 12 Hours). All ships remaining inport Mayport shall take all possible precautions to safeguard personnel and Continue actions specified for COR 3. No additional actions for afloat units.

m. Onset of Destructive Winds/Weather. All ships remaining inport Mayport shall:

SOPAMYPTINST 5440.3H
9 May 2017

- (1) Take all possible precautions to safeguard personnel and material.
- (2) Report emergencies which may require outside assistance to SOPA Mayport.
- (3) Make attainment reports to Mayport Port Control via Bridge-to-Bridge channel 12 for relay to SOPA MAYPORT.

n. Sortie sequence of events. The following is a typical sequence of events for a sortie evolution. The sequence is subject to change based on the situation.

- (1) Sortie Condition CHARLIE ordered by SOPA MAYPORT/CTG 883.2.
- (2) Sortie meeting is held discussing potential sortie and issues.
- (3) CTG 883.2 designates Sortie Task Unit Commander (CTU 883.2.1).
- (4) CTG 883.2 issues TG promulgation with sortie units and members.
- (5) COR 4 is ordered by CO NAVSTA MAYPORT.
- (6) Sortie Condition BRAVO is ordered by CTG 883.2.
- (7) CTU 883.2.1 issues OPGEN.
- (8) CTG 883.2 issues Sortie Order message.
- (9) COR 3 is ordered by CO NAVSTA MAYPORT.
- (10) Sortie Condition ALPHA is ordered by CTG 883.2.
- (11) Sortie.
- (12) CTU 883.2.1 issues Sortie Completion message to (CTG 883.0), info CTF 883 and CTG 883.2.
- (13) COR 2 is ordered by CO NAVSTA MAYPORT.
- (14) COR 1 is ordered by CO NAVSTA MAYPORT.
- (15) CO NAVSTA MAYPORT reports Port safe to RTHP. Sets post storm COR as appropriate.
- (16) CTF 883 orders Return to Port.

SOPAMYPTINST 5440.3H
9 May 2017

(17) CTU 883.2.1 submits consolidated LOGREQ and returns to port. CTU 883.2.1 issues Sortie Complete message to CTG 883.0.

(18) Submit Sortie Lessons Learned to CTG 883.2.

o. Tropical Cyclone COR Relaxation. Although it may be apparent that the hazardous weather conditions have ceased in Mayport, ships will not relax conditions of readiness and allow people to leave the ship until directed by SOPA MAYPORT. Uncontrolled personnel movement immediately following a hurricane or tropical storm may interfere with emergency restoration efforts and damage or casualty control. SOPA MAYPORT/CTG 883.2 will relax conditions as soon as feasible.

505. HEAVY WEATHER PREPARATIONS FOR SHIPS REMAINING INPORT. SOPA MAYPORT will be responsible for all Fleet assets that remain in Mayport or the local area during destructive weather. In the event the decision is made to ride out the destructive weather in port or a ship is unable to get underway for storm avoidance, the following guidance should be followed per NSTM Chapter 582. Additional guidance will be provided by NAVSTA MAYPORT Port Operations, as required.

a. The key to increasing the security of a moor is rigging additional breast lines correctly to ensure load equalization. NSTM 582 discusses placement, types, mixing, lengths and angles to the berth in detail and should be reviewed prior to rigging for heavy weather.

b. In general, only bollards should be used. Whenever possible, heavy weather mooring lines should be spread out to fittings across the berth to maximize the horizontal component of line pull. In addition, limit the use of a berth fitting to a single ship whenever practical.

c. When forced to use cleats, they should be used for spring lines only.

d. Normally only one mooring line should be secured to a ship's bitt or to a berth fitting. It is permissible for opposing spring lines to be led to the same berth fitting because they cannot be loaded at the same time. Do not mix line types, i.e., wire, nylon, and/or polyester. Wire rope lines should be avoided if significant surge is expected.

e. Per NSTM 582, whether using aramid or nylon lines, a standard mooring configuration with doubled lines require an eye with a bight passed around the cleat or bollard. Two eyes around the bollard or cleat is not acceptable.

f. Safety of Personnel. When sustained gale force winds are forecasted, ships will be directed to remove brow stands and move brow from amidships to the fantail. Coordinate with NAVSTA MAYPORT Port Operations to arrange crane support. In addition to risk management, shifting brows to the fantail early will significantly shorten time required to sortie if directed.

SOPAMYPTINST 5440.3H
9 May 2017

g. Ships are responsible for removing or lashing down missile hazards such as tents, scaffolding, garbage cans, etc., from the berth and surrounding areas.

506. PERSONNEL STAYING BEHIND DURING SORTIES. Personnel left behind during a sortie must be assigned to CNSS 14. They will be tasked to assist with base destructive weather preps. A complete list of personnel left behind, including leave, active duty out-of-area TAD, and expected new arrivals must be provided to CNSS 14 prior to sortie. An accurate list will eliminate confusion as personnel return to the NAVSTA MAYPORT post-destructive weather.

507. PARKING DURING HURRICANES/TROPICAL STORMS AND SORTIES.

a. DO NOT PARK INSIDE THE BASIN. Cars parked inside the basin are a potential hazard and will be towed. Damage is not the responsibility of NAVSTA MAYPORT.

b. The Basin areas will flood if storm intensity reaches a strong Category 1 level, depending on tidal conditions and storm direction. Cars may become a hazard and end up in the basin.

c. Designated parking zone for ship sortie is in front of BLDG 365 (Fire & Emergency Station). None of these parking areas are deemed “storm safe” and afford a certain amount of risk.

d. CO NAVSTA MAYPORT recommends that POVs be left at home or in off base locations during sorties where the potential for flooding is less likely to occur.

508. HURRICANE EVACUATION CONDITIONS. Hurricane Conditions of Readiness should not be confused with Aircraft HUREVAC conditions.

509. READINESS. During destructive weather, no condition of battle readiness is prescribed, although optimum watertight integrity shall be maintained compatible with existing conditions. All ships shall be prepared and organized to man their emergency stations (including assembly of rescue and assistance parties) as quickly and completely as personnel and conditions of the ship permit. All ships are expected to follow heavy weather mooring configuration and coordinate with NAVSTA MAYPORT Port Operations as to the availability of heavy weather mooring line to be rigged in accordance with heavy weather mooring studies.

SOPAMYPTINST 5440.3H
9 May 2017

APPENDIX A
REFERENCES

- (a) NAVSTAMYPTINST 8020.1 (Control of Hazards of Electromagnetic Radiation to Ordnance Emissions Control Bill)
- (b) NAVSTAMYPTINST 5560.2 (Vehicle and Safety Regulations)
- (c) NAVSTAMYPTINST 3120.32 (Standard Organization and Regulations Manual (SORM) Of Naval Station (NAVSTA) Mayport)
- (d) OPNAVINST 5090.1 (Environmental Readiness Program)
- (e) SOPA(ADMIN)MYPTINST 5090.4 (Fuel/Oil And Hazardous Substance Release/Spill Response)
- (f) SOPA(ADMIN)MYPTINST 5090.2 (Oily Waste/Bilgewater Management)
- (g) COMNAVREG SE JAX 04 1742Z Apr 17 (Operation Guidance for Small Unmanned Aerial Systems on Navy Region Southeast Installations)
- (h) NAVSTAMYPTINST 3700.1 (Operation of Unmanned Aircraft within or Near Installation Property)
- (i) SECNAVINST 7042.7 (Guidelines for Use of Official Representation Funds)
- (j) NAVSTAMYPTINST 4100.2 (Consolidated Hazardous Material Reutilization and Inventory Management Program (CHRIMP) and Hazardous Material Control and Management (HMC&M))
- (k) NAVNETTOPSCOMINST 2066.1 (Navy Base Communication Manual)
- (l) NAVSTAMYPTINST 5530.1 (Naval Station Mayport Auxiliary Security Force)
- (m) DoDI 2000.16 (DoD Antiterrorism (AT) Program Implementation: DoD AT Standards)
- (n) OPNAVINST 5510.36 (Department of the Navy (DON) Information Security Program Instruction)
- (o) NAVSTAMYPTINST 11320.4 (First Coast Navy Fire and Emergency Services Fire Prevention Program)
- (p) NSTM CHAPTER 555 VOL 1 (Surface Ship Firefighting)

SECTION 01 11 00 APPENDIX A

SOPAMYPTINST 5440.3H
9 May 2017

- (q) NAVSEA S0570-AC-CCM-010/8010 (Industrial Ship Safety Manual for Fire Prevention and Response)
- (r) OPNAVINST 3140.24 (Warnings and Conditions of Readiness Concerning Hazardous or Destructive Weather Phenomena)
- (s) NAVSTAMYPTINST 7220.1 (Single Basic Allowance for Housing (SBAH))
- (t) NAVAIRINST 10340.3 (Maintaining Quality and Limiting Contamination of Aircraft Fuels)
- (u) NAVSUP PUBLICATION 485 (Navy Supply Procedures)
- (v) NAVSUP PUBLICATION 487 (Ship's Store Afloat)
- (w) NAVMEDCOMINST 5360.1 (Decedent Affairs Manual)
- (x) OPNAVINST 3120.32 (Standard Organization Regulations of the U.S. Navy)
- (y) SECNAVINST 1640.9 (Department of the Navy Corrections Manual)
- (z) SECNAVINST 5820.8 (Release of Official Information for Litigation Purposes and Testimony by Department of the Navy (DON) Personnel)
- (aa) JAGINST 5800.7 (Manual of the Judge Advocate General)
- (ab) COMNAVLEGSVCCOMINST 5800.1 (Naval Legal Service Command (NLSC) Manual)
- (ac) FACSFACJAXINST 3000.1 (Operations Manual)
- (ad) NAVSTAMYPTINST 8023.1 (Explosive Standards and Ammunition Handling Procedures at Naval Station (NAVSTA) Mayport)
- (ae) NAVSEA OP5, VOL 1 (Ammunition and Explosives Safety Ashore)
- (af) SOPA(ADMIN)MYPTINST 8027.1 (Information and Procedures Concerning Explosive Ordnance Disposal (EOD))
- (ag) OPNAVINST 3591.1 (Small Arms Training and Qualification)

APPENDIX B
SOPA MANUAL ACRONYMS

A	Ampere, Amp
AC	Alternating Current
ADA	Americans with Disabilities Act
AEL	Allowance Equipage List
AGL	Above Ground Level
AMHAZ	Ammunition and Hazardous Material
AMMO	Ammunition
APHIS	Animal and Plant Health Inspection Service
APL	Barracks Craft
ASC	Automotive Skills Center
ASF	Auxiliary Security Force
ATC	Air Traffic Control
ATCF	Air Traffic Control Facility
ATF	Fleet Tug
ATG	Afloat Training Group
AWT	Advanced Warfare Training
BAH	Basic Allowance for Housing
BESEP	Basic Electronic Systems Engineering Plans
BH	Bachelor Housing
BOS	Base Operating Services
BPA	Blanket Purchase Agreement
BQ	Bachelor Quarters
CAAC	Counseling and Assistance Center
CAC	Common Access Card
CACO	Casualty Assistance Calls Officer
CAFSU	Carrier and Field Service Unit
CAMS	Commercial Air Movements
CANTRAC	Catalog of Navy Training Courses
CARQUALS	Carrier Qualifications
CART	Command Assessment of Readiness and Training
CASREP	Casualty Report
CCU	Correctional Custody Unit
C/D	Class/Division
CDC	Child Development Center
CDO	Command Duty Officer
CDWS	Civil Defense Warning System
CHT	Collection, Holding and Transfer
CICA	Competition in Contracting Act
CMC	Command Master Chief
CMIO	Communications Security Material Systems Issuing Office

CMS	Communications Security Material System
CNIC	Commander, Navy Installations Command
CNO	Chief of Naval Operations
CNRSE	Commander, Navy Region Southeast
CNSS	Commander, Naval Surface Squadron
CO	Commanding Officer
COD	Carrier Onboard Delivery
COG	Chief of the Guard
COMCARSTRKGRU	Commander, Carrier Strike Group
COMHSMWINGLANT	Commander, Helicopter Maritime Strike Wing, Atlantic Fleet
COMNAVIDFOR	Commander, Navy Information Dominance Forces
COMNAVAIRLANT	Commander, Naval Air Force, U.S. Atlantic Fleet
COMNAVSURFRON	Commander, Naval Surface Squadron
COMRATS	Commuted Rations
COMSEC	Communications Security
COMSUBLANT	Commander, Submarine Force, U.S. Atlantic Fleet
COMTACSUPWING	Commander, Tactical Support Wing
CONUS	Continental United States
COMUSNAVSO	Commander, U.S. Naval Forces Southern Command
COR	Condition of Readiness
CPO	Chief Petty Officer
CREDO	Chaplains' Religious Enrichment and Development Operation
CRT	Cathode Ray Tube
CS	Culinary Specialist
CSCS	Center for Surface Combat Systems
CSMTT	Combat Systems Mobile Team Trainer
CTF	Commander, Task Force
CTG	Commander, Task Group
CTU	Commander, Task Unit
CVIS	Commercial Vehicle Inspection Station
CW	Continuous Wave
DAP	Disaster Assistance Party
DAPA	Drug and Alcohol Programs Advisor
DCS	Defense Courier Service
DDG	Guided Missile Destroyer
DEFCON	Defense Condition
DESRON	Destroyer Squadron
DHS	Department of Homeland Security
DISP	Directed Import Security Plan
DLA	Defense Logistics Agency
DLR	Depot Level Repairables
DMAHC	Defense Mapping Agency Hydrographic Center
DMR	Disabled Machinery Requests

SECTION 01 11 00 APPENDIX A

SOPAMYPTINST 5440.3H

9 May 2017

DOD	Department of Defense
DON	Department of the Navy
DRMO	Defense Reutilization and Marketing Office
DSO	Defense Service Office
DV	Domestic Violence
EA	Executive Agent
ECC	Emergency Communication Center
ECP	Entry Control Point
ELMR	Enterprise Land Mobile Radio System
EMCON	Emission Control
EMS	Emergency Medical Services
eNTRS	Enterprise Navy Training Reservation System
EOC	Emergency Operating Center
EOD	Explosive Ordnance Disposal
EPA	Environmental Protection Agency
ESO	Explosive Safety Officer
ESQD	Explosive Safety Quantity Distance
ETA	Estimated Time of Arrival
FACSFAC	Fleet Area Control and Surveillance Facility
FEP	Final Evaluation Period
FFSC	Fleet and Family Support Center
FISC	Fleet Industrial Supply Center
FLC	Fleet Logistics Center
FM	Frequency Modulation
FOD	Foreign Object Debris
FOSIC	Fleet Ocean Surveillance Information Center
FPCON	Force Protection Condition
FSC	Fleet Support Center
FSG	Foreign Source Garbage
FTC	Fleet Training Center
FTCA	Federal Tort Claims Act
FWO	Fleet Watch Officer
GBL	Government Bill of Lading
GLS	Global Logistics Support
GMT	General Military Training
GPETE	General Purpose Electronics Test Equipment
GVW	Gross Vehicle Weight
H&ME	Hull, Mechanical and Electrical
HAZMAT	Hazardous Material
HAZMIN	Hazardous Material Minimization
HERO	Hazards of Electromagnetic Radiation to Ordnance
HF	High Frequency
HW	Hazardous Waste
HUREVAC	Aircraft Hurricane Evacuation

HWSF	Hazardous Waste Storage Facility
ID	Identification
IDC	Independent Duty Corpsman
IFF	Identification Friend or Foe
INSURV	The Board of Inspection and Survey
ISE	Independent Steaming Exercise
ISIC	Immediate Superior in Command
IT	Information Technology
ITT	Information, Tickets and Travel
JAGMAN	Manual of the Judge Advocate General
JAX	Jacksonville
JSO	Jacksonville Sheriff's Office
JTA	Jacksonville Transportation Authority
KMI	Key Management Infrastructure
LCS	Littoral Combat Ship
LCSRON	Littoral Combat Ship Squadron
LHD	Landing Helicopter Dock
LOGREQ	Logistics Requirement
LPD	Landing Platform Dock
LSC	Logistics Support Center
LSD	Dock Landing Ship
LTF	Littoral Combat Ship Training Facility
MAC	Military Airlift Command
MCM	Mine Countermeasures
MDAS	Material Documented as Safe
MEDEVAC	Medical Evacuation
MET DET	Meteorology Detachment
MHz	Megahertz
MLW	Mean Level Water
MOA	Memorandum of Agreement
MOE	Millioersted
MOPS	Mothers of Pre-Schoolers
MPPEH	Material Potentially Presenting an Explosive Hazard
MPO	Military Post Office
MSC	Military Sealift Command
MTAC	Multiple Threat Alert Center
MWD	Military Working Dog
MWR	Morale, Welfare and Recreation
NALO	Naval Air Logistics Office
NAS	Naval Air Station
NAVCOMT	Navy Comptroller
NAVELEXCEN	Naval Electronic Systems Engineering Center
NAVFAC	Naval Facilities Engineering Command
NAVHOSPJAX	Naval Hospital Jacksonville

SECTION 01 11 00 APPENDIX A

SOPAMYPTINST 5440.3H

9 May 2017

NAVPTO	Navy Passenger Transportation Office
NAVSTA	Naval Station
NAWAS	National Warning System
NBHC	Navy Branch Health Clinic
NCISRA	Naval Criminal Investigative Service Resident Agency
NCISFO	Naval Criminal Investigative Service Field Office
NCO	Navy Campus Office
NCPACE	Program for Afloat College Education
NCTAMS	Naval Computer and Telecommunications Area Master Station
NCTS	Naval Computer and Telecommunications Station
NETC	Naval Education Training Command
NEXCOM	Navy Exchange Service Command
NFPA	National Fire Protection Association
NJP	Non-Judicial Punishment
NKO	Navy Knowledge Online
NLMOD	Naval Atlantic Meteorology and Oceanography Detachment
NLMOF	Naval Atlantic Meteorology and Oceanography Facility
NLSO	Navy Legal Service Office
NMCLANT	Navy Munitions Command
NORAD	North American Aerospace Defense Command
NOSC	Navy On-Scene Coordinator
NSTM	Naval Ship's Technical Manual
OB/GYN	Obstetrics/Gynecology
OC	Oleoresin Capsicum
OCONUS	Outside the Contiguous United States
OFMLS	Operational Forces Medical Liaison Fleet Medical Liaison Services
OIC	Office in Charge
OMC	Operational Ministry Center
OOD	Officer of the Deck
OPAREA	Operational Area
OPGEN	Operational General Matters
OPORD	Operation Order
OPREP	Operational Report
OPTASK	Operational Task
OTC	Over-the-Counter
OWTP	Oily Waste Treatment Plant
OWWO	Oily Waste/Waste Oil
PAO	Public Affairs Officer
PC	Patrol Craft
PCM	Primary Care Manager
PCS	Permanent Change of Station
PERSUPPDET	Personnel Support Detachment
PHA	Periodic Health Assessment

SOPAMYPTINST 5440.3H
9 May 2017

PLAD	Plain Language Address Directory
PNOK	Primary Next of Kin
PO	Petty Officer
POD	Permissible Operating Distance
POV	Privately Owned Vehicle
PPE	Personal Protective Equipment
PPO	Personal Property Office
PPQ	Plant Protection and Quarantine Program
PPR	Pre-Planned Response
PRT	Physical Readiness Test
PSD	Personnel Support Detachment
PWD	Public Works Department
QRP	Qualified Recycling Program
RDD	Required Delivery Date
REGSUPGRU	Regional Support Group
RFMSS	Range Facility Management Support System
RFS	Ready-for-Sea
RLSO	Region Legal Service Office
RMC	Religious Ministry Center
RPP	Rental Partnership Program
RSO	Regimental Supply Officer
RTHP	Return to Home Port
RTV	Recreation Ticket Vehicle
SAAMS	Special Assigned Airlift Missions
SAG	Surface Action Group
SAR	Search and Rescue
SATO	Scheduled Airlines Ticket Office
SDO	Staff Duty Officer
SERMC	South East Regional Maintenance Center
SFRM	Surface Force Readiness Manual
SIGSEC	Signals Security
SIPR	Secret Internet Protocol Router Network
SITREP	Situational Report
SJA	Staff (or Station) Judge Advocate
SMDR	Senior Medical Department Representative
SOPA	Senior Officer Present Afloat
SORM	Standard Organization and Regulations Manual
SSAC	Ships Store Afloat Catalog
STJ	Buoy St. Johns
SUAS	Small Unmanned Aerial Systems
SUPSHIP	Supervisor of Shipbuilding, Conversion and Repair
SURFRON	Naval Surface Squadron
TACON	Tactical Control
TAD	Temporary Additional Duty

TCMD	Transportation Control and Movement Documents
TG	Task Group
TLO	Training Liaison Officer
TPU	Transient Personnel Unit
TSD	Training Support Department
TSTA	Tailored Ship Training Availability
TYCOM	Type Commander
UCMJ	Uniform Code of Military Justice
UH	Unaccompanied Housing
UHM	Used Hazardous Material
UNODIR	Unless Otherwise Directed
UNREP	Underway Replenishment
USCG	United States Coast Guard
USDA	United States Department of Agriculture
USFF	U.S. Fleet Forces Command
USMCR	U.S. Marine Corps Reserve
USNR	U.S. Navy Reserve
V	Volt
VHF	Very High Frequency
VIP	Very Important Person
VQ	Visitors Quarters
VRT	Voyage Repair Team
WPCS	Wireless Pier Connectivity System
WSOC	Waterfront Security Operation Center
XO	Executive Officer
YC	Open Lighter
YON	Fuel Oil Barge
YD	Floating Crane
YRBM	Repair, Berthing and Messing Barge

SOPAMYPTINST 5440.3H
9 May 2017

APPENDIX C
SOPA MANUAL PARKING



SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 20 00

PRICE AND PAYMENT PROCEDURES

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 CONTRACT COST BREAKDOWN
- 1.4 SINGLE JOB PAYMENT ITEMS
 - 1.4.1 Mobilization and Demobilization (Line Item 0001)
 - 1.4.1.1 Payment
 - 1.4.1.2 Unit of Measure
 - 1.4.2 Environmental and Water Quality Monitoring (Line Item 0007)
 - 1.4.2.1 Payment
 - 1.4.2.2 Unit of Measure
 - 1.4.3 Mobilization and Demobilization for Sea Turtle Trawling (Hopper Dredge Only, Line Item 0008)
 - 1.4.3.1 Payment
 - 1.4.3.2 Unit of Measure
- 1.5 UNIT PRICE PAYMENT ITEMS
 - 1.5.1 Excavation, Unclassified (Line Items 0002, 0003, 0004, 0005, and 0006)
 - 1.5.1.1 Payment
 - 1.5.1.2 Measurement
 - 1.5.1.3 Unit of Measure
 - 1.5.2 Sea Turtle Trawling (Hopper Dredge Only, Line Item 0009)
 - 1.5.2.1 Payment
 - 1.5.2.2 Measurement
 - 1.5.2.3 Unit of Measure
 - 1.5.3 Standby Time (Line Item 0010)
 - 1.5.3.1 Payment
 - 1.5.3.2 Measurement
 - 1.5.3.3 Unit of Measure

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

- 3.1 PAYMENT PROCEDURES
- 3.2 Requesting Progress Payment
 - 3.2.1 Options and Modification CLINS

-- End of Section Table of Contents --

SECTION 01 20 00

PRICE AND PAYMENT PROCEDURES

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 1110-2-1003

(2013) Hydrographic Surveying

1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES. Submit the following preconstruction submittal items no later than 30 calendar days after Notice to Proceed:

SD-01 Preconstruction Submittals

Cost Breakdown; G, R0

SD-07 Certificates

Request for Progress Payment; G, R0

1.3 CONTRACT COST BREAKDOWN

The Contractor must furnish within 30 days after the date of Notice to Proceed, and prior to the submission of its first partial payment estimate, a breakdown of its single job pay item or items which will be reviewed by the Contracting Officer as to propriety of distribution of the total cost to the various accounts. Any unbalanced items as between early and late payment items or other discrepancies will be revised by the Contracting Officer to agree with a reasonable cost of the work included in the various items. This Contract cost breakdown will then be utilized as the basis for progress payments to the Contractor.

1.4 SINGLE JOB PAYMENT ITEMS

Payment items for the work of this Contract for which Contract job payments will be made are listed in the CLIN SCHEDULE and described below. All costs for items of work, which are not specifically mentioned to be included in a particular job or unit price payment item, are included in the listed job item most closely associated with the work involved. The job price and payment made for each item listed constitutes full compensation for furnishing all plant, labor, materials, and equipment, and performing any associated Contractor quality control,

U.S. Naval Station Mayport
Duval County, Florida

environmental protection, meeting safety requirements, tests and reports, and for performing all work required for which separate payment is not otherwise provided.

1.4.1 Mobilization and Demobilization (Line Item 0001)

1.4.1.1 Payment

Payment will be made for costs associated with mobilization and demobilization of the Contractor's plant and associated equipment, and establishment of initial project management and coordination, as defined in Contract Clause PAYMENT FOR MOBILIZATION AND DEMOBILIZATION.

1.4.1.2 Unit of Measure

Unit of measure: Job.

1.4.2 Environmental and Water Quality Monitoring (Line Item 0007)

1.4.2.1 Payment

Payment will be made for costs associated with or incidental to environmental species monitoring and obtaining, analyzing and reporting the results of monitoring for turbidity. See Section 01 57 20.00 35 ENVIRONMENTAL PROTECTION and Section 01 57 25.00 35 TURBIDITY AND DISPOSAL MONITORING.

1.4.2.2 Unit of Measure

Unit of measure: Job.

1.4.3 Mobilization and Demobilization for Sea Turtle Trawling (Hopper Dredge Only, Line Item 0008)

1.4.3.1 Payment

Payment will be made for costs associated with and incidental to mobilization and demobilization for sea turtle trawling and relocation (Hopper Dredge Only). If performance of this line item is not required or implemented per the NOTES in the CLIN SCHEDULE, no payment will be made to the Contractor for this line item. See Section 01 57 20.00 35 ENVIRONMENTAL PROTECTION.

1.4.3.2 Unit of Measure

Unit of measure: Job.

1.5 UNIT PRICE PAYMENT ITEMS

Payment items for the work of this Contract on which the Contract unit price payments will be made are listed in the CLIN SCHEDULE and described below. The unit price and payment made for each item listed constitutes full compensation for furnishing all plant, labor, materials, and equipment, and performing any associated Contractor quality control, environmental protection, meeting safety requirements, tests and reports, and for performing all work required for each of the unit price items.

U.S. Naval Station Mayport
Duval County, Florida

1.5.1 Excavation, Unclassified (Line Items 0002, 0003, 0004, 0005, and 0006)

1.5.1.1 Payment

- a. Payment will be made for costs associated with and incidental to excavation, transportation, and disposal of materials; providing and maintaining access to the work site(s) and disposal area(s); noise control; and debris removal. See Section 35 20 23.00 35 DREDGING.
- b. Insofar as consistent with the paragraph CONTINUITY OF WORK of Section 35 20 23.00 35 DREDGING, monthly partial payments will be based on approximate quantities determined by soundings or sweepings performed by the Contractor behind the dredge. The term "area designated by the Contracting Officer" is also defined as "acceptance section" in paragraph CONTINUITY OF WORK of Section 35 20 23.00 35 DREDGING.
- c. Hydrographic surveys for final measurement and payment and for acceptance will be made by the Government. All surveys will be performed in accordance with USACE EM 1110-2-1003.

1.5.1.2 Measurement

- a. The maps and/or drawings already prepared (clause CONTRACT DRAWINGS AND SPECIFICATIONS) are believed to represent accurately average existing conditions, but the depths indicated thereon may be verified and corrected by soundings taken before dredging. Once quantities within a specified reach or section have been determined through the use of in-place measurement, they will not be reopened except on evidence of collusion, fraud, or obvious error.
- b. The total amount of material removed, and to be paid for under the contract, will be measured by the cubic yard in place. The volume computed will be between the bottom surface indicated by soundings taken within 3 weeks before dredging and the bottom surface indicated by the soundings taken within 2 weeks after the work specified in each acceptance section has been completed. Give 10 days advance notice, in writing, to the Contracting Officer of the need for a pre-dredging survey or after-dredging survey for final acceptance for each acceptance section. The quantity will include the volume within the limits of the side slopes described in subparagraph SIDE SLOPES of paragraph REQUIRED DEPTH, ALLOWABLE OVERDEPTH, AND SIDE SLOPES of Section 35 20 23.00 35 DREDGING, less any deductions that may be required for misplaced material described in subparagraph MISPLACED MATERIALS of paragraph DISPOSAL OF EXCAVATED MATERIAL of Section 35 20 23.00 35 DREDGING.

1.5.1.3 Unit of Measure

Unit of measure: Cubic Yard.

1.5.2 Sea Turtle Trawling (Hopper Dredge Only, Line Item 0009)

1.5.2.1 Payment

Payment will be made for costs associated with and incidental to the Sea Turtle Trawling. This price includes all costs for furnishing labor, equipment, fuel, oil, materials, and supplies required for trawling to clear and/or relocate sea turtles from in the vicinity of the dredge, and

U.S. Naval Station Mayport
Duval County, Florida

the cost of reporting as specified. See Section 01 57 20.00 35
ENVIRONMENTAL PROTECTION.

1.5.2.2 Measurement

Sea Turtle Trawling will be measured for payment based upon the duration of performance on a daily basis. A day of trawling is defined as 24 hours of continuous trawling. Conduct trawling as described in Section 01 57 20.00 35 ENVIRONMENTAL PROTECTION.

1.5.2.3 Unit of Measure

Unit of measure: Day.

1.5.3 Standby Time (Line Item 0010)

1.5.3.1 Payment

Payment will be made for costs associated with or incidental to standby of dredge plant only upon implementation of this line item. Standby time is defined as when, at the direction of the Contracting Officer, the dredge is put out of operation for reasons outside the requirements in the contract.

1.5.3.2 Measurement

Standby time, when exercised by the Contracting Officer, will be measured for payment by each hour, or part thereof, that the plant is not in operation.

1.5.3.3 Unit of Measure

Unit of measure: Hour.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.1 PAYMENT PROCEDURES

Upon receiving initial Resident Management System import file, go to PAY ACTIVITIES and establish a link between bid breakdown schedule of values of PAY ACTIVITIES to contract CLINs using SCHEDULE ACTIVITIES data entry page.

3.2 Requesting Progress Payment

For progress payments, ensure ACTIVITY SCHEDULE, FEATURE SCHEDULE, submittal register, and punchlists are all up to date. Use PROGRESS PAYMENTS to REQUEST ACTIVITY for both ACTIVITY EARNINGS data entry page and OTHER EARNING. Provide hard copies of supporting invoices and quantity measurements to support all requested earnings. Ensure that sums of payment activities do not exceed contract award CLIN funding amounts, or unbalanced CLINs error will prevent processing the payment. Submit a [request for progress payment](#) monthly in accordance with Clause 52.232-5 PAYMENTS UNDER FIXED-PRICE CONSTRUCTION CONTRACTS. In addition to the items listed in Clause 52.232-5(b)(1), submit the following in the request for progress payment:

U.S. Naval Station Mayport
Duval County, Florida

- a. Approved construction schedule.
- b. Up-to-date payroll records.
- c. Current submittal register.
- d. Professional Surveyor and Mapper (PSM) or Registered Land Surveyor (RLS) report and supporting data.
- e. Approved submittals on installed property.
- f. QA/QC deficiency tracking list.
- g. Environmental reports.
- h. Progress photographs.
- i. Working/current as-built drawings.

Once the complete pay application package is received to include the Contractor's signed ENG 93, the 14 calendar days will commence.

3.2.1 Options and Modification CLINS

When additional work is added by modification, existing CLINs funding amounts must be updated, or new CLINs for modification will be created. If contract has option CLINs not yet awarded, option CLINs will appear as zero dollar CLINs until option is awarded by modification. No payment may be requested for Options or Modification CLINs until contract modification has been funded and signed.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 30 00

ADMINISTRATIVE REQUIREMENTS

PART 1 GENERAL

- 1.1 SUMMARY
- 1.2 REFERENCES
- 1.3 SUBMITTALS
- 1.4 PROJECT COORDINATION
 - 1.4.1 Resident Management System (RMS)
 - 1.4.2 Coordination with Other Contracts
 - 1.4.3 Access by Others
 - 1.4.4 Inspections by Others
- 1.5 SUPERVISION
 - 1.5.1 Superintendent Qualifications
 - 1.5.2 Minimum Communication Requirements
 - 1.5.3 Duties
 - 1.5.4 Non-Compliance Actions
- 1.6 PRECONSTRUCTION CONFERENCE
 - 1.6.1 Preconstruction Conference Minutes and Phone
- 1.7 PARTNERING
 - 1.7.1 Team-Led (Informal) Partnering
- 1.8 NOTICE TO PROCEED
- 1.9 SIGNATURE AUTHORITY
- 1.10 DRUG-FREE WORK PLACE RECORD
- 1.11 MINIMUM INSURANCE REQUIREMENTS

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

-- End of Section Table of Contents --

SECTION 01 30 00

ADMINISTRATIVE REQUIREMENTS

PART 1 GENERAL

1.1 SUMMARY

Manage project and coordinate activities of own employees, subcontractors, suppliers and offsite fabricators. Use computers, E-mail, and Internet resources for administrative work. Notify Contracting Officer of important meetings, schedule events and activities. Furnish labor and materials and equipment required to plan and execute project management functions and coordination. Coordinate activities and manage resources to construct project conforming to contract, on time and within budget. Related Sections are 01 32 01.10 35 PROJECT SCHEDULE, and 01 45 05 DREDGING/BEACH FILL PLACEMENT - QUALITY CONTROL.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2024) Safety and Occupational Health
(SOH) Requirements

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Transmit submittal items in accordance with Section 01 33 00 SUBMITTAL PROCEDURES.

SD-07 Certificates

Signature Authority

Drug-Free Work Place Record

Insurance Requirements

1.4 PROJECT COORDINATION

1.4.1 Resident Management System (RMS)

Use Corps of Engineers' Resident Management System for construction information management. See Section 01 45 00.15 10 RESIDENT MANAGEMENT SYSTEM CONTRACTOR MODE (RMS CM).

1.4.2 Coordination with Other Contracts

Active maintenance dredging contracts in adjacent areas are indicated in Section 01 11 00 SUMMARY OF WORK.

1.4.3 Access by Others

Maintain access to power poles and other items owned or operated by utility companies. Coordinate with utility companies as required for line marking, hookups, and relocations as needed to perform work. Incoming electrical utilities become the responsibility of Contractor at transformers. Coordinate emergency incident response with local law enforcement and fire rescue authorities. Notify Contracting Officer in writing of coordination problems encountered affecting work.

1.4.4 Inspections by Others

During construction period, inspection or maintenance work will be performed by others within the construction limits. The Contractor is responsible for neatly mowing all grass and vegetation (not to include established trees or other protected vegetation designated as wetland areas) within the construction limits for the duration of the Contract. Areas not accessible to mowers must be edged or trimmed neatly. Mowing or trimming must be performed every three weeks or when vegetation exceeds twelve inches in height.

1.5 SUPERVISION

1.5.1 Superintendent Qualifications

Provide project superintendent with a minimum of 5 years experience in construction with at least 3 of those years as a superintendent on projects similar in size and complexity. The individual must be familiar with the requirements of EM 385-1-1 and have experience in the areas of hazard identification and safety compliance. The individual must be capable of interpreting a critical path schedule and construction drawings. The qualification requirements for the alternate superintendent are the same as for the project superintendent.

1.5.2 Minimum Communication Requirements

Have at least one qualified superintendent, or competent alternate, capable of reading, writing, and conversing fluently in the English language, on the job-site at all times during the performance of Contract work. In addition, if a Quality Control (QC) representative is required on the contract, then that individual must also have fluent English communication skills.

1.5.3 Duties

The project superintendent is primarily responsible for managing subcontractors and coordinating day-to-day production and schedule adherence on the project. The superintendent is required to attend Red Zone meetings, partnering meetings, and quality control meetings. The superintendent or qualified alternative must be on-site at all times during the performance of this contract until the work is completed and accepted.

1.5.4 Non-Compliance Actions

The Project Superintendent is subject to removal by the Contracting Officer for non-compliance with requirements specified in the contract and for failure to manage the project to ensure timely completion. Furthermore, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No part of the time lost due to such stop orders is acceptable as the subject of claim for extension of time for excess costs or damages by the Contractor.

1.6 PRECONSTRUCTION CONFERENCE

Contracting Officer will conduct a Preconstruction Conference for this project in accordance with Clause PRECONSTRUCTION CONFERENCE. Preconstruction Conference will be held within 30 to 60 calendar days after Notice to Proceed (NTP). Contracting Officer will notify Contractor of time, place, and agenda. At a minimum, Contractor attendees must include the Project Manager, Superintendent, Site Safety and Health Officer (SSHO), Quality Control Manager and key subcontractors and suppliers. Present and discuss basic approach to scheduling the construction work and any required phasing. Contracting Officer will discuss contract ground rules and general issues including:

- Lines of Contracting Officer authority
- Lines of Contractor authority
- Contract Clauses
- Special Contract Requirements
- Contract Administration
- Progress Payment
- Correspondence Procedures
- Project Schedule
- Submittal Register
- Labor requirements
- General Site Safety
- Permit Requirements

1.6.1 Preconstruction Conference Minutes and Phone

The Contracting Officer will take detailed minutes of Preconstruction Conference discussions and may record the meeting. Copies of typed minutes will be provided to the Contractor to review for accuracy, sign and return. Signed minutes become part of the contract file. Recordings, if used, will be made available for Contractor to review or copy.

1.7 PARTNERING

To most effectively accomplish this Contract, the Contractor and Government must form a cohesive partnership with the common goal of drawing on the strength of each organization in an effort to achieve a successful project without safety mishaps, conforming to the Contract, within budget and on schedule. The partnering team must consist of personnel from both the Government and Contractor including project level and corporate level leadership positions. Key Personnel from the supported command, end user, Contractor, key subcontractors and the Designer of Record are required to participate in the Partnering process.

1.7.1 Team-Led (Informal) Partnering

- a. The Contracting Officer will coordinate the initial Team-Led (Informal) Partnering Session with key personnel of the project team, including Contractor and Government personnel. The Partnering Session will be co-led by the Government Construction Manager and Contractor's Project Manager.
- b. The Initial Team-led Partnering session may be held concurrently with the Pre-Construction meeting. Partnering sessions will be held at a location mutually agreed to by the Contracting Officer and the Contractor, typically at a conference room on-base or at the Contractor's temporary trailer.
- c. The Initial Team-Led Partnering Session will be conducted and facilitated using electronic media (a video and accompanying forms) provided by Contracting Officer.
- d. The Partners will determine the frequency of the follow-on sessions.
- e. Participants will bear their own costs for meals, lodging and transportation associated with Partnering.

1.8 NOTICE TO PROCEED

Notice to proceed (NTP) will be issued 15 to 30 calendar days after award or after all required bonds are in place, whichever is later.

1.9 SIGNATURE AUTHORITY

Submit a power of attorney or a notarized letter of authority identifying local representatives authorized to sign contract documents.

1.10 DRUG-FREE WORK PLACE RECORD

Submit documentation of compliance with Clause DRUG-FREE WORKPLACE (see subparagraphs (b)(1) through (b)(7)).

1.11 MINIMUM INSURANCE REQUIREMENTS

Provide the minimum insurance coverage required by FAR 28.307-2 Liability during the entire period of performance under this contract. Provide other insurance coverage as required by state law.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 32 01.10 35

PROJECT SCHEDULE (Alternate)

PART 1 GENERAL

- 1.1 SUBMITTALS
- 1.2 PROJECT SCHEDULER QUALIFICATIONS
- 1.3 CONSTRUCTION SCHEDULES
 - 1.3.1 Initial Project Schedules
 - 1.3.2 Bar Chart with S Curve
 - 1.3.3 Project Milestones
 - 1.3.3.1 Project Start Date Milestone and Constraint
 - 1.3.3.2 End Project Finish Milestone and Constraint
- 1.4 PERIODIC SCHEDULE UPDATE
 - 1.4.1 Narrative Report
 - 1.4.2 Periodic Progress Meetings

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

-- End of Section Table of Contents --

SECTION 01 32 01.10 35

PROJECT SCHEDULE (Alternate)

PART 1 GENERAL

1.1 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES.

SD-01 Preconstruction Submittals

Project Scheduler Qualifications; G, R0

Initial Project Schedules; G, R0

SD-07 Certificates

Periodic Schedule Update; G, R0

1.2 PROJECT SCHEDULER QUALIFICATIONS

Designate an authorized representative to be responsible for the preparation of the schedule and all required updating and production of reports. The authorized representative will have a minimum of 2-years experience scheduling construction projects similar in size and nature to this project with scheduling software that meets the requirements of this specification. Representative will have a comprehensive knowledge of CPM scheduling principles and application. The authorized representative may also serve as the Project Manager or other staff assigned to the project.

1.3 CONSTRUCTION SCHEDULES

Construction schedules consist of Project Schedules or Periodic Schedule Updates, Bar Charts, and S Curve. Use work breakdown structure of CLINs, Features and Pay Activities. CLINs are Contract Line Item Numbers from bid schedule. Features are quality control grouping of like Pay Activities requiring similar quality control. Pay Activities are scheduled work activities or events. Most Pay Activities have resources (time, money, labor, material and equipment). Use Pay Activities with no dollars allocated to schedule administrative work activities or events.

1.3.1 Initial Project Schedules

See Section 01 45 05 DREDGING/BEACH FILL PLACEMENT - QUALITY CONTROL. Create the Initial Activity Schedule in RMS, in accordance with the terms of FAR 52.236-15 SCHEDULES FOR CONSTRUCTION CONTRACTS. Submit Initial Project Schedules not later than 30 calendar days after receipt of Notice to Proceed. Data can be entered manually or Contractor has option to use

standard data exchange format (SDEF.txt file). Cost allocate each Pay Activity to an assigned CLIN. Sum of Pay Activities will equal CLIN totals so there is no variance. Features duration is early start of first Pay Activity to late finish date of last Pay Activity. Primavera P6 v6.x and later versions (through 18.x) allow export of a file for conversion to standard data exchange format (SDEF.txt) file format. Primavera provides a utility that will convert the export file to an SDEF.txt file. An SDEF.txt file can then be used to directly import the Primavera schedule into RMS. Provide all programs and hardware necessary to prepare initial and progress construction schedules. The required Primavera P6 Activity Code Structure is as follows:

Field #	Code	Length	Field Description
1	WRKP	3	Workers per day
2	RESP	4	Responsibility
3	AREA	4	Area
4	MODF	6	Modification or Claim #
5	BIDI	6	Bid Item
6	PHAS	2	Phase
7	CATW	1	Category of Work
8	FOW	20	Feature of Work

1.3.2 Bar Chart with S Curve

Contractor may create a manual Bar Chart and S curve or use computer software of choice. Scheduling software allows for easier updates and changes. Submit the electronic schedule file in the file format used to prepare the schedule. The Bar Chart will include an appropriate level of detail including activities, submittals, Government review periods, material/equipment delivery, utility outages, on-site construction, inspection, testing, and closeout activities. Contractor's submissions will follow the direction of the Contracting Officer regarding reasonable activity durations. Reasonable durations are those that allow the progress of activities to be accurately determined between payment periods. All planned durations will take into account weather and climate data, work schedules, and contract restrictions on work periods. No single on-site construction activity can exceed 30 days in duration without further breakdown in detail.

Contractor has an option to schedule an early finish date with float until Contract required completion date. Schedules with early finish are for the benefit of the Contractor, and the Contracting Officer may elect to continue to make balanced progress payments based on funds availability. Neither Contractor nor Contracting Officer own float on early finish schedules. Contractor cannot claim added costs for delay for float used. When unit price CLINs have quantity variations and impact weighted percentage of a Pay Activity by five percent or more, revise bar chart and S Curve to accurately reflect impact of variation of estimated quantities.

1.3.3 Project Milestones

1.3.3.1 Project Start Date Milestone and Constraint

The first activity in the project schedule is a start milestone titled NTP ACKNOWLEDGED, which has a start on constraint date equal to the date that the NTP is acknowledged.

1.3.3.2 End Project Finish Milestone and Constraint

The last activity in the schedule is a finish milestone titled END PROJECT.

The project schedule is constrained to the Contract completion date in such a way that if the schedule calculates an early finish, then the float calculation for the END PROJECT milestone reflects a positive float on the longest path. If the project schedule calculates a late finish, then the END PROJECT milestone float calculation reflects a negative float on the longest path.

1.4 PERIODIC SCHEDULE UPDATE

Update the project schedule on a regular basis, monthly at a minimum. Provide a draft periodic schedule update for review as prescribed in the paragraph PERIODIC PROGRESS MEETINGS. These updates enable the Government to assess the Contractor's progress. Submit Bar Chart, S Curve schedule when schedule is revised. Reflect any changes occurring since the last update. Submit Periodic Schedule Updates not later than 4 working days after the periodic progress meeting. Submit copies of the purchase orders and confirmation of the delivery dates as directed.

When work is delayed by causes beyond the Contractor's control, notify the Contracting Officer within 10 days of the beginning of the delay. Provide a description of the event(s) that caused the delay or impact to the work. As part of the description, identify all schedule activities impacted. Show that the event that caused the delay or impact was the responsibility of the Government. Provide an analysis that demonstrates the effects of the delay or impact on the project completion date or interim completion date(s). Evaluate multiple impacts chronologically, each with its own justification of delay. Evaluate the delay or impact for concurrence with delays caused by the Contractor's own actions. With multiple impacts, consider any concurrence of delay. No time extension will be granted unless a delay occurs which consumes all available project float and extends the projected finish beyond the contract completion date. No delay damages will be paid unless the event is shown to be the responsibility of the Government.

1.4.1 Narrative Report

Provide a narrative report with each schedule submission which is expected to communicate to the Government the thorough analysis of the schedule output and the plans to compensate for any problems, either current or potential, which are revealed through that analysis. Include at a minimum the following information in the narrative report:

- a. The work scheduled to start in the next update period.
- b. A description of activities along the two most critical paths where the total float is less than or equal to 20 work days.
- c. A description of current and anticipated problem areas or delaying factors and their impact and an explanation of corrective actions taken or required to be taken.
- d. The reasons activities, based on their calculated late dates, were planned to have either started or finished during the update period but did not.

U.S. Naval Station Mayport
Duval County, Florida

- e. All schedule changes by activity ID and activity name, including what specifically changed, and why the change was needed. Include at a minimum new and deleted activities, logic changes, duration changes, calendar changes, lag changes, resource changes, and actual start and finish date changes.
- f. Out-of-sequence work.

1.4.2 Periodic Progress Meetings

Progress meetings to discuss payment will include an on-site meeting at regular intervals mutually agreed to at the Preconstruction Conference. During the meeting, the Contractor will describe, on an activity by activity basis, all proposed revisions and adjustments to the Project Schedule which are required to reflect the current status of the project. The Contracting Officer will approve activity progress, proposed revisions, and adjustments as appropriate.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 33 00

SUBMITTAL PROCEDURES

PART 1 GENERAL

- 1.1 SUMMARY
 - 1.1.1 Submittal Information
 - 1.1.2 Project Type
 - 1.1.3 Submission of Submittals
- 1.2 DEFINITIONS
 - 1.2.1 Submittal Descriptions (SD)
 - 1.2.2 Approving Authority
 - 1.2.3 Work
- 1.3 SUBMITTALS
- 1.4 SUBMITTAL CLASSIFICATION
 - 1.4.1 Government Approved (G)
 - 1.4.2 For Information Only
- 1.5 PREPARATION
 - 1.5.1 Transmittal Form
 - 1.5.2 Submittal Format
 - 1.5.2.1 Format of SD-01 Preconstruction Submittals
 - 1.5.2.2 Format for SD-02 Shop Drawings
 - 1.5.2.2.1 Drawing Identification
 - 1.5.2.3 Format of SD-03 Product Data
 - 1.5.2.3.1 Product Information
 - 1.5.2.3.2 Standards
 - 1.5.2.3.3 Data Submission
 - 1.5.2.4 Format of SD-04 Samples
 - 1.5.2.4.1 Sample Characteristics
 - 1.5.2.4.2 Sample Incorporation
 - 1.5.2.4.3 Comparison Sample
 - 1.5.2.5 Format of SD-05 Design Data
 - 1.5.2.6 Format of SD-06 Test Reports
 - 1.5.2.7 Format of SD-07 Certificates
 - 1.5.2.8 Format of SD-08 Manufacturer's Instructions
 - 1.5.2.8.1 Standards
 - 1.5.2.9 Format of SD-09 Manufacturer's Field Reports
 - 1.5.2.10 Format of SD-10 Operation and Maintenance Data (O&M)
 - 1.5.2.11 Format of SD-11 Closeout Submittals
 - 1.5.3 Source Drawings for Shop Drawings
 - 1.5.3.1 Source Drawings
 - 1.5.3.2 Terms and Conditions
 - 1.5.4 Electronic File Format
- 1.6 QUANTITY OF SUBMITTALS
 - 1.6.1 Number of SD-01 Preconstruction Submittal Copies
 - 1.6.2 Number of SD-04 Samples
- 1.7 INFORMATION ONLY SUBMITTALS
- 1.8 PROJECT SUBMITTAL REGISTER
 - 1.8.1 Submittal Management

U.S. Naval Station Mayport
Duval County, Florida

- 1.8.2 Preconstruction Use of Submittal Register
- 1.8.3 Contractor Use of Submittal Register
- 1.8.4 Approving Authority Use of Submittal Register
- 1.8.5 Action Codes
 - 1.8.5.1 Contractor Action Codes
- 1.8.6 Delivery of Copies
- 1.9 VARIATIONS
 - 1.9.1 Considering Variations
 - 1.9.2 Proposing Variations
 - 1.9.3 Warranting that Variations are Compatible
 - 1.9.4 Review Schedule Extension
- 1.10 SCHEDULING
- 1.11 GOVERNMENT APPROVING AUTHORITY
 - 1.11.1 Review Notations
- 1.12 DISAPPROVED SUBMITTALS
- 1.13 APPROVED SUBMITTALS
- 1.14 APPROVED SAMPLES
- 1.15 WITHHOLDING OF PAYMENT
- 1.16 CERTIFICATION OF SUBMITTAL DATA

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

ATTACHMENTS:

ENG Form 4025-R

Attachment A - Submittal Register

-- End of Section Table of Contents --

SECTION 01 33 00
SUBMITTAL PROCEDURES

PART 1 GENERAL

1.1 SUMMARY

1.1.1 Submittal Information

The Contracting Officer may request submittals in addition to those specified when deemed necessary to adequately describe the work covered in the respective sections. Each submittal is to be complete and in sufficient detail to allow ready determination of compliance with contract requirements.

Units of weights and measures used on all submittals are to be the same as those used in the contract drawings.

1.1.2 Project Type

The Contractor's Quality Control (CQC) System Manager are to check and approve all items before submittal and stamp, sign, and date indicating action taken. Proposed deviations from the contract requirements are to be clearly identified. Include within submittals items such as: Contractor's, manufacturer's, or fabricator's drawings; descriptive literature including (but not limited to) catalog cuts, diagrams, operating charts or curves; test reports; test cylinders; samples; O&M manuals (including parts list); certifications; warranties; and other such required submittals.

1.1.3 Submission of Submittals

Schedule and provide submittals requiring Government approval before acquiring the material or equipment covered thereby. Pick up and dispose of samples not incorporated into the work in accordance with manufacturer's Safety Data Sheets (SDS) and in compliance with existing laws and regulations.

1.2 DEFINITIONS

1.2.1 Submittal Descriptions (SD)

Submittal requirements are specified in the technical sections. Examples and descriptions of submittals identified by the Submittal Description (SD) numbers and titles follow:

SD-01 Preconstruction Submittals

Submittals that are required prior to or at the start of construction (work) or the next major phase of the construction on a multiphase contract.

Government approved Division 01 preconstruction submittals that are required prior to or commencing with the start of work must be submitted

within 30 calendar days of contract award unless specified elsewhere in the specifications. Contractor approved Division 01 submittals that are required prior to or commencing with the start of work must be submitted within 45 calendar days of contract award unless specified elsewhere in the specifications.

Preconstruction Submittals include schedules and a tabular list of locations, features, and other pertinent information regarding products, materials, equipment, or components to be used in the work.

Certificates Of Insurance

Surety Bonds

List Of Proposed Subcontractors

List Of Proposed Products

Baseline Network Analysis Schedule (NAS)

Submittal Register

Schedule Of Prices Or Earned Value Report

Accident Prevention Plan

Work Plan

Quality Control (QC) plan

Environmental Protection Plan

SD-02 Shop Drawings

Drawings, diagrams and schedules specifically prepared to illustrate some portion of the work.

Diagrams and instructions from a manufacturer or fabricator for use in producing the product and as aids to the Contractor for integrating the product or system into the project.

Drawings prepared by or for the Contractor to show how multiple systems and interdisciplinary work will be coordinated.

SD-03 Product Data

Catalog cuts, illustrations, schedules, diagrams, performance charts, instructions and brochures illustrating size, physical appearance and other characteristics of materials, systems or equipment for some portion of the work.

Samples of warranty language when the contract requires extended product warranties.

SD-06 Test Reports

Report signed by authorized official of testing laboratory that a material, product or system identical to the material, product or system to be provided has been tested in accord with specified

requirements. Unless specified in another section, testing must have been within three years of date of contract award for the project.

Report that includes findings of a test required to be performed on an actual portion of the work or prototype prepared for the project before shipment to job site.

Report that includes finding of a test made at the job site or on sample taken from the job site, on portion of work during or after installation.

Investigation reports

Daily logs and checklists

Final acceptance test and operational test procedure

SD-07 Certificates

Statements printed on the manufacturer's letterhead and signed by responsible officials of manufacturer of product, system or material attesting that the product, system, or material meets specification requirements. Must be dated after award of project contract and clearly name the project.

Document required of Contractor, or of a manufacturer, supplier, installer or Subcontractor through Contractor. The document purpose is to further promote the orderly progression of a portion of the work by documenting procedures, acceptability of methods, or personnel qualifications.

Confined space entry permits

Text of posted operating instructions

SD-11 Closeout Submittals

Documentation to record compliance with technical or administrative requirements or to establish an administrative mechanism.

Submittals required for Guiding Principle Validation (GPV) or Third Party Certification (TPC).

Special requirements necessary to properly close out a construction contract. For example, Record Drawings and as-built drawings. Also, submittal requirements necessary to properly close out a major phase of construction on a multi-phase contract.

1.2.2 Approving Authority

Office or designated person authorized to approve the submittal.

1.2.3 Work

As used in this section, on-site and off-site construction required by contract documents, including labor necessary to produce submittals, construction, materials, products, equipment, and systems incorporated or to be incorporated in such construction. In exception, excludes work to produce SD-01 submittals.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Submittal Register; G

1.4 SUBMITTAL CLASSIFICATION

1.4.1 Government Approved (G)

Government approval is required for extensions of design, critical materials, variations, equipment whose compatibility with the entire system must be checked, and other items as designated by the Government.

Within the terms of the Contract Clause SPECIFICATIONS AND DRAWINGS FOR CONSTRUCTION, submittals are considered to be "shop drawings."

1.4.2 For Information Only

Submittals not requiring Government approval will be for information only. Within the terms of the Contract Clause SPECIFICATIONS AND DRAWINGS FOR CONSTRUCTION, they are not considered to be "shop drawings."

1.5 PREPARATION

1.5.1 Transmittal Form

Use the ENG Form 4025-R transmittal form for submitting both Government-approved and information-only submittals. Submit in accordance with the instructions on the reverse side of the form. These forms or similar forms are included in the RMS CM software that the Contractor is required to use for this contract. Properly complete this form by filling out all the heading blank spaces and identifying each item submitted. Exercise special care to ensure proper listing of the specification paragraph and sheet number of the contract drawings pertinent to the data submitted for each item.

1.5.2 Submittal Format

1.5.2.1 Format of SD-01 Preconstruction Submittals

When the submittal includes a document that is to be used in the project, or is to become part of the project record, other than as a submittal, do not apply the Contractor's approval stamp to the document itself, but to a separate sheet accompanying the document.

Provide data in the unit of measure used in the contract documents.

1.5.2.2 Format for SD-02 Shop Drawings

Provide shop drawings not less than 8 1/2 by 11 inches nor more than 30 by 42 inches, except for full-size patterns or templates. Prepare drawings

to accurate size, with scale indicated, unless another form is required. Ensure drawings are suitable for reproduction and of a quality to produce clear, distinct lines and letters, with dark lines on a white background.

- a. Include the nameplate data, size, and capacity on drawings. Also include applicable federal, military, industry, and technical society publication references.
- b. Dimension drawings, except diagrams and schematic drawings. Prepare drawings demonstrating interface with other trades to scale. Use the same unit of measure for shop drawings as indicated on the contract drawings. Identify materials and products for work shown.

Submit an electronic copy of drawings in PDF format and native electronic format.

1.5.2.2.1 Drawing Identification

Include on each drawing the drawing title, number, date, and revision numbers and dates, in addition to information required in paragraph IDENTIFYING SUBMITTALS.

Number drawings in a logical sequence. Each drawing is to bear the number of the submittal in a uniform location next to the title block. Place the Government contract number in the margin, immediately below the title block, for each drawing.

Reserve a blank space, no smaller than **2 inches** on the right-hand side of each sheet for the Government disposition stamp.

1.5.2.3 Format of SD-03 Product Data

Present product data submittals for each section. Include a table of contents, listing the page and catalog item numbers for product data.

Indicate, by prominent notation, each product that is being submitted; indicate the specification section number and paragraph number to which it pertains.

1.5.2.3.1 Product Information

Supplement product data with material prepared for the project to satisfy the submittal requirements where product data does not exist. Identify this material as developed specifically for the project, with information and format as required for submission of SD-07 Certificates.

Provide product data in units used in the Contract documents. Where product data are included in preprinted catalogs with another unit, submit the dimensions in contract document units, on a separate sheet.

1.5.2.3.2 Standards

Where equipment or materials are specified to conform to industry or technical-society reference standards of such organizations as the American National Standards Institute (ANSI), ASTM International (ASTM), National Electrical Manufacturer's Association (NEMA), Underwriters Laboratories (UL), or Association of Edison Illuminating Companies (AEIC), submit proof of such compliance. The label or listing by the specified organization will be acceptable evidence of compliance. In lieu of the

label or listing, submit a certificate from an independent testing organization, competent to perform testing, and approved by the Contracting Officer. State on the certificate that the item has been tested in accordance with the specified organization's test methods and that the item complies with the specified organization's reference standard.

1.5.2.3.3 Data Submission

Collect required data submittals for each specific material, product, unit of work, or system into a single submittal that is marked for choices, options, and portions applicable to the submittal. Mark each copy of the product data identically. Partial submittals will not be accepted for expedition of the construction effort.

Submit the manufacturer's instructions before installation.

1.5.2.4 Format of SD-04 Samples

1.5.2.4.1 Sample Characteristics

Furnish samples in the following sizes, unless otherwise specified or unless the manufacturer has prepackaged samples of approximately the same size as specified:

- a. Sample of Equipment or Device: Full size.
- b. Sample of Materials Less Than 2 by 3 inches: Built up to 8 1/2 by 11 inches.
- c. Sample of Materials Exceeding 8 1/2 by 11 inches: Cut down to 8 1/2 by 11 inches and adequate to indicate color, texture, and material variations.
- d. Sample of Linear Devices or Materials: 10 inch length or length to be supplied, if less than 10 inches. Examples of linear devices or materials are conduit and handrails.
- e. Sample Volume of Nonsolid Materials: Pint. Examples of nonsolid materials are sand and paint.
- f. Color Selection Samples: 2 by 4 inches. Where samples are specified for selection of color, finish, pattern, or texture, submit the full set of available choices for the material or product specified. Sizes and quantities of samples are to represent their respective standard unit.
- g. Sample Panel: 4 by 4 feet.
- h. Sample Installation: 100 square feet.

1.5.2.4.2 Sample Incorporation

Reusable Samples: Incorporate returned samples into work only if so specified or indicated. Incorporated samples are to be in undamaged condition at the time of use.

Recording of Sample Installation: Note and preserve the notation of any area constituting a sample installation, but remove the notation at the

final clean-up of the project.

1.5.2.4.3 Comparison Sample

Samples Showing Range of Variation: Where variations in color, finish, pattern, or texture are unavoidable due to nature of the materials, submit sets of samples of not less than three units showing extremes and middle of range. Mark each unit to describe its relation to the range of the variation.

When color, texture, or pattern is specified by naming a particular manufacturer and style, include one sample of that manufacturer and style, for comparison.

1.5.2.5 Format of SD-05 Design Data

Provide design data and certificates on 8 1/2 by 11 inch paper.

1.5.2.6 Format of SD-06 Test Reports

By prominent notation, indicate each report in the submittal. Indicate the specification number and paragraph number to which each report pertains.

1.5.2.7 Format of SD-07 Certificates

Provide design data and certificates on 8 1/2 by 11 inch paper.

1.5.2.8 Format of SD-08 Manufacturer's Instructions

Present manufacturer's instructions submittals for each section. Include the manufacturer's name, trade name, place of manufacture, and catalog model or number on product data. Also include applicable federal, military, industry, and technical-society publication references. If supplemental information is needed to clarify the manufacturer's data, submit it as specified for SD-07 Certificates.

Submit the manufacturer's instructions before installation.

1.5.2.8.1 Standards

Where equipment or materials are specified to conform to industry or technical-society reference standards of such organizations as the American National Standards Institute (ANSI), ASTM International (ASTM), National Electrical Manufacturer's Association (NEMA), Underwriters Laboratories (UL), or Association of Edison Illuminating Companies (AEIC), submit proof of such compliance. The label or listing by the specified organization will be acceptable evidence of compliance. In lieu of the label or listing, submit a certificate from an independent testing organization, competent to perform testing, and approved by the Contracting Officer. State on the certificate that the item has been tested in accordance with the specified organization's test methods and that the item complies with the specified organization's reference standard.

1.5.2.9 Format of SD-09 Manufacturer's Field Reports

By prominent notation, indicate each report in the submittal. Indicate the specification number and paragraph number to which each report

pertains.

1.5.2.10 Format of SD-10 Operation and Maintenance Data (O&M)

Comply with the requirements specified in sections where O&M Data is required for O&M Data format.

1.5.2.11 Format of SD-11 Closeout Submittals

When the submittal includes a document that is to be used in the project or is to become part of the project record, other than as a submittal, do not apply the Contractor's approval stamp to the document itself, but to a separate sheet accompanying the document.

Provide data in the unit of measure used in the contract documents.

1.5.3 Source Drawings for Shop Drawings

1.5.3.1 Source Drawings

The entire set of source drawing files (DWG) will not be provided to the Contractor. Request the specific Drawing Number for the preparation of shop drawings. Only those drawings requested to prepare shop drawings will be provided. These drawings are provided only after award.

1.5.3.2 Terms and Conditions

Data contained on these electronic files must not be used for any purpose other than as a convenience in the preparation of construction data for the referenced project. Any other use or reuse is at the sole risk of the Contractor and without liability or legal exposure to the Government. The Contractor must make no claim, and waives to the fullest extent permitted by law any claim or cause of action of any nature against the Government, its agents, or its subconsultants that may arise out of or in connection with the use of these electronic files. The Contractor must, to the fullest extent permitted by law, indemnify and hold the Government harmless against all damages, liabilities, or costs, including reasonable attorney's fees and defense costs, arising out of or resulting from the use of these electronic files.

These electronic source drawing files are not construction documents. Differences may exist between the source drawing files and the corresponding construction documents. The Government makes no representation regarding the accuracy or completeness of the electronic source drawing files, nor does it make representation to the compatibility of these files with the Contractor hardware or software. The Contractor is responsible for determining if any conflict exists. In the event that a conflict arises between the signed and sealed construction documents prepared by the Government and the furnished source drawing files, the signed and sealed construction documents govern. Use of these source drawing files does not relieve the Contractor of the duty to fully comply with the contract documents, including and without limitation the need to check, confirm and coordinate the work of all contractors for the project. If the Contractor uses, duplicates or modifies these electronic source drawing files for use in producing construction data related to this contract, remove all previous indication of ownership (seals, logos, signatures, initials and dates).

1.5.4 Electronic File Format

Provide submittals in electronic format, with the exception of material samples required for SD-04 Samples items. In addition to electronic submittal, provide additional hard copies of submittals when requested by the Contracting Officer. Up to 2 additional hard copies of any submittal may be requested at the discretion of the Contracting Officer, at no additional cost to the Government. Compile the submittal file as a single, complete document, to include the Transmittal Form described within, and also separately attach the native files which were used to create PDF. The attached files should include the original digital files used to create the submittal. Name the electronic submittal file specifically according to its contents, and coordinate the file naming convention with the Contracting Officer. Electronic files must be of sufficient quality that all information is legible. Use PDF as the electronic format, unless otherwise specified or directed by the Contracting Officer. Generate PDF files from original documents with bookmarks so that the text included in the PDF file is searchable and can be copied. If documents are scanned, optical character resolution (OCR) routines are required. Index and bookmark files exceeding 30 pages to allow efficient navigation of the file. When required, the electronic file must include a valid electronic signature or a scan of a signature.

E-mail electronic submittal documents smaller than 10MB to an e-mail address as directed by the Contracting Officer. Provide electronic documents over 10 MB on an optical disc or through an electronic file sharing system such as the DOD SAFE Web Application located at the following website: <https://safe.apps.mil/>.

1.6 QUANTITY OF SUBMITTALS

1.6.1 Number of SD-01 Preconstruction Submittal Copies

Unless otherwise specified, submit two sets of administrative submittals.

1.6.2 Number of SD-04 Samples

- a. Submit two samples, or two sets of samples showing the range of variation, of each required item. One approved sample or set of samples will be retained by the approving authority and one will be returned to the Contractor.
- b. Submit one sample panel or provide one sample installation where directed. Include components listed in the technical section or as directed.
- c. Submit one sample installation, where directed.
- d. Submit one sample of nonsolid materials.

1.7 INFORMATION ONLY SUBMITTALS

Submittals without a "G" designation must be certified by the QC manager and submitted to the Contracting Officer for information-only. Provide information-only submittals to the Contracting Officer a minimum of 14 calendar days prior to the Preparatory Meeting for the associated Definable Feature of Work (DFOW). Approval of the Contracting Officer is not required on information only submittals. The Contracting Officer will mark "receipt acknowledged" on submittals for information and will return

only the transmittal cover sheet to the Contractor. Normally, submittals for information only will not be returned. However, the Government reserves the right to return unsatisfactory submittals and require the Contractor to resubmit any item found not to comply with the contract. This does not relieve the Contractor from the obligation to furnish material conforming to the plans and specifications; will not prevent the Contracting Officer from requiring removal and replacement of nonconforming material incorporated in the work; and does not relieve the Contractor of the requirement to furnish samples for testing by the Government laboratory or for check testing by the Government in those instances where the technical specifications so prescribe.

1.8 PROJECT SUBMITTAL REGISTER

A sample Project Submittal Register showing items of equipment and materials for when submittals are required by the specifications is provided as "Attachment A - Submittal Register."

1.8.1 Submittal Management

Prepare and maintain a submittal register, as the work progresses. Do not change data that is output in columns (c), (d), (e), and (f) as delivered by Government; retain data that is output in columns (a), (g), (h), and (i) as approved. As an attachment, provide a submittal register showing items of equipment and materials for which submittals are required by the specifications. This list may not be all-inclusive and additional submittals may be required. Maintain a submittal register for the project in accordance with Section 01 45 00.15 10 RESIDENT MANAGEMENT SYSTEM CONTRACTOR MODE(RMS CM). The Government will provide the initial submittal register in electronic format with the following fields completed, to the extent that will be required by the Government during subsequent usage.

Column (c): Lists specification section in which submittal is required.

Column (d): Lists each submittal description (SD Number. and type, e.g., SD-02 Shop Drawings) required in each specification section.

Column (e): Lists one principal paragraph in each specification section where a material or product is specified. This listing is only to facilitate locating submitted requirements. Do not consider entries in column (e) as limiting the project requirements.

Thereafter, the Contractor is to track all submittals by maintaining a complete list, including completion of all data columns and all dates on which submittals are received by and returned by the Government.

1.8.2 Preconstruction Use of Submittal Register

Submit the submittal register. Include the QC plan and the project schedule. Verify that all submittals required for the project are listed and add missing submittals. Coordinate and complete the following fields on the register submitted with the QC plan and the project schedule:

Column (a) Activity Number: Activity number from the project schedule.

Column (g) Contractor Submit Date: Scheduled date for the approving authority to receive submittals.

Column (h) Contractor Approval Date: Date that Contractor needs approval of submittal.

Column (i) Contractor Material: Date that Contractor needs material delivered to Contractor control.

1.8.3 Contractor Use of Submittal Register

Update the following fields in the Government-furnished submittal register program or equivalent fields in the program used by the Contractor with each submittal throughout the contract.

Column (b) Transmittal Number: List of consecutive, Contractor-assigned numbers.

Column (j) Action Code (k): Date of action used to record Contractor's review when forwarding submittals to QC.

Column (l) Date submittal transmitted.

Column (q) Date approval was received.

1.8.4 Approving Authority Use of Submittal Register

Update the following fields:

Column (b) Transmittal Number: List of consecutive, Contractor-assigned numbers.

Column (l) Date submittal was received.

Column (m) through (p) Dates of review actions.

Column (q) Date of return to Contractor.

1.8.5 Action Codes

1.8.5.1 Contractor Action Codes

DESIGN BID BUILD SUBMITTALS			
Submittal Classifications shown in UFGS Sections	Submittal Classification	Corresponding SpecsIntact Submittal Register Code which is populated in the SI Submittal Register. Software Limitations: (The software shows one character delineation in the SpecsIntact Submittal Register)	RMS - The following Submittal Classifications are populated in RMS when the SpecsIntact Submittal Data File is pulled into RMS)
G	Submittal requires Government Approval	G	GA
BLANK	Submittal is For Information Only (FIO)	BLANK	FIO
S	Submittal is for documentation of Sustainable requirements	S	S/FIO

1.8.6 Delivery of Copies

Submit an updated electronic copy of the submittal register to the Contracting Officer with each invoice request. Provide an updated Submittal Register monthly regardless of whether an invoice is submitted.

1.9 VARIATIONS

Variations from contract requirements require Contracting Officer approval pursuant to contract Clause FAR 52.236-21 Specifications and Drawings for Construction, and will be considered where advantageous to the Government.

1.9.1 Considering Variations

Discussion of variations with the Contracting Officer before submission of a variation submittal will help ensure that functional and quality requirements are met and minimize rejections and resubmittals. For variations that include design changes or some material or product substitutions, the Government may require an evaluation and analysis by a licensed professional engineer hired by the contractor.

Specifically point out variations from contract requirements in a variation submittal. Failure to point out variations may cause the Government to require rejection and removal of such work at no additional cost to the Government.

1.9.2 Proposing Variations

When proposing variation, deliver a submittal, clearly marked as a "VARIATION" to the Contracting Officer, with documentation illustrating the nature and features of the variation including any necessary technical submittals and why the variation is desirable and beneficial to Government. If lower cost is a benefit, also include an estimate of the cost savings. In addition to documentation required for variation, include the submittals required for the item. Clearly mark the proposed variation in all documentation.

The Contracting Officer will indicate an approval or disapproval of the variation request; and if not approved as submitted, will indicate the Government's reasons therefore. Any work done before such approval is received is performed at the Contractor's risk."

Specifically point out variations from contract requirements in a variation submittal. Failure to point out variations may cause the Government to require rejection and removal of such work at no additional cost to the Government.

Check the column "variation" of ENG Form 4025 for submittals that include variations proposed by the Contractor. Set forth in writing the reason for any variations and note such variations on the submittal. The Government reserves the right to rescind inadvertent approval of submittals containing unnoted variations.

1.9.3 Warranting that Variations are Compatible

When delivering a variation for approval, the Contractor warrants that this contract has been reviewed to establish that the variation, if incorporated, will be compatible with other elements of work.

1.9.4 Review Schedule Extension

In addition to the normal submittal review period, a period of 14 calendar days will be allowed for the Government to consider submittals with variations.

1.10 SCHEDULING

Schedule and submit concurrently product data and shop drawings covering component items forming a system or items that are interrelated. Submit pertinent certifications at the same time. No delay damages or time extensions will be allowed for time lost in late submittals. Allow an additional 14 calendar days for review and approval of submittals for food service equipment and refrigeration and HVAC control systems.

- a. Coordinate scheduling, sequencing, preparing, and processing of submittals with performance of work so that work will not be delayed by submittal processing. The Contractor is responsible for additional time required for Government reviews resulting from required resubmittals. The review period for each resubmittal is the same as for the initial submittal. Resubmit disapproved submittal items (codes C, D, E, G or X) within 10 calendar days after return.
- b. Submittals required by the contract documents are listed on the submittal register. If a submittal is listed in the submittal

register but does not pertain to the contract work, the Contractor is to include the submittal in the register and annotate it "N/A" with a brief explanation. Approval by the Contracting Officer does not relieve the Contractor of supplying submittals required by the contract documents but that have been omitted from the register or marked "N/A."

- c. Resubmit the submittal register and annotate it monthly with actual submission and approval dates. When all items on the register have been fully approved, no further resubmittal is required.
- d. Unless otherwise indicated in specific paragraphs governing each submittal, the Government is afforded 10 calendar days to review and respond to an Information Only FIO submittal item, and 30 calendar days to review and respond to a Government Approved GA submittal item, from the date of receipt by the approval authority.

1.11 GOVERNMENT APPROVING AUTHORITY

When the approving authority is the Contracting Officer, the Government will:

- a. Note the date on which the submittal was received.
- b. Review submittals for approval within the scheduling period specified and only for conformance with project design concepts and compliance with contract documents.
- c. Identify returned submittals with one of the actions defined in paragraph REVIEW NOTATIONS and with comments and markings appropriate for the action indicated.

Upon completion of review of submittals requiring Government approval, stamp and date submittals. Two copies of the submittal will be retained by the Contracting Officer and one copy of the submittal will be returned to the Contractor.

1.11.1 Review Notations

Submittals will be returned to the Contractor with the following notations:

- a. Submittals marked "approved" or "accepted" authorize proceeding with the work covered.
- b. Submittals marked "approved as noted" or "approved, except as noted, resubmittal not required," authorize proceeding with the work covered provided that the Contractor takes no exception to the corrections.
- c. Submittals marked "not approved," "disapproved," or "revise and resubmit" indicate incomplete submittal or noncompliance with the contract requirements or design concept. Resubmit with appropriate changes. Do not proceed with work for this item until the resubmittal is approved.
- d. Submittals marked "not reviewed" indicate that the submittal has been previously reviewed and approved, is not required, does not have evidence of being reviewed and approved by Contractor, or is not complete. A submittal marked "not reviewed" will be returned with an explanation of the reason it is not reviewed. Resubmit submittals

returned for lack of review by Contractor or for being incomplete, with appropriate action, coordination, or change.

- e. Submittals marked "receipt acknowledged" indicate that submittals have been received by the Government. This applies only to "information-only submittals" as previously defined.

1.12 DISAPPROVED SUBMITTALS

Make corrections required by the Contracting Officer. If the Contractor considers any correction or notation on the returned submittals to constitute a change to the contract drawings or specifications, give notice to the Contracting Officer as required under the FAR clause titled CHANGES. The Contractor is responsible for the dimensions and design of connection details and the construction of work. Failure to point out variations may cause the Government to require rejection and removal of such work at the Contractor's expense.

If changes are necessary to submittals, make such revisions and resubmit in accordance with the procedures above. Provide with resubmission, a response to each Government review comment received on previous submittal a discussion of the corrective and revision made. No item of work requiring a submittal change is to be accomplished until the changed submittals are approved.

1.13 APPROVED SUBMITTALS

The Contracting Officer's approval of submittals is not to be construed as a complete check, and indicates only that

Approval or acceptance by the Government for a submittal does not relieve the Contractor of the responsibility for meeting the contract requirements or for any error that may exist, because under the Quality Control (QC) requirements of this contract, the Contractor is responsible for ensuring information contained within each submittal accurately conforms with the requirements of the contract documents.

After submittals have been approved or accepted by the Contracting Officer, no resubmittal for the purpose of substituting materials or equipment will be considered unless accompanied by an explanation of why a substitution is necessary.

1.14 APPROVED SAMPLES

Approval of a sample is only for the characteristics or use named in such approval and is not to be construed to change or modify any contract requirements. Before submitting samples, provide assurance that the materials or equipment will be available in quantities required in the project. No change or substitution will be permitted after a sample has been approved.

Match the approved samples for materials and equipment incorporated in the work. If requested, approved samples, including those that may be damaged in testing, will be returned to the Contractor, at its expense, upon completion of the contract. Unapproved samples will also be returned to the Contractor at its expense, if so requested.

Failure of any materials to pass the specified tests will be sufficient cause for refusal to consider, under this contract, any further samples of

U.S. Naval Station Mayport
Duval County, Florida

the same brand or make as that material. The Government reserves the right to disapprove any material or equipment that has previously proved unsatisfactory in service.

Samples of various materials or equipment delivered on the site or in place may be taken by the Contracting Officer for testing. Samples failing to meet contract requirements will automatically void previous approvals. Replace such materials or equipment to meet contract requirements.

1.15 WITHHOLDING OF PAYMENT

Payment for materials incorporated in the work will not be made if required approvals have not been obtained.

1.16 CERTIFICATION OF SUBMITTAL DATA

Certify the submittal data as follows on Form ENG 4025: "I certify that the above submitted items had been reviewed in detail and are correct and in strict conformance with the contract drawings and specifications except as otherwise stated.

____NAME OF CONTRACTOR _____ SIGNATURE OF CONTRACTOR

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

-- End of Section --

SUBMITTAL REGISTER

CONTRACT NO.
W912EP26BA005

TITLE AND LOCATION

U.S. NAVAL STATION MAYPORT MAINTENANCE DREDGING

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASS S I F I C A T I O N A / E R E V I W R	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/ DATE RCD FROM CONTR	APPROVING AUTHORITY				MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION		DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 20 00	SD-01 Preconstruction Submittals														
			Cost Breakdown	1.3	G RO												
			SD-07 Certificates														
			Request for Progress Payment	3.2	G RO												
		01 30 00	SD-07 Certificates														
			Signature Authority	1.9													
			Drug-Free Work Place Record	1.10													
			Insurance Requirements	1.11													
		01 32 01.10 35	SD-01 Preconstruction Submittals														
			Project Scheduler Qualifications	1.2	G RO												
			Initial Project Schedules	1.3.1	G RO												
			SD-07 Certificates														
			Periodic Schedule Update	1.4	G RO												
		01 33 00	SD-01 Preconstruction Submittals														
			Submittal Register	1.8	G												
		01 35 26	SD-01 Preconstruction Submittals														
			Accident Prevention Plan (APP)	1.8	G DO												
			Dive Operations Plan	1.16	G DO												
			SD-06 Test Reports														
			Accident Reports	1.12.2	G RO												
			LHE Inspection Reports	1.12.3													
			Monthly Exposure Reports	1.5	G RO												
			SD-07 Certificates														
			Crane Operators, Signal Persons and Riggers	1.7.7	G RO												
			Activity Hazard Analysis (AHA)	1.9	G RO												

SUBMITTAL REGISTER

CONTRACT NO.
W912EP26BA005

TITLE AND LOCATION

U.S. NAVAL STATION MAYPORT MAINTENANCE DREDGING

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASS S I F I C A T I O N A / E R E V I O W N R	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/ DATE RCD FROM CONTR	APPROVING AUTHORITY				MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION		DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 35 26	Certificate of Compliance	1.12.4	G RO												
			Hot Work Permit	1.13													
			License Certificates	1.14	G RO												
			Portable Gauge Operations	1.14.1	G DO												
			Planning Worksheet														
			Radiography Operation Planning	1.14.1	G DO												
			Work Sheet														
			Standard Lift Plan	1.12.4	G RO												
		01 45 05	SD-01 Preconstruction Submittals														
			Contractor Quality Control (CQC)	1.5.2	G RO												
			Plan														
			SD-06 Test Reports														
			Verification Statement	1.12.3													
		01 50 00.00 35	SD-01 Preconstruction Submittals														
			Construction Site Plan	1.4.1	G RO												
			Mobilization/Demobilization Plan	1.4.2	G RO												
			Security Plan	1.4.3	G RO												
			Vegetation Protection and	1.4.4	G RO												
			Restoration Plan														
			Traffic Control Plan	3.3.1	G RO												
			Documentation Of Existing	1.4.5	G RO												
			Conditions														
			Haul Road Plan	2.2.1	G RO												
			SD-02 Shop Drawings														
			Temporary Electric Drawings	3.2.6													
			SD-03 Product Data														

SUBMITTAL REGISTER

CONTRACT NO.
W912EP26BA005

TITLE AND LOCATION

U.S. NAVAL STATION MAYPORT MAINTENANCE DREDGING

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASS SIF CATION REVIEW WORK	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 50 00.00 35	Project Identification And Safety Signs	2.1.2	G RO												
			SD-11 Closeout Submittals														
			Documentation Of Post Construction Restoration	3.10.1	G RO												
		01 57 20.00 35	SD-01 Preconstruction Submittals														
			Environmental Protection Plan	1.8	G DO												
			Whale Alert Text Message Address	3.1.5.1	G DO												
			SD-02 Shop Drawings														
			Turtle Deflector Device	3.1.5.6	G DO												
			SD-06 Test Reports														
			Protected Listed Species Sighting Reports	3.1.5.2	G DO												
			Weekly Logs/Summary Of FWC Hopper Dredge Reports	3.1.5.7	G DO												
			Weekly Protected Species Summary Report	3.1.5.3	G DO												
			SD-07 Certificates														
			Marine Animal Observer Qualifications	1.9.1	G DO												
			(Clamshell/Mechanical Only)														
			Endangered Species Observer and Trawl Supervisor Qualifications (For Hopper Dredges Only)	1.9.2	G DO												

SUBMITTAL REGISTER

CONTRACT NO.
W912EP26BA005

TITLE AND LOCATION

U.S. NAVAL STATION MAYPORT MAINTENANCE DREDGING

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 57 20.00 35	SD-11 Closeout Submittals														
			Protected Species Summary Report	3.1.5.3	G DO												
			Protected Species Summary Report	3.1.5.4	G DO												
			Comprehensive Trawling Daily Logs	3.1.5.7	G DO												
			Invasive And Nuisance Species Report	3.4.2.1	G DO												
			Project Environmental Summary Sheet	1.11	G DO												
		01 57 25.00 35	SD-01 Preconstruction Submittals														
			Turbidity Monitor Qualifications	1.3	G DO												
			Turbidity Monitoring Plan	1.4	G DO												
			Draft Turbidity Sampling Map	1.4	G DO												
			Testing Equipment And Procedure	3.1.1.1	G DO												
			SD-03 Product Data														
			Calibration Standard	3.1.1.1	G												
			SD-06 Test Reports														
			Turbidity Monitoring Reports	3.2.2	G DO												
			Turbidity Exceedance Report	3.3.1	G DO												
			Updated Turbidity Exceedance Report	3.3.2	G DO												
		35 20 23.00 35	SD-01 Preconstruction Submittals														
			Notice of Intent to Dredge	1.4													

CONTRACT NO.
W912EP26BA005

U.S. NAVAL STATION MAYPORT MAINTENANCE DREDGING

CONTRACTOR: SCHEDULE DATES	CONTRACTOR ACTION		APPROVING AUTHORITY		
-------------------------------	----------------------	--	---------------------	--	--

[illegible]

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 35 26

GOVERNMENTAL SAFETY REQUIREMENTS

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DEFINITIONS
 - 1.2.1 Site Safety and Health Officer (SSHO)
 - 1.2.1.1 Level One SSHO
 - 1.2.1.2 Level Two SSHO
 - 1.2.1.3 Level Three SSHO
 - 1.2.1.4 Alternate SSHO
 - 1.2.2 Competent Person (CP)
 - 1.2.3 Qualified Person (QP)
- 1.3 SUBMITTALS
- 1.4 PUBLIC HEALTH EMERGENCIES
- 1.5 MONTHLY EXPOSURE REPORTS
- 1.6 REGULATORY REQUIREMENTS
- 1.7 SITE QUALIFICATIONS, DUTIES, AND MEETINGS
 - 1.7.1 Site Safety and Health Officer (SSHO)
 - 1.7.1.1 Qualifications of SSHO
 - 1.7.1.2 Duties of SSHO
 - 1.7.1.3 Safety Meetings
 - 1.7.2 Roles and Responsibilities of Prime Contractor and SSHO
 - 1.7.3 Additional Requirements
 - 1.7.4 Contract Site Safety And Health Officer(s)(SSHOs) Minimum Requirements
 - 1.7.5 Dredging Contract Site Safety and Health Officer(s)(SSHOs) Requirements
 - 1.7.5.1 Dredging SSHO Personnel Requirements
 - 1.7.6 Competent Person for Confined Space Entry
 - 1.7.7 Crane Operators, Signal Persons and Riggers
 - 1.7.8 Competent Person for Scaffolding
 - 1.7.9 Preconstruction Conference
- 1.8 ACCIDENT PREVENTION PLAN (APP)
 - 1.8.1 Accident Prevention Plan (APP)
 - 1.8.2 Names and Qualifications
 - 1.8.3 Plans
- 1.9 ACTIVITY HAZARD ANALYSIS (AHA)
 - 1.9.1 AHA Management
 - 1.9.2 AHA Signature Log
- 1.10 SITE SAFETY REFERENCE MATERIALS
- 1.11 EMERGENCY MEDICAL TREATMENT
- 1.12 NOTIFICATIONS AND REPORTS
 - 1.12.1 Accident Notification
 - 1.12.2 Accident Reports
 - 1.12.3 LHE Inspection Reports
 - 1.12.4 Certificate of Compliance and Pre-lift Plan/Checklist for LHE and Rigging

U.S. Naval Station Mayport
Duval County, Florida

- 1.13 HOT WORK PERMIT
 - 1.13.1 Permit and Personnel Requirements
 - 1.13.2 Work Around Flammable Materials
- 1.14 RADIATION SAFETY REQUIREMENTS
 - 1.14.1 Radiography Operation Planning Work Sheet
 - 1.14.2 Site Access and Security
 - 1.14.3 Loss or Release and Unplanned Personnel Exposure
 - 1.14.4 Site Demarcation and Barricade
 - 1.14.5 Security of Material and Equipment
 - 1.14.6 Transportation of Material
 - 1.14.7 Schedule for Exposure or Unshielding
 - 1.14.8 Transmitter Requirements
- 1.15 CONFINED SPACE ENTRY REQUIREMENTS
 - 1.15.1 Rescue Procedures and Coordination with Local Emergency Responders
- 1.16 DIVE SAFETY REQUIREMENTS
- 1.17 SEVERE STORM PLAN

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

- 3.1 CONSTRUCTION AND OTHER WORK
 - 3.1.1 Worksite Communication
 - 3.1.2 Hazardous Material Exclusions
 - 3.1.3 Unforeseen Hazardous Material
- 3.2 UTILITY OUTAGE REQUIREMENTS
- 3.3 OUTAGE COORDINATION MEETING
- 3.4 CONTROL OF HAZARDOUS ENERGY (LOCKOUT/TAGOUT)
 - 3.4.1 Safety Preparatory Inspection Coordination Meeting with the Government or Utility
 - 3.4.2 Lockout/Tagout Isolation
 - 3.4.3 Lockout/Tagout Removal
- 3.5 FALL PROTECTION PROGRAM
 - 3.5.1 Fall Protection Equipment and Systems
 - 3.5.1.1 Additional Personal Fall Protection Measures
 - 3.5.1.2 Personal Fall Protection Equipment
- 3.6 EQUIPMENT
 - 3.6.1 Use of Explosives
- 3.7 ELECTRICAL
 - 3.7.1 Electrical Work
 - 3.7.2 Qualifications
 - 3.7.3 Arc Flash
 - 3.7.4 Grounding
 - 3.7.5 Testing

-- End of Section Table of Contents --

SECTION 01 35 26

GOVERNMENTAL SAFETY REQUIREMENTS

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ANSI/ASSP A10.34 (2021) Protection of the Public on or Adjacent to Construction Sites

ANSI/ASSP A10.44 (2020) Control of Energy Sources (Lockout/Tagout) for Construction and Demolition Operations

ASTM INTERNATIONAL (ASTM)

ASTM F855 (2023) Standard Specifications for Temporary Protective Grounds to Be Used on De-energized Electric Power Lines and Equipment

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 1048 (2016) Guide for Protective Grounding of Power Lines

IEEE C2 (2023) National Electrical Safety Code

INTERNATIONAL SAFETY EQUIPMENT ASSOCIATION (ISEA)

ANSI/ISEA Z89.1 (2014; R 2019) American National Standard for Industrial Head Protection

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 51B (2024) Standard for Fire Prevention During Welding, Cutting, and Other Hot Work

NFPA 70 (2023; ERTA 1 2024; TIA 24-1) National Electrical Code

NFPA 70E (2024) Standard for Electrical Safety in the Workplace

NFPA 241 (2022) Standard for Safeguarding Construction, Alteration, and Demolition Operations

NFPA 306 (2024) Standard for the Control of Gas Hazards on Vessels

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2024) Safety and Occupational Health (SOH) Requirements

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

10 CFR 20 Standards for Protection Against Radiation

29 CFR 1910 Occupational Safety and Health Standards

29 CFR 1915 Confined and Enclosed Spaces and Other Dangerous Atmospheres in Shipyard Employment

29 CFR 1926 Safety and Health Regulations for Construction

49 CFR 173 Shippers - General Requirements for Shipments and Packagings

CPL 2.100 (1995) Application of the Permit-Required Confined Spaces (PRCS) Standards, 29 CFR 1910.146

1.2 DEFINITIONS

The following definitions are for the convenience of the reader. If there is a referenced document in the text of this specification section, that is the document that should define terms for that paragraph. If further clarification is needed, contact the Contracting Officer.

1.2.1 Site Safety and Health Officer (SSHO)

A Contractor Employee that is responsible for overseeing and ensuring implementation of the prime Contractor's Safety and Occupational Health (SOH) program according to the Contract, EM 385-1-1, applicable federal, state, and local requirements.

1.2.1.1 Level One SSHO

A designated employee with full-time SOH responsibility that meets and follows the requirements of EM 385-1-1.

1.2.1.2 Level Two SSHO

A designated employee with Level Two SSHO responsibility that meets and follows the requirements of EM 385-1-1. Level Two SSHOs cannot be assigned to projects that have a residual Risk Assessment Code (RAC) of high or extremely high.

1.2.1.3 Level Three SSHO

A designated Qualified Person or Competent Person with SOH responsibility that meets and follows the requirements of EM 385-1-1. Level 3 SSHOs cannot be assigned to projects that have a residual RAC of high or

extremely high.

1.2.1.4 Alternate SSHO

An employee that meets the definition of the contract-required level SSHO, but is not the primary SSHO.

1.2.2 Competent Person (CP)

The CP is a person designated in writing, who, through training, knowledge and experience, is capable of identifying, evaluating, and addressing existing and predictable hazards in the working environment or working conditions that are unsanitary, hazardous, or dangerous to personnel, and who has authorization to take prompt corrective measures to eliminate them.

1.2.3 Qualified Person (QP)

The QP is a person designated in writing, who, by possession of a recognized degree, certificate, or professional standing, or extensive knowledge, training, and experience, has successfully demonstrated their ability to solve or resolve problems related to the subject matter, the work, or the project.

1.3 SUBMITTALS

Government Acceptance or Approval does not remove responsibility from the Contractors for their actions or liability.

Government approval is required for submittals with a "G" classification. Submittals not having a "G" are for information only. When used, a code following the "G" identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Accident Prevention Plan (APP); G, DO

Dive Operations Plan; G, DO

SD-06 Test Reports

Accident Reports; G, RO

LHE Inspection Reports

Monthly Exposure Reports; G, RO

SD-07 Certificates

Crane Operators, Signal Persons and Riggers; G, RO

Activity Hazard Analysis (AHA); G, RO

Certificate of Compliance; G, RO

Hot Work Permit

License Certificates; G, RO

Portable Gauge Operations Planning Worksheet; G, D0

Radiography Operation Planning Work Sheet; G, D0

Standard Lift Plan; G, R0

1.4 PUBLIC HEALTH EMERGENCIES

In the event of a declared public health emergency, follow safety precautions as required by the Occupational Safety and Health Administration (OSHA) www.osha.gov, the Centers for Disease Control and Prevention (CDC) www.cdc.gov, and as required by federal, state and local requirements.

1.5 MONTHLY EXPOSURE REPORTS

Provide a Monthly Exposure Report by the fifth of each month. This report is a compilation of employee-hours worked each month for all site workers, both Prime and subcontractor. Failure to submit the report may result in retention of up to 10 percent of the progress payment.

1.6 REGULATORY REQUIREMENTS

In addition to the detailed requirements included in the provisions of this Contract, comply with the most recent edition of USACE EM 385-1-1, and the following federal, state, and local laws, ordinances, criteria, rules and regulations at the date of the Solicitation for this Contract. Submit matters of interpretation of standards to the appropriate administrative agency for resolution before starting work. Where the requirements of this specification, applicable laws, criteria, ordinances, regulations, and referenced documents vary, the most stringent requirements govern.

1.7 SITE QUALIFICATIONS, DUTIES, AND MEETINGS

1.7.1 Site Safety and Health Officer (SSHO)

1.7.1.1 Qualifications of SSHO

All SSHOs will have met the training, experience requirements identified in the EM 385-1-1 and this Contract.

1.7.1.2 Duties of SSHO

All SSHOs will carry out the roles and responsibilities as identified in this Contract and the EM 385-1-1. All SSHOs will be designated on an ENG Form 6282, provided by the Contracting Officer. Superintendent, QC Manager, and SSHO are subject to dismissal if their required duties are not being effectively carried out. If either the Superintendent, QC Manager, or SSHO are dismissed, project work will be stopped and will not be allowed to resume until a suitable replacement is approved and the above duties are again being effectively carried out.

1.7.1.3 Safety Meetings

Conduct safety meetings to review past activities, plan for new or changed operations, review pertinent aspects of appropriate AHA (by trade), establish safe working procedures for anticipated hazards, and provide

pertinent Safety and Occupational Health (SOH) training and motivation. Conduct meetings at least once a month for all supervisors at the project location. The SSHO, supervisors, or foremen must conduct meetings at least once a week for the trade workers. Document meeting minutes to include the date, persons in attendance, subjects discussed, and names of individual(s) who conducted the meeting. Maintain documentation on-site and furnish copies to the Contracting Officer on request. Notify the Contracting Officer of all scheduled meetings 7 calendar days in advance.

1.7.2 Roles and Responsibilities of Prime Contractor and SSHO

The Prime Contractor and SSHO must ensure that the requirements of all applicable OSHA and EM 385-1-1 are met for the project. The Prime Contractor must ensure an SSHO or an equally qualified Alternate SSHO(s) is at the worksite at all times to implement and administer the Contractor's safety program and Government accepted Accident Prevention Plan. The utilization of a Level 2 SSHO as identified and outlined in EM 385-1-1 does not apply in this contract. If the primary SSHO is off-site for any reason an equally qualified Alternate SSHO must be at the worksite.

a. Prime contractor must ensure all SSHOs will:

- (1) Are designated on an ENG Form 6282.
- (2) Meet minimum training and experience requirements identified in EM 385-1-1.
- (3) Execute roles and responsibilities identified in EM 385-1-1.

1.7.3 Additional Requirements

The Level 1 SSHO may not serve as the Quality Control Manager. The Level 1 SSHO may not serve as the Superintendent.

1.7.4 Contract Site Safety And Health Officer(s)(SSHOs) Minimum Requirements

Provide a minimum of one Level One SSHO that meets the requirements of EM 385-1-1 for this project.

1.7.5 Dredging Contract Site Safety and Health Officer(s)(SSHOs) Requirements

1.7.5.1 Dredging SSHO Personnel Requirements

- a. Provide a minimum of one primary Level One SSHO assigned for the primary shift.
- b. For a project involving multiple work shifts, provide a minimum of a Level One SSHO for each additional shift.
- c. For individual dredging projects the prime contractor will designate additional Level Three SSHOs at locations where the primary Level One SSHO is not located (example: on dredge, tug, material placement site).

Examples of one dredging project site is reflected in each of the following:

- (1) a mechanical dredge, tug(s) and scow(s), scow route, and material placement site; or
 - (2) a hydraulic pipeline dredge, attendant plant, and material placement site; or,
 - (3) a hopper dredge (include land-based material placement site - if applicable.)
- d. Designated SSHOs must be present at the project site, located so that they have full mobility and reasonable access to all major work operations and must be available during their shift for immediate verbal consultation and notification.
- e. Designated Level One and Two SSHOs must have direct report authority to a senior project (or corporate) management official.
- f. Designated Level Three SSHOs must report potential safety and occupational health hazards, incidents, and concerns to the Level One or Two SSHO on shift.
- g. Level One and Level Two SSHOs for dredging must have a minimum of 3 years experience in one of the following areas:
- (1) Supervising/managing dredging activities.
 - (2) Supervising/managing marine construction activities.
 - (3) Supervising/managing land-based construction activities.
 - (4) Work managing safety programs or processes.
 - (5) Conducting hazard analyses and developing controls in activities or environments with similar hazards.

1.7.6 Competent Person for Confined Space Entry

Provide a CP for Confined Space Entry who meets the requirements of [EM 385-1-1](#) and herein. The CP for Confined Space Entry must supervise the entry into each confined space in accordance with [EM 385-1-1](#).

Since this work involves operations that handle combustible or hazardous materials, this person must have the ability to understand and follow through on the air sampling, Personal Protective Equipment (PPE), and instructions of a Marine Chemist, Coast Guard authorized persons, or Certified Industrial Hygienist. Confined space and enclosed space work must comply with [NFPA 306](#), [29 CFR 1915](#), "Confined and Enclosed Spaces and Other Dangerous Atmospheres in Shipyard Employment," or as applicable in [29 CFR 1910](#) for general industry, [29 CFR 1926](#) for construction.

1.7.7 Crane Operators, Signal Persons and Riggers

Provide Operators, Signal Persons, and Riggers meeting the requirements in [EM 385-1-1](#).

1.7.8 Competent Person for Scaffolding

Provide a Competent Person for Scaffolding who meets the requirements in [EM 385-1-1](#).

1.7.9 Preconstruction Conference

- a. Contractor representatives who have a responsibility or significant role in accident prevention on the project must attend the preconstruction conference. This includes the project superintendent, Site Safety and Occupational Health Officer, quality control manager, or any other assigned safety and health professionals who participated in the development of the APP (including the Activity Hazard Analyses (AHAs) and special plans, program and procedures associated with it).
- b. Discuss the details of the submitted APP to include incorporated plans, programs, procedures and a listing of anticipated AHAs that will be developed and implemented during the performance of the Contract. This list of proposed AHAs will be reviewed and an agreement will be reached between the Contractor and the Contracting Officer as to which phases will require an analysis. In addition, establish a schedule for the preparation, submittal, and Government review of AHAs to preclude project delays. The creation of the APP and Schedule will be created after being given Notice to Proceed.
- c. Deficiencies in the submitted APP, identified during the Contracting Officer's review, must be corrected, and the APP re-submitted for review prior to the start of construction. Work is not permitted to begin until an APP is established that is acceptable to the Contracting Officer.

1.8 ACCIDENT PREVENTION PLAN (APP)

1.8.1 Accident Prevention Plan (APP)

Submit the Accident Prevention Plan (APP) for review and acceptance by the Government within 30 days of the Notice to Proceed. A competent person must prepare the written site-specific APP. Prepare the APP in accordance with the format and requirements of **EM 385-1-1**, ENG Form 6293, and herein. The APP must be job-specific and address any unusual or unique aspects of the project or activity for which it is written. The APP must interface with the Contractor's overall safety and occupational health program referenced in the APP in the applicable APP element, and made site-specific. Describe the methods to evaluate past safety performance of potential subcontractors in the selection process. Also, describe innovative methods used to ensure and monitor safe work practices of subcontractors. The Government considers the Prime Contractor to be the "controlling employer" for all worksite safety and health of the subcontractors. Contractors are responsible for informing their subcontractors of the safety provisions under the terms of the Contract and the penalties for noncompliance, coordinating the work to prevent one craft from interfering with or creating hazardous working conditions for other crafts, and inspecting subcontractor operations to ensure that accident prevention responsibilities are being carried out. The APP must be signed in accordance with the APP and ENG Form 6293 Accident Prevention Plan Worksheet. The SSHO must provide and maintain the APP and a log of signatures by each subcontractor foreman, attesting that they have read and understand the APP, and make the APP and log available on-site to the Contracting Officer. If English is not the foreman's primary language, the Prime Contractor must provide an interpreter.

Work cannot proceed without an accepted APP. Once reviewed and accepted by the Contracting Officer, the APP and attachments will be enforced as

part of the Contract. Disregarding the provisions of this Contract or the accepted APP is cause for stopping of work, at the discretion of the Contracting Officer, until the matter has been rectified. Continuously review and amend the APP, as necessary, throughout the life of the Contract. Changes to the accepted APP must be made with the knowledge and concurrence of the Contracting Officer, project superintendent, SSHO and Quality Control Manager. Incorporate unusual or high-hazard activities not identified in the original APP as they are discovered. Should any severe hazard exposure (i.e., imminent danger) become evident, stop work in the area, secure the area, and develop a plan to remove the exposure and control the hazard. Notify the Contracting Officer within 24 hours of discovery. Eliminate and remove the hazard. In the interim, take all necessary action to restore and maintain safe working conditions in order to safeguard onsite personnel, visitors, the public (as defined by ANSI/ASSP A10.34), and the environment.

1.8.2 Names and Qualifications

Provide plans in accordance with the requirements outlined in EM 385-1-1, including the following:

- a. Names and qualifications (resumes including education, training, experience and certifications) of site safety and health personnel designated to perform work on this project to include the designated Site Safety and Health Officer and other competent and qualified personnel to be used. Specify the duties of each position.
- b. As a minimum, designate and submit qualifications of Competent Persons (CP) for each of the following major areas: excavation; scaffolding; fall protection; hazardous energy; confined space; health hazard recognition, evaluation and control of chemical, physical and biological agents; and personal protective equipment and clothing to include selection, use and maintenance. Designate and submit qualifications for additional CPs as applicable to the work performed under this Contract.

1.8.3 Plans

Provide plans in the APP in accordance with the requirements outlined in EM 385-1-1, including the following:

1.9 ACTIVITY HAZARD ANALYSIS (AHA)

Before beginning each activity, task or Definable Feature of Work (DFOW) involving a type of work presenting hazards not experienced in previous project operations, or where a new work crew or subcontractor is to perform the work, the Contractor(s) performing that work activity must prepare an AHA. AHAs must be developed by the Prime Contractor, subcontractor, or supplier performing the work, and provided for Prime Contractor review and approval before submitting to the Contracting Officer. AHAs must be signed by the SSHO, Superintendent, QC Manager and the subcontractor Foreman performing the work. Format the AHA in accordance with EM 385-1-1 or as directed by the Contracting Officer. Submit the AHA for review at least 15 working days prior to the start of each activity task, or DFOW. The Government reserves the right to require the Contractor to revise and resubmit the AHA if it fails to effectively identify the work sequences, specific anticipated hazards, site conditions, equipment, materials, personnel and the control measures to be implemented.

AHAs must identify competent persons required for all activities.

1.9.1 AHA Management

Review the AHA list periodically (at least monthly) at the Contractor supervisory safety meeting, and update as necessary when procedures, scheduling, or hazards change. Use the AHA during daily inspections by the SSHO to ensure the implementation and effectiveness of the required safety and health controls for that work activity.

1.9.2 AHA Signature Log

Each employee performing work as part of an activity, task or DFOV must review the AHA for that work and sign a signature log specifically maintained for that AHA prior to starting work on that activity. The SSHO must maintain a signature log on site for every AHA. Provide employees whose primary language is other than English, with an interpreter to ensure a clear understanding of the AHA and its contents.

1.10 SITE SAFETY REFERENCE MATERIALS

Maintain safety-related references applicable to the project, including those listed in paragraph REFERENCES. Maintain applicable equipment manufacturer's manuals.

1.11 EMERGENCY MEDICAL TREATMENT

Contractors must arrange for their own emergency medical treatment in accordance with [EM 385-1-1](#). The Government has no responsibility to provide emergency medical treatment.

1.12 NOTIFICATIONS AND REPORTS

1.12.1 Accident Notification

Notify the Contracting Officer in accordance with the [EM 385-1-1](#) Accident Reporting Timeline.

Table Accident Reporting Required Timeline		
Accident Type	Notify KO or COR	Complete Final Accident Report on ENG 3394 and provide to KO or COR
Fatality, in-patient hospitalization, amputation, eye loss, or property damage over \$600,000.	Immediately, no later than (NLT) 8 Hours	Within 7 Days
All other accidents and near misses	Immediately, no later than (NLT) 24 Hours	Within 7 Days

Within notification, provided by both phone call and email within

notification timeframe above, include Contractor name; Contract title; type of Contract; name of activity, installation or location where accident occurred; date and time of accident; names of personnel injured; extent of property damage, if any; extent of injury, if known, and brief description of accident (for example, type of construction equipment used and PPE used). Preserve the conditions and evidence on the accident site until the Government investigation team arrives on-site and Government investigation is conducted. Assist and cooperate fully with the Government's investigation(s) of any accident or near miss.

1.12.2 Accident Reports

- a. Conduct an accident investigation for recordable injuries and illnesses, property damage, and near misses as defined in EM 385-1-1, to establish the root cause(s) of the accident. All accidents are reportable, regardless of whether or not it is recordable. Complete the applicable USACE Accident Report, ENG Form 3394, and provide the report to the Contracting Officer within 7 calendar days of the accident. The Contracting Officer will provide copies of any required or special forms. All accidents are reportable, regardless of whether or not it is recordable.
- b. Near Misses: Near miss reports are considered positive and proactive Contractor safety management actions.

1.12.3 LHE Inspection Reports

Submit LHE inspection reports required in accordance with EM 385-1-1 and as specified herein with Daily Reports of Inspections.

1.12.4 Certificate of Compliance and Pre-lift Plan/Checklist for LHE and Rigging

Provide a Certificate of Compliance for LHE entering an activity under this Contract and in accordance with EM 385-1-1. Post certifications on the crane.

Develop a Standard Lift Plan (SLP) in accordance with EM 385-1-1 and using Standard Pre-Lift Crane Plan/Checklist for each lift planned. Submit SLP to the Contracting Officer for approval within 30 calendar days in advance of planned lift.

1.13 HOT WORK PERMIT

1.13.1 Permit and Personnel Requirements

Submit and obtain a written permit prior to performing "Hot Work" (i.e., welding or cutting) or operating other flame-producing/spark producing devices, from the Local Fire Division. A permit is required from the Explosives Safety Office for work in and around where explosives are processed, stored, or handled. Contractors are required to meet all criteria before a permit is issued. Provide at least two 20 pound 4A:20 BC rated extinguishers for normal "Hot Work". The extinguishers must be current inspection tagged, and contain an approved safety pin and tamper resistant seal. It is also mandatory to have a designated FIRE WATCH for any "Hot Work" done at this activity. The Fire Watch must be trained in accordance with NFPA 51B and remain on-site for a minimum of 30 minutes after completion of the task or as specified on the hot work permit.

When starting work in the facility, require personnel to familiarize themselves with the location of the nearest fire alarm boxes and place in memory the emergency Local Fire Division. Report any fire, no matter how small, to the responsible Local Fire Division immediately.

1.13.2 Work Around Flammable Materials

Obtain permit approval from a NFPA Certified Marine Chemist, or Certified Industrial Hygienist for "Hot Work" within or around flammable materials (such as fuel systems or welding/cutting on fuel pipes) or confined spaces (such as sewer wet wells, manholes, or vaults) that have the potential for flammable or explosive atmospheres.

Whenever these materials, except beryllium and chromium (VI), are encountered in indoor operations, local mechanical exhaust ventilation systems that are sufficient to reduce and maintain personal exposures to within acceptable limits must be used and maintained in accordance with manufacturer's instruction and supplemented by exceptions noted in [EM 385-1-1](#).

1.14 RADIATION SAFETY REQUIREMENTS

Submit [License Certificates](#), employee training records, and Leak Test Reports for radiation materials and equipment to the Contracting Officer and Radiation Safety Office (RSO) for all specialized and licensed material and equipment proposed for use on the construction project (excludes portable machine sources of ionizing radiation including moisture density and X-Ray Fluorescence (XRF)). Maintain on-site records whenever licensed radiological materials or ionizing equipment are on Government property.

Protect workers from radiation exposure in accordance with [10 CFR 20](#), ensuring any personnel exposures are maintained As Low As Reasonably Achievable.

1.14.1 Radiography Operation Planning Work Sheet

Submit a Gamma and X-Ray [Radiography Operation Planning Work Sheet](#) to Contracting Officer 14 days prior to commencement of operations involving radioactive materials or radiation generating devices. For portable machine sources of ionizing radiation, including moisture density and XRF, use and submit the [Portable Gauge Operations Planning Worksheet](#) instead. The Contracting Officer will review the submitted worksheet and provide questions and comments.

Contractors must use primary dosimeters process by a National Voluntary Laboratory Accreditation Program (NVLAP) accredited laboratory.

1.14.2 Site Access and Security

Coordinate site access and security requirements with the Contracting Officer for all radiological materials and equipment containing ionizing radiation that are proposed for use on a government facility. For gamma radiography materials and equipment, a Government escort is required for any travels on the Installation. The Contracting Officer will meet the Contractor at a designated location outside the Installation, ensure safety of the materials being transported, and will escort the Contractor for gamma sources onto the Installation, to the job site, and off the Installation. For portable machine sources of ionizing radiation,

including moisture density and XRF, the Contracting Officer will meet the Contractor at the job site.

Provide a copy of all calibration records, and utilization records to the Contracting Officer for radiological operations performed on the site.

1.14.3 Loss or Release and Unplanned Personnel Exposure

Loss or release of radioactive materials, and unplanned personnel exposures must be reported immediately to the Contracting Officer, RSO, and Base Security Department Emergency Number.

1.14.4 Site Demarcation and Barricade

Properly demark and barricade an area surrounding radiological operations to preclude personnel entrance, in accordance with [EM 385-1-1](#), Nuclear Regulatory Commission, and Applicable State regulations and license requirements, and in accordance with requirements established in the accepted Radiography Operation Planning Work Sheet.

Do not close or obstruct streets, walks, and other facilities occupied and used by the Government without written permission from the Contracting Officer.

1.14.5 Security of Material and Equipment

Properly secure the radiological material and ionizing radiation equipment at all times, including keeping the devices in a properly marked and locked container, and secondarily locking the container to a secure point in the Contractor's vehicle or other approved storage location during transportation and while not in use. While in use, maintain a continuous visual observation on the radiological material and ionizing radiation equipment. In instances where radiography is scheduled near or adjacent to buildings or areas having limited access or one-way doors, make no assumptions as to building occupancy. Where necessary, the Contracting Officer will direct the Contractor to conduct an actual building entry, search, and alert. Where removal of personnel from such a building cannot be accomplished and it is otherwise safe to proceed with the radiography, position a fully instructed employee inside the building or area to prevent exiting while external radiographic operations are in process.

1.14.6 Transportation of Material

Comply with [49 CFR 173](#) for Transportation of Regulated Amounts of Radioactive Material. Notify Local Fire authorities and the site Radiation Safety Officer (RSO) of any Radioactive Material use.

1.14.7 Schedule for Exposure or Unshielding

Actual exposure of the radiographic film or unshielding the source must not be initiated until after 5 p.m. on weekdays.

1.14.8 Transmitter Requirements

Adhere to the base policy concerning the use of transmitters, such as radios and cell phones. Obey Emissions control (EMCON) restrictions.

1.15 CONFINED SPACE ENTRY REQUIREMENTS

Confined space entry must comply with EM 385-1-1, 29 CFR 1926, 29 CFR 1910, and Directive CPL 2.100. Any potential for a hazard in the confined space requires a permit system to be used.

1.15.1 Rescue Procedures and Coordination with Local Emergency Responders

Develop and implement an on-site rescue and recovery plan and procedures. The rescue plan must not rely on local emergency responders for rescue from a confined space.

1.16 DIVE SAFETY REQUIREMENTS

Develop a Dive Operations Plan, AHA, emergency management plan, and personnel list that includes qualifications, for each separate diving operation. Submit these documents to the District Dive Coordinator (DDC) via the Contracting Officer, for review and approval at least 15 working days prior to commencement of diving operations. These documents must be at the diving location at all times. Provide each of these documents as a part of the project file.

1.17 SEVERE STORM PLAN

In the event of a severe storm warning, the Contractor must comply with the applicable Storm Plan and:

- a. Secure outside equipment and materials and place materials that could be damaged in protected areas.
- b. Check surrounding area, including roof, for loose material, equipment, debris, and other objects that could be blown away or against existing facilities.
- c. Ensure that temporary erosion controls are adequate.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.1 CONSTRUCTION AND OTHER WORK

Comply with EM 385-1-1, NFPA 70, NFPA 70E, NFPA 241, the APP, the AHA, Federal and State OSHA regulations, and other related submittals and activity fire and safety regulations. The most stringent standard prevails.

PPE is governed in all areas by the nature of the work the employee is performing. Use personal hearing protection at all times in designated noise hazardous areas or when performing noise hazardous tasks. Safety glasses must be worn or carried/available on each person. Mandatory PPE includes:

- a. Head Protection that meets ANSI/ISEA Z89.1
- b. Long Pants
- c. Appropriate Safety Footwear

d. Appropriate Class Reflective Vests

3.1.1 Worksite Communication

Employees working alone in a remote location or away from other workers must be provided an effective means of emergency communications (i.e., cellular phone, two-way radios, land-line telephones or other acceptable means). The selected communication must be readily available (easily within the immediate reach) of the employee and must be tested prior to the start of work to verify that it effectively operates in the area/environment. Develop an employee check-in/check-out communication procedure to ensure employee safety.

3.1.2 Hazardous Material Exclusions

Notwithstanding any other hazardous material used in this Contract, radioactive materials or instruments capable of producing ionizing/non-ionizing radiation (with the exception of radioactive material and devices used in accordance with EM 385-1-1 such as nuclear density meters for compaction testing and laboratory equipment with radioactive sources) as well as materials which contain asbestos, mercury or polychlorinated biphenyls, di-isocyanates, lead-based paint, and hexavalent chromium, are prohibited. The Contracting Officer, upon written request by the Contractor, may consider exceptions to the use of any of the above excluded materials. Low mercury lamps used within fluorescent lighting fixtures are allowed as an exception without further Contracting Officer approval. Notify the Radiation Safety Officer (RSO) prior to excepted items of radioactive material and devices being brought on base.

3.1.3 Unforeseen Hazardous Material

Contract documents identify materials such as PCB, lead paint, and friable and non-friable asbestos and other OSHA regulated chemicals (i.e., 29 CFR Part 1910.1000). If material(s) that may be hazardous to human health upon disturbance are encountered during demolition, repair, renovation, or construction operations. Stop that portion of work and notify the Contracting Officer immediately. Within 14 calendar days the Government will determine if the material is hazardous. If material is not hazardous or poses no danger, the Government will direct the Contractor to proceed without change. If material is hazardous and handling of the material is necessary to accomplish the work, the Government will issue a modification.

3.2 UTILITY OUTAGE REQUIREMENTS

Apply for utility outages in sufficient time as to not result in impacts or delays to the project schedule. At a minimum, the written request must include the location of the outage, utilities being affected, duration of outage, any necessary sketches, and a description of the means to fulfill energy isolation requirements in accordance with EM 385-1-1. In accordance with EM 385-1-1, where outages involve Government or Utility personnel, coordinate with the Government on all activities involving the control of hazardous energy.

These activities include, but are not limited to, a review of Hazardous Energy Control Program (HECP) and HEC procedures, as well as applicable Activity Hazard Analyses (AHAs). In accordance with EM 385-1-1 and NFPA 70E, work on energized electrical circuits must not be performed

without prior Government authorization. Government permission is considered through the permit process and submission of a detailed AHA. Energized work permits are considered only when de-energizing introduces additional or increased hazard or when de-energizing is infeasible.

3.3 OUTAGE COORDINATION MEETING

After the utility outage request is approved and prior to beginning work on the utility system requiring shut-down, conduct a pre-outage coordination meeting in accordance with EM 385-1-1. This meeting must include the Prime Contractor, the Prime and subcontractors performing the work, the Contracting Officer, and the Installation representative, and the Public Utilities representative. All parties must fully coordinate HEC activities with one another. During the coordination meeting, all parties must discuss and coordinate on the scope of work, HEC procedures (specifically, the lock-out/tag-out procedures for worker and utility protection), the AHA, assurance of trade personnel qualifications, identification of competent persons, and compliance with HEC training in accordance with EM 385-1-1. Clarify when personal protective equipment is required during switching operations, inspection, and verification.

3.4 CONTROL OF HAZARDOUS ENERGY (LOCKOUT/TAGOUT)

Provide and operate a Hazardous Energy Control Program (HECP) in accordance with EM 385-1-1, 29 CFR 1910, 29 CFR 1915, ANSI/ASSP A10.44, NFPA 70E.

3.4.1 Safety Preparatory Inspection Coordination Meeting with the Government or Utility

For electrical distribution equipment that is to be operated by Government or Utility personnel, the Prime Contractor and the subcontractor performing the work must attend the safety preparatory inspection coordination meeting, which will also be attended by the Contracting Officer, and required by EM 385-1-1. The meeting will occur immediately preceding the start of work and following the completion of the outage coordination meeting. Both the safety preparatory inspection coordination meeting and the outage coordination meeting must occur prior to conducting the outage and commencing with lockout/tagout procedures.

3.4.2 Lockout/Tagout Isolation

Where the Government or Utility performs equipment isolation and lockout/tagout, the Contractor must place their own locks and tags on each energy-isolating device and proceed in accordance with the HECP. Before any work begins, both the Contractor and the Government or Utility must perform energy isolation verification testing while wearing required PPE detailed in the Contractor's AHA and required by EM 385-1-1. Install personal protective grounds, with tags, to eliminate the potential for induced voltage in accordance with EM 385-1-1.

3.4.3 Lockout/Tagout Removal

Upon completion of work, conduct lockout/tagout removal procedure in accordance with the HECP. In accordance with EM 385-1-1, each lock and tag must be removed from each energy isolating device by the authorized individual or systems operator who applied the device. Provide formal notification to the Government (by completing the Government form if provided by Contracting Officer), confirming that steps of de-energization

and lockout/tagout removal procedure have been conducted and certified through inspection and verification. Government or Utility locks and tags used to support the Contractor's work will not be removed until the authorized Government employee receives the formal notification.

3.5 FALL PROTECTION PROGRAM

Establish a fall protection program, for the protection of all employees exposed to fall hazards. Within the program include company policy, identify roles and responsibilities, education and training requirements, fall hazard identification, prevention and control measures, inspection, storage, care and maintenance of fall protection equipment and rescue and evacuation procedures in accordance with EM 385-1-1.

3.5.1 Fall Protection Equipment and Systems

Enforce use of personal fall protection equipment and systems designated (to include fall arrest, restraint, and positioning) for each specific work activity in the Site Specific Fall Protection and Prevention Plan and AHA at all times when an employee is exposed to a fall hazard. Protect employees from fall hazards as specified in EM 385-1-1.

Provide personal fall protection equipment, systems, subsystems, and components that comply with EM 385-1-1 and 29 CFR 1926.

3.5.1.1 Additional Personal Fall Protection Measures

In addition to the required fall protection systems, other protective measures such as safety skiffs, personal floatation devices, and life rings, are required when working above or next to water in accordance with EM 385-1-1. Personal fall protection systems and equipment are required when working from an articulating or extendible boom, swing stages, or suspended platform. In addition, personal fall protection systems are required when operating other equipment such as scissor lifts. The need for tying-off in such equipment is to prevent ejection of the employee from the equipment during raising, lowering, travel, or while performing work.

3.5.1.2 Personal Fall Protection Equipment

Only a full-body harness with a shock-absorbing lanyard or self-retracting lanyard is an acceptable personal fall arrest body support device. The use of body belts is not acceptable. Harnesses must have a fall arrest attachment affixed to the body support (usually a Dorsal D-ring) and specifically designated for attachment to the rest of the system. Snap hooks and carabineers must be self-closing and self-locking, capable of being opened only by at least two consecutive deliberate actions and have a minimum gate strength of 3,600 lbs in all directions. Use webbing, straps, and ropes made of synthetic fiber. The maximum free fall distance when using fall arrest equipment must not exceed 6 feet, unless the proper energy absorbing lanyard is used. Always take into consideration the total fall distance and any swinging of the worker (pendulum-like motion), that can occur during a fall, when attaching a person to a fall arrest system. Equip all full body harnesses with Suspension Trauma Preventers such as stirrups, relief steps, or similar in order to provide short-term relief from the effects of orthostatic intolerance in accordance with EM 385-1-1.

3.6 EQUIPMENT

3.6.1 Use of Explosives

Explosives must not be used or brought to the project site without prior written approval from the Contracting Officer. Such approval does not relieve the Contractor of responsibility for injury to persons or for damage to property due to blasting operations.

Storage of explosives, when permitted on Government property, must be only where directed and in approved storage facilities. These facilities must be kept locked at all times except for inspection, delivery, and withdrawal of explosives.

3.7 ELECTRICAL

Perform electrical work in accordance with [EM 385-1-1](#).

3.7.1 Electrical Work

As described in [EM 385-1-1](#), electrical work is to be conducted in a de-energized state unless there is no alternative method for accomplishing the work. In those cases obtain an energized work permit from the Contracting Officer. The energized work permit application must be accompanied by the AHA and a summary of why the equipment/circuit needs to be worked energized. Underground electrical spaces must be certified safe for entry before entering to conduct work. Cables that will be cut must be positively identified and de-energized prior to performing each cut. Attach temporary grounds in accordance with [ASTM F855](#) and [IEEE 1048](#). Perform all high voltage cable cutting remotely using hydraulic cutting tool. When racking in or live switching of circuit breakers, no additional person other than the switch operator is allowed in the space during the actual operation. Plan so that work near energized parts is minimized to the fullest extent possible. Use of electrical outages clear of any energized electrical sources is the preferred method.

When working in energized substations, only qualified electrical workers are permitted to enter. When work requires work near energized circuits as defined by [NFPA 70](#), high voltage personnel must use personal protective equipment that includes, as a minimum, electrical hard hat, safety footwear, insulating gloves and electrical arc flash protection for personnel as required by [NFPA 70E](#). Insulating blankets, hearing protection, and switching suits may also be required, depending on the specific job and as delineated in the Contractor's AHA. Ensure that each employee is familiar with and complies with these procedures and [29 CFR 1910](#).

3.7.2 Qualifications

Electrical work must be performed by QP with verifiable credentials who are familiar with applicable code requirements. Verifiable credentials consist of State, National and Local Certifications or Licenses that a Master or Journeyman Electrician may hold, depending on work being performed, and must be identified in the appropriate AHA. Journeyman/Apprentice ratio must be in accordance with State, Local requirements applicable to where work is being performed.

3.7.3 Arc Flash

Conduct a hazard analysis/arc flash hazard analysis whenever work on or near energized parts greater than 50 volts is necessary, in accordance with NFPA 70E.

All personnel entering the identified arc flash protection boundary must be QPs and properly trained in NFPA 70E requirements and procedures. Unless permitted by NFPA 70E, no Unqualified Person is permitted to approach nearer than the Limited Approach Boundary of energized conductors and circuit parts. Training must be administered by an electrically qualified source and documented.

3.7.4 Grounding

Ground electrical circuits, equipment and enclosures in accordance with NFPA 70 and IEEE C2 to provide a permanent, continuous and effective path to ground unless otherwise noted by EM 385-1-1.

3.7.5 Testing

Temporary electrical distribution systems and devices must be inspected, tested and found acceptable for Ground-Fault Circuit Interrupter (GFCI) protection, polarity, ground continuity, and ground resistance before initial use, before use after modification and at least monthly. Monthly inspections and tests must be maintained for each temporary electrical distribution system, and signed by the electrical CP or QP.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 45 00.15 10

RESIDENT MANAGEMENT SYSTEM CONTRACTOR MODE (RMS CM)

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 MEASUREMENT AND PAYMENT
- 1.3 CONTRACT ADMINISTRATION
 - 1.3.1 Correspondence and Electronic Communications
 - 1.3.2 Other Factors
- 1.4 RMS SOFTWARE
- 1.5 CONTRACT DATABASE - GOVERNMENT
- 1.6 CONTRACT DATABASE - CONTRACTOR
 - 1.6.1 Administration
 - 1.6.1.1 Contractor Information
 - 1.6.1.2 Subcontractor Information
 - 1.6.1.3 Correspondence
 - 1.6.1.4 Equipment
 - 1.6.1.5 Reports
 - 1.6.1.6 Request For Information (RFI)
 - 1.6.1.7 Payroll Records
 - 1.6.2 Finances
 - 1.6.2.1 Pay Activity Data
 - 1.6.2.2 Payment Requests
 - 1.6.3 Quality Control (QC)
 - 1.6.3.1 Quality Control (QC) Reports
 - 1.6.3.2 Deficiency Tracking
 - 1.6.3.3 Three-Phase Control Meetings
 - 1.6.3.4 Labor and Equipment Hours
 - 1.6.3.5 Accident/Safety Reporting
 - 1.6.3.6 Definable Features of Work
 - 1.6.3.7 Activity Hazard Analysis
 - 1.6.4 Submittal Management
 - 1.6.5 Schedule
 - 1.6.6 Closeout
- 1.7 IMPLEMENTATION
- 1.8 NOTIFICATION OF NONCOMPLIANCE

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

-- End of Section Table of Contents --

SECTION 01 45 00.15 10

RESIDENT MANAGEMENT SYSTEM CONTRACTOR MODE (RMS CM)

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this section to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2024) Safety and Occupational Health
(SOH) Requirements

1.2 MEASUREMENT AND PAYMENT

The work of this section is not measured for payment. The Contractor is responsible for the work of this section, without any direct compensation other than the payment received for contract items.

1.3 CONTRACT ADMINISTRATION

The Government will use the Resident Management System (RMS) to assist in its monitoring and administration of this contract. The Government accesses the system using the Government Mode of RMS (RMS GM) and the Contractor accesses the system using the Contractor Mode (RMS CM). The term RMS will be used in the remainder of this section for both RMS GM and RMS CM. The joint Government-Contractor use of RMS facilitates electronic exchange of information and overall management of the contract. The Contractor accesses RMS to record, maintain, input, track, and electronically share information with the Government throughout the contract period in the following areas:

- Administration
- Finances
- Quality Control
- Submittal Monitoring
- Scheduling
- Closeout
- Import/Export of Data

The Contractor must be prepared to transition from the Resident Management System (RMS) to the new Construction Management Platform (CMP) at any time during the period of performance of this contract, once the USACE enterprise implements this change. The Contractor's obligation to transition from RMS to the CMP system is a requirement of this contract. Therefore, additional time or cost increase to complete this transition will not be considered. In the event of the transition, RMS project information will be migrated to CMP by the government and available in the CMP system. After the transition, the Contractor must utilize the CMP system to perform all contract administration functions previously required in RMS, including all areas listed above. After the transition,

"RMS" and "CMP" will be synonymous for all "Resident Management System" and "RMS" references in this contract's specification. To facilitate the transition from RMS to CMP, online training for the CMP system will be made available to the Contractor with enough time to train staff, prior to the transition occurring.

1.3.1 Correspondence and Electronic Communications

For ease and speed of communications, exchange correspondence and other documents in electronic format to the maximum extent feasible. Some documents, such as correspondence, including pay requests and payrolls, may be required in paper format with original signatures. Paper documents will govern in the event of discrepancy with the electronic version. Contracting Officer will determine which documents require submission of paper copies.

1.3.2 Other Factors

Other portions of this document have a direct relationship to the reporting accomplished through RMS. Particular attention is directed to FAR 52.236-15 Schedules for Construction Contracts; FAR 52.232-27 Prompt Payment for Construction Contracts; FAR 52.232-5 Payments Under Fixed-Priced Construction Contracts; Section 01 32 01.10 35 PROJECT SCHEDULE; Section 01 33 00 SUBMITTAL PROCEDURES; Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS; and 01 45 05 DREDGING/BEACH FILL PLACEMENT - QUALITY CONTROL.

1.4 RMS SOFTWARE

RMS is a web based application. Download, install and be able to utilize the latest version of RMS within 7 calendar days of receipt of the Notice to Proceed. RMS software, user manuals, access and installation instructions, program updates and training information are available from the RMS website (<https://rms.usace.army.mil>). The Government and the Contractor will have different access authorities to the same contract database through RMS. The common database will be updated automatically each time a user finalizes an entry or change.

1.5 CONTRACT DATABASE - GOVERNMENT

The Government will enter the basic contract award data in RMS prior to granting the Contractor access. The Government entries into RMS will generally be related to submittal reviews, correspondence status, and Quality Assurance(QA)comments, as well as other miscellaneous administrative information.

1.6 CONTRACT DATABASE - CONTRACTOR

Contractor entries into RMS establish, maintain, and update data throughout the duration of the contract. Contractor entries generally include prime and subcontractor information, daily reports, submittals, RFI's, schedule updates and payment requests. RMS includes the ability to import attachments and export reports in many of the modules, including submittals. The Contractor responsibilities for entries in RMS typically include the following items:

1.6.1 Administration

1.6.1.1 Contractor Information

Enter all current Contractor administrative data and information into RMS within 7 calendar days of receiving access to the contract in RMS. This includes, but is not limited to, Contractor's name, address, telephone numbers, management staff, and other required items.

1.6.1.2 Subcontractor Information

Enter all missing subcontractor administrative data and information into RMS CM within seven calendar days of receiving access to the Contract in RMS or within seven calendar days of the signing of the subcontractor agreement for agreements signed at a later date. This includes name, trade, address, phone numbers, and other required information for all subcontractors. A subcontractor is listed separately for each trade to be performed.

1.6.1.3 Correspondence

Identify all Contractor correspondence to the Government with a serial number. Prefix correspondence initiated by the Contractor's site office with "S". Prefix letters initiated by the Contractor's home (main) office with "H". Letters are numbered starting from 0001. (e.g., H-0001 or S-0001). The Government's letters to the Contractor will be prefixed with "C" or "RFP".

1.6.1.4 Equipment

Enter and maintain a current list of equipment planned for use or being used on the jobsite, including the most recent and planned equipment inspection dates in accordance with Section 01 45 05 DREDGING/BEACH FILL PLACEMENT - QUALITY CONTROL.

1.6.1.5 Reports

Track the status of the project using the reports available in RMS. The value of these reports reflects the quality of the data input. These reports include but are not limited to the Progress Payment Request worksheet, Quality Control (QC) comments, Deficiency Tracking Log, Submittal Register Status, and Three-Phase Control worksheets in accordance with Sections 01 20 00 PRICE AND PAYMENT PROCEDURES, 01 32 01.10 35 PROJECT SCHEDULE, 01 33 00 SUBMITTAL PROCEDURES, and QUALITY CONTROL 01 45 05 DREDGING/BEACH FILL PLACEMENT - QUALITY CONTROL.

1.6.1.6 Request For Information (RFI)

Create and track all Requests For Information (RFI) in the RMS Administration Module for Government review and response. Provide status at weekly progress meeting in accordance with Section 01 32 01.10 35 PROJECT SCHEDULE.

1.6.1.7 Payroll Records

Submit and maintain all payroll records in the RMS Administration Module for Government review and acceptance. Payroll records must be submitted at a minimum monthly for the duration of the contract.

1.6.2 Finances

1.6.2.1 Pay Activity Data

Develop and enter a list of pay activities in conjunction with the project schedule in accordance with Section 01 32 01.10 35 PROJECT SCHEDULE. The sum of pay activities equals the total Contract amount, including modifications. Each pay activity must be assigned to a Contract Line Item Number (CLIN). The sum of the activities assigned to a CLIN equals the amount of each CLIN.

1.6.2.2 Payment Requests

Prepare all progress payment requests using RMS. Update the work completed under the Contract at least monthly, measured as percent or as specific quantities. After the update, generate a payment request and prompt payment certification using RMS. Submit the signed prompt payment certification and payment request as well as supporting data, either electronically or by hard copy.

1.6.3 Quality Control (QC)

Enter and track implementation of the 3-phase QC Control System, QC testing, transferred and installed property and warranties in RMS. Prepare daily reports, identify and track deficiencies, document progress of work, and support other Contractor QC requirements in RMS. Maintain all data on a daily basis. Ensure that RMS reflects all quality control methods, tests and actions contained within the Contractor Quality Control (CQC) Plan and Government review comments. Provide within seven calendar days of Government acceptance of the CQC Plan.

1.6.3.1 Quality Control (QC) Reports

The Contractor's Quality Control (QC) Daily Report in RMS is the official report. The Contractor can use other supplemental formats to record QC data, but information from any supplemental formats are to be consolidated and entered into the RMS QC Daily Report. Any supplemental information may be entered into RMS as an attachment to the report. QC Daily Reports must be finalized and signed in RMS within 24 hours after the date covered by the report. Provide the Government a printed signed copy of the QC Daily Report, unless waived by the Contracting Officer.

1.6.3.2 Deficiency Tracking

Use the QC Daily Report Module to enter and track deficiencies. Deficiencies identified and entered into RMS by the Contractor or the Government will be sequentially numbered with a QC or QA prefix for tracking purposes. Enter each deficiency into RMS the same day that the deficiency is identified. Monitor, track and resolve all QC and QA entered deficiencies. A deficiency is not considered to be corrected until the Government indicates concurrence in RMS.

1.6.3.3 Three-Phase Control Meetings

Maintain scheduled and actual dates and times of preparatory and initial control meetings in RMS, and upload meeting minutes of the meetings. Worksheets for the three-phase control meetings are generated within RMS.

1.6.3.4 Labor and Equipment Hours

Enter labor and equipment exposure hours on a daily basis. Roll up the labor and equipment exposure data into a monthly exposure report.

1.6.3.5 Accident/Safety Reporting

Both the Contractor and the Government enter safety related comments in RMS as a deficiency. The Contractor must monitor, track and show resolution for safety issues in the QC Daily Report area of the RMS QC Module. In addition, follow all reporting requirements for accidents and incidents as required in EM 385-1-1, Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS and as required by any other applicable Federal, State or local agencies.

1.6.3.6 Definable Features of Work

Enter each feature of work, as defined in the approved CQC Plan, into the RMS QC Module. A feature of work may be associated with a single or multiple pay activities, however a pay activity is only to be linked to a single feature of work.

1.6.3.7 Activity Hazard Analysis

Import activity hazard analysis electronic document files into the RMS QC Module utilizing the document package manager. Prepare activity hazard analysis in accordance with Section 01 35 26 GOVERNMENT SAFETY REQUIREMENTS.

1.6.4 Submittal Management

Enter all current submittal register data and information into RMS within seven calendar days of receiving access to the Contract in RMS. Contractor is responsible for entering Activity Numbers for each submittal item corresponding to their Schedule. The Contractor will import the information shown in the submittal register following the specification Section 01 33 00 SUBMITTAL PROCEDURES unless otherwise directed by the Contracting Officer. Group electronic submittal documents into transmittal packages to send to the Government, except very large electronic files, samples, spare parts, mock ups, color boards, or where hard copies are specifically required. Track transmittals and update the submittal register in RMS on a daily basis throughout the duration of the Contract.

Contractor may request submittals to be added to RMS throughout the lifecycle of the project. Approval/disapproval of Contractor requests is at the discretion of the Contracting Officer. Submittals entry into RMS must be updated daily or as practical for performance of the Contract, as approved by the Contracting Officer.

1.6.5 Schedule

Enter and update the contract project schedule in RMS by either manually entering all schedule data or by importing the Standard Data Exchange Format (SDEF) file, based on the requirements in Section 01 32 01.10 35 PROJECT SCHEDULE.

U.S. Naval Station Mayport
Duval County, Florida

1.6.6 Closeout

Closeout documents, processes and forms are managed and tracked in RMS by both the Contractor and the Government. Ensure that all closeout documents are entered, completed and documented within RMS.

1.7 IMPLEMENTATION

Use of RMS as described in the preceding paragraphs is mandatory. Ensure that sufficient resources are available to maintain contract data within the RMS system. RMS is an integral part of the Contractor's required management of quality control.

1.8 NOTIFICATION OF NONCOMPLIANCE

Take corrective action within seven calendar days after receipt of notice of RMS non-compliance by the Contracting Officer.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 45 05

DREDGING/BEACH FILL PLACEMENT - QUALITY CONTROL

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 PAYMENT
- 1.3 SUBMITTALS
- 1.4 GENERAL REQUIREMENTS
- 1.5 QUALITY CONTROL (QC) PROGRAM REQUIREMENTS
 - 1.5.1 Meetings
 - 1.5.1.1 Quality Control Plan Meeting
 - 1.5.1.2 Coordination and Mutual Understanding Meeting
 - 1.5.1.2.1 Purpose
 - 1.5.1.2.2 Coordination of Activities
 - 1.5.1.2.3 Attendees
 - 1.5.1.3 Quality Control (QC) Meetings
 - 1.5.2 Contractor Quality Control (CQC) Plan
 - 1.5.2.1 Content of Contractor Quality Control (CQC) Plan
 - 1.5.3 Acceptance of the Quality Control (QC) Plan
 - 1.5.4 Notification of Changes
- 1.6 QUALITY CONTROL (QC) ORGANIZATION
 - 1.6.1 Personnel Requirements
 - 1.6.2 Quality Control (QC) Manager
 - 1.6.2.1 Duties
 - 1.6.2.2 Qualifications
 - 1.6.2.3 Construction Quality Management Training
 - 1.6.3 Organizational Changes
 - 1.6.4 Alternate Quality Control (QC) Manager Duties and Qualifications
 - 1.6.5 Quality Control (QC) Specialists
 - 1.6.5.1 Professional Surveyor and Mapper (PSM)
- 1.7 SUBMITTAL AND DELIVERABLES REVIEW AND APPROVAL
- 1.8 THREE PHASES OF CONTROL
 - 1.8.1 Preparatory Phase
 - 1.8.2 Initial Phase
 - 1.8.3 Follow-Up Phase
 - 1.8.4 Additional Preparatory and Initial Phases
 - 1.8.5 Deficiency/Rework Items List
- 1.9 TESTING
 - 1.9.1 Laboratory Accreditation Authorities
 - 1.9.2 Capability Check
 - 1.9.2.1 Capability Recheck
 - 1.9.3 Test Results
- 1.10 COMPLETION INSPECTIONS
 - 1.10.1 Post-Construction Measurement and Inspection
 - 1.10.2 Pre-Final Inspection
 - 1.10.3 Final Acceptance Inspection
- 1.11 QUALITY CONTROL (QC) CERTIFICATIONS

U.S. Naval Station Mayport
Duval County, Florida

- 1.11.1 Contractor Quality Control (CQC) Report Certification
 - 1.12 DOCUMENTATION AND INFORMATION FOR THE CONTRACTING OFFICER
 - 1.12.1 Construction Documentation
 - 1.12.2 Quality Control Activities
 - 1.12.3 Verification Statement
 - 1.12.4 Reports from the Quality Control (QC) Specialist(s)
 - 1.12.5 Quality Control Validation
 - 1.12.6 Testing Plan and Log
 - 1.13 NOTIFICATION ON NON-COMPLIANCE
 - 1.14 DELIVERY, STORAGE, AND HANDLING
- PART 2 PRODUCTS (NOT USED)
- PART 3 EXECUTION (NOT USED)

-- End of Section Table of Contents --

SECTION 01 45 05

DREDGING/BEACH FILL PLACEMENT - QUALITY CONTROL

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM D3740 (2023) Minimum Requirements for Agencies Engaged in the Testing and/or Inspection of Soil and Rock as Used in Engineering Design and Construction

ASTM E329 (2023) Standard Specification for Agencies Engaged in Construction Inspection, Testing, or Special Inspection

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2024) Safety and Occupational Health (SOH) Requirements

1.2 PAYMENT

Separate payment will not be made for providing and maintaining an effective Quality Control program. Include all associated costs in the applicable Line Items and Pricing Schedule item.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES.

SD-01 Preconstruction Submittals

Contractor Quality Control (CQC) Plan; G, R0

SD-06 Test Reports

Verification Statement

1.4 GENERAL REQUIREMENTS

Establish and maintain an effective quality control (QC) system that complies with FAR 52.246-12 Inspection of Construction. QC is comprised of plans, procedures, and organization necessary to produce an end product

that complies with the Contract requirements. The QC system covers all construction operations, both onsite and offsite, and must be keyed to the proposed construction sequence. The Quality Control Manager, Superintendent, Site Safety and Health Officer (SSHO), and all on-site supervisors are responsible for the quality of work and are subject to removal by the Contracting Officer for non-compliance with the quality requirements specified in the Contract. The Quality Control Manager must maintain a physical presence at the work site at all times and is the primary individual responsible for all quality control.

1.5 QUALITY CONTROL (QC) PROGRAM REQUIREMENTS

Establish and maintain a QC program as described in this section. The QC program consists of a QC Organization, QC Plan, QC Plan Meeting(s), a Coordination and Mutual Understanding Meeting, QC meetings, three phases of control, submittal review and approval, testing, completion inspections, QC certifications, and documentation necessary to provide materials, equipment, workmanship, fabrication, construction and operations that comply with the requirements of this Contract. The QC program must cover on-site and off-site work and be keyed to the work sequence. No construction work or testing may be performed unless the QC Manager is on the work site. The QC Manager must report to an officer of the firm and not be subordinate to the Project Superintendent or the Project Manager. The QC Manager, Project Superintendent and Project Manager must work together effectively. Although the QC Manager is the primary individual responsible for quality control, all individuals will be held responsible for the quality of work on the job.

1.5.1 Meetings

1.5.1.1 Quality Control Plan Meeting

Prior to submission of the QC Plan, the Contractor may request a meeting with the Contracting Officer to discuss the QC Plan requirements of this Contract.

The purpose of this meeting is to develop a mutual understanding of the QC Plan requirements prior to plan development and submission and to agree on the Contractor's list of Definable Feature of Work (DFOW).

1.5.1.2 Coordination and Mutual Understanding Meeting

After the Preconstruction Conference, before start of construction, and prior to acceptance by the Government of the CQC Plan, meet with the Contracting Officer and discuss the Contractor's quality control system. During the meeting, a mutual understanding of the system details must be developed, including the forms for recording the CQC operations, control activities, testing, administration of the system for both onsite and offsite work, and the interrelationship of Contractor's Management and control with the Government's Quality Assurance. Minutes of the meeting will be prepared by the Government and signed by the Contractor and the Government. Provide a copy of the signed minutes to all attendees and include in the QC Plan. At a minimum the Coordination and Mutual Understanding Meeting must be repeated when a new QC Manager is appointed. There can be other occasions when subsequent conferences will be called by either party to reconfirm mutual understandings or address deficiencies in the CQC system or procedures which can require corrective action by the Contractor.

1.5.1.2.1 Purpose

The purpose of this meeting is to develop a mutual understanding of the QC details, including documentation, administration for on-site and off-site work, design intent, environmental requirements and procedures, coordination of activities to be performed, and the coordination of the Contractor's management, production, and QC personnel. At the meeting, the Contractor must explain in detail how three phases of control will be implemented for each DFO, as well as how each DFO will be affected by each management plan or requirement as listed below:

- a. QC reporting and documentation.
- b. Procedures for noise and acoustics management.
- c. Environmental Protection Plan.
- d. Environmental regulatory requirements.
- e. Safety.
- f. Turbidity monitoring.
- g. Plant and equipment location and positioning.
- h. Endangered species monitoring.

1.5.1.2.2 Coordination of Activities

Coordinate activities included in various sections to assure efficient and orderly installation of each component. Coordinate operations included under different sections that are dependent on each other for proper installation and operation.

1.5.1.2.3 Attendees

As a minimum, the Contractor's personnel required to attend include an officer of the firm, the Project Manager, Project Superintendent, QC Manager, Alternate QC Manager, QC Specialists, Environmental Manager, and subcontractor representatives. Each subcontractor who will be assigned QC responsibilities must have a principal of the firm at the meeting.

1.5.1.3 Quality Control (QC) Meetings

After the start of construction, conduct weekly QC meetings led by the QC Manager at the work site with the Project Superintendent, the QC Specialists, and the other personnel as necessary. The QC Manager is to prepare the minutes of the meeting and provide a copy to the Contracting Officer within 2 working days after the meeting. The Contracting Officer may attend these meetings. As a minimum, accomplish the following at each meeting:

- a. Review the minutes of the previous meeting.
- b. Review the schedule and the status of work and deficiencies/rework. Review the most current approved schedule (in accordance with schedule specification) and the status of work and deficiencies/rework.
- c. Review the status of submittals and Request For Information (RFIs).

- d. Review the work to be accomplished in the next 3 weeks as defined by the schedule section paragraph WEEKLY PROGRESS MEETINGS in Section 01 32 01.10 35 PROJECT SCHEDULE and all documentation required for that work.
- e. Review Testing Plan and Log including status of tests performed since last QC Meeting.
- f. Resolve QC and production problems. Discuss status of pending change orders.
- g. Address items that may require revising the QC Plan.
- h. Review Accident Prevention Plan (APP) and effectiveness of the safety program.
- i. Review environmental requirements and procedures.
- j. Review Environmental Management Plan.
- k. Review Waste Management Plan.
- l. Review the status of training completion.

1.5.2 Contractor Quality Control (CQC) Plan

Submit no later than 30 days after receipt of notice to proceed, the CQC Plan proposed to implement the requirements FAR 52.246-12 Inspection of Construction. The Government will consider an interim plan for the first 30 days of operation. Construction may be permitted to begin only after acceptance of the CQC Plan and other Contract requirements or acceptance of an interim plan applicable to the particular feature of work to be started. Work outside of the accepted interim plan will not be permitted to begin until acceptance of a CQC Plan or another interim plan containing the additional work.

1.5.2.1 Content of Contractor Quality Control (CQC) Plan

Provide a CQC Plan, prior to start of construction that includes a table of contents, with major sections identified, pages numbered sequentially, and that documents the proposed methods and responsibilities for accomplishing quality control during the construction of the project. The CQC Plan must at a minimum include the following sections:

- a. A description of the quality control organization and acknowledgment that the CQC staff will implement the three phase control system for all aspects of the work specified.
- b. An organizational chart showing the quality control organization with individual names and job titles and lines of authority up to an executive of the company at the home office.
- c. NAMES AND QUALIFICATIONS: Names and qualifications, in resume format, (including position titles and durations for qualifying experiences) for each person in the QC organization. Include the Construction Quality Management (CQM) for Contractors course certifications for the QC personnel as required by the paragraph CONSTRUCTION QUALITY MANAGEMENT TRAINING.

- d. DUTIES, RESPONSIBILITY AND AUTHORITY OF QC PERSONNEL: Duties, responsibilities, and authorities of each person in the QC organization.
- e. OUTSIDE ORGANIZATIONS: A listing of outside organizations, such as architectural and consulting engineering firms, that will be employed by the Contractor and a description of the services these firms will provide.
- f. APPOINTMENT LETTERS: Letters signed by an officer of the firm appointing the QC Manager and Alternate QC Manager, and stating that they are responsible for implementing and managing the QC program as described in this Contract. Include in this letter the responsibility of the QC Manager and Alternate QC Manager to implement and manage the three phases of control, and their authority to stop work that is not in compliance with the Contract. Letters of direction are to be issued by the QC Manager to all other QC Specialists or quality control representatives outlining their duties, authorities, and responsibilities. Include copies of the letters in the QC Plan.
- g. SUBMITTAL PROCEDURES AND INITIAL SUBMITTAL REGISTER: Procedures for reviewing, approving, scheduling, and managing submittals, including those of subcontractors, offsite fabricators, suppliers, and purchasing agents. Provide the name(s) of the person(s) in the QC organization authorized to review and certify submittals prior to approval. Provide the initial submittal of the Submittal Register as specified in Section 01 33 00 SUBMITTAL PROCEDURES.
- h. TESTING LABORATORY INFORMATION: Testing laboratory information required by the paragraph ACCREDITATION REQUIREMENTS, as applicable.
- i. TESTING PLAN AND LOG: A Testing Plan and Log that includes the tests required, associated feature of work required, referenced by the specification paragraph number requiring the test, the frequency, and the person responsible for each test.
- j. Procedures to complete construction deficiencies from identification through acceptable corrective action. Establish verification procedures that identified deficiencies have been corrected. This phase is performed prior to beginning work on each definable feature of work, after all required plans, documents, materials are approved, and after copies are at the work site.
- k. Reporting procedures, including proposed reporting formats.
- l. Procedures for submitting and reviewing design changes/variations prior to submission to the Contracting Officer.
- m. LIST OF DEFINABLE FEATURES: A Definable Feature of Work (DFOW) is a task that is separate and distinct from other tasks and has control requirements and work crews unique to that task. A DFOW is identified by different trades or disciplines, or it is work by the same trade in a different environment. A DFOW is by definition any item or activity on the construction schedule, and the schedule specification provides direction regarding how the DFOWs are to be structured. Include in the list of DFOWs for all activities on the Construction Schedule. Although each section of the specifications can generally be considered as a definable feature of work, there are frequently more

than one definable features under a particular section. Identify the specification section number and schedule activity ID for each DFW listed. The DFW list will be reviewed in coordination with the construction schedule and agreed upon during the Coordination of Mutual Understanding Meeting.

- n. PROCEDURES FOR PERFORMING AND TRACKING THE THREE PHASES OF CONTROL: Identify procedures used to ensure the three phases of control to manage the quality on this project. For each Definable Feature of Work (DFOW), a Preparatory and Initial phase checklist will be filled out during the Preparatory and Initial phase meetings. Conduct the Preparatory and Initial Phases and meetings with a view towards obtaining quality construction by planning ahead and identifying potential problems for each DFW.
- o. PROCEDURES FOR COMPLETION INSPECTION: Procedures for identifying and documenting the completion inspection process. Include in these procedures the responsible party for punch out inspection, pre-final inspection, and final acceptance inspection.
- p. TRAINING PROCEDURES AND TRAINING LOG: Procedures for coordinating and documenting the training of personnel required by the Contract.
- q. ORGANIZATION AND PERSONNEL CERTIFICATIONS LOG: Procedures for coordinating, tracking and documenting all certifications required for entities such as subcontractors, testing laboratories, suppliers, and personnel. The QC Manager will ensure that certifications are current, appropriate for the work being performed, and will not lapse during any period of the Contract that the work is being performed.

1.5.3 Acceptance of the Quality Control (QC) Plan

The Contracting Officer's acceptance of the Contractor QC Plan, or interim plan applicable to the particular feature of work to be started, is required prior to the start of construction. The Government may consider an interim plan for the first 30 days of operation. Work outside of the accepted interim plan will not be permitted to begin until acceptance of a CQC Plan or another interim plan containing the additional work. The Contracting Officer reserves the right to require changes in the QC Plan and operations as necessary, including removal or addition of personnel, to ensure the specified quality of work. The Contracting Officer reserves the right to interview any member of the QC organization at any time to verify the submitted qualifications. All QC organization personnel are subject to acceptance by the Contracting Officer. The Contracting Officer may require the removal of any individual for non-compliance with quality requirements specified in the Contract.

1.5.4 Notification of Changes

Notify the Contracting Officer, in writing, of any proposed changes in the QC Plan or changes to the QC organization personnel. Proposed changes are subject to acceptance by the Contracting Officer.

1.6 QUALITY CONTROL (QC) ORGANIZATION

1.6.1 Personnel Requirements

The requirements for the CQC organization are a Site Safety and Health Officer (SSHO), QC Manager, and enough qualified personnel to ensure

safety and contract quality control for workmanship and materials, including environmental compliance. The SSHO reports directly to a senior project (or corporate) official independent from the QC Manager. The SSHO will also serve as a member of the CQC Staff Personnel identified in the technical provisions as requiring specialized skills to assure the required work is being performed properly. The CQC staff always maintains a presence at the site during progress of the work and have complete authority and responsibility to take any action necessary to ensure Contract compliance. The CQC staff will be subject to acceptance by the Contracting Officer. Provide adequate office space, filing systems and other resources as necessary to maintain an effective and fully functional CQC organization. Promptly complete and furnish all letters, material submittals, shop drawing submittals, schedules, and all other project documentation to the CQC organization. The CQC organization is responsible for always maintaining these documents and records at the site, except as otherwise acceptable to the Contracting Officer.

1.6.2 Quality Control (QC) Manager

1.6.2.1 Duties

Provide a QC Manager at the work site to implement and manage the QC program. The QC Manager must be employed by the Prime Contractor. The only duties and responsibilities of the QC Manager are to manage and implement the QC program on this Contract. The QC Manager must attend the Coordination and Mutual Understanding Meeting, perform the three phases of control, perform submittal review and approval, ensure testing is performed and provide QC certifications and documentation required in this Contract. The QC Manager is responsible for managing and coordinating the three phases of control and documentation performed by the QC Specialists, testing laboratory personnel and any other inspection and testing personnel required by this Contract. The QC Manager is the manager of all QC activities.

1.6.2.2 Qualifications

The QC Manager must be an individual with a minimum of 3 years combined experience in the following positions: Project Superintendent, QC Manager, Project Manager, Project Engineer or Construction Manager on similar size and type construction Contracts which included the major trades that are part of this Contract. The individual must be familiar with the requirements of [EM 385-1-1](#) and have experience in the areas of hazard identification, safety compliance, and sustainability.

The QC Manager and all members of the QC organization must be capable of reading, writing, and conversing fluently in the English language.

1.6.2.3 Construction Quality Management Training

In addition to the above experience and education requirements, the QC Manager must have completed the CQM for Contractors course. If the QC Manager does not have a current certification, obtain the CQM for Contractors course certification within 90 days of award. This course is periodically offered by the Naval Facilities Engineering Systems Command and the Army Corps of Engineers. Contact the Contracting Officer for information on the next scheduled class.

The Construction Quality Management Training certificate expires after 5 years. If the QC Manager's certificate has expired, retake the course to

remain current.

1.6.3 Organizational Changes

Maintain the QC staff with personnel as required by the specification section at all times. When it is necessary to make changes to the QC staff, revise the CQC Plan to reflect the changes and submit the changes to the Contracting Officer for acceptance.

1.6.4 Alternate Quality Control (QC) Manager Duties and Qualifications

Designate an alternate for the QC Manager at the work site to serve in the event of the designated QC Manager's absence. The qualification requirements for the Alternate QC Manager must be the same as for the QC Manager.

1.6.5 Quality Control (QC) Specialists

Provide a separate QC Specialist at the work site for each of the areas as listed in the Matrix listed below, who must assist and report to the QC Manager and who may perform production related duties but must be allowed sufficient time to perform their assigned quality control duties. These individuals or specialized technical companies are employees of the Prime or subcontractor. QC Specialists must be physically present at the work site with frequency as indicated in the Experience Matrix below, to participate in the QC Meetings, perform the three phases of control, including participation in Preparatory and Initial Phase meetings, and to perform and document Follow-up inspections as an extension of the QC Manager for each definable feature of work in their area of responsibility. QC Specialist must assist and be present for training events, and Critical System Acceptance inspections by the Government. Qualification, experience, Area of Responsibility, and frequency of QC surveillance are provided in Matrix listed herein.

Experience Matrix		
1. Area	2-1. Qualification 2-2. Experience	3-1. Area of Responsibility 3-2. Frequency
Surveyor	Persons currently working under direct supervision of Professional Surveyor and Mapper (PSM).	Note: See paragraph Professional Surveyor and Mapper (PSM).

1.6.5.1 Professional Surveyor and Mapper (PSM)

Contract requires survey quality control, quantity surveys for progress payments, and as-built surveys to be certified by a PSM registered in the State the work is being performed. This person must directly supervise contractor surveys in accordance with FAR Clause 52.236-16, QUANTITY SURVEYS. PSM must certify field notes, computations, and other records relating to Quantity or As-Built surveys and layout of work.

1.7 SUBMITTAL AND DELIVERABLES REVIEW AND APPROVAL

Procedures for submission, review and approval of submittals are described in Section 01 33 00 SUBMITTAL PROCEDURES. Procedures must include field

verification of relevant dimensions and component characteristics by the QC organization prior to submittal being sent to the Contracting Officer. The CQC organization is responsible for certifying that all submittals and deliverables are in compliance with the Contract.

1.8 THREE PHASES OF CONTROL

CQC enables the Contractor to ensure that the construction, including that of subcontractors and suppliers, complies with the requirements of the Contract. At least three phases of control must be conducted by the QC Manager to adequately cover both on-site and off-site work for each definable feature of the construction work as follows:

1.8.1 Preparatory Phase

Document the results of the preparatory phase actions by separate minutes prepared by the QC Manager and attach to the daily CQC report. Instruct applicable workers as to the acceptable level of workmanship required to meet Contract specifications.

Notify the Contracting Officer at least 2 business days in advance of each preparatory phase meeting. The meeting will be conducted by the QC Manager and attended by the QC Specialists, the Project Superintendent, SSHO, Competent Person(s), and the foreman responsible for the DFO. When the DFO will be accomplished by a subcontractor, that subcontractor's foreman must attend the preparatory phase meeting. This phase is performed prior to beginning work on each definable feature of work, after all required plans/documents/materials are approved/accepted, and after copies are at the work site. Perform the following prior to beginning work on each DFO:

- a. Review each paragraph of the applicable specification sections, reference codes, and standards. Make available during the preparatory inspection a copy of those sections of referenced codes and standards applicable to that portion of the work to be accomplished in the field. Maintain and make available in the field for use by Government personnel until final acceptance of the work.
- b. Review the Contract drawings.
- c. Verify that field measurements are as indicated on construction or shop drawings or both before confirming product orders, to minimize waste due to excessive materials.
- d. Verify that appropriate shop drawings and submittals for materials and equipment have been submitted and approved.
- e. Review the testing plan and ensure that provisions have been made to provide the required QC testing.
- f. Examine the work area to ensure that the required preliminary work has been completed and complies with the Contract and ensure any deficiencies/rework items in the preliminary work have been corrected and confirmed by the Contracting Officer.
- g. Review coordination of equipment/material delivery to designated prepared areas to execute the work.
- h. Examine the required materials, equipment and sample work to ensure

that they are on hand and conform to the approved shop drawings and submitted data and are properly stored.

- i. Check to assure that all materials and equipment have been inspected, tested, submitted, and approved.
- j. Discuss specific controls to be used, construction methods, construction tolerances, workmanship standards, and the approach that will be used to provide quality construction by planning ahead and identifying potential problems for each DFO. Ensure any portion of the plan requiring separate Contracting Officer acceptance has been approved.
- k. Review the APP and appropriate Activity Hazard Analysis (AHA) to ensure that applicable safety requirements are met, and that required Safety Data Sheets (SDS) are submitted.
- l. Discussion of procedures for controlling quality of the work including repetitive deficiencies. Document construction tolerances and workmanship standards for that feature of work.
- m. Discuss schedule and execution of the initial control phase and confirmation or construction quality compliance.
- n. Before and After Dredge Survey Requirements and Procedures.

1.8.2 Initial Phase

Notify the Contracting Officer at least 2 business days in advance of each initial phase. When construction crews are ready to start work on a DFO, conduct the initial phase with the QC Specialists, the Project Superintendent, SSHO, Competent Person(s), and the foreman responsible for that DFO. Observe the initial segment of the DFO to ensure that the work complies with Contract requirements. Document the results of the initial phase in the daily CQC Report and in the Initial Phase Checklist. Repeat the initial phase for each new crew to work on-site when acceptable levels of specified quality are not being met. Indicate the exact location of initial phase for definable feature of work for future reference and comparison with follow-up phases. Perform the following for each DFO:

- a. Check work to ensure that it is in full compliance with Contract requirements. Review minutes of the preparatory meeting.
- b. Verify adequacy of controls to ensure full Contract compliance. Verify required control inspection and testing comply with the Contract.
- c. Establish level of workmanship and verify that it meets the minimum acceptable workmanship standards. Compare with required sample panels as appropriate.
- d. Resolve any issues with environmental compliance observations and testing, quantity survey, and/or control.
- e. Ensure that testing is performed by the approved laboratory.
- f. Check work procedures for compliance with the APP and the appropriate AHA to ensure that applicable safety requirements are met.

- g. Review project specific work plans to ensure all preparatory work items have been completed and documented.

1.8.3 Follow-Up Phase

Perform the following for on-going DFOW daily, or more frequently as necessary, until the completion of each DFOW. The Final Follow-Up for any DFOW will clearly note in the daily report the DFOW is completed, and all deficiencies/rework items have been completed in accordance with the paragraph DEFICIENCY/REWORK ITEMS LIST. Each DFOW that has completed the Initial Phase and has not completed the Final Follow-up must be included on each daily report. If no work was performed on that DFOW for the period of that daily report, it must be so noted. Document all Follow-Up activities for DFOWs in the daily CQC Report:

- a. Ensure the work including control testing complies with Contract requirements until completion of that particular work feature. Record checks in the CQC documentation.
- b. Maintain the quality of workmanship required.
- c. Ensure that testing is performed by the approved laboratory.
- d. Ensure that deficiencies/rework items are being corrected. Conduct final follow-up checks and correct all deficiencies prior to the start of additional features of work which may be affected by the deficient work.
- e. Do not build upon nor conceal non-conforming work.
- f. Assure manufacturers' representatives have performed necessary inspections if required and perform safety inspections.

1.8.4 Additional Preparatory and Initial Phases

Conduct additional preparatory and initial phases on the same DFOW if the quality of on-going work is unacceptable, if there are changes in the applicable QC organization, if there are changes in the on-site production supervision or work crew, if work on a DFOW has not started within 45 days of the initial preparatory meeting or has resumed after 45 days of inactivity, or if other problems develop.

1.8.5 Deficiency/Rework Items List

The QC Manager must maintain a list of work that does not comply with the Contract, identifying what items need to be corrected, the activity ID number associated with the item, the date the item was originally discovered, the date the item will be corrected by, and the date the item was corrected.

The list shall be reviewed at each weekly QC Meeting:

- a. There is no requirement to report a deficiency/rework item that is corrected the same day it is discovered.
- b. No successor task may be advanced beyond the preparatory phase meeting until all deficiencies/rework items have been cleared by the QC Manager and concurred with by the Contracting Officer. This must be

confirmed as part of the Preparatory Phase activities.

- c. Attach a copy of the "Deficiency/Rework Items List" to the last daily CQC Report of each month.
- d. The Contractor is responsible for including those items identified by the Contracting Officer.
- e. All deficiencies/rework items must be confirmed as corrected by the QC Manager, and concurred by the Contracting Officer, prior to commencement of any completion inspections per paragraph COMPLETION INSPECTIONS unless specifically exempted by the Contracting Officer.
- f. Non-Compliance with these requirements shall be grounds for removal in accordance with paragraph ACCEPTANCE OF THE QUALITY CONTROL (QC) PLAN.
- g. All delays, concurrent or related to failure to manage, monitor, control, and correct deficiencies/rework items are entirely the responsibility of the Contractor and shall not be made the subject, or any component of any request for additional time or compensation.

1.9 TESTING

Perform specified or required tests to verify that control measures are adequate to provide a product which conforms to Contract requirements. Upon request, furnish to the Government duplicate samples of test specimens for possible testing by the Government. Testing includes operation and acceptance tests when specified. Procure the services of an U.S. Army Corps of Engineers approved testing laboratory or establish an approved testing laboratory. Perform the following activities and record and provide the following data:

- a. Verify that testing procedures comply with Contract requirements.
- b. Verify that facilities and testing equipment are available and comply with testing standards.
- c. Check test instrument calibration data against certified standards.
- d. Verify that recording forms and test identification control number system, including all test documentation requirements, have been prepared.
- e. Record results of all tests taken, both passing and failing on the CQC report for the date taken. Specification paragraph reference, location where tests were taken, and the sequential control number identifying the test. If approved by the Contracting Officer, actual test reports are submitted later with a reference to the test number and date taken. Provide an information copy of tests performed by an offsite or commercial test facility directly to the Contracting Officer. Failure to submit timely test reports as stated results in nonpayment for related work performed and disapproval of the test facility for this Contract.

1.9.1 Laboratory Accreditation Authorities

All testing laboratories must be validated by the USACE Material Testing Center (MTC) for the tests to be performed. Information on the USACE MTC with web-links to both a list of validated testing laboratories and for

the laboratory inspection request for can be found at:
<https://mtc.erdcdren.mil>

1.9.2 Capability Check

The Contracting Officer retains the right to check laboratory equipment in the proposed laboratory and the laboratory technician's testing procedures, techniques, and other items pertinent to testing, for compliance with the standards set forth in this Contract. Laboratories utilized for testing soils, concrete, asphalt, and steel must meet criteria detailed in [ASTM D3740](#) and [ASTM E329](#).

1.9.2.1 Capability Recheck

If the selected laboratory fails the capability check, the Contractor will bear all costs associated with each succeeding recheck of the laboratory or the checking of a subsequent laboratory.

1.9.3 Test Results

Cite applicable Contract requirements, tests or analytical procedures used. Provide actual results and include a statement that the item tested or analyzed conforms or fails to conform to specified requirements. If the item fails to conform, notify the Contracting Officer immediately. Conspicuously stamp the cover sheet for each report in large red letters "CONFORMS" or "DOES NOT CONFORM" to the specification requirements, whichever is applicable. Test results must be signed by a testing laboratory representative authorized to sign certified test reports. Furnish the signed reports, certifications, and other documentation to the Contracting Officer via the QC Manager.

1.10 COMPLETION INSPECTIONS

1.10.1 Post-Construction Measurement and Inspection

Near end of work on each acceptance section, notify Contracting Officer to conduct post-construction survey and inspection to ensure contract conformance. Contracting Officer reserves right to witness Contractor performed survey to conduct quality assurance or pay on Contractor conducted survey. Submit certified survey data and correct contract drawings to show as-built information. Spot check grading for berm for beach placement and confirm beach fill is within allowable tolerances. Notify Contracting Officer to schedule a joint inspection of each acceptance section once deficiencies have been corrected.

1.10.2 Pre-Final Inspection

Near completion of project Contracting Officer and Contractor will conduct a Pre-Final Inspection. In addition to completion of quantity survey, Contracting Officer will inspect for demobilization of temporary facilities and cleanup of staging areas used to ensure contract conformance. Contractor and Contracting Officer will inspect for final grading, tilling, and removal of escarpments. Contracting Officer will perform Pre-Final Inspection to verify work is complete and ready for Owner acceptance. Contracting Officer Pre-Final Inspection may result in additional work such as beach tilling to be done. Contractor's CQC System Manager must ensure items are corrected before notifying Contracting Officer so that a Final Inspection with the Customer can be scheduled. Correct any items noted on Pre-Final Inspection in a timely manner.

Pre-Final Inspection and deficiency corrections must be accomplished within project completion period including time for establishment of dune vegetation.

1.10.3 Final Acceptance Inspection

Final Inspection will be scheduled by Contracting Officer based upon results of Pre-Final Inspection. Notify the Contracting Officer in writing when ready for Final Inspection and propose a date for the Final Inspection. Include a statement in the notification assuring the Contracting Officer that deficiencies will be corrected and the work will be acceptable by the date proposed for Final Inspection. The Contracting Officer will schedule the Final Inspection within 14 working days of Contractor's notification. The Contracting Officer will notify Owner or Sponsor to attend a Final Inspection. Contractor, Project Manager/Superintendent, and CQC System Manager must attend the Final Inspection. In addition to Contracting Officer and Owner or Sponsor, persons from local government or other agencies may attend. Contractor's failure to have contract work completed for this inspection will be cause for Contracting Officer to bill the Contractor for Contracting Officer's additional inspection cost in accordance with the Clause INSPECTION OF CONSTRUCTION.

1.11 QUALITY CONTROL (QC) CERTIFICATIONS

1.11.1 Contractor Quality Control (CQC) Report Certification

Contain the following statement within the CQC Report: "On behalf of the Contractor, I certify that this report is complete and correct and equipment and material used, and work performed during this reporting period is in compliance with the Contract drawings and specifications to the best of my knowledge, except as noted in this report."

1.12 DOCUMENTATION AND INFORMATION FOR THE CONTRACTING OFFICER

Maintain current and complete records of on-site and off-site QC program operations and activities.

Contact the Contracting Officer for sample forms or print from RMS-QCS as needed. Prior to commencing work on construction, the Contractor must obtain a copy set of the current report forms. The report forms will consist of the Contractor Quality Control (CQC) Report, CQC Report (Continuation Sheet), Contractor Production Report, Contractor Production Report (Continuation Sheet), Preparatory Phase Checklist, Initial Phase Checklist, Testing Plan and Log, and Rework Items List. Unless otherwise provided by the Contracting Officer, Contractor may use the forms provided as related material located at <https://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/ufgs-01-45-00>.

1.12.1 Construction Documentation

Reports are required for each day that work is performed and must be attached to the Contractor Quality Control Report prepared for the same day. Maintain current and complete records of on-site and off-site QC program operations and activities. Reports are required for each day work is performed. Account for each calendar day throughout the life of the Contract.

The Project Superintendent and the QC Manager must prepare and sign the

Contractor Production and CQC Reports, respectively. Every space on the forms must be filled in. Use N/A if nothing can be reported in one of the spaces. The reporting of work must be identified by terminology consistent with the construction schedule. In the "Remarks" sections of the reports, enter pertinent information including directions received, problems encountered during construction, work progress and delays, conflicts or errors in the drawings or specifications, field changes, safety hazards encountered, instructions given and corrective actions taken, delays encountered, a record of visitors to the work site, quality control problem areas, deviations from the QC Plan, construction deficiencies encountered, and meetings held. For each entry in the report(s), identify the Schedule Activity No. that is associated with the entered remark.

1.12.2 Quality Control Activities

CQC and Contractor Production reports will be prepared daily to maintain current records providing factual evidence that required quality control activities and tests have been performed. Include in these records the work of subcontractors and suppliers on an acceptable form that includes, as a minimum, the following information:

- a. The name and area of responsibility of the Contractors and any subcontractors.
- b. Operating plant/equipment with hours worked, idle, or down for repair.
- c. Work performed each day, giving location, description, and by whom. When a Network Analysis Schedule (NAS) is used, identify each item of work performed each day by NAS activity number.
- d. Control phase activities performed. Preparatory, and Initial phase Checklists associated with the DFOW referenced to the construction schedule. Follow-up phase activities identified to the DFOW. If testing or specific QC Specialist activities are associated with the Follow-up phase activities for a specific DFOW note this and include those reports.
- e. Test and control activities performed with results and references to specifications and drawings requirements. Identify the control phase (Preparatory, Initial, Follow-up). List of deficiencies noted, along with corrective action in accordance with the paragraph DEFICIENCY/REWORK ITEMS LIST.
- f. Quantity of materials received at the site with statement as to acceptability, storage, and reference to specifications and drawings requirements.
- g. Submittals and deliverables reviewed, with Contract reference, by whom, and action taken.
- h. Offsite surveillance activities, including actions taken.
- i. Job safety evaluations stating what was checked, results, and instructions or corrective actions.
- j. Instructions given/received and conflicts in plans and specifications.

1.12.3 Verification Statement

Indicate a description of trades working on the project; the number of personnel working; weather conditions encountered; and any delays encountered. Cover both conforming and deficient features and include a statement that equipment and materials incorporated in the work and workmanship comply with the Contract.

Furnish the original and one copy of these records in report form to the Government by 10:00 AM the next working day after the date covered by the report. As a minimum, prepare and submit one report for every 7 days of no work and on the last day of a no work period. All calendar days need to be accounted for throughout the life of the Contract. The first report following a day of no work will be for that day only. Reports need to be signed and dated by the QC Manager. Include copies of test reports and copies of reports prepared by all subordinate quality control personnel within the QC Manager Report.

1.12.4 Reports from the Quality Control (QC) Specialist(s)

Document inspection results on a QC specialist report prepared each day work is performed in their area of responsibility. The report must include a description of the visual inspection or observation performed, a written summary of findings, a conclusion on compliance with the Contract documents, and signature of the QC Specialist. In person inspections must be documented with Video/photographs. Video/photographic documentation of deficiencies must include before and after conditions and physical measurements, as necessary. Forward the QC daily report to the QC Manager who must include the report with the submission of their daily QC Report to the Government each day. Every site visit by the QC Specialist must be documented on a QC Specialist daily report.

1.12.5 Quality Control Validation

Establish and maintain the following in an electronic folder. Divide folder into a series of tabbed sections as shown below. Ensure folder is updated at each required progress meeting.

- a. CQC Meeting minutes in accordance with paragraph QUALITY CONTROL (QC) MEETINGS.
- b. All completed Preparatory and Initial Phase Checklists, arranged by specification section, further sorted by DFOW referenced to the construction schedule. Submit each individual Phase Checklist the day the phase event occurs as part of the CQC daily report.
- c. All milestone inspections, arranged by Activity Number referenced to the construction schedule.
- d. An up-to-date copy of the Testing Plan and Log with supporting field test reports, arranged by specification section referenced to the DFOW to which individual reports results are associated. Individual field test reports will be submitted within 2 working days after the test is performed in accordance with the paragraph QUALITY CONTROL ACTIVITIES.
- e. Copies of all Contract modifications, arranged in numerical order. Also include documentation that modified work was accomplished.
- f. An up-to-date copy of the paragraph DEFICIENCY/REWORK ITEMS LIST.

- g. Upon commencement of Completion Inspections of the entire project or any defined portion, maintain up-to-date copies of all punch lists issued by the QC staff to the Contractor and subcontractors and all punch lists issued by the Government in accordance with the paragraph COMPLETION INSPECTIONS.

1.12.6 Testing Plan and Log

As tests are performed, the QC Manager will record on the "Testing Plan and Log" the date the test was performed and the date the test results were forwarded to the Contracting Officer. Attach a copy of the updated "Testing Plan and Log" to the last daily CQC Report of each month. Provide a copy of the final "Testing Plan and Log" to the preparer of the Operation & Maintenance (O&M) documentation.

1.13 NOTIFICATION ON NON-COMPLIANCE

The Contracting Officer will notify the Contractor of any detected non-compliance with the Contract. Take immediate corrective action after receipt of such notice. Such notice, when delivered to the Contractor at the work site, is deemed sufficient for the purpose of notification. If the Contractor fails or refuses to comply promptly, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No part of the time lost due to such stop orders will be made the subject of a claim for extension of time for excess costs or damages by the Contractor.

1.14 DELIVERY, STORAGE, AND HANDLING

Designate receiving/storage areas for incoming material to be delivered according to installation schedule and to be placed convenient to work area in order to minimize waste due to excessive materials handling and misapplication. Store and handle materials in a manner as to prevent loss from weather and other damage. Keep materials, products, and accessories covered and off the ground, and store in a dry, secure area. Prevent contact with material that may cause corrosion, discoloration, or staining. Protect all materials and installations from damage by the activities of other trades.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 50 00.00 35

TEMPORARY CONSTRUCTION FACILITIES AND CONTROLS

PART 1 GENERAL

- 1.1 SUMMARY
- 1.2 REFERENCES
- 1.3 SUBMITTALS
- 1.4 PRECONSTRUCTION REQUIREMENTS
 - 1.4.1 Construction Site Plan
 - 1.4.2 Mobilization/Demobilization Plan
 - 1.4.3 Security Plan
 - 1.4.4 Vegetation Protection and Restoration Plan
 - 1.4.5 Documentation of Existing Conditions
- 1.5 EMERGENCY PLANNING
- 1.6 EXISTING UTILITIES
 - 1.6.1 Water
 - 1.6.2 Electricity
- 1.7 WORK AND ACCESS DATA
 - 1.7.1 Staging and Access Areas
 - 1.7.2 Contractor Responsibilities
 - 1.7.3 Transport of Government Personnel
- 1.8 ADJACENT PROPERTY AND STRUCTURES

PART 2 PRODUCTS

- 2.1 TEMPORARY SIGNAGE
 - 2.1.1 Bulletin Board
 - 2.1.2 Project Identification and Safety Signs
 - 2.1.3 Construction Forms and Details
 - 2.1.4 Warning Signs
 - 2.1.5 Haul Road Traffic and Construction Warning Signs
- 2.2 TEMPORARY TRAFFIC CONTROL
 - 2.2.1 Haul Roads
 - 2.2.2 Barricades
- 2.3 TEMPORARY FENCING
 - 2.3.1 Polyethylene Mesh Safety Fencing
 - 2.3.2 Chain Link Panel Fencing
 - 2.3.3 Post-Driven Chain Link Fencing
- 2.4 TEMPORARY WIRING
- 2.5 STORAGE CONTAINERS

PART 3 EXECUTION

- 3.1 GENERAL REQUIREMENTS
 - 3.1.1 Identification of Employees
 - 3.1.2 Employee Parking
 - 3.1.3 Onsite Information
 - 3.1.4 Janitorial and Other Services

U.S. Naval Station Mayport
Duval County, Florida

- 3.2 AVAILABILITY AND USE OF UTILITY SERVICES
 - 3.2.1 Lighting
 - 3.2.2 Heating and Cooling
 - 3.2.3 Water
 - 3.2.4 Fire Extinguisher
 - 3.2.5 Utility Lines
 - 3.2.6 Electric Power
- 3.3 TRAFFIC PROVISIONS
 - 3.3.1 Maintenance of Traffic
 - 3.3.2 Protection of Traffic
 - 3.3.3 Rush Hour Restrictions
 - 3.3.4 Dust Control
- 3.4 MAINTENANCE OF HAUL AND ACCESS ROADS
- 3.5 CONTRACTOR'S TEMPORARY FACILITIES
 - 3.5.1 Contractor Field Office
 - 3.5.1.1 Office Space for Government Personnel
 - 3.5.2 Appearance of Trailer(s)
 - 3.5.3 Storage Area
 - 3.5.3.1 Maintenance of Storage Area
 - 3.5.4 Waste Storage
 - 3.5.5 Fuel Storage and Fueling Operations
- 3.6 SITE SECURITY
 - 3.6.1 Visitors
- 3.7 ACCOMMODATIONS AND MEALS FOR INSPECTORS
- 3.8 CLEANUP
- 3.9 RESTORATION OF STORAGE AREA
- 3.10 RESTORATION OF STAGING AND ACCESS AREAS
 - 3.10.1 Documentation of Post Construction Restoration

-- End of Section Table of Contents --

SECTION 01 50 00.00 35

TEMPORARY CONSTRUCTION FACILITIES AND CONTROLS

PART 1 GENERAL

1.1 SUMMARY

In addition to Construction Facilities this Section covers:

Temporary Utilities
Construction Aids
Vehicular Access and Parking
Project Identification

See Section 01 57 20.00 35 ENVIRONMENTAL PROTECTION for requirements including silt control, trailer placement, fueling restrictions, dust control, solid waste, and cleanup. Upon completion of project, clean up and restore area in accordance with Clause CLEANING UP.

a. Construction Facilities include, but are not limited to, the following:

- (1) Contractor Offices
- (2) Information Bulletin Board
- (3) Material and Equipment Storage Area
- (4) Fueling Area
- (5) Secured Storage Area
- (6) Employee Parking Area
- (7) Debris Container (dumpster)
- (8) Construction Signage to include Project Sign; Safety Sign; and, Construction Warning Signs

b. Temporary Utilities include, but are not limited to, the following:

- (1) Water
- (2) Electric
- (3) Sewage
- (4) Communications
 - (i) Telephone
 - (ii) Internet (10 Mbps or higher speed)
- (5) Lighting

1.2 REFERENCES

The publications listed below form a part of this specification to extent referenced. The publications are referred to in text by basic designation only.

FLORIDA DEPARTMENT OF TRANSPORTATION (FDOT)

FDOT Specifications

(2024) Standard Specifications for Road
and Bridge Construction

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE C2 (2023) National Electrical Safety Code

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023; ERTA 1 2024; TIA 24-1) National
Electrical Code

NFPA 241 (2022) Standard for Safeguarding
Construction, Alteration, and Demolition
Operations

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2024) Safety and Occupational Health
(SOH) Requirements

1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification; submittals not having a "G" classification are for information only. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES.

SD-01 Preconstruction Submittals

Construction Site Plan; G, R0

Mobilization/Demobilization Plan; G, R0

Security Plan; G, R0

Vegetation Protection and Restoration Plan; G, R0

Traffic Control Plan; G, R0

Documentation Of Existing Conditions; G, R0

Haul Road Plan; G, R0

SD-02 Shop Drawings

Temporary Electric Drawings

SD-03 Product Data

Project Identification And Safety Signs; G, R0

SD-11 Closeout Submittals

Documentation Of Post Construction Restoration; G, R0

1.4 PRECONSTRUCTION REQUIREMENTS

Submit the following preconstruction submittal items no later than 30 calendar days after Notice to Proceed.

1.4.1 Construction Site Plan

Prior to the start of work, submit for Government approval a site plan showing the locations and dimensions of temporary facilities (including layouts and details, equipment and material storage area (onsite and offsite), and access and haul routes, avenues of ingress/egress to the fenced area and details of the fence installation. Identify any areas which may have to be graveled to prevent the tracking of mud. Indicate if the use of a supplemental or other staging area is desired. Show locations of safety and construction fences, site trailers, construction entrances, trash dumpsters, temporary sanitary facilities, and worker parking areas.

1.4.2 Mobilization/Demobilization Plan

Plan must include, but not be limited to, the following:

a. Mobilization Requirements:

- (1) Methods, equipment and materials
- (2) Connection of utilities
- (3) Placement of site facilities and temporary controls
- (4) Construction of facilities

b. Demobilization Requirements (methods, equipment and materials required to clean-up and restore site at project conclusion):

- (1) Collection, recycle and disposal of solid waste
- (2) Contract-generated material
- (3) Utility disconnection
- (4) Removal of Contractor facilities
- (5) Repair and restoration of site (i.e., fences, roads, or permanent facilities)

1.4.3 Security Plan

Prepare a Security Plan for Contracting Officer describing site security as follows:

- a. Day and night security
- b. Weekend and holiday security
- c. General security duties

1.4.4 Vegetation Protection and Restoration Plan

Prior to clearing for access, submit a Vegetation Protection and Restoration Plan describing the methods to be used for protection of vegetation that is not to be disturbed or otherwise impacted. Include restoration measures for vegetation cleared from access areas and in the event of the accidental damage to, or destruction of, said vegetation.

1.4.5 Documentation of Existing Conditions

Before beginning any mobilization or work, survey the site and photo-document the work area including all staging areas, haul routes, parking areas, and vegetation to show existing limits that could potentially be impacted. Include in the record a description of each picture taken as well as the conditions of roadways, utilities, and other site features, and the location and extent of existing damage prior to

starting work. It is the Contractor's responsibility to document all pertinent existing conditions that may affect the subject property.

1.5 EMERGENCY PLANNING

Based on the weather forecast for sustained winds 50 knots (93 km/hr 58 mph) or greater. Contact the Contracting Officer for the current Emergency Action Plan. Monitor weather conditions a minimum of twice a day and take appropriate actions according to the approved Emergency Plan in the accepted Accident Prevention Plan, [EM 385-1-1](#) Emergency Plan requirements and the instructions below. Unless otherwise directed by the Contracting Officer, comply with:

- a. Condition FOUR (Sustained winds of 93 km/hr 58 mph or greater expected within 72 hours): Normal daily jobsite cleanup and good housekeeping practices. Collect and store in piles or containers scrap lumber, waste material, and rubbish for removal and disposal at the close of each work day. Maintain the construction site including storage areas, free of accumulation of debris. Stack form lumber in neat piles less than one m 3.3 feet high. Remove all debris, trash, or objects that could become missile hazards. Review requirements pertaining to "Condition THREE" and continue action as necessary to attain "Condition FOUR" readiness. Contact Contracting Officer for weather and COR updates and completion of required actions.
- b. Condition THREE (Sustained winds of 93 km/hr 58 mph or greater expected within 48 hours): Maintain "Condition FOUR" requirements and commence securing operations necessary for "Condition ONE" which cannot be completed within 18 hours. Cease all routine activities which might interfere with securing operations. Commence securing and stow all gear and portable equipment. Make preparations for securing buildings. Reinforce or remove formwork and scaffolding. Secure machinery, tools, equipment, materials, or remove from the jobsite. Expend every effort to clear all missile hazards and loose equipment from general base areas. Contact Contracting Officer for weather and COR updates and completion of required actions. Review requirements pertaining to "Condition TWO" and continue action as necessary to attain "Condition THREE" readiness.
- c. Condition TWO (Sustained winds of 93 km/hr 58 mph or greater expected within 24 hours): Secure the jobsite, and leave Government premises.
- d. Condition ONE. (Sustained winds of 93 km/hr 58 mph or greater expected within 12 hours): Contractor access to the jobsite and Government premises is prohibited.

1.6 EXISTING UTILITIES

Investigate the availability of utilities at the project site. Provide potable water, sanitary sewer, electricity, communications and optional utilities require at the project site. Provide internet services with 10 Mbps minimum speed and wireless capabilities to the Contractor's Field Office complex to include Government office space.

1.6.1 Water

In addition to the above, provide and maintain an adequate supply of water for construction use at no additional cost to the Government. Install and maintain necessary supply connections and piping for same, but only at

such locations and in such manner as may be approved by the Contracting Officer. Provide and maintain temporary toilet and washing facilities. Install and maintain toilet and washing facilities in a location approved by the Contracting Officer. Refer to paragraph AVAILABILITY AND USE OF UTILITY SERVICES below.

1.6.2 Electricity

In addition to the above, furnish all electric current at no additional cost to the Government. Install, connect, and maintain all temporary lines in a workmanlike manner satisfactory to the Contracting Officer. Remove temporary lines in like manner at no additional cost to the Government prior to completion of the construction. Refer to paragraph AVAILABILITY AND USE OF UTILITY SERVICES below.

1.7 WORK AND ACCESS DATA

1.7.1 Staging and Access Areas

Obtain and coordinate staging areas and submit for Contracting Officer's approval. Access areas are shown on the contract drawings for the Contractor's use. The Contracting Officer will field-determine the final limits of the access areas indicated on the drawings in coordination with the Local Sponsor and the Contractor. Investigate and obtain any additional areas which may be necessary for construction operations at the Contractor's expense. The additional areas will be subject to the approval of the Contracting Officer.

1.7.2 Contractor Responsibilities

Exclude the public from the work area in the immediate vicinity of Contractor operations including but not limited to fill placement, grading, and transport. Minimize areas closed to the public. Install warning signs to warn the public and all commercial recreational boats of all construction activities. Provide and maintain all water and land access routes necessary for his equipment and plant to and from the work sites. Ascertain the environmental conditions which can affect access, such as climate, terrain, and winds.

1.7.3 Transport of Government Personnel

Provide safe access for the Contracting Officer's personnel and dedicated government personnel at the project site, to all points ashore and afloat, as required by the EM 385-1-1. Provide a dedicated four passenger UTV with an operator dedicated for Government personnel project access which meets the requirements of the EM 385-1-1.

1.8 ADJACENT PROPERTY AND STRUCTURES

Immediately repair any damage to private or public property within the project boundaries, including staging site(s) and work and access areas/roads. Repair any damage as a result of the Contractor's operations at no cost to the Government.

PART 2 PRODUCTS

2.1 TEMPORARY SIGNAGE

2.1.1 Bulletin Board

Prior to commencement of work activities, provide a clear weatherproof covered bulletin board not less than 915 by 1220 mm (36 by 48 inches) in size for displaying the Equal Employment Opportunity poster, a copy of the wage decision contained in the contract, Wage Rate Information poster, WH1090 Paid Sick Leave for Contractors poster (<https://www.dol.gov/whd/regs/compliance/wh1090.pdf>), Safety and Health Information as required by EM 385-1-1 and other information approved by the Contracting Officer. Display legible copies of the aforementioned data until work is completed. Remove the bulletin board upon completion of work. The bulletin board will remain the property of the Contractor. Coordinate requirements herein with Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS.

2.1.2 Project Identification and Safety Signs

Provide signs, their content, and location as indicated on the web site provided in paragraph "Construction Forms and Details" below. Erect the signs within 15 days after receipt of the Notice to Proceed. Correct the data required by the safety sign, with light colored metallic or non-metallic numerals. Upon completion of the project, remove the signs from the site.

2.1.3 Construction Forms and Details

See the CONSTRUCTION FORMS & DETAILS module at the following web address: <https://www.saj.usace.army.mil/About/Divisions-Offices/Engineering/Design-Branch/Specifications-Section/>

2.1.4 Warning Signs

Post temporary signs, tags, and labels to give workers and the public adequate warning and caution of construction hazards according to the EM 385-1-1 Chapter 4. Attach signs to the perimeter fencing every 150 feet or as determined by the Contracting Officer warning the public of the presence of construction hazards. Install signs requiring unauthorized persons to keep out of the construction site. Update the data required by the safety signs daily. Post signs at all points of entry designating the construction site as a hard hat area.

2.1.5 Haul Road Traffic and Construction Warning Signs

Use 1/2 inch HDO plywood, or aluminum sheet, conforming to FDOT Specifications Section 700 Uniform Traffic Manual Standard Signs, non-standard signage high visibility orange with black lettering. Orange fabric mesh is acceptable for temporary survey crew use.

2.2 TEMPORARY TRAFFIC CONTROL

During construction, provide access and temporary relocated roads as necessary to maintain traffic. Maintain and protect traffic on all affected roads during the construction period except as otherwise approved by the Contracting Officer. Install measures for the protection and diversion of traffic, including the provision of watchmen and flagmen,

erection of barricades, placing of lights around and in front of equipment and the work, and the erection and maintenance of adequate warning, danger, and direction signs, as required by the State and local authorities having jurisdiction. Protect the traveling public from damage to person and property. Limit the interface between the Contractor's traffic on roads selected for hauling material to and from the site with public traffic. Investigate the adequacy of existing roads and the allowable load limit on these roads. Repair any damage to roads caused by construction operations.

2.2.1 Haul Roads

Construct access and haul roads necessary for proper prosecution of the work under this contract in accordance with EM 385-1-1 Chapter 4. Construct with suitable grades and widths; sharp curves, blind corners, and dangerous cross traffic are to be avoided. Submit haul road plan for approval. Provide necessary lighting, signs, barricades, and distinctive markings for the safe movement of traffic. Provide an method of dust control to ensure safe operation at all times. Location, grade, width, and alignment of construction, and hauling roads area subject to approval by the Contracting Officer. Provide adequate lighting to assure full and clear visibility for full width of haul road and work areas during any night work operations.

2.2.2 Barricades

Erect and maintain temporary barricades to limit public access to hazardous areas. Whenever safe public access to paved areas such as roads, parking areas or sidewalks is prevented by construction activities or as otherwise necessary to ensure the safety of both pedestrian and vehicular traffic barricades will be required. Securely place barricades clearly visible with adequate illumination to provide sufficient visual warning of the hazard during both day and night.

2.3 TEMPORARY FENCING

Provide fencing along the construction site and at all open excavations and tunnels as indicated in the Contractor's Temporary Facilities shop drawings to control access by unauthorized personnel. Provide highly visible safety fencing to be seen by pedestrians and vehicular traffic. Specific fencing requirements are as described herein. All fencing will meet the requirements of EM 385-1-1.

2.3.1 Polyethylene Mesh Safety Fencing

Temporary safety fencing must be a high visibility orange colored, high density polyethylene grid, a minimum of 48 inches high and maximum mesh size of 2 inches. Extend fencing from the grade to a minimum of 48 inches above the grade and be tightly secured to T-posts spaced as necessary to maintain a rigid and taut fence.

2.3.2 Chain Link Panel Fencing

Temporary panel fencing must be galvanized steel chain link panels 6 feet high. Multiple fencing panels may be linked together at the bases to form long spans as needed. Weight down each panel base using sand bags or other suitable materials in order for the fencing to withstand anticipated winds while remaining upright. Fencing must remain rigid and taut with a minimum of 200 pounds of force exerted on it from any direction with less

than 4 inches of deflection.

2.3.3 Post-Driven Chain Link Fencing

Temporary post-driven fencing must be galvanized chain link fencing 6 feet high supported by a tightly secured to galvanized steel posts driven below grade. Locate fence posts on minimum 10 foot centers. Posts may be set in various surfaces such as sand, soil, asphalt or concrete as necessary. Chain link fencing must remain rigid and taut with a minimum of 200 pounds of force exerted on it from any direction with less than 4 inches of deflection. Completely remove fencing and posts at the completion of construction. Restore any surfaces disturbed or damaged to its original condition. Locate and identify underground utilities prior to setting fence posts. Equip fence with a lockable gate. Gate must remain locked when construction personnel are not present.

2.4 TEMPORARY WIRING

Provide temporary wiring in accordance with EM 385-1-1 Chapter 11, NFPA 241 and NFPA 70. Include monthly inspection and testing of all equipment and apparatus.

2.5 STORAGE CONTAINERS

Welded steel construction, locking, shipping containers or equal.

PART 3 EXECUTION

3.1 GENERAL REQUIREMENTS

3.1.1 Identification of Employees

Furnish identification as approved by the Contracting Officer to each employee and require each employee engaged on the work to display them. Immediately deliver prescribed identification to the Contracting Officer for cancellation upon release of any employee. When required, obtain and provide fingerprints of persons employed on the project. Wear identifying markings on hard hats clearly identifying the company for whom the employee works.

3.1.2 Employee Parking

- a. Park employee's vehicles in areas designated by Contractor, away from construction traffic, within reasonable walking distance of site. Maintain area free of ruts, mud holes and puddles. Place gravel where required by deteriorated conditions.
- b. Protect unattended equipment as it may be subject to vandalism.
- c. Storage trailers and storage area with Government material will be locking type with lighting.

3.1.3 Onsite Information

Keep copy of contract drawings, specifications, and other contract documents at Contractor's Office onsite, available for use at all times.

3.1.4 Janitorial and Other Services

- a. Daily -- Sweep and mop floors; empty wastebaskets; service and sanitize bathrooms and other common areas; restock cleaning supplies and toiletries such as, but not limited to, hand soap, hand sanitizer, paper towels, and toilet paper.
- b. Weekly -- Wash floors, windows (inside and out).
- c. Cut grass as needed to keep below 3 inches high.
- d. Pest control for all field offices.

3.2 AVAILABILITY AND USE OF UTILITY SERVICES

Install temporary facilities and utilities in accordance with [IEEE C2](#), [EM 385-1-1](#), [NFPA 70](#). Obtain necessary construction, building, zoning, or soil erosion and sediment control approvals required by local authorities and utility companies. Equip trailer(s) with wind tie downs in accordance with local wind and building code requirements.

3.2.1 Lighting

Electric light, non-glare type luminaries to provide a minimum illumination, level of 30-foot candles at desk height level. Comply with temporary lighting, wiring and Ground Fault Circuit Interrupter (GFCI) requirements found in [EM 385-1-1](#).

3.2.2 Heating and Cooling

Adequate equipment to maintain internal ambient air temperature of 75 degrees F in 105 degrees F and 70 degrees F in 30 degrees F. Equip trailer(s) with smoke alarms. If gas is used to heat, equip trailer(s) with carbon monoxide alarms.

3.2.3 Water

Non-potable well water may be used to flush toilets, and for dish and hand washing. Provide bulk potable water in storage tanks or bottled water for drinking.

3.2.4 Fire Extinguisher

Refer to [EM 385-1-1](#). Non-toxic, dry chemical, fire extinguisher meeting Underwriters Laboratories, Inc., approval for Class A, Class B, and Class C fires with a minimum rating of 2A; 10B; and 10C.

3.2.5 Utility Lines

Install, connect and modify temporary lines as coordinated with owning utility. Conform requirements in accordance with [IEEE C2](#) and [NFPA 70](#) for Temporary Electric Lines. Remove temporary line at completion of project.

3.2.6 Electric Power

Submit [temporary electric drawings](#) including electricity supply and lighting signed by licensed electrician- from electrical source to layout locations, fixtures, and materials, to include outdoor lighting. Provide uninterrupted electrical service sufficient to operate office systems. Directly purchase electrical power.

3.3 TRAFFIC PROVISIONS

3.3.1 Maintenance of Traffic

- a. Conduct operations in a manner that will not close a thoroughfare or interfere with traffic on railways or highways except with written permission of the Contracting Officer at least 15 calendar days prior to the proposed modification date, and provide a [Traffic Control Plan](#) for Government approval detailing the proposed controls to traffic movement for approval. The plan must be in accordance with State and local regulations and the MUTCD, Part VI. Make all notifications and obtain all permits required for modification to traffic movements outside Station's jurisdiction. Contractor may move oversized and slow-moving vehicles to the worksite provided requirements of the highway authority have been met.
- b. Conduct work so as to minimize obstruction of traffic, and maintain traffic on at least half of the roadway width at all times. Obtain approval from the Contracting Officer prior to starting any activity that will obstruct traffic.
- c. Provide, erect, and maintain, at Contractor's expense, lights, barriers, signals, passageways, detours, Life Safety Signage, overhead protection, and other items that may be required by the authority having jurisdiction.
- d. Provide cones, signs, barricades, lights, or other traffic control devices and personnel required to control traffic. Do not use foil-backed material for temporary pavement marking because of its potential to conduct electricity during accidents involving downed power lines.

3.3.2 Protection of Traffic

Maintain and protect traffic on all affected roads during the construction period except as otherwise specifically directed by the Contracting Officer. Measures for the protection and diversion of traffic, including the provision of watchmen and flagmen, erection of barricades, placing of lights around and in front of equipment the work, and the erection and maintenance of adequate warning, danger, and direction signs, will be as required by the State and local authorities having jurisdiction. Provide self-illuminated (lighted) barricades during hours of darkness. All personnel working in roadways will wear brightly-colored vests or other high visibility apparel in accordance with [EM 385-1-1](#). Protect the traveling public from damage to person and property. Minimize the interference with public traffic on roads selected for hauling material to and from the site. Investigate the adequacy of existing roads and their allowable load limit. Contractor is responsible for the repair of damage to roads caused by construction operations.

3.3.3 Rush Hour Restrictions

Do not interfere with the peak traffic flows preceding and during normal operations without notification to and approval by the Contracting Officer.

3.3.4 Dust Control

Dust control methods and procedures must be approved by the Contracting Officer. Coordinate dust control methods with [01 57 20.00 35 ENVIRONMENTAL](#)

PROTECTION.

3.4 MAINTENANCE OF HAUL AND ACCESS ROADS

Providing and maintain safe and passable means of ingress and egress to the site and to all project features on the site for Contractor, Government, and third party entities. Requirements herein are applicable to both new and existing haul and access roads. Repair and re-grade haul and access roads as necessary. Maintain roadway drainage, including maintaining roadway crown and profile and keeping drainage ditches clear. Repair potholes, mud holes, ruts, and/or depressions of 6 inches deep or more within 24 hours of discovery. Check condition of haul and access roads at least weekly and after any severe weather event. All requirements of EM 385-1-1 concerning access and haul roads also apply.

3.5 CONTRACTOR'S TEMPORARY FACILITIES

Place the field offices, staging areas, stockpile storage, and temporary buildings in areas as approved by the Contracting Officer and in accordance with Section 01 57 20.00 35 ENVIRONMENTAL PROTECTION. Make temporary movement or relocation of Contractor facilities when approved by the Contracting Officer. Provide a general layout sketch of temporary site facilities including, but not limited to, the following:

- a. Trailer locations
- b. Parking areas
- c. Material storage
- d. Equipment lay down area
- e. Areas for gravel
- f. Fuel areas
- g. Supplemental or other staging area
- h. Temporary well, water supply
- i. Septic field or holding tanks, port-a-lets
- j. Contaminated water handling
- k. Concrete cleaning area and methods of disposal
- l. Fences -- location and dimensions, entrance and exit points, and details of installation
- m. Utilities (poles, connections and services)
- n. Explosives storage

3.5.1 Contractor Field Office

Provide on-site field office of sufficient size and staff capability to manage project activities. Provide communications and computer capabilities to manage the work, including implementing RMS, facsimile, electronic mail, reproduction, to track schedule and cost. Include a conference area with table and chairs to accommodate a minimum of 8 people. Provide handicap access with a 1 on 12 ramp and gravel parking spot marked with handicap parking sign. Provide high-speed internet services (250 Mbps or higher speed rate) and wireless capabilities to the field office facilities/compound to include Government office space to facilitate RMS, SharePoint (file transfer) and visitor usage.

3.5.1.1 Office Space for Government Personnel

Provide a minimum of 80 square feet air conditioned office space (with locking door) for exclusive use by Government personnel with the following:

- a. Two office desks with lockable drawers (60 inches by 30 inches,

laminated top) and two swivel office chairs.

- b. Five drawer, legal size, lockable filing cabinet.
- c. A 1.5 cf microwave oven and 7 cf refrigerator.
- d. A minimum of two designated Government parking spaces per each active staging/access area.
- e. Temporary toilet and washing facilities, for Government personnel, either in or outside of the Government office space. Toilet and washing facilities are not considered part of the minimum 80 square feet office space.

3.5.2 Appearance of Trailer(s)

Maintain trailer(s) used for both office and material storage purposes in clean, neat exterior appearance and in good repair. Trailer(s) requiring exterior painting or maintenance will not be allowed on site until determined satisfactory by Contracting Officer.

3.5.3 Storage Area

Construct a temporary 6 foot high chain link fence around trailers and materials. Fence posts may be driven, in lieu of concrete bases, where soil conditions permit. Do not place or store trailers, materials, or equipment outside the fenced area unless such trailers, materials, or equipment are assigned a separate and distinct storage area by the Contracting Officer away from the vicinity of the construction site but within the installation boundaries. Do not stockpile materials outside the fence in preparation for the next day's work. Park mobile equipment, such as tractors, wheeled lifting equipment, cranes, trucks, and like equipment within the fenced area at the end of each work day.

3.5.3.1 Maintenance of Storage Area

Keep fencing in a state of good repair and proper alignment. Should the Contractor elect to traverse with construction equipment or other vehicles, cover grassed or unpaved areas which are not established roadways, with a layer of gravel as necessary to prevent rutting and the tracking of mud onto paved or established roadways; gravel gradation will be at the Contractor's discretion. Mow grass located within the boundaries of the construction site for the duration of the project. Edge or trim neatly, grass and vegetation along fences, buildings, under trailers, and in areas not accessible to mowers.

3.5.4 Waste Storage

Provide dumpsters or suitable debris containers. Prevent windblown trash; cover as needed. Dispose of offsite when needed. Refer to Section 01 57 20.00 35 ENVIRONMENTAL PROTECTION.

3.5.5 Fuel Storage and Fueling Operations

Refer to Section 01 57 20.00 35 ENVIRONMENTAL PROTECTION. Provide light when fueling at night.

3.6 SITE SECURITY

Site Security is the Contractor's responsibility. Provide site security person(s) at all times, even when there is not on-site activities. Maintain 24-hour security. Site security must include, but not be limited, to:

- a. Limit vehicular access to authorized vehicles and personnel only.
- b. Limit public access to travel along access road only. Keep access road closed at all times except for passage of authorized personnel and vehicles.
- c. Maintain a list of authorized personnel and subcontractors available at site offices.
- d. Maintain a sign-in log documenting visitors, deliveries, and security incidents. Include date, name, address, company, time in and time out for each employee and visitor.
- e. Check fenced areas, equipment, trailers on a daily basis. If damage is observed or vandalism is found report to Contracting Officer.

3.6.1 Visitors

No visitors will be allowed on site without knowledge of Contractor and permission of Contracting Officer. Direct visitors to report upon arrival to Contractor's Field Office for site safety and accident prevention briefing. Provide visitors appropriate protective equipment (i.e., ear plugs, safety glasses, etc.).

3.7 ACCOMMODATIONS AND MEALS FOR INSPECTORS

- a. Furnish regularly to inspectors, for office purposes, a suitable separate room on board the dredge or other craft upon which they are employed or, if not available thereon. Furnish suitable alternate accommodations ashore at a location approved by the Contracting Officer, suitable transportation between the alternate accommodations and the dredge or other craft upon which they are employed.
- b. If the Contractor maintains on this work an establishment for the subsistence of the Contractor's own employees, furnish to inspectors employed on the work, and to all Government agents who may visit the work on official business, meals of a quality satisfactory to the Contracting Officer. The meals furnished will be paid for by the Government employees at a rate no greater than the JTR meal rate per person for the county in which work is performed.

3.8 CLEANUP

Remove construction debris, temporary fencing, waste materials, packaging material and the like from the work site daily. Clean away any dirt or mud tracked onto paved or surfaced roadways. Store materials resulting from demolition activities which are salvageable within the fenced area described above or at the supplemental storage area. Neatly stack stored material not in trailers, whether new or salvaged, when stored. Refer to Section 01 57 20.00 35 ENVIRONMENTAL PROTECTION for solid waste and post construction cleanup.

3.9 RESTORATION OF STORAGE AREA

Remove the fence upon completion of the project. After removal of trailers, materials, and equipment from within the fenced area, the area will become the property of the Contractor. Restore areas used by the Contractor for the storage of equipment or material, or other use, to the original or better condition. Remove gravel used to traverse grassed areas and restore the area to its original condition, including top soil and seeding as necessary.

3.10 RESTORATION OF STAGING AND ACCESS AREAS

Restore areas used for staging or access to the original or better condition. Replace any fencing, signage, curbing, or any other structural features disturbed by the Contractor's activities. Provide revegetation of access and staging areas with sod or seeding. Include a survival warranty of 90 percent of the plant material for 90 days for revegetation. Install vegetation with fertilization and irrigation, at no additional expense to the Government. Replace shrubs and trees to preexisting conditions in accordance with the requirements of Section 01 57 20.00 35 ENVIRONMENTAL PROTECTION.

3.10.1 Documentation of Post Construction Restoration

Following the end of work and demobilization, survey the site and photo-document the post-construction condition of all areas surveyed for the Documentation of Existing Conditions. Include in the record conditions of roadways, utilities, and other site features, and the location and extent of any changes observed since documentation of pre-construction conditions. It is the Contractor's responsibility to document all pertinent post-construction conditions that may affect the subject property.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 57 20.00 35

ENVIRONMENTAL PROTECTION

PART 1 GENERAL

- 1.1 SCOPE
- 1.2 ENVIRONMENTAL LAWS, REGULATIONS AND DOCUMENTS
 - 1.2.1 Miscellaneous Environmental Laws And Regulations
 - 1.2.2 NEPA Reference
- 1.3 REFERENCES
- 1.4 QUALITY CONTROL
- 1.5 PERSONNEL TRAINING
- 1.6 PERMITS AND AUTHORIZATIONS
- 1.7 SUBMITTALS
- 1.8 ENVIRONMENTAL PROTECTION PLAN
- 1.9 QUALIFICATIONS
 - 1.9.1 Marine Animal Observer Qualifications (Clamshell/Mechanical Only)
 - 1.9.2 Endangered Species Observer and Trawl Supervisor Qualifications (For Hopper Dredges Only)
- 1.10 SUBCONTRACTORS
- 1.11 NOTIFICATION
- 1.12 CONTRACTOR PERSONNEL QUALIFICATIONS IN POLLUTION CONTROL

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

- 3.1 PROTECTION OF ENVIRONMENTAL RESOURCES
 - 3.1.1 General Project Environmental Design and Installation Criteria
 - 3.1.1.1 Sewage-Based Systems Environmental Design and Installation Criteria
 - 3.1.2 Protection of Land Resources
 - 3.1.2.1 Work Area Limits
 - 3.1.2.2 Contractor Facilities and Other Work Areas
 - 3.1.2.3 Solid Wastes
 - 3.1.2.4 Fuel, Oil, and Lubricants
 - 3.1.2.5 Hazardous Waste
 - 3.1.2.6 Hazardous Materials
 - 3.1.2.7 Disposal of Other Materials
 - 3.1.3 Preservation and Recovery of Historic, Archeological, and Cultural Resources
 - 3.1.3.1 Applicable Law
 - 3.1.3.2 Known Resources
 - 3.1.3.3 Inadvertent Discoveries
 - 3.1.3.4 Claims for Downtime due to Inadvertent Discoveries
 - 3.1.4 Protection of Water Resources
 - 3.1.4.1 Monitoring of Water Areas
 - 3.1.4.2 Turbidity
 - 3.1.4.3 Oil, Fuel, and Hazardous Substance Spill Prevention and

U.S. Naval Station Mayport
Duval County, Florida

Mitigation

- 3.1.5 Protection of Fish and Wildlife Resources
 - 3.1.5.1 Listed Species Protection
 - 3.1.5.2 Protected Listed Species Sighting Reports
 - 3.1.5.3 Protected Species Summary Report
 - 3.1.5.4 Endangered Species Observers (Hopper Dredge Only)
 - 3.1.5.5 Disposition of Endangered or Threatened Species Carcasses
 - 3.1.5.6 Hopper Dredge Equipment
 - 3.1.5.7 Sea Turtle Trawling and Relocation (For Hopper Dredges Only)
- 3.1.6 Protection of Air Resources
 - 3.1.6.1 Particulates
 - 3.1.6.2 Odors
- 3.1.7 Protection of Sound Intrusions
- 3.2 POSTCONSTRUCTION CLEANUP
- 3.3 PRESERVATION AND RESTORATION OF LANDSCAPE AND AQUATIC VEGETATION DAMAGES
- 3.4 INVASIVE AND NUISANCE SPECIES
 - 3.4.1 Prevention of Invasive and Nuisance Species Transfer
 - 3.4.2 Invasive and Nuisance Species Reporting
 - 3.4.2.1 Invasive And Nuisance Species Report
- 3.5 MAINTENANCE OF POLLUTION CONTROL FACILITIES
- 3.6 CONSTRUCTION FORMS AND DETAILS

-- End of Section Table of Contents --

SECTION 01 57 20.00 35

ENVIRONMENTAL PROTECTION

PART 1 GENERAL

1.1 SCOPE

This Section covers prevention of environmental damage as the result of construction operations under this contract and for those measures set forth in other Technical Requirements of these specifications. For the purpose of this specification, environmental damage is defined as the presence of hazardous, physical, chemical, or biological elements or agents which adversely affect human health or welfare; unfavorably alter ecological balances; affect other species, biological communities, or ecosystems; or degrade the quality of the environment for aesthetic, cultural, and/or historical purposes. The control of environmental damage requires consideration of land, water, and air, and includes management of visual aesthetics, noise, solid waste, radiant energy and radioactive materials, as well as other pollutants.

1.2 ENVIRONMENTAL LAWS, REGULATIONS AND DOCUMENTS

1.2.1 Miscellaneous Environmental Laws And Regulations

There are numerous environmental laws and regulations. At the Federal level, the applicable laws and regulations include compliance with the Clean Water Act (CWA); Clean Air Act (CAA); Coastal Zone Management Act (CZMA); Comprehensive Environmental Response, Compensation, and Liability Act (CERCLA); Endangered Species Act (ESA); Fish and Wildlife Coordination Act (FWCA); Marine Protection, Research, and Sanctuaries Act (MPRSA); Magnuson-Stevens Fishery Conservation and Management Act (MSFCMA); National Environmental Policy Act (NEPA); National Historic Preservation Act (NHPA); National Pollutant Discharge Elimination System (NPDES); Research and Sanctuaries Act (RSA); Native American Graves Protection and Repatriation Act (NAGPRA); Resource Conservation and Recovery Act (RCRA); Rivers and Harbors Act (R&H); Safe Drinking Water Act (SDWA); Toxic Substance Control Act (TSCA); Wild and Scenic Rivers Act (WSRA); Federal Insecticide, Fungicide, and Rodenticide Act (FIFRA); Marine Mammal Protection Act (MMPA); Migratory Bird Treaty Act (MBTA); Code of Federal Regulations (CFRs); Executive Orders (EO); and, Environmental Protection Agency (EPA) requirements.

1.2.2 NEPA Reference

Commitments for the Contractor made in the NEPA document have been incorporated into these specifications.

1.3 REFERENCES

Any publication listed below forms a part of this specification to the extent referenced. Publications are referred to in the text by basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 1110-1-1003

(2013) NAVSTAR Global Positioning System
Surveying

Code of Federal Regulations (CFR) indicated herein may be found at
<https://www.govinfo.gov/app/collection/cfr>.

1.4 QUALITY CONTROL

Establish and maintain quality control for environmental protection of all items set forth herein. Record on daily quality control reports or attachments thereto, any problems in complying with laws, regulations and ordinances, and corrective action taken.

1.5 PERSONNEL TRAINING

The Contractor is to train their personnel in all phases of environmental protection.

1.6 PERMITS AND AUTHORIZATIONS

Comply with all requirements under the terms and conditions set out in the following permit(s) and authorization(s) obtained by the Corps of Engineers listed below.

- a. Florida Department of Environmental Protection Permit No. 0298451-006-BE Exemption.
- b. DA RHA Section 10 Permit SAJ-2002-02052-(SP-TDS); effective 7/26/2024; expires 7/26/2034. DA MPRSA Section 103 Permit Modification Number 4 (SAJ-2002-2052-(SP-TDS); effective 11/06/24; expires 07/26/27.
- c. EPA MPRSA Section 103 Concurrence; effective 7/26/2024 and expires 7/26/2027.
- d. Jacksonville Ocean Dredged Material Disposal Site, Site Management and Monitoring Plan (SMMP); Issuance 09/03/2025 and Expiration 09/03/2035.
- e. NMFS South Atlantic Regional Biological Opinion (SARBO).

The above permits indicate issuing agency approval of work required by this contract. Permits are available at the following web address:
<https://www.saj.usace.army.mil/About/Divisions-Offices/Planning/Environmental-Branch/Environmental-Compliance/>

Obtain and comply with the requirements of all permits or licenses required for construction of this project in accordance with the Clause PERMITS AND RESPONSIBILITIES.

1.7 SUBMITTALS

Government approval is required for submittals with a "G" classification; submittals not having a "G" classification are for information only. When used, a designation following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES.

SD-01 Preconstruction Submittals

Environmental Protection Plan; G, D0

Whale Alert Text Message Address; G, D0

SD-02 Shop Drawings

Turtle Deflector Device; G, D0

SD-06 Test Reports

Protected Listed Species Sighting Reports; G, D0

Weekly Logs/Summary Of FWC Hopper Dredge Reports; G, D0

Weekly Protected Species Summary Report; G, D0

SD-07 Certificates

Marine Animal Observer Qualifications (Clamshell/Mechanical Only);
G, D0

Endangered Species Observer and Trawl Supervisor Qualifications
(For Hopper Dredges Only); G, D0

SD-11 Closeout Submittals

Protected Species Summary Report; G, D0

Comprehensive Trawling Daily Logs; G, D0

Invasive And Nuisance Species Report; G, D0

Project Environmental Summary Sheet; G, D0

1.8 ENVIRONMENTAL PROTECTION PLAN

Submit a draft [Environmental Protection Plan](#) no later than 5 calendar days after Notice to Proceed for review and acceptance by the Contracting Officer. The Government will consider an interim plan for the first 30 days of operations. However, furnish an acceptable final plan no later than 30 calendar days after receipt of Notice to Proceed. Acceptance of the plan will not relieve the Contractor of their responsibility for adequate and continuing control of pollutants and other environmental protection measures. Acceptance of the plan is conditional and predicated on satisfactory performance during construction. The Government reserves the right to require the Contractor to make changes to the Environmental Protection Plan or operations if the Contracting Officer determines that environmental protection requirements are not being met. No physical work at the site is to begin prior to acceptance of the plan or an interim plan covering the work to be performed.

The EPP is to have a Table of Contents. Pages of the EPP will be numbered, including appendices. Sections of the plan and all appendices are to be indexed in the table of contents. The Environmental Protection Plan is to include but not be limited to the following:

- a. A list of Federal, State, and local laws, regulations, and permits

concerning environmental protection, pollution control, and abatement that are applicable to the proposed operations and the requirements imposed by those laws, regulations, and permits.

- b. Methods for protection of features to be preserved within authorized work areas. Prepare a listing of methods to protect resources needing protection, i.e., trees, shrubs, vines, grasses and ground cover, landscape features, air and water quality, fish and wildlife, soil, historical, archeological, and cultural resources.
- c. Procedures to be implemented to provide the required environmental protection and to comply with the applicable laws and regulations. Provide written assurance that immediate corrective action will be taken to correct pollution of the environment due to accident, natural causes, or failure to follow the procedures set out in accordance with the environmental protection plan.
- d. A permit or license for and the location of the solid waste disposal area.
- e. Drawings showing locations of any proposed temporary excavations or embankments for haul roads, stream crossing, material storage areas, structures, sanitary facilities, and stockpiles of excess or spoil materials.
- f. Environmental monitoring plans for the job site, including land, water, air, and noise monitoring.
- g. Traffic control plan.
- h. Methods of protecting surface and ground water during construction activities.
- i. Spill prevention. Specify all potentially hazardous substances to be used on the job site and intended actions to prevent accidental or intentional introduction of such materials into the air, ground, water, wetlands, or drainage areas. Specify the provisions to be taken to meet Federal, State, and local laws and regulations regarding labeling, storage, removal, transport, and disposal of potentially hazardous substances. List the Corps environmental POC's with phone numbers and email addresses. Specify that the contractor will notify the Corps Environmental Branch POC's by phone the same day any spill if possible or no later than the morning of the next day(to include weekends).
- j. Spill contingency plan for hazardous, toxic, or petroleum material. There must be a secondary containment (110% of full tank capacity) for any fuel storage tanks (including pumps/generators). Fuel storage tanks must have over head cover to prevent rainwater from entering the secondary containment. The spill response material quantity must be sufficient to absorb/contain the maximum expected fuel storage.
- k. Work area plan showing the proposed activity in each portion of the area and identifying the areas of limited use or nonuse. Include measures for marking the limits of use areas.
- l. Include a statement as to the person who will be responsible for implementation of the Environmental Protection Plan. The Contractor personnel responsible is to report directly to the Contractor's top

management and is to have the authority to act for the Contractor in all environmental protection matters.

- m. Recycling and Waste Management Plan. Executive Order 12873 of 20 October 1993 requires a number of considerations in planning a project. Fallen trees should not be burned or buried to the maximum extent practical. Mulching, composting, and other uses for trees should be considered. Also, recovery of metals at the job site, including aluminum cans, should be considered with proceeds to be retained by the Contractor. Non-Federal recycling and waste minimization efforts are to also be incorporated into this plan.
- n. Operational plan to achieve protection of sea turtles during hopper dredge(s) operation.
- o. Endangered Species Watch Plan detailing how the requirements to minimize the risk of a North Atlantic right whale/dredge vessel interaction will be accomplished.
- p. Invasive and Nuisance Species Transfer Prevention Plan. The invasive and nuisance species transfer prevention plan is to identify specific transfer prevention procedures and designated cleaning sites/locations. The Contractor is to provide a report with photographs verifying that equipment brought on the site was cleaned. The prevention plan and report are to be submitted to the Invasive Species Management Branch, Operations Division.

1.9 QUALIFICATIONS

The individual(s) are to have a combination of education and experience that demonstrates they have the knowledge and background to identify the species of concern and be able to fulfill the requirements of the environmental specification. Resume/qualification submittals must specifically address the knowledge/experience with the species of concern. Any specific requirements for an environmental observer/monitor called out in environmental coordination or authorization documents (such as agency coordination letters, biological opinions, or permits) regarding the project area must be also addressed in the submittals.

1.9.1 Marine Animal Observer Qualifications (Clamshell/Mechanical Only)

Submit to the Contracting Officer for approval, the qualifications of the marine animal observer no later than 5 calendar days after Notice to Proceed. A resume of demonstrated experience monitoring manatees and marine turtles and their behaviors in association with in-water construction projects is required. Specifically, include a table with the names and locations of specific in-water construction projects that observers have worked on. Also, the table is to include for each in-water construction project the permit number(s), the estimated amount of time (hours) spent observing, the estimated number of manatees and marine turtles seen, and whether the work was ever halted due to manatees or marine turtles in the area. Complete the FWC Protected Marine Species Observer Information Form (see CONSTRUCTION FORMS AND DETAILS) for each marine animal observer to document their experience.

1.9.2 Endangered Species Observer and Trawl Supervisor Qualifications (For Hopper Dredges Only)

Submit the names of the endangered species observer-biologists (ESOs) to

U.S. Naval Station Mayport
Duval County, Florida

perform the monitoring for this contract no later than 5 calendar days after Notice to Proceed. Only NMFS pre-approved ESOs are authorized to monitor threatened/endangered species takes by hopper dredges. A pre-approved list of ESOs can be obtained by contacting NOAA Fisheries' Northeast Region, Protected Resources Division at 978-282-8460/(978)-281-9291.

1.10 SUBCONTRACTORS

Assurance of compliance with this section by subcontractors will be the responsibility of the Contractor.

1.11 NOTIFICATION

The Contracting Officer will notify the Contractor in writing of any observed noncompliance with the aforementioned Federal, State, or local laws or regulations, permits and other elements of the environmental protection specifications. After receipt of such notice, inform the Contracting Officer of proposed corrective action and take such action as may be approved. If the Contractor fails to comply promptly, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No time extensions are to be granted or costs or damages allowed to the Contractor for any such suspension.

Within 30 calendar days following project completion, the Contractor is to submit to the Contracting Officer and cc the Environmental Compliance POC the [Project Environmental Summary Sheet](#). The Project Environmental Summary Sheet is to provide absence or occurrence of environmental incidents, as indicated in paragraph CONSTRUCTION FORMS AND DETAILS below.

1.12 CONTRACTOR PERSONNEL QUALIFICATIONS IN POLLUTION CONTROL

Provide qualified personnel to perform all phases of environmental protection, including methods of detecting and avoiding pollution, familiarization with pollution standards, both statutory and contractual, and careful installation and monitoring of the project to ensure adequate and continuous environmental pollution control. Quality Control and supervisory personnel are to be thoroughly knowledgeable of Federal, State, and local laws, regulations, and permits as listed in the Environmental Protection Plan submitted by the Contractor. Identify the Quality Control personnel in the Quality Control Plan submitted in accordance with Section 01 45 05 DREDGING/BEACH FILL PLACEMENT - QUALITY CONTROL.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.1 PROTECTION OF ENVIRONMENTAL RESOURCES

For contract work, comply with all applicable Federal, State, or local laws and regulations. The environmental resources within the project boundaries and those affected outside the limits of permanent work under this Contract will be protected at least during the entire period of this Contract. Confine activities to areas defined by the drawings and specifications. Environmental protection is to be as stated in the following subparagraphs.

3.1.1 General Project Environmental Design and Installation Criteria

Some project sites have features that must not be impacted in any way, including cultural, historic, or archeological features. At all sites, minimize disturbance to existing features at the site to the extent possible, including vegetative, topographic, and drainage pattern features. Wetland, seagrass or hardbottoms impacts (temporary access, detours, staging areas, and other work area impacts) to project sites should be avoided and may require separate permitting action. Any wetlands temporarily impacted are to have its soil restored upon project completion. Expansion of previously permitted project footprints may likewise require separate permitting action.

In all cases, the installation of temporary systems to aid construction will provide for protection of the environment during handling, installing, storing, utilizing, transporting, servicing, testing, refilling, transferring, pumping, processing, removing waste products, repairing and maintaining systems and their components. Necessary protection is to also be considered that would prevent contamination of the environment from impacts to the systems caused by storm water runoff and flooding. Retrofit of temporary systems on project sites to modern environmental protection design standards will also be considered.

In the event environmental protection measures fail, implement procedures to control and correct environmental damage. Coordinate with the Contracting Officer if environmental protection measures fail.

3.1.1.1 Sewage-Based Systems Environmental Design and Installation Criteria

In general, there will be no waste or debris discharges of any kind for a project unless authorized by the Contracting Officer. This is to include the Contractor's providing sufficient temporary sanitary equipment and facilities for the project. The design and/or installation of temporary or permanent sewage systems are to ensure that waters will be free of effects of sewage discharges. Adhere to Federal, State, or local codes and requirements regarding sewage in the design, such as those of the EPA and, in the case of the State, Chapter 62-620 (Wastewater Facilities) of the FAC. Adhere to Best Management Practices from the applicable agencies in the design.

3.1.2 Protection of Land Resources

Prior to the beginning of any construction, identify all land resources to be preserved or avoided within the work area. Remove materials displaced into not cleared areas. Do not remove, cut, deface, injure, or destroy land resources including trees, shrubs, vines, grasses, topsoil, and land forms without special permission from the Contracting Officer. Engage a qualified tree surgeon to perform all tree surgery. Repair injuries to bark, trunk, branches, and roots of protected trees by dressing, cutting, and painting as specified for Class I Fine Pruning, of the National Arborist Association Pruning Standards for Shade Tree or as per State's Agricultural Extension Agency Guidelines, immediately as occurrences arise. No ropes, cables, or guys are to be fastened to or attached to any trees for anchorage unless specifically authorized. Where such special emergency use is permitted, provide effective protection for land and vegetation resources at all times as defined in the following subparagraphs.

3.1.2.1 Work Area Limits

Prior to any construction, mark the areas that are not required to accomplish all work to be performed under this Contract. Mark or fence isolated areas within the general work area to be saved and protected. Protect existing trees designated to remain. Provide protection of tree roots against noxious materials in solution caused by run-off or spillage. Fires are to be located outside the canopy of protected trees. Do not store materials, trailers, or equipment within the drip line of any protected tree. Protect monuments and markers before construction operations commence. Where construction operations are to be conducted during darkness, the markers will be visible. Convey to personnel the purpose of marking and/or protection of all necessary objects.

3.1.2.2 Contractor Facilities and Other Work Areas

Locate facilities, staging and other work areas as indicated in Section 01 50 00.00 35 TEMPORARY CONSTRUCTION FACILITIES AND CONTROLS, and in compliance with permits listed in the paragraph PERMITS AND AUTHORIZATIONS above and obtained for performance of work. Manage borrow or dredge material management (DMMA) areas to minimize erosion and to prevent sediment from entering nearby watercourses, wetlands, or lakes. Manage and control dredged material areas to limit spoil intrusion into areas designated on the drawings and to prevent erosion of soil or sediment from entering nearby watercourses, wetlands, or lakes. Develop dredged material areas in accordance with the grading plan indicated on the drawings. Control temporary excavation and embankments for plant and/or work areas to protect adjacent areas from despoilment. If there is suspicion that sediment may be unsuitable for disposal at a specified location, immediately take measures to contain the suspect sediment and notify the Contracting Officer.

3.1.2.3 Solid Wastes

The Contracting Officer will approve the type of solid waste containers, staging areas on land for these containers, the roads to haul the solid waste, and the final solid waste disposal site. In addition, comply with Federal, State and local regulations pertaining to the use of the solid waste disposal site.

Place solid wastes (excluding clearing debris) in containers which are emptied on a regular schedule. Conduct handling and disposal to prevent contamination.

3.1.2.4 Fuel, Oil, and Lubricants

Manage fuel, oil, and lubricants in order to prevent spills and evaporation. Provide a 4-foot square, 16-gauge metal pan with borders banded up and welded at corners right below the bibb for fuel dispensers to prevent spills. Edges of the pans are to be 8-inch minimum in depth to ascertain that no contamination of the ground takes place. Clean pans by an approved method immediately after every dispensing of fuel and wastes disposed of offsite in an approved area. If any spilling of fuel occurs, immediately recover the contaminated ground and dispose of it offsite in an approved area. Store petroleum waste generated in marked corrosion-resistant containers and recycled or disposed of in accordance with 40 CFR 279, State, and local regulations. Any fueling operation is to be conducted such that any accidental spills will be prevented from entering surface waters or wetlands.

3.1.2.5 Hazardous Waste

Hazardous wastes are defined in 40 CFR 261. Ensure that hazardous wastes are stored and disposed of in accordance with 40 CFR 261 and State and local regulations. Ensure that hazardous wastes are packed, labeled, and transported in accordance with 49 CFR 173 and State and local regulations.

3.1.2.6 Hazardous Materials

Ensure that hazardous materials are labeled, stored, and transported in accordance with 49 CFR 173, State and local regulations.

3.1.2.7 Disposal of Other Materials

Handle other materials than previously discussed (Construction and Demolition, vegetative waste, etc.) as directed by the Contracting Officer.

3.1.3 Preservation and Recovery of Historic, Archeological, and Cultural Resources

3.1.3.1 Applicable Law

A number of Federal laws require protection of cultural resources. Two laws, in particular, can be potentially involved with construction activities: (1) the National Historic Preservation Act, as amended; and, (2) the Abandoned Shipwreck Act.

3.1.3.2 Known Resources

If known historic, archeological, and cultural resources within the Contractor's work area(s) are present, it will be designated as an avoidance area on the contract drawings or other documents. If so designated, install protection for these resources and preserve during the Contract's duration. Do not dredge, spud, or anchor within the designated avoidance area. Do not distribute maps or other information on these resource locations except for distribution among the Contractor's staff with a need to know technical responsibility for protecting the resources.

3.1.3.3 Inadvertent Discoveries

If, during construction activities, items that may have historic or archaeological origin are observed, such observations are to be reported immediately to the Contracting Officer so that the USACE Project Archaeologist may be notified. Cease all activities adjacent to the discovery that may result in the destruction of these resources and prevent employees from further removing, or otherwise damaging, such resources. Once reported, USACE Project Archaeologist will initiate coordination with the appropriate federal, tribal and state agencies to determine if archaeological investigation is required. Additional work in the area of the discovery will be suspended at the site until all federal and state regulations have been successfully completed and the USACE staff members provide further directive. Project activities in the vicinity of the discovery may not resume until the Contracting Officer approves work to proceed.

The possibility of encountering submerged cultural resources is inherent in dredging and snagging operations. Such findings could include shipwrecks, shipwreck debris fields (such as steam engine parts, or wood

planks and beams), anchors, ballast rock, concreted iron objects, concentrations of coal, prehistoric watercraft (such as log dugouts), and other structural features intact or displaced. The materials may be deeply buried in sediment, resting in shallow sediments or above them, or protruding into water. Suspected cultural materials inadvertently gathered from a water-saturated context should be kept moist by re-immersion, spraying, or some other expedient means of wetting until the Contracting Officer provide further directives in coordination with the USACE Project Archaeologist. No interviews or other contact with the media or federal, tribal or state agencies are to occur without clear authorization from the Contracting Officer.

In the unlikely event that unmarked human remains are identified, they will be treated in accordance with Section 872.05 Florida Statutes. Immediately cease all work and ground disturbing activities within a 50-meter diameter of the unmarked human remains and notify the Contracting Officer or the USACE representative is to be immediately notified. The USACE Project Archaeologist will notify the local medical examiner, and the appropriate State Historic Preservation Office (SHPO) and Tribal Historic Preservation Office(s) (THPO). Based on the circumstances of the discovery, equity to all parties, and considerations of the public interest, the Contracting Officer may modify, suspend, or terminate work activities. Such activity is not to resume without written authorization from the State Archeologist and from the Contracting Officer.

3.1.3.4 Claims for Downtime due to Inadvertent Discoveries

Upon discovery and subsequent reporting of a possible inadvertent discovery of cultural resources, continue work well away from, or otherwise protectively avoiding, the area of interest, or in some other manner that strives to continue productive activities in keeping with the Contract. If an inadvertent discovery be of the nature that substantial impact(s) to the work schedule are evident, such delays are to be coordinated with the Contracting Officer. Contract adjustments resulting from compliance with this paragraph are to be determined in accordance with Clause DIFFERING SITE CONDITIONS.

3.1.4 Protection of Water Resources

Keep construction activities under surveillance, management, and control to avoid pollution of surface, ground waters, and wetlands. Implement special management techniques as set out below to control water pollution by the listed construction activities which are included in this Contract. Use construction methods that protect wetland and surface water areas from damage due to mechanical grading, erosion, sedimentation and turbid discharges. Storageing or stockpiling of equipment, tools, or materials within wetlands or along the shoreline within the littoral zone is not allowed unless specifically authorized.

3.1.4.1 Monitoring of Water Areas

Monitor all water areas affected by construction activities.

3.1.4.2 Turbidity

Conduct operations in a manner to minimize turbidity. Refer to Section 01 57 25.00 35 TURBIDITY AND DISPOSAL MONITORING for further instructions.

3.1.4.3 Oil, Fuel, and Hazardous Substance Spill Prevention and Mitigation

Prevent oil, fuel, or other hazardous substances from entering the air, ground, drainage, local bodies of water, or wetlands. This is to be accomplished by design and procedural controls. In the event that a spill occurs despite the design and procedural controls, the following is to occur:

- (1) Take immediate action to contain and cleanup any spill of oil, fuel or other hazardous substance.
 - (2) Immediately report spills to the Contracting Officer and any spills to surface water or wetlands to the National Response Center (NRC) at 1-800-424-8802. Any spills above 25 gallons to soils must be reported to the NRC.
 - (3) Spill contingency planning is to be strictly in accordance with the criteria of 40 CFR, Part 109.
 - (4) To control the spread of any potential spill, absorbent materials are to be readily available and capable of absorbing the contents of the single largest tank.
 - (5) To control the spread of any potential spill, provide a written certification of commitment of manpower, equipment, and materials required to expeditiously cleanup and dispose of spill materials.
- a. Spill Preventive Systems: System design and installation requirements have been discussed at the beginning of this Section. Temporary or portable tanks will conform to applicable Federal, State and local codes and requirements and is not to be placed where they may be affected by storm, flooding, or washout. Diversionary structures for spills are to be put in place in advance where practical. Fuel storage locations to include pumps and generators will be placed and have features such as earthen berms, to prevent any fuel spills to surface water or wetlands. Both spill preventive systems and any deviations from associated requirements will be approved by the Contracting Officer prior to implementation.
 - b. Liabilities: Liable in the amounts established in 40 CFR, Part 113 when it can be shown that oil was discharged as a result of willful negligence or willful misconduct. The penalty for failure to report the discharge of oil will be in accordance with the provision of 33 CFR, Part 153.

3.1.5 Protection of Fish and Wildlife Resources

Keep construction activities under surveillance, management, and control to minimize interference with, disturbance to, and damage of fish and wildlife. List species that require specific attention along with measures for their protection in the Environmental Protection Plan prior to the beginning of construction operation.

In the event that a threatened, endangered, or Marine Mammal Protection Act (MMPA) species is harmed as a result of construction activities, cease all work and notify the Contracting Officer. The order of contact within the Corps of Engineers will be as follows:

TABLE 1 - Order of Contact of Corps Personnel

U.S. Naval Station Mayport
Duval County, Florida

TITLE	TELEPHONE NUMBER
Corps, Inspector	Onsite/After hours to be provided
Corps, Project Engineer	Onsite/After hours to be provided
Contracting Officer Representative	To be provided
Administrative Contracting Officer	To be provided
Area Engineer for Area Office	To be provided

Information provided above will be updated during the pre-construction meeting.

3.1.5.1 Listed Species Protection

Endangered and Threatened Species and MMPA species that may be in the project area are listed in Table 2.

TABLE 2 - Listed Species

Common Name	Scientific Name	ESA Status
Florida manatee	Trichechus manatus latirostris	Threatened
North Atlantic Right whale	Eubalaena glacialis	Endangered
Sperm whale	Physeter macrocephalus	Endangered
Humpback whale	Megaptera novaeangliae	MMPA protected
Common Bottlenose Dolphin	Tursiops truncatus	Marine Mammal Protection Act Protected
Shortnose sturgeon	Acipenser brevirostrum	Endangered
Atlantic sturgeon	Acipenser oxyrinchus oxyrinchus	Endangered
Green sea turtle	Chelonia mydas	Threatened
Loggerhead sea turtle	Caretta caretta	Threatened
Kemp's ridley sea turtle	Lepidochelys kempii	Endangered
Leatherback sea turtle	Dermochelys coriacea	Endangered
Hawksbill sea turtle	Eretmochelys imbricata	Endangered
Smalltooth sawfish	Pristis pectinata	Endangered
Giant Manta Ray	Manta birostris	Threatened
Scalloped hammerhead	Sphyrna lewini	Threatened

U.S. Naval Station Mayport
Duval County, Florida

Common Name	Scientific Name	ESA Status
Oceanic Whitetip	Carcharhinus longimanus	Threatened

Instruct personnel associated with the project of the potential presence of listed species as noted in the Table 1 in the area and the need to avoid collisions with and/or harming these animals. All construction personnel are to be advised that there are civil and criminal penalties for harming, harassing, or killing, or attempting to harm, harass or kill or listed species as noted in the Table 2 in the area which are protected under the Marine Turtle Protection Act, Marine Mammal Protection Act of 1972, the Endangered Species Act of 1973, and/or the Florida Manatee Sanctuary Act. The Contractor will be held responsible for any listed species as noted in the Table 2 in the area harmed, harassed, or killed as a result of construction activities.

- a. Siltation Barriers: Made of material in which manatees cannot become entangled, properly secured, and regularly monitored to avoid manatee entrapment. Barriers must not block manatee entry to or exit from essential habitat.
- b. Special Operating Conditions:
 - (1) All vessels associated with the project will operate at no wake/idle speeds at all times while in waters where the draft of the vessel provides less than a 4-foot clearance from the bottom, and vessels will follow routes of deep water whenever possible. Boats used to transport personnel must be shallow-draft vessels, preferably of the light-displacement category, where navigational safety permits. Mooring bumpers must be placed on all barges, tugs, and similar large vessels wherever and whenever there is a potential for manatees to be crushed between two moored vessels. The bumpers are to provide a minimum stand-off distance of four feet under maximum designed compression.
 - (2) Manatee or Marine Turtle Sighting: If a manatee(s) or marine turtle(s) are sighted within 100 yards of the project area, all appropriate precautions will be implemented by the Contractor to ensure protection of the manatee or marine turtle. These precautions will include the operation of all moving equipment, including vessels, no closer than 50 feet of a manatee or marine turtle. If a manatee or marine turtle are closer than 50 feet to moving equipment or the project area, the equipment will be shut down and all construction activities are to cease within the waterway to ensure protection of the manatee or marine turtle. For unanchored vessels, operators are to disengage the propeller and drift out of the potential impact zone. If drifting would jeopardize the safety of the vessel then idle speed may be used to leave the potential impact zone. Construction activities will not resume until animal(s) has moved beyond the 50-foot radius of the project operation, or until 30 minutes elapses if the animal(s) has not reappeared within 50 feet of the operation. Animals will not be herded away or harassed into leaving. If construction activity will cease, notify the Contracting Officer.
 - (3) During the period November 1 through April 30, barges or dredges

moving through the designated critical habitat of the North Atlantic right whale (*Eubalaena glacialis*) will take the following precautions. Before the start of dredging, the Contractor will provide one whale observer email address to receive aerial survey-related notification (status, fly/no-fly plans, etc.) that will be immediately sent to all active vessels working in water for the project. The contractor is to include in this notification all crew member email addresses that should receive this email alert along with the "Bin" boundary and the project name. (See Whale Alert Quick Guide, Figure 5 for a map of "Bin" boundaries.) This information will be provided to nmfs.ser.rw.subscribe@noaa.gov and Contracting Officer.

Additionally, all captains of dredges, relocation trawlers, survey vessels, and support vessels over 33-ft in length will provide a text message address (that is capable of receiving short emails as text messages) to receive real-time whale alerts throughout the calving season. The text message address submittal must include the name of the phone carrier (e.g. Verizon, AT&T, etc.) and include the "Bin" boundary for alerts and the project name (See Whale Alert Quick Guide, Figure 5 for a map of "Bin" boundaries). The [Whale Alert Text Message Address](#) will be provided to the Contracting Officer and nmfs.ser.rw.subscribe@noaa.gov at least 14 days prior to the start of dredging.

Vessel Speed Requirements: Speed requirements must be followed if a North Atlantic right whale has been spotted or reported in the area. North Atlantic right whale presence may be determined by observers on the vessel, reports from aerial surveys, Established Right Whale Early Warning System (EWS), or confirmed public sighting reports. All captains are required to use daily available information and reports on the presence of North Atlantic right whales and aerial survey activities in the project area. These speed restrictions apply to all vessels associated with a project covered under SARBO.

Vessels over 65 ft in length: When a whale is observed or reported within 38 nmi of dredge or support vessels, vessels must slow to 10 knots or slowest safe navigable speed for 36 hours or until next North Atlantic right whale survey when no whales are observed, whichever is shorter.

Vessels 33- 65 ft: When a whale is observed or reported within 38 nmi of dredge or support vessels which are within the Southeast Seasonal Management Area (SMA), vessels must slow to 10 knots or slowest safe navigable speed until the next North Atlantic right whale survey when no whales are observed. Vessels may resume speed once outside the Southeast SMA. Outside of the Southeast SMA, but within the range of North Atlantic right whale habitat, the vessel must slow to 10 knots for 36 hours, or until the next North Atlantic right whale survey when no whales are observed.

Right whale sightings will be immediately communicated by marine radio to the dredging Contractor's dredge. In addition, the tug/barge or dredge operator is to maintain a 500-yard buffer between the vessel and any whale. The area designated as critical habitat in the southeastern United States is currently defined as occurring from the Virginia/North Carolina border south to Cape Canaveral, Florida. If a stranded/injured/incapacitated whale is

observed within the construction site, the Contractor is requested to immediately contact the NMFS Whale Stranding Network Hotline at 1-877-WHALE-HELP (1-877-94253-4357)(Note: This is a working phone number) for guidance and 888-404-FWCC (3922). Sightings of any large whale in the project area (without regard to species) are to be made to the NMFS Whale Help Hotline - 1-877-WHALE-HELP (1-877-94253-4357). All large whales should be assumed to be right whales. All on-site project personnel including the vessel captain, crew, and PSO on dredge vessels, survey vessels, and all supporting vessels will be instructed on the presence of North Atlantic right whale and other ESA-listed species and the requirements to observe, avoid, and report North Atlantic right whale in the area.

All project vessels will carry operational automatic identification system transmitters as required by the U.S. Coast Guard. Transmitters will be powered on and transmitting while vessels are underway and NMFS will be provided the vessel name and vessel tracking number (maritime mobile service identities) so that all vessels operating can be tracked and confirm compliance with this Plan. Vessel tracking numbers will be recorded in ODESS and emailed to NMFS at SERO.DREDGE@NOAA.GOV for all vessels over 33-ft in length operating from the Virginia/North Carolina border south to Cape Canaveral, Florida, during the North Atlantic right whale migration and calving season from November 1 through April 30.

Two Endangered Species Observers (ESOs) are required on Hopper Dredges between November 1 and April 30th. The ESO observer coverage requirement is 100% (24hr/day) monitoring for hopper dredging and relocation trawling.

- (4) Report interactions with threatened or endangered (T&E) species to the Contracting Officer. Cease dredging operations each time an T&E species is potentially taken until the Contracting Officer provide notification to resume dredging. Promptly following the potential take of an T&E species, the Government will perform a risk assessment to include a review of all potential circumstances which contributed to the take, a review of the DQM data, and a physical inspection of the dredge and its operating procedures in accordance with the hopper dredge inspection checklist, project specifications and applicable biological opinions and permits. The Government, in coordination with the Contractor, will develop a risk management plan containing specific actions to minimize risk of additional potential take. The Government may terminate dredging if it is determined that take numbers are at or near the incidental take statement limits or additional take jeopardizes other Government approved regional dredging projects authorized by the South Atlantic/Gulf of Mexico Regional Biological Opinion and the risk of additional take by the project is determined to be high.
 - (5) If a sea turtle or sturgeon is taken by hopper dredge (dead or alive), a submittal of incident information is to be made into the Operations and Dredging Endangered Species System (ODESS).
- c. Marine Animal Monitoring (Clamshell/Mechanical Only): During clamshell dredging operations, a qualified dedicated observer is to monitor for the presence of manatees and marine turtles. The

dedicated observer must be in a location with direct line of sight for all in-water construction activity (i.e. direct line of sight of clamshell bucket or excavator arm entering water). The dedicated observer is to have experience in manatee and marine turtle observation and be equipped with polarized sunglasses to aid in observing. Nighttime lighting of waters within and adjacent to the work area is to be illuminated, using shielded or low-pressure sodium-type lights, to a degree that allows the dedicated observer to sight any manatee or marine turtle on the surface within 200 feet of the operation. The dredge operator must gravity-release the clamshell bucket only at the water surface, and only after confirmation that there are no manatees or marine turtles within the safety distance identified in the standard construction conditions.

- d. Manatee Signs: Post temporary signs concerning manatees prior to and during all in-water project activities at sufficient locations to be regularly and easily viewed by all personnel engaged in water-related activities. Remove all signs upon completion of the project. Two temporary signs that have already been approved for this use by the FF&WCC must be used at each location. One sign which reads CAUTION BOATERS - WATCH FOR MANATEES must be posted. A second sign measuring at least 8 ½ inch by 11 inch must explain the requirements for Idle Speed/No Wake and the shutdown of in-water operations. These signs can be viewed at MyFWC.com/manatee.

3.1.5.2 Protected Listed Species Sighting Reports

Report any take concerning any Listed Species in the Table 3.1.5.1; or sighting of any injured or incapacitated of Listed Species to the Contracting Officer by notifying the personnel indicated in the table ORDER OF CONTACT OF CORPS PERSONNEL above.

Provide a copy of the incidental take report in PDF format with photographs to the Contracting Officer within 24 hours of the incident. Report any collision with and/or injury to a manatee to the Florida Fish and Wildlife Conservation Commission Manatee Hotline 1-888-404-FWCC (3922) and imperiledspecies@myFWC.com, as well as the U.S. Fish and Wildlife Service, Jacksonville Field Office 904-731-3336 for North Florida. Report any collision with (and/or injury to) a marine turtle to the Sea Turtle Stranding and Salvage Network (STSSN) at SeaTurtleStranding@myfwc.com. Report any take of Smalltooth sawfish to 1-844-SAW-FISH or email Sawfish@MyFWC.com. Report any take of a Giant manta ray to manta.ray@noaa.gov.

3.1.5.3 Protected Species Summary Report

If a hopper dredge is used, submit all details regarding potential takes in ODESS and the Quality Control Report. Any incidents, including sightings, collisions with, injuries, or killing of manatees, sea turtles (not associated with a take via the dredge) or whales occurring during the contract period are to be recorded in a log that will be turned into the Contracting Officer at the end of the project. Maintain paper copies of all forms entered into the ODESS until the project is deemed complete by the Contracting Officer. An electronic copy of all incident reports/photos associated with take via the dredge is to be included in the summary report. [Weekly Protected Species Summary Report](#) is to be submitted weekly in electronic spreadsheet format to the Contracting Officer and the Project Biologist. An example spreadsheet is available upon request.

Maintain a log detailing all incidents, including sightings, collisions with, injuries, or killing of Listed Species occurring during the contract period. The data is to be recorded on forms provided by the Contracting Officer (sample forms are on the web site indicated in paragraph CONSTRUCTION FORMS AND DETAILS below). Within 30 calendar days following project completion, a [Protected Species Summary Report](#) summarizing the above incidents and sightings is to be submitted to the Contracting Officer and cc the Environmental Compliance POC.

3.1.5.4 Endangered Species Observers (Hopper Dredge Only)

During dredging operations, observers approved by the National Oceanic and Atmospheric Administration - Fisheries (NOAA-Fisheries/NMFS) must be aboard to monitor for the presence of the listed species. Observer coverage must be 100 percent (24hr/day) and is to be conducted year round. During transit to and from the disposal area, the observer is to monitor from the bridge during daylight hours for the presence of T&E species, especially the North Atlantic right whale, during the period November 1 through April 30. The ESO(s) are responsible for personally monitoring, handling, and reporting all captured listed species at all times when the hopper dredge is operating. The ESO on-duty will stand watch to detect listed species in the area and to alert the captain of their presence to avoid vessel collision whenever the vessel is moving. The on-duty ESO will only be responsible for standing watch and not performing other tasks such as inspecting or handling captures when the vessel is in motion. During dredging operations, while dragheads are submerged, the observer is to continuously monitor the inflow and/or overflow screening for turtles and/or turtle parts and sturgeon and/or sturgeon parts. Upon completion of each load cycle, dragheads should be monitored as the draghead is lifted from the sea surface and is placed on the saddle in order to assure that listed species that may be impinged within draghead are not lost and un-accounted for. Observers are to physically inspect dragheads and inflow and overflow screening/boxes for threatened and endangered species take. Hopper dredge operators will not open the hydraulic doors on the inflow boxes prior to inspection by the ESO for evidence of listed species take. The screened area must be accessible to the ESO to ensure 100% observer coverage. The ESO must inspect the contents of all inflow screening boxes after every load, including opening the box (where applicable and safely accessible) and looking inside at all contents for evidence of listed species entrainment. If the contents are not clearly visible and identifiable from a location outside of the box, then in limited instances, the ESO may be required to enter the inflow box to identify contents. If the inflow box cannot be observed due to clogging, the box contents cannot be dumped or flushed unless overflow screening that captures contents for observation by the ESO is operational and monitored for evidence of take. ESOs are to visually monitor box contents as they are dumped or flushed into the hopper.

- a. Monitoring Reports: The results of the monitoring are to be recorded on the appropriate observation sheets and entered into the ODESS system (See Appendix A attached after the end of this Section) and Quality Control Daily Report. There is a form for each load and a form for an incident involving a listed species, specific to the species (turtles, sturgeon, etc.). Complete load forms regardless of whether any incidents with sturgeon (Gulf, Shortnose or Atlantic), whales, or sea turtles occur. In the event of any sea turtle or sturgeon (Gulf, Atlantic or Shortnose) is potentially taken by the dredge, appropriate

incident reporting forms are to be completed and entered into the ODESS system. Additionally, photograph all specimens with the digital camera built into the ODESS tablet. Upload these photographs to the respective incident reports on the ODESS within 4-hours from observing the incident. Do not commence dredging of subsequent loads until all appropriate reports are completed from the previous dredging load to ensure completeness and thoroughness of documentation associated with the incidental take exports. Submit reports to the Contracting Officer within 4-hours of the take. Observer forms can be accessed on the web site indicated in paragraph CONSTRUCTION FORMS AND DETAILS below.

- b. Digital Photographs: Use a digital camera embedded in the ODESS tablet, with an image resolution capability of at least 300 dpi, in order to photographically report all incidental takes, without regard to species, during dredging operations. Upload photos taken associated with the take to the ODESS with the incident information.
- c. A copy of all incident forms, load forms, and photographs from incidents with listed species is to be included in the [Protected Species Summary Report](#).

3.1.5.5 Disposition of Endangered or Threatened Species Carcasses

a. Listed Species Taken by Hopper Dredge:

- (1) Dead ESA-listed species including Turtles or sturgeon (regardless of species): Upon removal of carcass and/or parts from the draghead or screening, observers are to take photographs as to sufficiently document major characteristics of the carcass or parts including but not limited to dorsal, ventral, anterior, and posterior views. For all photographs taken, a backdrop is to be prepared to document the dredge name, observer company name, contract title, time, date, species, load number, location of dredging, and specific location taken (draghead, screening, etc.). Carcass/parts are to be scanned for flipper and Passive Integrated Transponder (PIT) tags. Any identified tags are to be recorded on the ODESS Turtle Incident Form or ODESS Sturgeon Incident Form (as appropriate) that is included in the ODESS Program Forms available through the ODESS tablet (or on the ODESS website if the tablet is not available). Also, fill out the ODESS Dredge Load Form and take photos for submission. Animal parts which cannot be positively identified to species, on board the dredge or barge(s) are to be preserved by the observer(s) for later identification. A tissue sample must be collected from any lethally taken ESA species submitted under the process stated in the Protocol for Collecting Tissue Samples from Turtles for Genetic Analysis found in the CONSTRUCTION FORMS AND DETAILS below. All genetic samples collected are to be submitted to NMFS within 30 days of collection, and verification of submittal to NMFS is to be provided to the Contracting Officer and USACE Project Biologist. After all data collection is complete, the carcass/parts should be marked (spray paint works well), weighted down and disposed of, or provided to a state agency staff member, in accordance with the direction of the Contracting Officer.
- (2) Live ESA-listed species including Turtles: Observer(s) is to measure, weigh, scan for PIT tags, tag (Iconnel flipper and PIT tags (if PIT tag not located during scan, and only if observer is

qualified to tag using PIT tags)), and photograph any live listed species incidentally taken by the dredge. Observer(s) (or their authorized representative) is to coordinate with the Contracting Officer and Environmental branch staff prior to transport. Handling of any marine turtles captured during hopper dredging is to be conducted only by persons with prior experience and training in these activities, such as a NMFS-approved marine turtle observer, or by persons who have submitted documentation to the Contracting Officer of meeting the FWC Marine Turtle Conservation Guidelines specific to stranding activities. The Contracting Officer or their designee who transport live or dead marine turtles or marine turtle parts into, out of, or within, the state of Florida will notify FWC in writing specifying the number, species of turtle, type of specimen, and the destination after transport is complete. Before transport, the Contracting Officer or their designee must coordinate with FWC at SeaTurtleStranding@myfwc.com to determine the appropriate facility to receive live marine turtles for rehabilitation. The Contracting Officer or their designee will abide by the State of Florida's FWC Marine Turtle Conservation Guidelines (<https://myfwc.com/media/3133/fwc-mtconservationhandbook.pdf>) specific to transport of live stranded turtles.

b. Turtles Taken by Relocation Trawler:

- (1) Live Turtles: At least one (1) crewmember, who possesses the required permits for handling endangered species, experienced in sea turtle capture or is a NMFS-approved observer, is to be on board the trawler during the trawl and act as the sea turtle trawling and relocation supervisor. Only a NMFS-approved observer or observer candidate in training under the direct supervision of a NMFS-approved observer is to conduct the sampling operations of tagging, measuring, weighing and collecting tissue samples from turtles. Each turtle that is captured is to be identified to species and age class (juvenile, sub-adult, adult), digitally photographed, scanned for Passive Integrated Transponder (PIT) tags, measured, tagged (Inconel flipper tag and PIT tag, if one not located by scan, and only if observer is qualified to tag using PIT tags), and released and the data recorded on the Sea Turtle Tagging and Relocation Report found in the CONSTRUCTION FORMS AND DETAILS below. External tags are to also be noted and data recorded into the observer's log. Turtles are to be released at locations at least 3 nautical miles (nmi) away from the project area, at time intervals designated by the Sea Turtle Trawling and Relocation Supervisor who is proficient in the viable handling and relocating of sea turtles.

Weight/Size Measurements and Tissue Collection. Turtles must be measured, tagged, and weighed when safely possible, prior to release. Turtle measurements is to be recorded and must include, at a minimum, weight when possible, straight-line length, straight-line width, body depth, and tail length. Turtles are to be tagged with NMFS No. 681 Inconel tags in each of the front flippers and PIT tags according to NMFS protocol (found at https://www.sefsc.noaa.gov/turtles/TM_579_SEFSC_STRTM.pdf and on the ODESS website under the Information tab). Aseptic conditions are maintained for tags and tag attachment. A tissue sample is to be collected from any trawl captured turtle and submitted under the process stated in the Protocol for Collecting Tissue Samples

from Turtles for Genetic Analysis found in the CONSTRUCTION FORMS AND DETAILS below. All genetic samples collected are to be submitted to NMFS within 30 days of collection, and verification of submittal to NMFS is to be provided to the Contracting Officer. Obtain permits related to trawling from the appropriate State agencies.

- (2) Dead Turtles: Any turtle found dead in the nets during relocation trawling efforts is to be digitally photographed, measured, weighed (if possible), scanned for tags. The carcass should then be placed on ice and kept cold (but not frozen) as quickly as possible and the Contracting Officer is to be contacted immediately. The Contracting Officer will contact the Florida State Sea Turtle Salvage and Stranding network (SeaTurtleStranding@myFWC.com) to arrange for the carcass to be transferred for necropsy. The turtle is to be kept on ice (but not frozen) until the appropriate state agency can be contacted. If the state agency declines to recover the carcass for necropsy, the carcass is to be marked (spray paint works well), weighted down and disposed of in accordance with the direction of the Contracting Officer.

Weight/Size Measurements and Tissue Collection. Turtles are to be measured, and weighed. Measurements are to be recorded and must include, at a minimum, weight when possible, straight-line length, straight-line width, body depth, and tail length. The Carcass is to also be scanned for flipper and Passive Integrated Transponder (PIT) tags. Any identified tags are to be recorded on the Sea Turtle Incidental Take Form that is included in the Endangered Species Observer Program Forms located on the web site indicated in paragraph CONSTRUCTION FORMS AND DETAILS below. A tissue sample is to be collected from any lethally taken sea turtle and submitted under the process stated in the Protocol for Collecting Tissue Samples from Turtles for Genetic Analysis found in the CONSTRUCTION FORMS AND DETAILS below. All genetic samples collected are to be submitted to NMFS within 30 days of collection, and verification of submittal to NMFS is to be provided to the Contracting Officer. Obtain permits related to trawling from the appropriate State agencies.

3.1.5.6 Hopper Dredge Equipment

Hopper dredge dragheads are to be equipped with rigid sea turtle deflectors which are rigidly attached. Deflectors are to be solid with no openings in the face. Alternative designs will be considered provided sufficient information is included indicating a particular modification is effective in minimizing potential turtle takes. The Government technical staff will coordinate with NMFS on the effectiveness of this alternate design. A recommendation from the Government technical staff and NMFS will be provided to the Contracting Officer for final approval or disapproval of the alternate design. Do not presume that a decision on an alternative design will be provided during the contracting period. The Contractor's unit price is to be based on the original, solid faced deflector design. No adjustment in unit price will be made for the approval or denial of an alternate deflector design. No dredging is to be performed by a hopper dredge without an installed turtle deflector device approved by the Contracting Officer. Sample Turtle Deflector Design Details are on the first web site indicated in paragraph CONSTRUCTION FORMS AND DETAILS below. For each dredge operating on the project,

dredging operations must not begin until the USACE Sea Turtle Deflector Checklist for Hopper Dredges for USACE and USACE/Army-permitted projects has been completed by a USACE-approved inspector. The completed checklist for each dredge is to be submitted by the USACE-approved inspector to the Contracting Officer and forwarded to the Project Biologist. The checklist can be downloaded from the ODESS website - <https://dqm.usace.army.mil/odess/documents/ODESS%20Sea%20Turtle%20Deflector%20Checklist%20for%20Hopper%20Dredges%20-%2007-11-16.pdf>.

If a hopper dredge is used for this work, submit detail drawings showing the proposed **Turtle Deflector Device** and its attachment to the equipment no later than 30 calendar days after notice to proceed. Include the approach angle for any depths to be dredged during this Contract on the submitted drawing. A copy of the approved drawings and calculations are to be available on the vessel during the life of this Contract. Do not commence any dredging work until approval of the installed turtle deflector device.

a. Deflector Design:

- (1) The leading vee-shaped portion of the deflector is to have an included angle of less than 90 degrees. Internal reinforcement is to be installed in the deflector to prevent structural failure of the device. The leading edge of the deflector is to be designed to have a plowing effect of at least 6 inch depth when the drag head is being operated. Appropriate instrumentation or indicator is to be used and kept in proper calibration to insure the critical approach angle. (Information Only Note: The design approach angle or the angle of lower drag head pipe relative to the average sediment plane is very important to the proper operation of a deflector. If the lower drag head pipe angle in actual dredging conditions varies tremendously from the design angle of approach used in the development of the deflector, the 6 inch plowing effect does not occur. Therefore, every effort should be made to ensure this design approach angle is maintained with the lower drag pipe.)
- (2) If adjustable depth deflectors are installed, they must be rigidly attached to the drag head using either a hinged aft attachment point or an aft trunnion attachment point in association with an adjustable pin front attachment point or cable front attachment point with a stop set to obtain the 6 inch plowing effect. This arrangement allows fine-tuning the 6 inch plowing effect for varying depths. After the deflector is properly adjusted there is to be no openings between the deflector and the drag head that are more than 4 inches by 4 inches.
- (3) Draghead Screening: If the Contractor has the draghead(s) screened, the screening for the draghead production opening must be 8 inches by 8 inches or larger. Pumps employed must be sufficient to operate efficiently with this screening as a minimum.

b. InFlow/Overflow Basket Design:

- (1) Install baskets or screening over the hopper inflow/overflow(s) with no greater than 4 inch by 4 inch openings. The method selected is to depend on the construction of the dredge used and is to be approved by the Contracting Officer prior to commencement of dredging. The screening is to provide 100 percent coverage of

the hopper inflow and overflow(s). The screens and/or baskets are to remain in place throughout the performance of the work. 100 percent inflow and overflow screening is to be maintained throughout the project. If 100 percent inflow and overflow screening cannot be maintained due to clogging of the baskets by dredged material, or other material including reeds, shell, etc, coordinate with the Contracting Officer to determine an appropriate path forward.

- (2) Install and maintain floodlights suitable for illumination of the baskets or screening to allow the observer to safely monitor the hopper basket(s) during non-daylight hours or other periods of poor visibility. Safe access is to be provided to the inflow baskets or screens to allow the observer to inspect for turtles, turtle parts or damage.

c. Hopper Dredge Operation:

- (1) Operate the hopper dredge to minimize the possibility of taking sea turtles or sturgeon and to comply with the requirements stated in the Incidental Take Statement provided by the NMFS in the appropriate Biological Opinion.
- (2) The turtle deflector device and inflow and overflow screens are to be maintained in operational condition for the entire dredging operation.

When initiating dredging, suction through the drag heads is to be allowed just long enough to prime the pumps, then the drag heads are to be placed firmly on the bottom. When lifting the drag heads from the bottom, suction through the drag heads must be allowed just long enough to clear the lines, and then must cease. Pumping water through the drag heads is to cease while maneuvering or during travel to/from the disposal area. (Information Only Note: Optimal suction pipe densities and velocities occur when the deflector is operated properly. If the required dredging section includes compacted fine sands or stiff clays, a properly configured arrangement of teeth may enhance dredge efficiency which reduces total dredging hours and turtle takes. The operation of a drag head with teeth is to be monitored for each dredged section to ensure that excessive material is not forced into the suction line.

- (3) When excess high-density material enters the suction line, suction velocities drop to extremely low levels causing conditions for plugging of the suction pipe. Dredge operators should configure and operate their equipment to eliminate all low level suction velocities. Pipe plugging cannot be corrected by raising the drag head off the bottom. Arrangements of teeth and/or the reconfiguration of teeth should be made during the dredging process to optimize the suction velocities.)
- (4) Raising the drag head off the bottom to increase suction velocities is not acceptable. The primary adjustment for providing additional mixing water to the suction line should be through water ports. To ensure that suction velocities do not drop below appropriate levels, monitor production meters throughout the job and adjust primarily the number and opening sizes of water ports. Water port openings on top of the drag head

or on raised stand pipes above the drag head is to be screened before they are utilized on the dredging project. If a dredge section includes sandy shoals on one end of a tract line and mud sediments on the other end of the tract line, adjust the equipment to eliminate drag head pick-ups to clear the suction line.

- (5) Near the completion of each payment section, perform sufficient surveys to accurately depict those portions of the acceptance section requiring cleanup. Keep the drag head buried a minimum of 6 inches in the sediment at all times. Although the over depth prism is not the required dredging prism, achieve the required prism by removing the material from the allowable over depth prism.
 - (6) During turning operations the pumps are to either be shut off or reduced in speed to the point where no suction velocity or vacuum exists.
 - (7) These operational procedures are intended to stress the importance of balancing the suction pipe densities and velocities in order to keep from taking sea turtles. Develop a written operational plan to minimize turtle takes and submit it as part of the Environmental Protection Plan.
 - (8) Comply with all requirements of this specification and the accepted Environmental Protection Plan. The contents of this specification and the Environmental Protection Plan is to be shared with all applicable crew members of the hopper dredge.
 - (9) To prevent impingement or entrainment of listed species within the water column, dredging pumps will be disengaged by the operator when the dragheads are not actively dredging and therefore working to keep the draghead firmly on the bottom. Pumps will be disengaged when lowering dragheads to the bottom to start dredging, turning, or lifting dragheads off the bottom at the completion of dredging. Hopper dredges may utilize a bypass or other system that would allow pumps to remain engaged, but result in no suction passing through the draghead. Pumping water through the dragheads is not allowed while maneuvering or during travel to/from the disposal or pumpout area. The dredge operator will ensure the draghead is embedded in sediment when pumps are operational, to the maximum extent practicable.
 - (10) All waterport or other openings on the hopper dredge are required to be screened to prevent listed species from entering the dredge.
- d. Hopper Dredge Lighting Requirements: From May 1st through October 31st, sea turtle nesting and emergence season, all lighting aboard hopper dredges and hopper dredge pumpout barges operating within three nautical miles of sea turtle nesting beaches is to be limited to the minimal lighting necessary to comply with U.S. Coast Guard and Occupational Safety and Health Administration (OSHA) requirements. All non-essential lighting on the dredge and pumpout barge is to be minimized through reduction, shielding, lowering and appropriate placement of lights to reduce potential disorientation effects on female sea turtles approaching the nesting beaches and sea turtle hatchlings heading seaward.

3.1.5.7 Sea Turtle Trawling and Relocation (For Hopper Dredges Only)

- a. Sea Turtle Trawling and Relocation: Initiation of sea turtle relocation via trawling will be determined by the Contracting Officer following the take of two sea turtles, without regard to species, and continue until as directed by the Contracting Officer. The results of each trawl is to be recorded on Sea Turtle Trawling Report available from the ODESS website (<https://dqm.usace.army.mil/odess/#/download>). An interim trawling report is to be submitted within 24 hours after the third turtle take by the dredge. A final report is to be prepared after the completion of all trawling efforts. Both reports are to be submitted in hard copy and electronic format to the Contracting Officer, and must summarize the results of the trawling (with all forms and including total trawling times, number of trawls and number of captures). Any turtles captured during the trawling effort are to be measured and tagged in accordance with standard biological sampling procedures with sampling data recorded on Sea Turtle Tagging and Relocation Report available from the ODESS website (<https://dqm.usace.army.mil/odess/#/download>). Any captured sea turtles are to be relocated from the work area no less than 3 miles away from the location recorded on the Sea Turtle Tagging and Relocation Report form.
- b. Sea Turtle Trawling Procedures: A NMFS approved sea turtle trawling and relocation supervisor is to provide sea turtle trawl observers and nets to capture and relocate sea turtles and is to conduct any initiated sea turtle trawling. Turtles are to be captured with trawl nets to relocate them away from the project area to reduce the potential of sea turtle take. Methods and equipment is to be standardized including nets, trawling direction to tide, length of station, length of tow, and number of tows per station. Data on each tow is to be recorded using Sea Turtle Trawling Report available from the ODESS website (<http://dqm.usace.army.mil/odess/#/download>). The trawler is to be equipped with two 60-foot nets constructed from 8-inch mesh (stretch) fitted with mud rollers and flats as specified in Turtle Trawl Nets Specifications on the first web site indicated in paragraph CONSTRUCTION FORMS AND DETAILS below. Paired net tows are to be made for 10 to 12 hours per day or night. Trawling is to be conducted with the tidal flow using repetitive 15-30 minute (total time) tows in the channel or borrow area. Tows are to be made as close to the operating dredge as the trawler captain determines is safe. Positions at the beginning and end of each tow are to be determined from GPS Positioning equipment. Tow speed must be recorded at the approximate midpoint of each tow. Refer to EM 1110-1-1003, paragraph 5.3 and Table 5-1, for acceptable GPS criteria.
- c. Water Quality and Physical Measurements: Water temperature measurements are to be taken at the water surface each day using a laboratory thermometer or a thermometer built into the trawling vessel. Weather conditions are to be recorded from visual observations and instruments on the trawler. Weather conditions, air temperature, wind velocity and direction, sea state-wave height, and precipitation is to be recorded on the Sea Turtle Trawling Report on the second web site indicated in paragraph CONSTRUCTION FORMS AND DETAILS below. High and low tides are to be recorded.
- d. Initiation of Trawling: Trawling will be initiated at the discretion of the Contracting Officer. Initiate trawling and relocation activity in the dredging area as quickly as possible (within 48-72 hours) from

the notification of the Contracting Officer. Continue trawling until suspended by the Contracting Officer.

- e. **Approved Trawling Supervisor:** Trawling is to be conducted under the supervision of a biologist approved by the NMFS. A letter of approval from NMFS is to be provided to the Contracting Officer prior to commencement of trawling. Approved trawl supervisors can be verified through NMFS-NERO (978)-281-9291.
- f. **Turtle Excluder Devices:** Approval for trawling for sea turtles without Turtle Excluder Devices (TEDs) must be obtained from NMFS. Approval for capture and relocation of sea turtles must be obtained from the Florida Fish and Wildlife Conservation Commission (FF&WCC). Approvals must be submitted to the Contracting Officer prior to trawling.
- g. **Daily Reporting:** A daily log is to be kept for each capture trawling operation. The results of each trawl are to be recorded on the Sea Turtle Trawling Report posted on the second web site indicated in paragraph CONSTRUCTION FORMS AND DETAILS below. Positions at the beginning and end of each tow are to be determined from GPS positioning equipment. Tow speed is to be recorded at the approximate midpoint of each tow. See [EM 1110-1-1003](#), paragraph 5.3 and Table 5-1, for acceptable GPS criteria. Data to be recorded in the comments field must also include:
- h. **Weekly Logs/Summary Of FWC Hopper Dredge Reports:** The Contractor is to e-mail the Contracting Officer, (MTP@MyFWC.com), and cc the Environmental Compliance POC weekly reports to the Imperiled Species Management section on Friday each week that trawling is conducted in Florida waters. These weekly reports are to include the species and number of turtles captured in Florida waters, general health, and release information. The Contractor is to provide a summary (on FWC provided Excel spreadsheet) of the following: all trawling activity, all turtles captured in Florida waters, all measurements, the latitude and longitude (in decimal degrees) of captures and tow start-stop points, and times for the start-stop points of the tows (including those tows on which no turtles are captured).
- i. **Comprehensive Trawling Daily Logs:** Within 30 calendar days of completion of the project, submit a copy of the logs regarding capture trawling to the Contracting Officer, (MTP@MyFWC.com), and cc the Environmental Compliance POC.

3.1.6 Protection of Air Resources

Keep construction activities under surveillance, management, and control to minimize pollution of air resources. All activities, equipment, processes and work operated or performed by the Contractor in accomplishing the specified construction is to be in strict accordance with the applicable air pollution standards of the State of Florida (Florida Statute, Chapter 403 and others and Chapters 200 series of the FAC) and all Federal emission and performance laws and standards, including the U.S. Environmental Protection Agency's Ambient Air Quality Standards.

3.1.6.1 Particulates

Particulates, such as dust, is to be controlled at all times, including

weekends, holidays, and hours when work is not in progress. Maintain excavations, stockpiles, haul roads, permanent and temporary access roads, plant sites, spoil areas, borrow areas, and work areas within or outside the project boundaries free from particulates that would cause air pollution standards to be exceeded or that would cause a hazard or nuisance. Provide equipment and approved methods to control particulates as the work proceeds and before a problem develops.

3.1.6.2 Odors

Odors are to be controlled at all times for all construction activities.

3.1.7 Protection of Sound Intrusions

Keep construction activities under surveillance and control to minimize damage to the environment by noise.

3.2 POSTCONSTRUCTION CLEANUP

Clean up any area(s) used for construction.

3.3 PRESERVATION AND RESTORATION OF LANDSCAPE AND AQUATIC VEGETATION DAMAGES

Restore all landscape features and aquatic vegetation damaged or destroyed during construction operations outside the limits of the approved work areas. Such restoration is to be a part of the Environmental Protection Plan as defined in subparagraph ENVIRONMENTAL PROTECTION PLAN of paragraph SUBMITTALS above. This work is to be accomplished at the Contractor's expense.

3.4 INVASIVE AND NUISANCE SPECIES

3.4.1 Prevention of Invasive and Nuisance Species Transfer

Thoroughly clean equipment prior to and following work on the project site to ensure that items/materials including, but not limited to, soil, vegetative debris, eggs, mollusk larvae, seeds, and vegetative propagules are not transported from a previous work location to this project site, nor transported from this project site to another location. Prevention protocols require cleaning all equipment surfaces, including but not limited to, undercarriages, tires, and sheet metal. All equipment, including but not limited to, heavy equipment, vehicles, trailers, ATV's, and chippers must be cleaned. Smaller equipment, including, but not limited to, chainsaws, loppers, shovels, and backpack sprayers, must be cleaned and inspected to ensure they are free of eggs, vegetative debris, vegetative propagules, etc. Utilize any method accepted by the Government; common accepted methods include pressure washing and steam cleaning/washing equipment. Prevention protocols are to also address clothing and personal protective equipment.

Prior to the commencement of work, complete and provide an invasive and nuisance species transfer prevention plan to the Corps for approval. This plan is to be part of the Environmental Protection Plan as defined in subparagraph ENVIRONMENTAL PROTECTION PLAN above. The invasive and nuisance species transfer prevention plan is to identify specific transfer prevention procedures and designated cleaning sites/locations. Prevention protocols may vary depending upon the nature of the project site. Ensure all equipment coming onto and leaving the project site is inspected and

not harboring materials that would spread, or potentially spread, invasive and nuisance species onto or off the project site. Provide a report with photographs, verifying equipment brought on site was cleaned and provide a report verifying equipment was cleaned prior to removal from the project site.

Work delays due to the fault and/or negligence of the Contractor or the Contractor's failure to comply with this specification is not to be compensable.

3.4.2 Invasive and Nuisance Species Reporting

Report sightings of invasive and nuisance species not identified and documented in the pre-construction condition within 24 hours. The reporting is to include the date, time, location (latitude and longitude preferred), environmental conditions, circumstances surrounding the sighting, disposition/behavior of the species, and any other notable observations. Pictures of the reported species are desirable. Reporting is to be made to the Jacksonville District Corps of Engineers Planning Division, Environmental Branch and Operations Division, Invasive Species Management Branch, InvasivePreventionPlan@usace.army.mil. Points of contact will be provided in the Preconstruction Conference.

3.4.2.1 Invasive And Nuisance Species Report

Within 30 calendar days following completion of work, provide a report to the Contracting Officer and cc the Environmental Compliance POC with photographs verifying equipment brought on site was cleaned. Provide a report with photographs verifying equipment was cleaned prior to removal from the project site.

3.5 MAINTENANCE OF POLLUTION CONTROL FACILITIES

Maintain all constructed facilities and pollution control facilities and devices for the duration of the contract or for that length of time construction activities create the particular pollutant.

3.6 CONSTRUCTION FORMS AND DETAILS

See the Construction Forms & Details module at the following web address:
<https://www.saj.usace.army.mil/About/Divisions-Offices/Engineering/Design-Branch/Specifications-Section/>

Also, see web site address <https://dqm.usace.army.mil/odess/#/download> to obtain the Sea Turtle/Post-Hopper Dredging Project Checklist, Endangered Species Observer Program Forms, Sea Turtle Tagging and Relocation Report, and Sea Turtle Trawling Report. Links to these forms are under the heading Turtle Information.

-- End of Section --



Operations and Dredging Endangered Species System (ODESS) Guidance

MONITORING ENDANGERED SPECIES

Maintenance of US waterways for navigation is essential for national and international trade, job creation, and national security as well as for hydropower, flood protection, municipal water supply, agricultural irrigation, recreation, and regional development.

Typically, cutterhead pipeline, hopper, and mechanical dredges are used to maintain navigation depths in these channels and to construct new waterways. However, hopper dredging in the southeastern US potentially impacts five species of threatened or endangered sea turtles (Kemp's ridley, leatherback, loggerhead, green, hawksbill) and three species of sturgeon (shortnose, gulf, and atlantic).

To monitor these impacts and assist in the evaluation of these data, the US Army Corps of Engineers (USACE) Engineer Research and Development Center (ERDC) created the Sea Turtle Data Warehouse (STDW) in 1992. The database was designed as a central repository for current and historical data on impacted sea turtles.

In 2014 the US Army Corps of Engineers, National Dredging Quality Management Program (DQM), at the request of USACE Headquarters (HQUSACE), partnered with ERDC; the US Army Corps of Engineers, Mobile District, Spatial Data Branch (CESAM-OP-J) and Bowhead to develop a new and enhanced STDW database and website. This new Program is referred to as the Operations & Dredging Endangered Species System (ODESS) Program. The use of ODESS will facilitate enhanced monitoring and data collection, enable faster transmittal of information, and meet threatened and endangered species reporting requirements to the National Marine Fisheries Service (NMFS).

In order to monitor dredging impacts on threatened and endangered aquatic species, the dredge shall be equipped with a dedicated tablet computer running ODESS software to track and document the presence of sea turtle, sturgeon, and marine mammal species during dredging operations.

The ODESS system, which consists of a tablet computer with an Internet connection, shall be a standalone system, exclusive to other systems, and shall have USACE ODESS data



collection and reporting software, referred to as the ODESS Field Collector (FC) tool, installed by USACE ODESS support personnel. In the event hardware or software problems prevent the storage or transmission of the collected data, paper copies of the latest ODESS forms and information shall be maintained and submitted to ODESS Support and the USACE Inspector or Contracting Officer according to the schedule outlined in the contract specifications.

OBSERVER QUALIFICATIONS AND TRAINING

Prior to the initiation of the project, Observers shall be familiar with the operation of the ODESS FC tool and proficient in its use so as to be able to prepare and transmit the results of their observations. ODESS system webinar training can be requested by contacting ODESS Support at ODESS@usace.army.mil or 1.877.840.8024.

Depending on the target audience (Observer, dredging Contractor, USACE District personnel, or other Federal agencies), ODESS training could, in addition to the webinar training, consist of demonstrating the steps involved in setting up the FC tool on the dredge, loading Observer-collected data and attachments into the FC tool, submitting these data and attachments to the ODESS database, and/or navigating around the ODESS public website to view and pull down data and/or decision-making information for later analysis.

OBSERVER DATA COLLECTION AND REPORTING

Observers shall record the results of the threatened and endangered species monitoring (described in Section 3.1.5.1, "Endangered Species Protection" of Section 01 57 20, "Environmental Protection," of this specification) in the ODESS system by filling in the appropriate electronic forms on the ODESS FC tool and transmitting the data to the ODESS database. If there is an issue with recording data straight to the FC tool due the logistical nature of how the Observer is collecting this data, paper copies of these forms can be downloaded from the ODESS public website (<https://dqm.usace.army.mil/odess/>) and later entered into the FC tool when the Observer has the best opportunity.

Start of the Project

Prior to the start of dredging, Observers shall verify that the ODESS FC tool is installed and operational on a dredge's dedicated tablet computer and that a viable Internet connection is available. In addition, before a project is initiated, on the ODESS FC tool homepage Observers shall retrieve (or "pull down") project-specific information from the ODESS database and perform a one-time setup of the dredging project by establishing the dredge name and time zone.

During the Project

The following forms shall be used in the FC tool and submitted to the ODESS database at the indicated reporting frequency.

Load Data Form

Observers shall complete the Load Data Form, including a description of screen contents and sea conditions, based on their observations. This form shall be completed and

transmitted to the ODESS database for each load. At the end of each Observer shift, or when an Internet signal is available (not to exceed 24 hours from the start of the shift), the Observer shall submit all of his/her Load Data Forms. If this is not possible due to hardware or software problems, the Observer shall revert to email submission of the forms to ODESS@usace.army.mil.

Sea Turtle Incidental Data Form

If a sea turtle or its remains are identified during a load inspection, after the appropriate parties are notified (according to the requirements identified in the endangered species compliance section of the contract specification), a Sea Turtle Incidental Data Form shall be completed and submitted to the ODESS database as soon as possible (not to exceed 12 hours after the incident). Any applicable documentation (scanned copies of the paper Observer load and incident forms, species photos, etc.) shall be included as electronic attachments (.JPG or .PDF) and submitted using the FC tool.

Sturgeon Incidental Data Form

If a sturgeon or sturgeon parts are identified during a load, after the appropriate parties are notified (according to the requirements identified in the endangered species compliance section of the contract specification), a Sturgeon Incidental Data Form shall be completed and submitted to the ODESS database as soon as possible (not to exceed 12 hours after the incident). Any applicable documentation (scanned copies of the paper Observer load and incident forms, species photos, etc.) shall be included as electronic attachments (.JPG or .PDF) and submitted using the FC tool.

Marine Mammal Observation Data Form

If a large whale is observed, both the Dredge Load and the Marine Mammal Observation Data Forms shall be completed and submitted (not to exceed 12 hours after the observation) to ODESS Support at ODESS@usace.army.mil consistent with the endangered species compliance section of the contract specification (Section 3.1.5.1, "Endangered Species Protection" of Section 01 57 20, "Environmental Protection").

Observers are required to use the FC tool to send all incident attachments and any necessary documentation (i.e., pictures, etc). Do not send attachments via personal email unless the FC tool is unavailable. Also, the paper forms, if needed, should be used as either a "scratch pad" for data collection notes or used any time the FC tool becomes unavailable. The FC tool is the **primary** means of observer data collection and reporting **not** the paper forms. It is not required to scan, attach and submit a copy of the load and incident paper forms as part of the electronic incident record unless it is needed to support the electronic incident record (e.g., species diagram markups).

End of the Project

At the completion of project, the dredging Contractor shall coordinate with a designated USACE district QA/QC point of contact (POC) to determine whether electronic or paper copies of all applicable Observer paper forms will be submitted for the project. Information previously entered on the Post Hopper Dredging Checklist will be available on the ODESS public website (<https://dqm.usace.army.mil/odess>) for the dredging project.

HARDWARE REQUIREMENTS

The dredge shall be equipped and the Contractor is responsible for an ODESS hardware system consisting of a tablet computer, wireless keyboard, wireless mouse and data modem (or equivalent onboard internet connection) along with a proper tote bag and setup location for the afore mentioned hardware components. If a hardware problem occurs, or if a part of the system is physically damaged, the Contractor shall be responsible for repairing it within 48 hours of determination of the condition. The Contractor shall also keep ODESS personnel updated on the status of the onboard ODESS system and the progress of any repairs.

Computer

The Contractor shall provide a dedicated onboard tablet computer for use by the observers and shall have ODESS software installed on it prior to project initiation. This computer shall be located and oriented to allow data entry and data viewing. It must meet or exceed the following specifications:

Tablet Hardware Component	Specification
CPU	Intel or AMD processor with a (non-overclocked) clock speed of at least 2.4 gigahertz (GHz)
Hard Disk	128 gigabytes (GB); solid state internal storage
RAM	4 gigabytes (GB)
Network Adapter	Internal wired or wireless network hardware to match internet connection
Video Adapter	Support for 1024x768 resolution at 16-bit color depth
Display	>= 10.8 in.
Integrated Camera	2MP HD webcam (front); 8MP (back)
Ports	1 free USB port

Internet Access

The Contractor shall maintain an Internet connection capable of transmitting data to the ODESS database. The telemetry system shall always be available and have connectivity in the contract area. If connectivity is lost, unsent data shall be stored locally within the FC tool and transmitted upon restoration of connectivity. The Contractor shall acquire and install all necessary hardware and software to make the Internet connection available for data transmission to the ODESS database. The hardware and software must be configured to allow remote access to the computer by USACE ODESS personnel. Coordination between the dredging company's IT and ODESS Support may be required in order to configure remote access through any security, firewall, router, and telemetry systems. Telemetry systems must be capable of meeting these minimum reporting requirements in all operating conditions.

SOFTWARE REQUIREMENTS

ODESS personnel shall be responsible for installing and testing all ODESS software tools on the dedicated onboard ODESS tablet computer. No other software which conflicts with the ODESS function of recording and transmitting data shall be installed on the tablet computer. The Contractor shall be responsible for installing and/or maintaining any necessary manufacturer-provided software for the installed hardware. If any software problem occurs, the Contractor shall contact ODESS Support at ODESS@usace.army.mil or 1.877.840.8024.

The ODESS tablet computer shall have the following minimum software installed in support of the ODESS system.

Software Component	Specification
Operating System	Windows 10, Contractor-installed
Browser**	Chrome, Internet Explorer, Contractor-installed
ODESS Software	Field Collector (FC) tool, USACE ODESS Support-installed
Remote Access Software	Team Viewer, USACE ODESS Support-installed
**Latest version recommended, Chrome is preferred.	

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 57 25.00 35

TURBIDITY MONITORING

PART 1 GENERAL

- 1.1 SCOPE
- 1.2 SUBMITTALS
- 1.3 TURBIDITY MONITOR QUALIFICATIONS
- 1.4 Turbidity Monitoring Plan

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

- 3.1 MONITORING REQUIREMENTS
 - 3.1.1 General
 - 3.1.1.1 Turbidity Monitoring Equipment
 - 3.1.2 Dredging Locations
 - 3.1.2.1 Station Descriptions
 - 3.1.2.2 Frequency
 - 3.1.2.3 Unanticipated Discharge
- 3.2 TURBIDITY TESTS
 - 3.2.1 Testing
 - 3.2.2 Reporting
 - 3.2.2.1 Report Contents
- 3.3 NOTIFICATION
 - 3.3.1 Turbidity Exceedance
 - 3.3.2 Re-Commencing Construction
- 3.4 WORK DELAY
- 3.5 CONSTRUCTION FORMS AND DETAILS

-- End of Section Table of Contents --

SECTION 01 57 25.00 35

TURBIDITY MONITORING

PART 1 GENERAL

1.1 SCOPE

The work covered by this section consists of furnishing all labor, materials, and equipment, and performing all work required to obtain, analyze, and report the results of turbidity monitoring.

1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification; submittals not having a "G" classification are for information only. When used, a designation following the "G" classification identifies the office that will review the submittal for the Government. The following is to be submitted in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Turbidity Monitor Qualifications; G, D0

Turbidity Monitoring Plan; G, D0

Draft Turbidity Sampling Map; G, D0

Testing Equipment And Procedure; G, D0

SD-03 Product Data

Calibration Standard; G

SD-06 Test Reports

Turbidity Monitoring Reports; G, D0

Turbidity Exceedance Report; G, D0

Updated Turbidity Exceedance Report; G, D0

1.3 TURBIDITY MONITOR QUALIFICATIONS

No later than 10 calendar days after NTP, and prior to hiring, the Contractor is to furnish to the Contracting Officer for approval, the names and qualifications of those individuals performing turbidity monitoring, along with 24-hour contact information. Appropriate qualifications for a turbidity monitor must be familiar with the dredging techniques being used and professional experience monitoring water quality without a record of permit violations. Each monitor is to submit a resume which includes the names and locations of dredging projects worked on, permit number (if known), and dates spent monitoring. In addition, monitor needs to certify they have read and are familiar with the FDEP Standard Operating Procedures (SOPs) for sampling and analysis.

1.4 Turbidity Monitoring Plan

The Contractor is to submit, no later than 10 calendar days after NTP, the following:

- a. A statement as to the person who is to be responsible for turbidity monitoring and reporting to be performed.
- b. Method of turbidity monitoring and reporting that will be performed.
- c. A [Draft Turbidity Sampling Map](#) is to have a georeferenced map, overlaid on an aerial photograph, indicating the location of the current construction activity, the sampling locations (background and compliance), the visible plume pattern of the applicable mixing zone, and the direction of flow. Also include on the map the boundaries of any benthic resources or location of nearby Outstanding Florida Waters, if applicable.
- c. The turbidity monitoring requirements of any special areas such as outstanding Florida waters.
- d. Documentation that the person(s) who will be conducting the turbidity monitoring meets the following requirements:
 - i. is independent of the construction contractor;
 - ii. has formal training in water quality monitoring; and
 - iii. has professional experience in monitoring turbidity for coastal construction projects; and submit a scope of work for the turbidity monitoring to ensure that the appropriate equipment is available to conduct the monitoring correctly at any location, and under any safe conditions.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.1 MONITORING REQUIREMENTS

3.1.1 General

Conducted sampling in accordance with techniques described in the latest edition of Standard Methods published by the American Public Health Association (APHA), American Water Works Association (AWWA), and Water Pollution Control Federation (WPCF), and other current techniques recognized by the scientific community and approved by the Jacksonville District, Corps of Engineers. Analyze samples obtained for turbidity analysis within 30 minutes of collection. Take samples with a sampler obtaining samples uncontaminated by water from any other depth. Additionally, any additional requirement for turbidity monitoring called out in the FDEP permit must be complied with. If there is any conflict with the afore mentioned references with the FDEP permit, the FDEP requirements have precedence.

3.1.1.1 Turbidity Monitoring Equipment

Monitoring required for turbidity is to be measured in Nephelometric

Turbidity Units (NTU) using a Nephelometer. Global Positioning System (GPS) is also required to record sampling stations. Provide the Government with a certification, attesting to the accuracy of [Testing Equipment And Procedure](#) no later than 10 calendar days after NTP. Provide a duplicate of the [Calibration Standard](#) used to calibrate the testing instrument as well as a complete set of operating instructions for the turbidity testing equipment no later than 10 calendar days after NTP. This standard will be used throughout the project to maintain the calibration of the equipment.

Provide instruments used to measure turbidity that are fully calibrated with primary standards within one month of the commencement of the project, and at least once a month throughout the project. Verify calibration with secondary standards each morning prior to use, after each time the instrument is turned on, and after field sampling using two secondary turbidity standards that bracket the anticipated turbidity samples. If the post-sampling calibration value deviates more than 8 percent from the previous calibration value, report results as estimated and include a description of the problem in the field notes.

Perform analysis of turbidity samples in accordance with DEP-SOP-001/01 FT 1600 Field Measurement of Turbidity:
<https://floridadep.gov/dear/quality-assurance/content/dep-sops>

3.1.2 Dredging Locations

Obtain maps and electronic files (Shapefiles, KML, or similar) for all Outstanding Florida Waters (OFW) within the project area. Maps and electronic files can be downloaded from the DEP GIS website at the following link: <https://ca.dep.state.fl.us/mapdirect/?focus-standard>.

Conduct monitoring at dredging locations provided sea, wind, visibility, marine traffic and other conditions allow for safe performance of the activity. Document all instances where turbidity monitoring within the bounds of the above parameters could not be performed. If weather related, provide National Weather Service (NWS) advisories for the day and time turbidity monitoring was not conducted.

Routine monitoring must be collected at surface and mid-depth at the following locations:

3.1.2.1 Station Descriptions

- a. Station 1 (Compliance): No more than 150 meters downcurrent of the dredge or point of discharge and within the densest portion of any project-related turbidity plume.
- b. Station 2 (Background): At least 300 meters upcurrent from the dredge or point of discharge and outside of any turbidity generated by the project.

Turbidity monitoring is not required at the ODMDS.

3.1.2.2 Frequency

Samples shall be collected approximately every four (4) hours during daylight dredging, beginning approximately 30 minutes after the commencement of dredging or placement activities, and also whenever a substantial plume approaches the edge of the mixing zone (i.e., 150 meters

from dredge) such that a water quality violation may exist.

3.1.2.3 Unanticipated Discharge

The 150 meters mixing zone only applies at the dredge site during the time of dredging. Any project-associated discharge occurring within State Waters that is observed to extend beyond the established mixing zone (e.g. scow leakage during transit to the ODMDS) shall be monitored as close to the source as possible until turbidity levels meet the state standard, or until otherwise directed by the Contracting Officer. When this type of unanticipated discharge causes a turbidity exceedance, as compared to a corresponding Background site at least 300 meters upcurrent of the discharge and outside any obvious plume, the contractor shall adhere to the turbidity compliance actions outlined below.

3.2 TURBIDITY TESTS

3.2.1 Testing

Whenever there is doubt as to the adequacy of the testing or validity of the results, the Contracting Officer may direct that additional tests be performed at no additional cost to the Government.

3.2.2 Reporting

Record the monitoring data on summary forms, and as a daily QC test in the RMS system, that contain the pertinent information in the following paragraphs. If there is no construction activity requiring turbidity on a given day, the contractor is to fill out the daily turbidity monitoring report and explain why no monitoring occurred on that day. A standardized form meeting the criteria below, such as DEP's 'Turbidity Monitoring Form' and/or 'Turbidity Monitoring Form - Additional Monitoring Stations' is to be utilized. If one of the DEP forms is used, follow the associated instructions for guidance on naming, filing and distributing the reports. DEP's standardized forms and instructions can be found at <https://floridadep.gov/rcp/beaches-inlets-ports/content/tools-applicants>. Other example forms are on the web site indicated .../content/tools-applicants. Other example forms are on the web site indicated in paragraph CONSTRUCTION FORMS AND DETAILS below. Submit other data in the form supplied by the laboratory chosen to do the analysis. Electronically submit all [Turbidity Monitoring Reports](#) to the Contracting Officer within 24 hours after completion of each test and submit weekly to the Environmental Compliance POC and FDEP within 7 days of collection. Electronic mail addresses of the Corps and FDEP personnel to receive these reports are provided below. Reports are to be provided in Adobe (.pdf) or Spreadsheets (.xls) files.

NAME	ORGANIZATION	E-MAIL ADDRESS
To be determined	USACE ACO	To be determined
To be determined	CESAJ-PD-E POC	To be determined
JCP Compliance Officer	FDEP	jcpcompliance@dep.state.fl.us

Information to be determined will be provided in the Preconstruction Conference. Names and contact information for the Corps personnel to receive these reports will be provided at the Preconstruction Conference. See Section 01 57 20.00 35 ENVIRONMENTAL PROTECTION.

3.2.2.1 Report Contents

Monitoring reports are to include the following information for each day during the construction period:

- a. Permit number.
- b. A statement describing the methods used in collection, handling, storage, and quality control methods used in the analysis of the samples.
- c. Turbidity meter calibration documentation.
- d. A statement by the individual responsible for implementation of the sampling program concerning the authenticity, precision, limits of detection, and accuracy of the data.
- e. Specific Parameters:
 - (1) Time of day and date samples were taken.
 - (2) Depth of water body.
 - (3) Depth of each sample.
 - (4) Antecedent weather conditions including wind direction and velocity.
 - (5) Tidal stage and direction of flow.
 - (6) Water temperature, C degrees.
 - (7) Water sample location.
 - (8) Sampling results, the net difference between compliance and background results, and whether the turbidity level is in compliance.
 - (9) Provide GPS coordinates (x and y and/or Latitude/Longitude) for all sampling stations along with the coordinates of the dredge and discharge pipe and the distance between the sampling station and dredge/discharge pipe for each sampling event to demonstrate compliance with the stated sampling distances.
- f. A description of any factors influencing the dredging operation or the sampling program. Furnish reports daily even when no sampling is conducted. When sampling is not conducted, give a brief statement in the report explaining the reason for not conducting the sampling, such as dredge not working due to mechanical problems or no sampling taken due to high seas. If unable to collect samples due to severe weather conditions, include a copy of a current weather report from a reliable and independent source, such as an online weather service.
- g. A georeferenced map, overlaid on an aerial photograph, indicating the location of the current construction activity, the sampling locations (background and compliance), the visible plume pattern of the applicable mixing zone, and the direction of flow. Also include on the map the boundaries of any benthic resources or location of nearby Outstanding Florida Waters, if applicable. When possible, coordinates should be provided in decimal degrees with a 5-decimal level of precision (i.e., 0.000001). Please also indicate the datum.

3.3 NOTIFICATION

3.3.1 Turbidity Exceedance

If turbidity exceeds background levels by more than 29 NTU, immediately

notify via email the Contracting Officer, FDEP, and Corps Compliance staff member. In addition, cease all dredging activity immediately and take all measures to reduce turbidity. Do not resume dredging until corrective measures have been taken and turbidity has returned to acceptable levels as determined by proper testing described in subparagraph DREDGING LOCATIONS above. Provide a [Turbidity Exceedance Report](#) to include the following to the Contracting Officer and FDEP within 24 hours of the time when the Contractor first becomes aware of the discharge (email acceptable).

- a. the Project Name;
- b. the Permit Number;
- c. location of the turbidity exceedance (shown on map with GPS coordinates);
- d. Level (NTUs above background) of the turbidity exceedance;
- e. the time and date that the exceedance occurred; and
- f. the time and date that construction ceased.

3.3.2 Re-Commencing Construction

Prior to re-commencing construction, an [Updated Turbidity Exceedance Report](#) is to be emailed to the Government Quality Assurance Representative (QAR), Contracting Officer, Administrative Contracting Officer (ACO), and the Environmental Compliance POC in addition to FDEP. with the same information that was included in the Exceedance Report, plus the following information:

- a. turbidity monitoring data collected during the shutdown documenting the decline in turbidity levels and achievement of acceptable levels;
- b. corrective measures that were taken; and
- c. cause of the exceedance.

3.4 WORK DELAY

Delays in work due to the fault or negligence of the Contractor or the Contractor's failure to comply with this specification are not compensable. Any adjustments to the contract performance period or price that are required as a result of compliance with this section are to be made in accordance with the provisions of the Clause SUSPENSION OF WORK.

3.5 CONSTRUCTION FORMS AND DETAILS

See the Construction Forms & Details module at the following web address:
<https://www.saj.usace.army.mil/About/Divisions-Offices/Engineering/Design-Branch/Specifications-Section/>

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 35 - WATERWAY AND MARINE CONSTRUCTION

SECTION 35 20 23.00 35

DREDGING

PART 1 GENERAL

- 1.1 SCOPE
- 1.2 REFERENCES
- 1.3 SUBMITTALS
- 1.4 NOTICE OF INTENT TO DREDGE
- 1.5 DREDGING RESTRICTIONS
 - 1.5.1 Order of Work
 - 1.5.2 Transportation of Material
 - 1.5.3 "Charlie" Pier Restriction
 - 1.5.4 Mayport Vessel Movements
- 1.6 SPECIAL INSTRUCTIONS
- 1.7 PUMPING OF BILGES
- 1.8 HISTORICAL PERIOD SHIPWRECK SITES
- 1.9 UTILITY CROSSINGS
 - 1.9.1 General
 - 1.9.2 Subaqueous Cable Crossings
 - 1.9.2.1 Known Utility Crossings
- 1.10 PERMITS
- 1.11 FUEL OIL TRANSFER OPERATIONS
- 1.12 SIGNAL LIGHTS
- 1.13 NOTICE TO MARINERS -- DREDGING CONTRACTS
- 1.14 FINAL CLEANUP
- 1.15 WORK VIOLATIONS
- 1.16 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM SYSTEM (DREDGES AND DUMP SCOWS)
 - 1.16.1 System Requirements

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

- 3.1 DAILY REPORT OF OPERATIONS
- 3.2 NOTIFICATION OF COAST GUARD
 - 3.2.1 Relocation of Navigation Aids
 - 3.2.2 Dredging Aids
 - 3.2.3 Surveillance of Ocean Disposal
- 3.3 WORK AREA
 - 3.3.1 Access
 - 3.3.2 Protection of Existing Waterways
 - 3.3.3 Adjacent Property and Structures
- 3.4 CHARACTER OF MATERIALS
- 3.5 DISPOSAL OF EXCAVATED MATERIAL
 - 3.5.1 General
 - 3.5.2 Load Restrictions

U.S. Naval Station Mayport
Duval County, Florida

- 3.5.3 Ocean Dredged Material Disposal Site (ODMDS)
 - 3.5.3.1 Post-Dump Closure of Release Mechanisms
 - 3.5.3.2 Spillage/Leakage During Transport to Disposal Site
- 3.5.4 Placing of Dredged Material
- 3.5.5 Barges
- 3.5.6 Dredge Pipelines
 - 3.5.6.1 Dredge Discharge Pipeline
 - 3.5.6.2 Submerged Pipeline
 - 3.5.6.3 Floating Pipeline
- 3.5.7 Booster Pumps
- 3.5.8 Misplaced Material
- 3.5.9 Cumulative Summary Disposal Log
- 3.6 REQUIRED DEPTH, ALLOWABLE OVERDEPTH, AND SIDE SLOPES
 - 3.6.1 Acceptance Sections
 - 3.6.2 Required Depth
 - 3.6.3 Allowable Overdepth
 - 3.6.4 Side Slopes
 - 3.6.5 Excessive Dredging
 - 3.6.6 Areas to be Dredged
- 3.7 SURVEYS
 - 3.7.1 General
 - 3.7.2 Contractor Representative
 - 3.7.3 Survey Certification
 - 3.7.4 Positioning - Horizontal Location and Vertical/Tidal Determination
 - 3.7.4.1 Real Time Kinematic (RTK) GPS
 - 3.7.4.2 Kinematic Tidal Datum (KTD)
- 3.8 ENCOUNTER OF MUNITIONS AND EXPLOSIVES OF CONCERN (MEC) IN PROJECT AREA
 - 3.8.1 Munitions and Explosives of Concern (MEC)
 - 3.8.2 Procedure Upon MEC Encounter
 - 3.8.3 Munition Use and Disposal Throughout Florida
- 3.9 INSPECTION
 - 3.9.1 Quality Assurance Representative (QAR)
 - 3.9.2 Failure to Comply
- 3.10 FINAL EXAMINATION AND ACCEPTANCE
 - 3.10.1 Final Examination of Work
 - 3.10.2 Bed Leveling
 - 3.10.3 Bed Leveling - Reporting and Documenting
 - 3.10.4 Final Acceptance
- 3.11 SHOALING
- 3.12 CONTINUITY OF WORK
- 3.13 NOISE CONTROL
- 3.14 GENERAL DREDGING SAFETY REQUIREMENTS
 - 3.14.1 Dredge Plant Restriction Definitions
 - 3.14.1.1 Certification Required Line - Coast Guard Inspected (CRL-C)
 - 3.14.1.2 Certification Required Line - Load Line (CRL-L)
 - 3.14.1.3 Certification Required Line - Hopper (CRL-H)
 - 3.14.2 Certification of Attendant and Auxiliary Plant
 - 3.14.3 Minimum Tow-Boat Requirements
 - 3.14.4 Harbor Specific Dredging Restrictions
 - 3.14.4.1 Jacksonville Harbor
- 3.15 CONSTRUCTION FORMS AND DETAILS

-- End of Section Table of Contents --

SECTION 35 20 23.00 35

DREDGING

PART 1 GENERAL

1.1 SCOPE

The work covered by this section consists of providing labor, materials, and equipment, and performing excavation and disposal of material as specified herein and indicated on the drawings. This scope also includes necessary measures for protection of the environment. Environmental protection requirements under this Contract are as important to overall completion of the work as other technical aspects. Failure to meet the requirements of these specifications for environmental protection may result in work stoppages or termination for default. No part of the time lost due to such work stoppages can be made the subject of claims for extensions of time or for excess costs or damages by the Contractor. If the Contractor fails or refuses to promptly repair any damage caused by violation of the provisions of these specifications, the Contracting Officer may have the necessary work performed and charge the cost thereof to the Contractor.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2024) Safety and Occupational Health (SOH) Requirements
EM 1110-1-1000	(2015) Photogrammetric and LiDAR Mapping
EM 1110-1-1002	(2012) Survey Markers and Monumentations
EM 1110-1-1003	(2013) NAVSTAR Global Positioning System Surveying
EM 1110-1-1005	(2007) Control and Topographic Surveying
EM 1110-1-2909	(2012) Geospatial Data and Systems
EM 1110-2-1003	(2013) Hydrographic Surveying
ERDC/ITL TR-19-7	(2021) A/E/C CAD Standard - Release 6.1

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

33 CFR 81	72 COLREGS: Implementing Rules
33 CFR 84	Annex I: Positioning and Technical Details of Lights and Shapes

33 CFR 89 Inland Navigation Rules: Implementing Rules
33 CFR 156.120 Requirements for transfer

FLORIDA BOARD OF PROFESSIONAL SURVEYORS AND MAPPERS (FBPSM)

FBPSM Minimum Technical Standards, Chapters 177,
472, 61G17

Code of Federal Regulations (CFR) indicated herein may be found at
<https://www.govinfo.gov/app/collection/cfr>.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Notice of Intent to Dredge

Relocation of Navigation Aids

Marine Survey Inspection Certificates

SD-02 Shop Drawings

Drag Bar

Acceptance Sections Plan; G, D0

SD-07 Certificates

Historical Period Shipwreck Sites

Surveys

Daily Report of Operations

Misplaced Material

Declaration of Inspection

Surveillance of Ocean Disposal

1.4 NOTICE OF INTENT TO DREDGE

Prior to commencement of work on this contract, notify the Commander, Seventh Coast Guard District of intended operations to dredge and request that it be published in the Local Notice to Mariners. This notification must be given in enough time so that it appears in the Notice to Mariners at least two weeks prior to the commencement of this dredging operation. Provide a copy of the notification to the Contracting Officer.

1.5 DREDGING RESTRICTIONS

1.5.1 Order of Work

There is no specific order of work for this project. A reach is defined as a single acceptance section unless otherwise approved by the Contracting Officer. Complete a reach in its entirety before dredging begins on any subsequent reach. If multiple dredges are utilized, then multiple reaches may be worked simultaneously.

1.5.2 Transportation of Material

Water and dredge material are not permitted to overflow or spill out of barges or hopper dredges during transport to the disposal site.

1.5.3 "Charlie" Pier Restriction

Dredging around "Charlie" pier is restricted during weapons loading and offloading operations. Dates, times and distances of restriction will be provided at least one week in advance.

1.5.4 Mayport Vessel Movements

Naval vessels arrive and depart at times on short notice, any movement of dredging operations required to allow unimpeded movements of the ships will be directed by the Contracting Officer.

1.6 SPECIAL INSTRUCTIONS

Dredging of insitu rock is not required; but if encountered, report the location and extent of any insitu rock above grade or required depth to the Contracting Officer.

1.7 PUMPING OF BILGES

Contractors are warned that pumping oil or bilge water containing oil into navigable waters, or into areas which would permit the oil to flow into such waters, is prohibited by Section 13 of the River and Harbor Act of 1899, approved 3 March 1899 (30 Stat. 1152; 33 U.S.C. 407). Violation of this prohibition is subject to the penalties under the referenced Acts.

1.8 HISTORICAL PERIOD SHIPWRECK SITES

If any shipwreck, artifact, or other objects of antiquity that have scientific or historical value, or are of interest to the public, are discovered, located, and/or recovered, the Contractor acknowledges:

- a. The site(s), articles, or other materials are the property of the State of Florida, with title vested in the Department of State, Division of Historical Resource.
- b. To immediately notify the Contracting Officer.

1.9 UTILITY CROSSINGS

1.9.1 General

Investigate the location of utility crossings. Take precautions against damages which might result from operations in the vicinity of the utility

crossings. Suspend dredging until the damage is repaired and approved by the Contracting Officer if damage occurs as a result of dredging operations. Costs of such repairs and downtime of the dredge and attendant plant is part of the Contract.

1.9.2 Subaqueous Cable Crossings

Verify the locations and depths of all utility crossings and take precautions against damages which might result from the operations, especially the sinking of dredge spuds and/or anchors into the channel bottom, in the vicinity of utility crossings. If any damage occurs as a result of the operations, suspend dredging until the damage is repaired and approved by the Contracting Officer. Costs of such repairs and downtime of the dredge and attendant plant will be at the Contractor's expense.

1.9.2.1 Known Utility Crossings

Two utility crossings are located within the project limits and one area of communication cables. Both crossings lie below the allowable dredging depth but may be susceptible to damage caused by the placing of spuds or other deep penetrations of the bottom. The first lies near the junction of the Entrance Channel and the Turning Basin. The second lies inside the Turning Basin. Their approximate locations and depths are shown in the drawings. The area of telecommunication cables is near the utility crossing at the junction of the entrance channel and the turning basin.

In addition to the utility crossings, there are five areas within the project limits where no dredging can occur which contain submerged sensitive apparatus. These areas are shown in the drawings. Do not dredge, anchor or make any other type of bottom contact.

1.10 PERMITS

Refer to the Clause PERMITS AND RESPONSIBILITIES and the paragraph PERMITS AND AUTHORIZATIONS of Section 01 57 20.00 35 ENVIRONMENTAL PROTECTION.

1.11 FUEL OIL TRANSFER OPERATIONS

In accordance with U.S. Coast Guard regulations ([33 CFR 156.120](#)), couplings used in fuel oil transfer operations on any vessel with a capacity of 250 or more barrels of oil must be either a bolted or full-threaded connection; or a quick-connect coupling approved by the Commandant; or an automatic back-pressure shutoff nozzle used to fuel the vessel. Submit an executed fuel oil transfer (Declaration) form signed by the tanker operator to the Contracting Officer for each refueling operation. Notify the U.S. Coast Guard prior to any refueling. A copy of the [Declaration of Inspection](#) for Refueling is on the web site indicated in the paragraph CONSTRUCTION FORMS AND DETAILS.

1.12 SIGNAL LIGHTS

Display signal lights and conduct operations in accordance with the General Regulations of the Department of the Army and of the Coast Guard governing lights and day signals to be displayed by towing vessels with tows on which no signals can be displayed, vessels working on wrecks, dredges, and vessels engaged in laying cables or pipe or in submarine or bank protection operations, lights to be displayed on dredge pipe lines, and day signals to be displayed by vessels of more than 65 feet in length

moored or anchored in a fairway or channel, and the passing by other vessels of floating plant working in navigable channels, as set forth in Commandant U.S. Coast Guard Instruction M16672.2, Navigation Rules: International-Inland (COMDTINST M16672.2), or 33 CFR 81 Appendix A (International) and 33 CFR 84 through 33 CFR 89 (Inland) as applicable.

1.13 NOTICE TO MARINERS -- DREDGING CONTRACTS

If objects are encountered on the channel bottom which could be a hazard to navigation during dredging operations, immediately notify the Contracting Officer of the location of said object and provide other pertinent information necessary for the Contracting Officer to prepare and issue a Notice to Mariners.

1.14 FINAL CLEANUP

Final cleanup, as stated in the paragraph COMMENCEMENT, PROSECUTION, AND COMPLETION OF WORK, includes the removal of the Contractor's plant and equipment either for disposal or reuse. Dispose of plant, equipment, and materials in a manner and at locations approved by the Contracting Officer. Unless otherwise approved by the Contracting Officer, do not abandon equipment in the disposal area or other areas adjacent to the worksite. The Government considers failure to promptly remove plant, pipeline, equipment, and materials upon completion of the dredging a delay in the completion of the final cleanup and demobilization work. In such case, the Government will exercise its right as stated in Clause DEFAULT (FIXED-PRICE CONSTRUCTION) to remove plant, equipment, and materials at the Contractor's expense.

1.15 WORK VIOLATIONS

Work done in violation of these specifications or a verbal or written stop order of the Contracting Officer will be considered as unsatisfactory progress for purposes of progress payments in accordance with Clause PAYMENTS UNDER FIXED-PRICE CONSTRUCTION CONTRACTS.

1.16 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM SYSTEM (DREDGES AND DUMP SCOWS)

1.16.1 System Requirements

See Section 35 20 23.13 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM - SCOW ULLAGE PROFILE and Section 35 20 23.33 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM - PIPELINE HYDRAULIC DREDGE and 35 20 23.23 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM - HOPPER DREDGE. However, in the event of NDQMP System failure (not fully operational), notify the Contracting Officer. If the NDQMP System cannot be used within 48 hours of failure, cease all dredging operations for the vessel until the NDQMP System is fully operational. Delays resulting from NDQMP System failure will be at the Contractor's expense.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.1 DAILY REPORT OF OPERATIONS

Prepare in RMS and submit the [Daily Report of Operations](#), using either ENG Form No. 27A or ENG Form No. 4267, for each dredge and unloader working.

U.S. Naval Station Mayport
Duval County, Florida

Submit report daily and not in groups (groups = multi-days reports packaged together at one time).

3.2 NOTIFICATION OF COAST GUARD

3.2.1 Relocation of Navigation Aids

Navigation aids located within or near the areas required to be dredged will be relocated, if necessary, by the U.S. Coast Guard (USCG). Within 7 calendar days following receipt of Notice to Proceed, contact the USCG D7 Command Center via telephone at 305-415-6800, to acquire the appropriate USCG Sector ATON Officer contact information (name, number, and email). Notify the USCG Sector ATON Officer of plans to dredge adjacent to aids which require relocation. USCG requires 45 days notice prior to movement of ATONs to schedule buoy tender availability. Submit a formal written request to the USCG Sector ATON Officer within 3 calendar days after the telephone notification and provide a copy to the Contracting Officer. Do not remove, change the location of, obstruct, willfully damage, make fast to, or interfere with aid of navigation.

3.2.2 Dredging Aids

Obtain approval from the U.S. Coast Guard for buoys, dredging aid markers to be placed in the water, and dredging aid markers affixed with a light prior to the installation. Do not color or place dredging aid markers and lights in a manner that they will obstruct or be confused with navigation aids.

3.2.3 Surveillance of Ocean Disposal

Notify via certified mail the local Coast Guard Captain of the Port at least 5 calendar days prior to the first ocean disposal and provide a copy to the Contracting Officer. Include the following information in the notification:

- (1) Project designation; Corps of Engineers' Contracting Officer's name and contract number; and, the Contractor's name, address, and telephone number.
- (2) Port of departure.
- (3) Location of ocean disposal area.
- (4) Quantity of material to be deposited in ocean.
- (5) Schedule for ocean disposal, giving date and time proposed for first ocean disposal.

3.3 WORK AREA

The Contractor will be permitted to exclude the public from the work areas in the immediate vicinity of his dredging, transporting, and disposal operations. Enforcement is the Contractor's responsibility at no additional cost to the Government. Coordinate enforcement with local enforcement agencies and will be subject to approval of the Contracting Officer.

3.3.1 Access

Provide and maintain access for equipment and plant to and from the work site, mooring area, and disposal area. Ascertain the environmental conditions which can affect the access such as climate, winds, currents, waves, depths, shoaling, and scouring tendencies.

3.3.2 Protection of Existing Waterways

Conduct operations in such a manner that material or other debris are not pushed outside of dredging limits or otherwise deposited in existing side channels, basins, docking areas, or other areas being utilized by vessels. Change the method of operations as may be required to comply with the above requirements. Should any bottom material or other debris be pushed into areas described above, as a result of the Contractor's operations, the same must be promptly removed by and at the expense of the Contractor to the satisfaction of the Contracting Officer.

3.3.3 Adjacent Property and Structures

No dredging will be permitted within 25 feet of any structure. Any damage to private or public property or structures resulting from the Contractor's negligence will result in suspension of dredging and require prompt repair at the Contractor's expense as a prerequisite to the resumption of dredging. The approximate location of the offset is indicated on the drawings. Verify the location of all structures prior to dredging, and adjust the dredging limits as necessary to ensure the offset distance.

3.4 CHARACTER OF MATERIALS

- a. Materials to be removed under this Contract are described in the GEOTECHNICAL DATA REPORT as referred to in Section 01 11 00 SUMMARY OF WORK.
- b. Due to typical harbor activities, debris such as trash, rope, chain, cable, tires, miscellaneous scrap metal, woody debris, etc., may be encountered in the channel. Bring such materials to the surface and dispose of in a disposal area obtained by the Contractor in a manner approved by the Contracting Officer.

3.5 DISPOSAL OF EXCAVATED MATERIAL

3.5.1 General

Transport and deposit material excavated in the disposal areas designated on the drawings. The average distance to which the material will have to be transported is approximately 4.9 nautical miles and the maximum distance will be approximately 5.4 nautical miles.

3.5.2 Load Restrictions

Disposal of material consisting of dredged material and water are restricted to a maximum of 20,000 cubic yards for any single dump from a Hopper Dredge and 15,000 cubic yards for any single dump from a dump scow/barge.

3.5.3 Ocean Dredged Material Disposal Site (ODMDS)

Transport material excavated to and deposit in Release Zone "A" West of the Jacksonville ODMDS as indicated on the drawings. Do not place material higher than a depth of 30 feet MLLW. The use of bottom dump barges or bottom dump dredges and hydraulic unloading barges and hydraulic unloading hopper dredges to dispose of dredged material in the offshore disposal area will be permitted.

3.5.3.1 Post-Dump Closure of Release Mechanisms

Close and seal all hopper doors, dump scow doors, or split hull dumping mechanisms prior to exiting the ODMDS site boundaries. In the case where the Contractor encounters a mechanical issue preventing hopper doors, dump scow doors, or split hull dumping mechanisms from fully closing, circle the operator within the boundaries of the ODMDS a minimum of three times prior to exiting the site. If a dump vessel exits the ODMDS with open doors, notify the Contracting Officer immediately of the occurrence. Resolve all mechanical issues prior to the vessel returning to service under this Contract and inform the Contracting Officer of the actions taken.

3.5.3.2 Spillage/Leakage During Transport to Disposal Site

Water and excavated material are not permitted to overflow, leak out, or spill out of the barges, dump scows, or hopper dredges while in transit to the disposal site. Transit to the disposal site begins as soon as dredged material loading into the disposal vessel is completed and the vessel begins moving to the disposal site. Failure to repair leaks or change the method of operation which is resulting in the overflow, leakage, or spillage will result in suspension of dredging operations and require prompt repair or change of operation to prevent overflow, leakage, or spillage as a prerequisite to the resumption of dredging. Appropriate measures may include, but are not limited to, up-to-date U.S. Coast Guard and/or American Bureau of Shipping certification of all disposal-related vessels, maintenance (inspection and/or replacement) of gaskets on barge doors, minimization of excess free liquids in barge loads, pre-transit testing of barge door hydraulics, and pre-transport verification of appropriate weather and sea state conditions. Notify the Contracting Officer within 24 hours if any apparent leaking or spilling of dredged material occurs as indicated by an average loss of draft during transit from the dredging area to the disposal site (forward draft loss plus aft draft loss divided by 2) in excess of 1.5 feet. Excessive leakage may be classified as a mis-dump and be subject to the actions described below in paragraph "Misplaced Material" below.

3.5.4 Placing of Dredged Material

During placement of dredged material in the disposal areas, provide an attendant at each disposal area to maintain constant radio contact with the dredge. Immediately notify the dredge in the event of dike or pipeline failure. In the event of dike or pipeline failure, suspend the dredging operations and promptly repair of the dike or pipeline as a prerequisite to the resumption of dredging.

3.5.5 Barges

Do not allow water and dredged materials to overflow or spill out of barges or dump scows during transport to the disposal site(s). Failure to repair leaks or change the method of operation which is resulting in overflow or spillage will result in suspension of dredging operations and require prompt repair or change of operation to prevent overflow or spillage as a prerequisite to the resumption of dredging.

3.5.6 Dredge Pipelines

3.5.6.1 Dredge Discharge Pipeline

Plainly mark the pipeline access routes with conspicuous stakes, targets and/or buoys to be maintained throughout the contract operations. Maintain a tight dredge discharge pipeline to prevent spilling of dredged material or dredge water outside of the disposal area. Provide and maintain radio communication between the dredge and the disposal areas and the dredge and the Contracting Officer. Inspect the pipeline at least twice daily for leaks. Failure to immediately repair leaks in the discharge pipeline will result in suspension of dredging operations and require prompt repair of pipeline as a prerequisite to the resumption of dredging. Repair any damage to private or public property resulting from the Contractor's operations at the Contractor's expense.

3.5.6.2 Submerged Pipeline

In the event the Contractor elects to submerge the pipeline, rest the pipeline on the bottom, and the top of the submerged pipeline and any anchor securing the submerged pipeline cannot be higher than the required project depth for the channel in which the submerged pipeline is placed. If a pipeline material which is buoyant or semi-buoyant, such as PVC pipe or similar low-density materials is used, securely anchor the pipeline to prevent the pipeline from lifting off the bottom under any conditions. Conduct daily inspections of the submerged pipeline to ensure buoyancy has not loosened the anchors. Remove all anchors when the submerged pipeline is removed. Mark the location of the entire length of submerged pipeline with signs, buoys, lights, and flags conforming to U.S. Coast Guard regulations.

3.5.6.3 Floating Pipeline

Should the Contractor's pipeline not rest on the bottom, it will be considered a floating pipeline and will be visible on the surface and clearly marked. In no case will the Contractor's pipeline be allowed to fluctuate between the surface and the bottom or lie partly submerged. Install lights on the floating pipeline as required in paragraph SIGNAL LIGHTS above. Support the lights either by buoys or by temporary piling, provided by the Contractor and approved by the Contracting Officer. Where the pipeline does not cross a navigable channel, space the flashing yellow all-around lights not over 200 feet apart, unless closer spacing is required by U.S. Coast Guard personnel, in which case the requirements of the U.S. Coast Guard will govern, at no additional cost to the Government.

3.5.7 Booster Pumps

Install booster pumps at least 300 feet from any residential-type building or house. Provide booster pumps, their prime movers, and any auxiliary equipment fitted or equipped with mufflers, noise control enclosures, or other engineering noise control methods, measures, and features such that steady noise emanating from this equipment does not exceed 85 decibels on the A scale at slow response, and impulsive noise does not exceed 140 decibels. Maintain such items throughout the course of the work.

3.5.8 [Misplaced Material](#)

Any scow or hopper dredge load, or portion thereof, that is released outside the boundaries of the Target Zone indicated on the drawings for

the Nearshore Disposal Area or the Offshore Dredge Material Disposal Area (ODMDS) will be classified as a misplaced material. Cease dredging operations and immediately notify the Contracting Officer of any misplaced material. Dredging and disposal operations will not be permitted to resume until satisfactory corrective action is approved by the Contracting Officer. Additionally, within 24 hours of the discovery of misplaced material, provide notification to EPA Region 4 at OceanDumpingR4@epa.gov. Redredging of misplaced materials will be required as a prerequisite to the resumption of dredging unless the Contracting Officer, at their discretion, determines that redredging of such material is not practical. If redredging of such material is not required, then the quantity of such misplaced material will be deducted from the Contractor's pay quantity. If the quantity for each misplaced load to be deducted cannot initially be agreed to by both the Contractor and Contracting Officer, then an average hopper/scow load quantity for the entire contract will be used in the determination. Materials deposited above the maximum indicated elevation or outside of the disposal area template indicated will require the degrading or removal of such materials at the Contractor's expense. The Contractor will not be held responsible for erosion caused by waves after the material has been satisfactorily placed.

3.5.9 Cumulative Summary Disposal Log

Maintain and provide a cumulative summary disposal log which contains the following information:

- a. Load Number.
- b. Disposal Vessel Name and Type (e.g. scow).
- c. Estimated Volume of Load.
- d. Description of Material Disposed or Placed.
- e. Source of Dredged Material.
- f. Disposal/Placement Area (i.e. beach, nearshore, ODMDS, etc.)
- g. Vessel Transit Draft Change.

Provide a pdf of the cumulative summary disposal log to the Contracting Officer within 24 hours of collection as part of the Contractor's Quality Control (CQC) daily report indicated in paragraph DAILY REPORT OF OPERATIONS above.

3.6 REQUIRED DEPTH, ALLOWABLE OVERDEPTH, AND SIDE SLOPES

3.6.1 Acceptance Sections

For purposes of payment, divide the work to be done into acceptance sections. Prior to dredging, prepare and provide a proposed [Acceptance Sections Plan](#) for approval by the Contracting Officer. Generally, divide the work into areas that equate to approximately 30 days of dredging or that constitute separable or definable features of work.

3.6.2 Required Depth

The material actually removed from the designated areas to be dredged, to a depth of not more than the required depth shown on the drawings, will be estimated and paid for in accordance with the provisions contained in the subparagraphs EXCAVATION, UNCLASSIFIED of Section 01 20 00 PRICE AND PAYMENT PROCEDURES.

3.6.3 Allowable Overdepth

To cover the inaccuracies of the dredging process, material actually removed from the designated areas to be dredged, to a depth below the required depth of not more than the allowable overdepth shown on the drawings, will be measured and paid for in accordance with the provisions contained in the subparagraph EXCAVATION UNCLASSIFIED of Section 01 20 00 PRICE AND PAYMENT PROCEDURES.

3.6.4 Side Slopes

Although dredging of side slope material may be required to provide the required project channel dimensions (depth and width), the side slopes shown on the drawings are provided for payment purposes only. Side slopes may be formed by box cutting, step cutting, or dredging along the side slope. Material actually removed, within the limits approved by the Contracting Officer, to provide for final side slopes not flatter than that shown on the contract drawings, but not in excess of the amount originally lying above this limiting side slope, will be measured and paid for in accordance with the provisions contained in subparagraphs EXCAVATION UNCLASSIFIED of Section 01 20 00 PRICE AND PAYMENT PROCEDURES. Such amount will be estimated and paid for whether dredged in original position or by box cut dredging whereby a space is dredged below the allowable side slope plane on the bottom of the slope for upslope material capable of falling into the cut. End slopes and transition slopes will not be estimated or paid for under this contract. In such cases, a 0 horizontal on 1 vertical will be used with no upslope allowance provision applied outside the required prism.

3.6.5 Excessive Dredging

It is recognized that the limits of dredging need to extend beyond the pay prism in order to remove material from within the pay prism. Therefore, the limits of active dredging may exceed the limits of the pay prism in vertical dimension as well as horizontal dimension to the extent required to remove material from the pay prism. Indirect impacts (sideslope sloughing, etc.) may extend even beyond the limits of active dredging. Nevertheless, material taken from beyond the limits of the pay prism as described in subparagraphs ALLOWABLE OVERDEPTH and SIDE SLOPES above will be deducted from the total amount dredged as excessive overdepth dredging, or excessive sideslope dredging, for which payment will not be made. Nothing herein will be construed to prevent payment for the removal of shoals performed in accordance with the applicable provisions of the paragraphs FINAL EXAMINATION AND ACCEPTANCE or SHOALING of this Section.

3.6.6 Areas to be Dredged

Based on information currently available to the Government, areas known to require dredging are depicted on the drawings as shaded areas. The actual areas to be dredged may vary from the shaded areas shown in the drawings. In order to provide the required project dimensions within and throughout the project limits shown on the drawings, remove material located within the project limits as directed by the Contracting Officer, regardless of whether the material is in a shaded area or not. Payment for all dredged material dredged from a shaded area will be made at the applicable contract unit price. Payment for all material dredged from a non-shaded area may be made at the applicable contract unit price; however, the Contractor may request an adjustment to the contract unit price in accordance with DFAR Clause 252.243-7002 REQUESTS FOR EQUITABLE ADJUSTMENT

and/or FAR Clause 52.243-4 CHANGES.

3.7 SURVEYS

3.7.1 General

Surveys will be performed in accordance with the paragraph QUANTITY SURVEYS; paragraph LAYOUT OF WORK of Section 01 11 00 SUMMARY OF WORK; Section 01 45 05 DREDGING/BEACH FILL PLACEMENT - QUALITY CONTROL; EM 1110-1-1000, EM 1110-1-1002, EM 1110-1-1003, EM 1110-1-1005, EM 1110-1-2909, and EM 1110-2-1003; FBPSM; and, ERDC/ITL TR-19-7. A copy of the EM's can be downloaded from the following web site: <http://www.publications.usace.army.mil/USACEPublications/EngineerManuals.aspx>. A copy of the ERDC/ITL TR-19-7 can be downloaded from the following web site: <https://cadbimcenter.erd.dren.mil/default.aspx?p=a&t=1&i=2>.

3.7.2 Contractor Representative

The Government will provide notification when soundings and sweepings are conducted and will be offered the opportunity to be on board the Government survey vessel, but may decline the offer. In-place measurement surveys and final acceptance sweep surveys will be performed by the Government. The Contractor's representative must be fully knowledgeable in offshore construction subsurface surveying procedures, techniques, equipment, and horizontal and vertical calibration methods, and state-of-the-art horizontal and vertical accuracy limitations. The Contractor's representative may observe and review, in progress, the adequacy and accuracy of the survey for in-place payment purposes, and for the potential existence of collusion, fraud, or obvious error in the data.

3.7.3 Survey Certification

- a. Immediately upon completion of any survey, the Contractor's representative, based on their on-site review of the survey execution, determine that the survey contains no evidence of collusion, fraud, obvious error, and that subsequent horizontal and vertical corrections are accurately annotated on the subsurface record.
- b. The Contractor's authorized representative must bring aboard the survey vessel a blank copy of the Certification Statement and attest to an acceptable survey by signing the Certification Statement before leaving the vessel. Sample copy of the Certification Statement is on the web site indicated in paragraph CONSTRUCTION FORMS AND DETAILS below.
- c. If the Contractor's authorized representative observes (and quantifies) specific documentary evidence of either fraud, collusion, or obvious error, the survey will be immediately rerun. Resurveys will totally supersede any previously run survey and will be run over the full reach of any Acceptance Section.
- d. If acceptability is not acquired after performing one resurvey of an Acceptance Section, a meeting will be held between the Contractor and the Contracting Officer to expeditiously resolve the issue causing rejection of the survey. Contractor equipment and personnel standby time to resolve acceptability of the survey will be at the Contractor's expense.
- e. In no case will a previously unacceptable survey be later judged

acceptable by the Contractor; unless such a reassessment/reevaluation is performed within 24 hours after the original survey, and prior to initiating any resurvey action based upon identifiable collusion, fraud, or obvious error.

- f. Should the Contractor or their authorized representative refuse to certify to the acceptability of a survey for contract payment without identifiable collusion, fraud, or obvious error, then the following actions will follow:
 - (1) Preconstruction (pre-dredging) Survey: Do not commence excavation until representatives of the Contractor and Contracting Officer have met and resolved the basis for refusal of certification. Should the Contractor commence excavation prior to obtaining an acceptable survey, the Contractor will be liable for any excavation performed. If a resurvey is performed, and accepted, prior excavation will not be measured, estimated, or paid for.
 - (2) Post-Construction (after-dredging) Survey: The 2-week survey window allowed under subparagraph EXCAVATION, UNCLASSIFIED of Section 01 20 00 PRICE AND PAYMENT PROCEDURES will be indefinitely extended until a final survey is accepted. Any material accretion which might occur due to such a time extension will neither be measured, estimated, or paid for.
 - (3) Refusal to Certify: Contractor equipment and personnel standby time to resolve refusal to certify to the acceptability of a survey when there is no identifiable collusion, fraud, or obvious error will be at the Contractor's expense and resultant delays will not be the basis for time extensions of the contract.
- g. Intermediate surveys taken between the pre-dredging and post-dredging surveys will not be considered for the purposes of determining quantities for final payment and acceptance of the area dredged.

3.7.4 Positioning - Horizontal Location and Vertical/Tidal Determination

3.7.4.1 Real Time Kinematic (RTK) GPS

The Government will use RTK GPS for horizontal positioning and the determination of Real Time water levels (tide corrections) for measurement and payment purposes. All data will be collected relative to the mapping datums and horizontal coordinate system(s) as noted within the survey notes on the accompanying plan sheets. All surveying and mapping will be exclusively tied, either by occupation or calibration, to the Project Survey Control Points as identified by tabulation and called for by individual survey notes on the plan sheets. The Kinematic Tidal Datum (KTD) model file (next section) employed for the design surveys/site investigations contributing to the design of this project will be the sole method to relate GPS surveyed heights (NAVD88) to the chart datum, be it Mean Lower Low Water (MLLW) or pool datum. The Contractor is responsible for providing all applicable RTK equipment for their surveying and dredging operations.

3.7.4.2 Kinematic Tidal Datum (KTD)

A file listing the separations between the North American Vertical Datum of 1988 (NAVD88) and the Chart Datum (MLLW or Pool) will be provided to the Contractor for entry into the hydrographic survey software. Obtain

the Kinematic Tidal Datum file from the Contracting Officer prior to commencing surveying and dredging operations.

3.8 ENCOUNTER OF MUNITIONS AND EXPLOSIVES OF CONCERN (MEC) IN PROJECT AREA

See the "Naval Station Mayport Probability Assessment For Munitions and Explosives of Concern, Duval County, FL" in Appendix A after the end of this section.

3.8.1 Munitions and Explosives of Concern (MEC)

Munition and Explosives of Concern (MEC) are military munitions that may pose unique explosives safety risks and include Unexploded Ordnance (UXO), Discarded Military Munitions (DMM), and Munitions Constituents (MC) present in high enough concentrations to pose an explosive hazard.

3.8.2 Procedure Upon MEC Encounter

Use due caution when working in coastal areas. In the event a suspect MEC is encountered, follow the 3Rs of explosive safety: Recognize, Retreat, Report. Recognize that munitions are dangerous; Retreat - do not approach, touch, move or disturb it, but carefully leave the area; and Report immediately what you saw and where you saw it to local law enforcement - call 911 and then inform the Contracting Officer. Cease all work efforts until the suspect MEC has been evaluated by emergency response personnel. In the event that the suspect MEC is positively identified as MEC, cease further work efforts until a new MEC probability assessment has been conducted.

3.8.3 Munition Use and Disposal Throughout Florida

The State of Florida was used extensively for war preparations in the 1940s and is home to a number of Formerly Used Defense Site (FUDS). More information on the FUDS program and sites throughout Florida can be found at the below web address.

<https://www.usace.army.mil/Missions/Environmental/Formerly-Used-Defense-Sites/FUDS-GIS>

FUDS are not the only source of MEC along the coastline of Florida. MEC may be found along the entire coastline of Florida, both on land and in State and Federal waters. The militaries of the U.S. and other countries have conducted live-fire training and combat operations at sea for centuries, and commonly disposed of excess, obsolete, and unserviceable munitions in coastal waters.

3.9 INSPECTION

3.9.1 Quality Assurance Representative (QAR)

The QAR will be notified prior to the establishment of horizontal control work (baseline layout, ranges, station flags, shore-based control for EPS/RPS, etc.) and vertical control work (tide staff(s), upland cross sections, construction elevations top/invert, maximum/minimum elevations of dredged materials within disposal area(s), etc.), but the presence or absence of the QAR will not relieve the Contractor of their responsibility for proper execution of the work in accordance with the specifications. The Contractor will be required:

- a. To furnish, on the request of the Contracting Officer or any QAR, the use of such boats, boatmen, laborers, and material forming a part of

the ordinary and usual equipment and crew of the dredging plant as may be reasonably required in inspecting and supervising the work.

- b. To furnish, on the request of the Contracting Officer or any QAR, suitable transportation from all points on shore designated by the Contracting Officer to and from the various pieces of plant, and to and from the disposal area.

3.9.2 Failure to Comply

In conjunction with the Clause INSPECTION OF CONSTRUCTION, should the Contractor refuse, neglect, or delay compliance with these requirements, the specific facilities may be furnished and maintained by the Contracting Officer and the cost thereof will be deducted from any amounts due or to become due the Contractor.

3.10 FINAL EXAMINATION AND ACCEPTANCE

3.10.1 Final Examination of Work

As soon as practicable and no later than two (2) weeks after the completion of the entire work or any section thereof (if the work is divided into sections) as in the opinion of the Contracting Officer will not be subject to damage by further operations under the contract, such work will be thoroughly examined at the cost and expense of the Government by sounding or by sweeping, or both, as determined by the Contracting Officer. Should any shoals, lumps, or other lack of contract depth be disclosed by this examination, the Contractor will be required to remove the same by dragging the bottom in accordance with the subparagraph BED LEVELING below or by dredging at the contract rate of dredging.

The Contractor or authorized representative will be notified when soundings and/or sweepings are to be made and will be permitted to accompany the survey party, but may decline. When the area is found to be in a satisfactory condition, it will be accepted finally. Should more than two sounding or sweeping operations by the Government over an area be necessary by reason of work for the removal of shoals disclosed at a prior sounding or sweeping, the cost of such third and any subsequent soundings or sweeping operations will be charged against the Contractor at the rate of \$11,600 for each day in which the Government plant is engaged in sounding or sweeping and/or is in route to or from the site or held at or near the said site for such operation.

3.10.2 Bed Leveling

Bed leveling by dragging the bottom with a drag bar or other apparatus will be allowed only in the designated dredging areas shown on the drawings. Dragging in areas outside of the designated dredging areas shown on the drawings is specifically prohibited without written approval of the Contracting Officer. Submit drawings and one photograph showing [drag bar](#) equipment used for final leveling work.

3.10.3 Bed Leveling - Reporting and Documenting

Fully document all bed leveling activity including date and time for initiation and completion of bed leveling in the Contractor's Quality Control Report (QCR).

3.10.4 Final Acceptance

Final acceptance of the whole or a part of the work and the deductions or corrections of deductions made thereon will not be reopened after having once been made, except on evidence of collusion, fraud or obvious error, and the acceptance of a completed section will not change the time of payment of the retained percentages of the whole or any part of the work.

3.11 SHOALING

If, before the Contract is completed, shoaling occurs in any section previously accepted, including shoaling in the finished channel because of the natural lowering of the side slopes, redredging at contract price, within the limits of available funds may be done if agreeable to both the Contractor and the Contracting Officer.

3.12 CONTINUITY OF WORK

Full payment for, and formal acceptance of acceptance sections, will not be rendered until the acceptance section is excavated to the full depth and extent required by this contract, except by decision of the Contracting Officer. The Contracting Officer reserves the right to withhold the invoiced amounts necessary to protect the Government's interests until acceptance of an acceptance section is achieved.

3.13 NOISE CONTROL

Equip all hauling and excavating equipment and dredge/barges, boats, and tugs used on this work with satisfactory mufflers or other noise abatement devices. Conduct the operations to comply with all Federal, State and local laws pertaining to noise. Limit the use of horns and whistle signals to the minimum necessary in order to ensure as quiet an operation as possible.

3.14 GENERAL DREDGING SAFETY REQUIREMENTS

3.14.1 Dredge Plant Restriction Definitions

To increase the safety of dredges on Federal contracts, the contract plans, and specifications may identify a location within the Federal channel (usually indicated by a channel station or navigation aid) requiring minimum equipment standards. This is a Certification Required Line (CRL). The following are specific designations that may apply to this Contract.

3.14.1.1 Certification Required Line - Coast Guard Inspected (CRL-C)

A designation of Certification Required Line - Coast Guard inspected (CRL-C) means only dredges with a current U.S. Coast Guard Certificate of Inspection for the intended use of the vessel will be allowed to work seaward of this line under this Contract. The inspections will include both topside and drydock. The inspection of the topside would have been conducted with 15 months prior to the date that dredging commences. The drydock inspection would have occurred within 3 years prior to the date that dredging commences.

3.14.1.2 Certification Required Line - Load Line (CRL-L)

A designation of Certification Required Line - Load Line (CRL-L) means

only dredges having a current Load Line Certification issued pursuant to 46 CFR 41-47 will be allowed to work seaward of this line under this Contract. Load Line inspections may be performed by either the American Bureau of Shipping (ABS) or Des Norkte Veritas (DNV) on behalf of the U.S. Coast Guard. The statutory requirement for Load Line Certification is 46 U.S.C. 5101-5116.

3.14.1.3 Certification Required Line - Hopper (CRL-H)

A designation of Certification Required Line - Hopper (CRL-H) means only hopper dredges with a current U.S. Coast Guard Certificate of Inspection for the intended use of the vessel will be allowed to work seaward of this line under this contract. Include both topside and drydock in the inspections. The topside inspection should have occurred with 15 months prior to the date that dredging commences. The drydock inspection should have occurred within 3 years prior to the date that dredging commences.

3.14.2 Certification of Attendant and Auxiliary Plant

In addition to the requirements of [EM 385-1-1](#), all supporting floating plant for dredging operations (to include, but not limited to, anchor barges, deck barges, deck/work barges, tug boats, tenders, workboats, pump-out barges, derrick barges, and spider barges) must have a current marine survey inspection certificate. The only exceptions to this requirement include pipeline support pontoons or floats and outboard skiffs under 16 feet in length. The certification must be issued by a licensed and accredited marine surveyor within the previous 12 months. The marine surveyor must be accredited by either the National Association of Marine Surveyors or the Society of Accredited Marine Surveyors. No attendant plant that does not have a valid certification will be used on this Contract. The certifications and inspection must be appropriate for the intended use of the plant in all locations specified in this Contract. Prior to commencement of work, submit [marine survey inspection certificates](#) of all floating plant and equipment.

3.14.3 Minimum Tow-Boat Requirements

In addition to the requirements of [EM 385-1-1](#), when dredging with non-self-propelled dredges seaward of the CRL-L or CRL-C, provide the following.

- a. One primary towboat of adequate size and horsepower to move all attendant and support plant against prevailing tides, currents, and winds during normal and severe weather conditions. Always make the towboat available within the project work area.
- b. A towboat(s) of adequate size and horsepower to move the dredge(s) against prevailing tides, currents, and winds during normal and severe weather conditions must be able to mobilize to the work site, and move the dredge(s) to safe harbor with 4 to 6 hours of first notification.

3.14.4 Harbor Specific Dredging Restrictions

3.14.4.1 Jacksonville Harbor

A CRL-L is established at Cut-14/15, Station 0+00, and applies seaward of this point to Bar Cut-3, Station 191+75. A CRL-C is established at Bar Cut-3, Station 191+75 (located at the easterly junction of the Mayport Navy Basin) and applies seaward. Dredging westward of Cut-14/15, Station

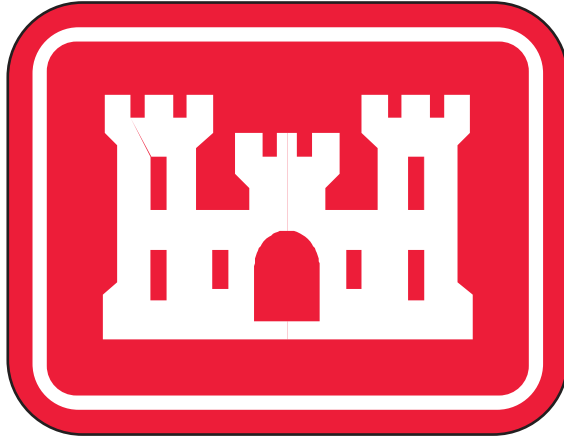
U.S. Naval Station Mayport
Duval County, Florida

0+00 will have no CRL restriction.

3.15 CONSTRUCTION FORMS AND DETAILS

See the CONSTRUCTION FORMS & DETAILS module at the following web address:
<https://www.saj.usace.army.mil/About/Divisions-Offices/Engineering/Design-Branch/Specifications-Section/>

-- End of Section --



Naval Station Mayport Probability Assessment For Munitions and Explosives of Concern

Duval County, FL

20 August 2024

Prepared by
**Environmental and Munitions Design Center
Baltimore District
U.S. Army Corps of Engineers**

TABLE OF CONTENTS

Section	Page
1.0 INTRODUCTION.....	1
2.0 ASSESSMENT AREA BACKGROUND.....	1
3.0 EVALUATION FOR POTENTIAL MEC.....	1
4.0 SUMMARY AND CONCLUSION.....	1
5.0 RECOMMENDATIONS.....	2

APPENDIX A.....FIGURES

1.0 Introductions

This munitions and explosive of concern (MEC) probability assessment has been prepared for the U.S. Army Corps of Engineers (USACE) Jacksonville District (CESAJ) Coastal Storm Risk Management (CSRM), Navigation, Operations & Maintenance Branch in support to their dredging and beach replenishment operations. Planned dredging activity requires an assessment for the potential for MEC to be encountered.

This document was prepared by the CENAB Environmental and Munitions Design Center (EMDC), Engineering Division, and summarizes historical activities involving military munitions and assesses the probability of encountering MEC. This assessment documents the findings and recommendations for Naval Station Mayport Maintenance Dredging, See **Figure's**.

2.0 Assessment Area Background

This project is a navigational project, Maintenance dredging of the navigable channel at Naval Station Mayport, Jacksonville, FL. This maintenance has been ongoing from FY13 - FY21. All dredged material will be going to an Offshore Dredged Material Disposal Area.

Historical records reviewed.

- Formerly Used Defense Sites (FUDS)
- USACE Archives
- Joint Explosive Ordnance Disposal (JEOD) Portal – EOD Incident reports

3.0 Evaluation for Potential Munitions

3.1 FUDS search

A review of FUDS records found no results for military munitions use/activity.

3.2 USACE Archives

A review of USACE archives found no results or reports of MEC or munitions debris (MD) indicative of MEC were made during any past operations.

3.3 EOD Incident Reports

A search of the JEOB Portal found no records of EOD incident reports for the Naval Station Mayport.

There are no known military munitions use, range fans or dump sites within the boundaries of this project.

4.0 Summary and Conclusion

After reviewing all historical documents and conducting records search for military munitions use, there is no evidence indicative of MEC to be present within the construction footprint of this site. In addition, no MEC has been reported during the multiple years of maintenance dredging in this location since FY13 – FY21.

5.0 Recommendations

It is the recommendation of CENAB Military Munitions Design Center that Naval Station Mayport O&M Navigational dredging project be classified as “**No Probability**” of encountering MEC. In accordance with DESR 6055-09 Edition 1.

It is further recommended that 3R’s awareness be provided to all site workers that are involved with soil disturbance. The following website is a good resource for the 3 R’s briefing, <https://www.denix.osd.mil/uxo/>.

If MEC is encountered, it should be reported by calling 911 and all work must stop until a MEC re-assessment is completed.

Point of contact for this MEC Probability Assessment is David Holland, Ordnance & Explosives Safety Specialist, U.S. Army Corps of Engineers, Baltimore District.

Signature: 96 HOLLAND.DAVID.L.10964502
Digitally signed by
HOLLAND.DAVID.L.1096450296
Date: 2024.12.12 16:17:34 -05'00'

Ordnance and Explosive Safety Specialist, David Holland

Signature: 90972 HOLMES.MARTY.ALAN.10170
Digitally signed by
HOLMES.MARTY.ALAN.1017090972
Date: 2024.12.12 18:24:43 -05'00'

Ordnance and Explosive Safety Chief, Marty Holmes

USACE CESAJ Commander or Designee

Name: Milan A. Mora

Signature: MORA.MILLAN.A.1367741470 Digitally signed by MORA.MILLAN.A.1367741470
Date: 2024.12.13 09:40:28 -05'00'

APPENDIX A

Area of Dredging

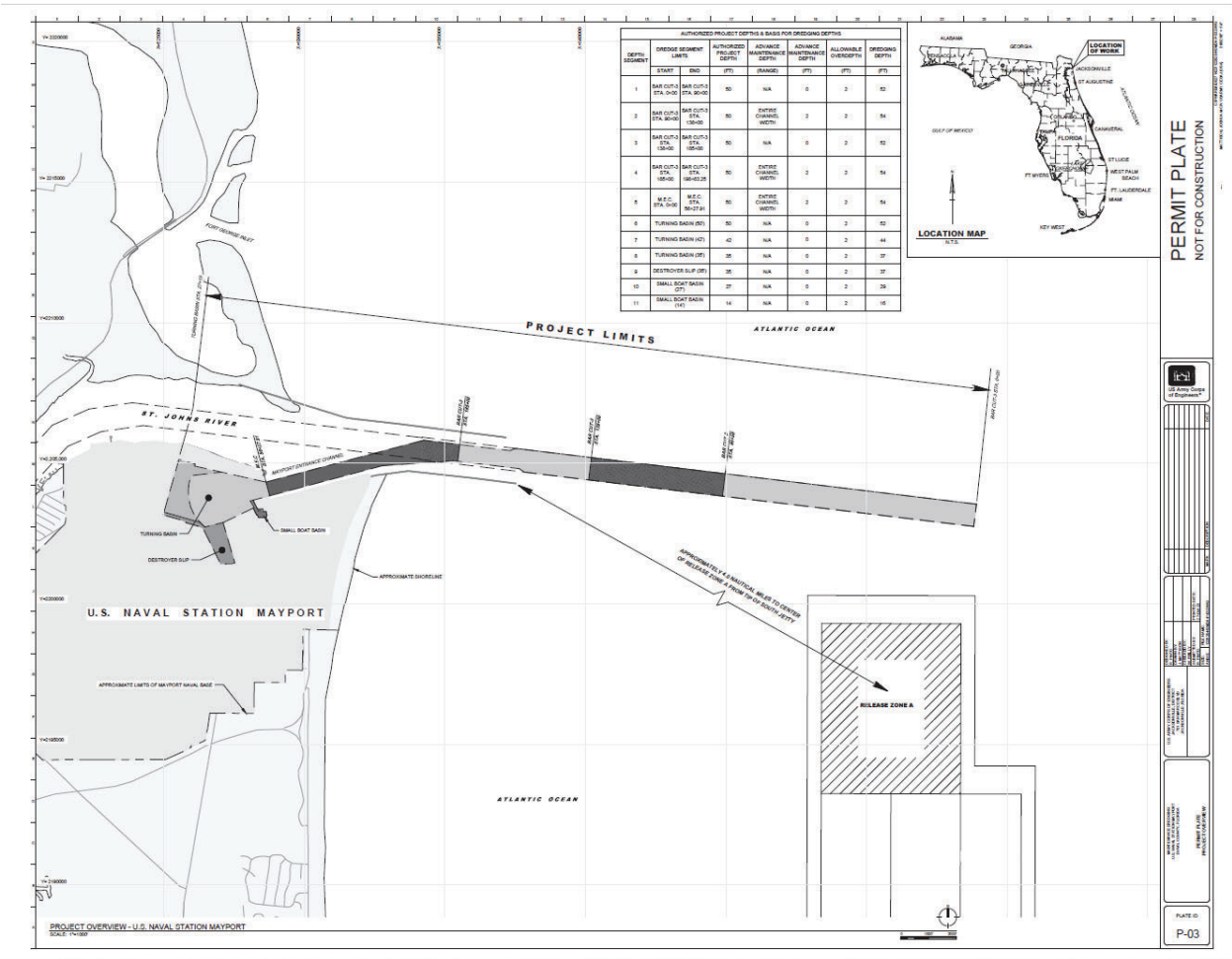
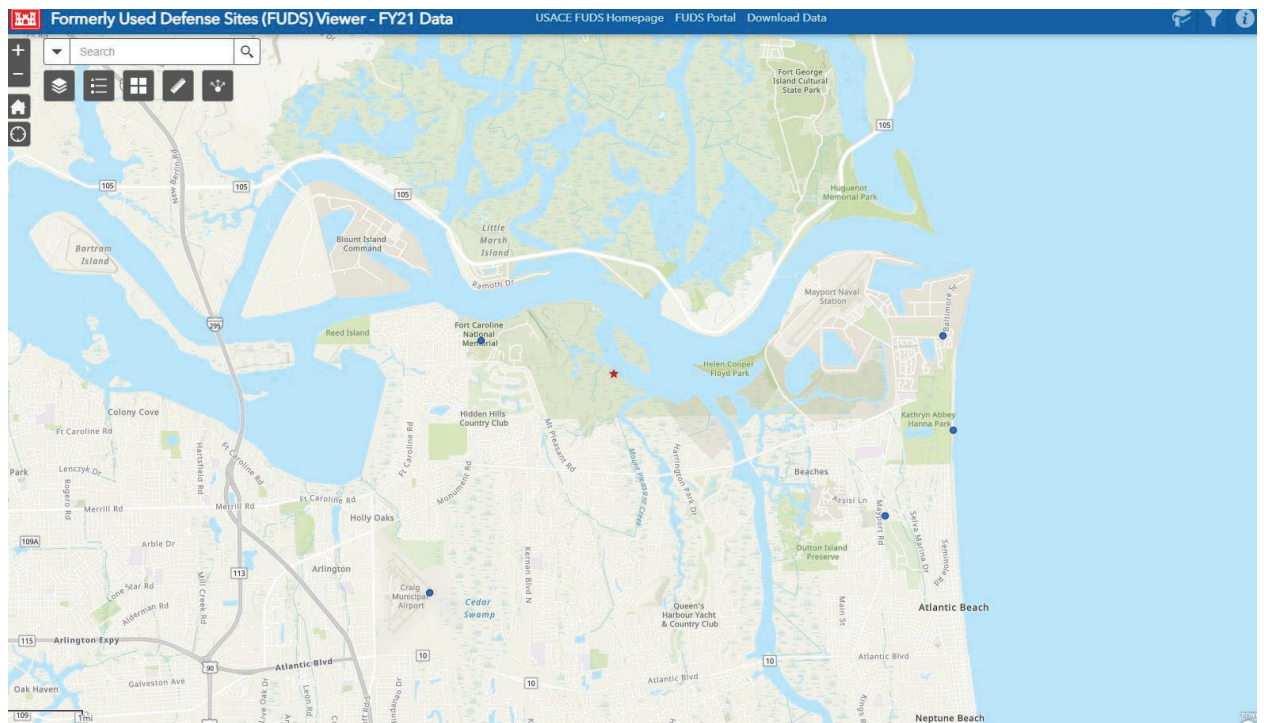


Figure 1



FUDS
Figure 2

SECTION TABLE OF CONTENTS

DIVISION 35 - WATERWAY AND MARINE CONSTRUCTION

SECTION 35 20 23.13

NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM - SCOW ULLAGE PROFILE

PART 1 GENERAL

- 1.1 DESCRIPTION
- 1.2 SUBMITTALS
- 1.3 PAYMENT
- 1.4 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM CERTIFICATION
 - 1.4.1 Certification
 - 1.4.2 Quality Assurance (QA)
 - 1.4.3 Recertification
- 1.5 DREDGE PLANT INSTRUMENTATION PLAN (DPIP)

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

- 3.1 REQUIREMENTS FOR REPORTED DATA
 - 3.1.1 Scow Name
 - 3.1.2 Contract Number
 - 3.1.3 Load Number
 - 3.1.4 Horizontal Positioning
 - 3.1.5 Date and Time
 - 3.1.6 Hull Status
 - 3.1.6.1 Open-Water Disposal
 - 3.1.7 Offloading
 - 3.1.8 Course
 - 3.1.9 Speed
 - 3.1.10 Heading
 - 3.1.11 Draft
 - 3.1.12 Displacement
 - 3.1.13 Bin Ullage Sounding
 - 3.1.14 Bin Volume
- 3.2 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM SYSTEM REQUIREMENTS
 - 3.2.1 Telemetry
 - 3.2.2 Data Reporting Frequency
 - 3.2.3 Data Transmission to the Web Service
 - 3.2.4 XML-Formatted Sensor Data String
 - 3.2.5 Contractor Data Backup
- 3.3 PERFORMANCE REQUIREMENTS
- 3.4 LIST OF ITEMS TO BE PROVIDED BY THE CONTRACTOR

-- End of Section Table of Contents --

SECTION 35 20 23.13

NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM - SCOW ULLAGE PROFILE

PART 1 GENERAL

1.1 DESCRIPTION

The work under this contract requires use of the National Dredging Quality Management Program (DQM) to monitor the dredge's status at all times during the contract and to manage data history. For the purpose of these specifications, a scow is defined as any non-self-propelled vessel used to transport dredged material. This includes, but is not limited to, split-hull scows, pocket scows, hopper barges, and deck barges.

This performance-based specification section identifies the minimum required output and the precision and instrumentation requirements. The requirements may be satisfied using equipment and technical procedures selected by the Contractor.

1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-07 Certificates

National Dredging Quality Management Program Certification,
Jacksonville District District

1.3 PAYMENT

Separate payment for installation, operation, and maintenance of the DQM-certified system as specified herein for the duration of the dredging operations is not allowed; all costs in connection therewith are considered a subsidiary obligation of the Contractor and are covered under the contract unit price for dredging in the bidding schedule.

1.4 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM CERTIFICATION

1.4.1 Certification

The Contractor is required to have a current certification from DQM for the scow instrumentation system to be used under this contract. Criteria for certification is based on the most recent specification posted on the DQM website <https://go.mil/4gicirhsvc>, Verify compliance with these criteria by onsite quality assurance (QA) checks conducted by the DQM Support Center Data Acquisition and Analysis Team and by periodic review of the transmitted data. If a system is installed specifically for this contract, in order to ensure that it is capable of transmitting quality data to the DQM database, the QA checks should take place either prior to the start of the contract or, with prior approval of the local USACE District, as soon as practical after dredging commences. DQM

Certification is valid for one year from the date of certification and is contingent upon the system's ability to meet the performance requirements as outlined in paragraph PERFORMANCE REQUIREMENTS. If issues with data quality are not corrected within 48 hours, the system certification will be revoked and additional QA checks by the Data Acquisition and Analysis Team may be necessary. Annual DQM Certification must be based on the following:

A series of QA checks as outlined on the DQM website
<https://go.mil/4gicirhsvc>

Verification of data acquisition and transfer as described in paragraph PERFORMANCE REQUIREMENTS

Review of the Dredge Plant Instrumentation Plan (DPIP) as described in paragraph DREDGE PLANT INSTRUMENTATION PLAN (DPIP)

1.4.2 Quality Assurance (QA)

The Dredging Contractor must have personnel who are familiar with the system instrumentation and who have the ability to recalibrate the sensors on site during the QA process. The Dredging Contractor must coordinate pickup times and locations and provide transportation to and from any platform with a DQM system to team personnel in a timely manner. The Dredging Contractor must also have on site for the QA checks a tug capable of towing the scow. As a general rule, DQM Data Acquisition and Analysis Team personnel will come with personal protective equipment (PPE) consisting of hardhats, steel toe boots, and life jackets. If additional safety equipment is needed, such as eye protection, safety harnesses, work gloves, or personal location beacons, provide these items to the team while on site. Submit a test data package to the DQM database from the system on each scow and have it accepted by the DQM Support Center prior to scow compliance checks. Also submit data collected during the QA Checks from the scow monitoring system to the DQM database and the Data Acquisition and Analysis Team personnel while on site. It is the Dredging Contractor's obligation to inform the QA team if the location designated for the QA checks has any site-specific safety concerns prior to their arrival on site.

The owner or operator of the scow must contact DQM at <https://go.mil/4gicirhsvc> on an annual basis, or at least three weeks prior to the proposed beginning of dredging, to schedule QA checks. This notification is meant to make the Data Acquisition and Analysis Team aware of a target date and the contract on which the plant will be used. At least one week prior to the target date, the Dredging Contractor must contact the Data Acquisition and Analysis Team and verbally coordinate a specific date and location. The Contractor must then follow up this conversation with a written email confirmation. The owner/operator must coordinate the QA checks with all local authorities including, but not limited to, the local USACE Contracting Officer's Representative (COR).

1.4.3 Recertification

Recertification is required for any yard work which produces modification to displacement (for example, a change in scow lines, or repositioning or repainting hull marks), modification to bin volume (change in bin dimensions or addition or subtraction of structure), or changes in sensor type or location; report these changes in the sensor log section of the DPIP. A system does not have to be transmitting data between jobs;

however, in order to retain certification during this period, the system sensors or hardware should not be disconnected or removed from the scow. If the system is powered down, retain calibration coefficients.

1.5 DREDGE PLANT INSTRUMENTATION PLAN (DPIP)

The Contractor must have a digital copy of the DPIP on file with the DQM Support Center. While working on site, the Contractor must also maintain on the dredge a copy of the DPIP which is easily accessible to Government personnel at all times. This document must describe the sensors used, configuration of the system, how sensor data will be collected, how quality control on the data will be performed, and how sensors/data reporting equipment will be calibrated and repaired if they fail. A description of computed scow-specific data and how the sensor data will be transmitted to the DQM database must also be included. The Contractor must submit to the DQM Support Center any addendum or modifications made to the plan, subsequent to its original submission, prior to start of work.

A complete list of the required DPIP contents is provided on the DQM website <https://go.mil/4gicirhsvc>. Submit to the DQM Support Center any addendum or modifications made to the plan, subsequent to its original submission, prior to the start of work. Any changes to the computation methods must be approved by the DQM Support Center prior to their implementation.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.1 REQUIREMENTS FOR REPORTED DATA

Provide, operate, and maintain all hardware and software to meet these specifications. The Contractor is responsible for replacement, repair, and calibration of sensors and other necessary data acquisition equipment needed to supply the required data. Complete repairs within 48 hours of any sensor failure. Notify the Contracting Officer's Representative (COR) upon completion of a repair, replacement, installation, modification, or calibration. The COR may request recalibration of sensors or other hardware components at any time during the contract as deemed necessary.

Keep a log of sensor repair, replacement, installation, modification, and calibration in the onsite copy of the DPIP. The log must contain a three-year history of sensor maintenance, including the time of sensor failures (and subsequent repairs), the time and results of sensor calibrations, the time of sensor replacements, and the time that backup sensor systems were initiated to provide the required data. It must also contain the name of the person responsible for the sensor work. Install sensors that are capable of collecting parameters within specified accuracies and resolutions indicated in the following subparagraphs.

With the exception of position and any value calculated, reported sensor values should represent a weighted average with the highest and lowest values not included in the calculated average for the given interval. The averaging routine used should be consistent across all event triggers. This information should be documented in the DPIP sections that say "Calculations done external to the instrumentation." These data-reporting requirements cover the collection of electronic data on a scow through the entire dredging cycle. Disposal events can consist of both open-water disposal and offloading. Open-water disposal is the placement of material

via bottom doors or split hull. Offloading is the placement of material via either hydraulic or mechanical means.

3.1.1 Scow Name

Assign a unique name for each scow that will remain constant from one dredging operation to the next.

3.1.2 Contract Number

The USACE-assigned contract number for the project will be reported.

3.1.3 Load Number

A DQM load number must document the end of a disposal event for a given SCOW.

3.1.4 Horizontal Positioning

Record horizontal positioning as the geographic coordinates of the vessel as indicated by the location of the Global Positioning System (GPS) antenna. Obtain all locations using a positioning system operating with a minimum accuracy level of **3 to 10 feet** horizontal Circular Error Probable (CEP). Report positions as Latitude/Longitude WGS 84 in decimal degrees. West Longitude and South Latitude values are reported as negative.

3.1.5 Date and Time

Report the date and time to the nearest second and referenced to Universal Time Coordinated (UTC) based on a 24-hour format: yyyy-mm-dd hh:mm:ss.

3.1.6 Hull Status

Hull status is meant to reflect a condition when material could be removed or released from the scow. For this contract, hull status must register closed prior to leaving the disposal area.

3.1.6.1 Open-Water Disposal

Indicate an open split hull or open bottom door of a scow by reporting an "OPEN" value. Indicate a closed split hull or closed bottom door of a scow by reporting a "CLOSED" value. An open status must be indicated as the bin starts to open, and a closed status must be indicated only once the bin is fully closed. For pocket scows, the open/closed status must correspond to the compartment which is first to open and last to close.

3.1.7 Offloading

Offloading occurs when the scow is being unloaded, either by hydraulic or mechanical means. When offloading is occurring, a value of "true" must be reported; otherwise, a "false" value must be reported. The only permissible values are "true" and "false".

3.1.8 Course

Provide scow course-over-ground (COG) using industry-standard equipment. Provide scow course-over-ground (to the nearest whole degree) with values from 000 (true north) to 359 degrees referenced to a clockwise positive direction convention.

3.1.9 Speed

Provide scow speed-over-ground in knots using industry-standard equipment with a minimum accuracy of 1.0 knot and resolution to the nearest 0.1 knot.

3.1.10 Heading

Provide scow heading using industry-standard equipment. The scow heading must be accurate to within 5 degrees and reported to the nearest whole degree with values from 000 (true north) to 359 degrees referenced to a clockwise positive direction convention.

3.1.11 Draft

Report all draft measurements in feet, tenths, and hundredths with an accuracy of plus or minus 0.1 foot relative to observed physical draft readings. Report the measurements at a resolution of two decimal places (hundredths of a foot). The reported forward draft value must be equal to the sum of the visual forward port and starboard draft mark readings divided by two. The reported aft draft value must be equal to the sum of the visual aft port and starboard draft mark readings divided by two. Forward draft, aft draft, and average draft will be reported. Place sensors at an optimum location on the scow to be reflective of observed physical draft mark readings at any trim or list. Minimum accuracies are conditional to relatively calm water. The reported sensor value is an average of at least ten samples per event, with at least one maximum value and one minimum value removed, and the minimum eight remaining values averaged. When the average draft is calculated for the purpose of determining displacement, maintain significant digits for average draft such that if forward draft were 0.15 and aft draft were 0.1, then the average draft would be 0.125.

3.1.12 Displacement

Report scow displacement in long tons, based on the most accurate method available for the scow. The minimum standard of accuracy for displacement is interpolation from the displacement table, based on the average draft. For this contract the density of water used to calculate displacement is 64.2* lbs/cubic foot and is used for an additional interpolation between the fresh and salt water tables.

*The water density used is project-/location-specific. Enter the appropriate water density in the blank:

Fresh Water: 62.43 lbs/cubic foot
Salt Water: 64.11 - 64.30 lbs/cubic foot

3.1.13 Bin Ullage Sounding

Report all ullage soundings in feet, tenths, and hundredths with an accuracy of plus or minus 0.1 foot with respect to the combing and be representative of the forward and aft extents of the hopper as close to the centerline as is possible. Report the measurements at a resolution of two decimal places (hundredths of a foot). If sensors must be offset from the centerline of the bin, they should be offset to opposite sides of the vessel. Forward ullage, aft ullage, and average ullage soundings will be reported. The reported sensor value is an average of at least ten samples per event, with at least one maximum value and one minimum value removed,

and the minimum eight remaining values averaged. When the average ullage is calculated for the purpose of determining the hopper volume, maintain significant digits for the average ullage such that if the forward ullage were 0.15 and aft ullage were 0.1, then the average ullage would be 0.125. Special arrangements for pocket scows may be made in consultation with the DQM Support Center.

3.1.14 Bin Volume

Report scow bin volume in cubic yards based on the most accurate method available for the scow. The minimum standard of accuracy for bin volume is interpolation from the bin volume table based on the average ullage soundings.

3.2 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM SYSTEM REQUIREMENTS

The Contractor's DQM system must be capable of collecting, displaying, and transmitting information to the DQM database. The parameters which must be reported to the DQM database include trip number, date and time, hull status, offloading status, scow course, scow speed, scow heading, draft, displacement, ullage, and bin volume. Provide an easily accessible, permanent visual display on the scow to show in real time the parameters collected by the system in the same units as the data submitted to the DQM database. In the event a reported parameter is calculated based on multiple sensors, the sensor values as used in the equation must be able to be viewed in addition to the required parameter. If a hardware problem occurs, or if a part of the system is physically damaged, the Contractor is responsible for repairing it within 48 hours of determination of the condition.

3.2.1 Telemetry

The Contractor may select any commercial satellite, cellular phone, or other data communications systems available, as long as it is capable of transmitting real-time data as well as enough additional bandwidth to clear historically queued data when a connection is reobtained. If connectivity is lost, queue and transmit unsent data upon restoration of connectivity. Delays in pushing real-time data to the DQM database should not exceed four hours. Exceptions to these requirements may be granted by the DQM Center on a case-by-case basis with consideration for contract-specific requirements, site-specific conditions, and extreme weather events.

The data transmission process from the scow to the DQM database must be automated. The data may be sent from the scow directly to the DQM database or to a shore-based system. Data transmitted to the DQM database should be raw data; use repeatable automated software or programming routine to process any shoreside data. Include a description of this process in the DPIIP.

3.2.2 Data Reporting Frequency

Log disposal activities with high temporal and spatial resolution. Log data as a series of events. Each set of measurements (time, position, etc.) will be considered an event. Collect any required information in paragraph REQUIREMENTS FOR REPORTED DATA, that is not an averaged variable (that is, draft and ullage) within 1 second of the reported time. Measure data with sufficient frequency by the scow system to resolve the events to the accuracy specified in the following table. Any averaged variable must

U.S. Naval Station Mayport
Duval County, Florida

be collected and computed within this sampling interval. Reporting intervals must be consistent and not change for the data collected on a given scow. This interval should be documented by the Contractor in the DPIP.

Event Type	Event Trigger Descriptions	Event Time Resolution	Event Position Resolution
Loading	No change in position with hull status closed An elapsed time of 1 hour since the last event. NONCLOSURE In the event a scow has completed an open water disposal and transited back to a holding station without closing the hull, the sampling must be changed to once per hour.	1 minute	N/A
Sailing	Change in position with hull status closed Time from the last sample equals 1 minute.	1 second	plus or minus 10 ft
Offloading	Offloading material, hull status reported as open A position must be recorded within 1 minute arrival at the offload location and within one second of the material starting to be removed from scow. The time from the last sample equals 1 minute. STANDBY OFFLOADING In the even a scow is not being actively offloaded at the offload location for a time equal to one hour, the sampling interval must be equal to once an hour.	1 second 1 minute	plus or minus 10 ft

Event Type	Event Trigger Descriptions	Event Time Resolution	Event Position Resolution
Open Water Disposal	Hull status open A data point must be recorded within 1 second of the hull status going from closed to open and again within 1 second of the hull status going from open to closed. Between these events, report the data at equal interval from 6 to 12 seconds. This interval must always remain consistent for the dredge plant.	1 second	plus or minus 10 ft

Example: The scow is stationary for 1 hour and 15 minutes, and then it sails to the disposal area. You should have a "Loading/Stationary" event at time 0, time 1 hour, and time 1 hour and 15 minutes. Then, for "Sailing," within 1 second of an elapsed time of 1 minute from the 1 hour and 15 minutes event, another event occurs.

3.2.3 Data Transmission to the Web Service

Use a Simple Object Access Protocol (SOAP) web service to report sensor data to the DQM database. Transmit data as it is collected in real time and pushed to the DQM web service. If the web service is not available or returns an error message, store the data in a queue and transmit upon re-establishment of the connection, starting with the oldest data in the queue and continuing until real-time transmission is restored. Delays in pushing real-time data to the DQM database should not exceed four hours. Exceptions to these requirements may be granted by the DQM Support Center on a case-by-case basis with consideration for contract-specific requirements, site-specific conditions, and extreme weather events.

Contact dqm-support@usace.army.mil to obtain the web service URL and the appropriate key credentials and communication protocol.

The data transmission method call takes two arguments: a string containing the plant identifier assigned by the DQM Support Center and a second string containing the XML-formatted sensor data. The method returns the string "OK" if the data is received. If the data is not received, either the web service or the client application throws an error.

3.2.4 XML-Formatted Sensor Data String

Pass each scow event as a string on one continuous line of data. The example below is broken up by variable for ease of reading:

```
<?xml version="1.0"?>
```

U.S. Naval Station Mayport
Duval County, Florida

```
<SCOW_DREDGING_DATA version="2.5">
  <SCOW_NAME>AU1994</SCOW_NAME>
  <PLANT_IDENTIFIER>1999</PLANT_IDENTIFIER>
  <CONTRACT>W123BA-09-D-0087_RL01</CONTRACT>
  <TRIP_NUMBER>34</TRIP_NUMBER>
  <X_POSITION>-81.670632</X_POSITION>
  <Y_POSITION>41.528987</Y_POSITION>
  <DATE_TIME>2010-08-14 10:50:15</DATE_TIME>
  <SCOW_SPEED>0.0</SCOW_SPEED>
  <SCOW_COURSE>0.0</SCOW_COURSE>
  <HULL_STATUS>OPEN</HULL_STATUS>
  <OFFLOADING>FALSE</OFFLOADING>
  <SCOW_HEADING></SCOW_HEADING>
  <SCOW_FWD_DRAFT></SCOW_FWD_DRAFT>
  <SCOW_AFT_DRAFT></SCOW_AFT_DRAFT>
  <SCOW_AVG_DRAFT></SCOW_AVG_DRAFT>
  <ULLAGE_FWD></ULLAGE_FWD>
  <ULLAGE_AFT></ULLAGE_AFT>
  <ULLAGE_AVG></ULLAGE_AVG>
  <SCOW_BIN_VOLUME></SCOW_BIN_VOLUME>
  <SCOW_DISPLACEMENT></SCOW_DISPLACEMENT>
  <ADDITIONAL_DATA>Some more scow info, if needed</ADDITIONAL_DATA>
</SCOW_DREDGING_DATA>
```

Format DATE_TIME values as YYYY-MM-DD HH:MM:SS, as shown above. If, for any reason, a field has no value, the enclosing XML tags should be sent with nothing between them (for example, <DRAFT_AFT></DRAFT_AFT>). The web service cannot handle a "null" value or any other indicators of no value collected.

3.2.5 Contractor Data Backup

Maintain an archive of all data sent to the DQM database during the dredging contract. The Contracting Officer's Representative (COR) may require, at no increase in the contract price, that the Contractor provide a copy of these data covering specified time periods. Provide the data in the HTML format which would have been transmitted to the DQM database. Submit data via storage medium acceptable to the COR.

At the end of the dredging contract, contact the DQM Support Center prior to discarding the data. The DQM Support Center will verify that all data has been received and appropriately archived before giving the Contractor discard permission. Record in a separate section at the end of the scow's onsite copy of the DPIP the following information:

- Person who made the call
- Date of the call
- DQM representative who gave permission to discard

3.3 PERFORMANCE REQUIREMENTS

The Contractor's DQM system must be fully operational at the start of dredging operations and fully certified prior to moving dredge material on the contract (see paragraph NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM CERTIFICATION). To meet contract requirements for operability, in addition to certification, the Contractor's system must provide a data string with values for all parameters while operating, as described within the specifications. Additionally, all hardware must be compliant with DPIP requirements (see paragraph DREDGE PLANT INSTRUMENTATION PLAN

U.S. Naval Station Mayport
Duval County, Florida

(DPIP)). Quality data strings are considered to be those providing values for all parameters reported when operating according to the specification. Make repairs necessary to restore data return compliance within 48 hours. Failure by the Contractor to report the required data within the specified time window for scow measurements (see paragraph DATA REPORTING FREQUENCY, and paragraph DATA TRANSMISSION TO THE WEB SERVICE) and failure to receive DQM certification prior to dredging will result in withholding of up to 10 percent of the contract progress payment.

3.4 LIST OF ITEMS TO BE PROVIDED BY THE CONTRACTOR

DPIP - <https://dqm.usace.army.mil>

Paragraph DREDGE PLANT INSTRUMENTATION PLAN (DPIP)

DQM SYSTEM

Sensor instrumentation - Paragraph REQUIREMENTS FOR REPORTED DATA

SCOW DATA

Event documentation - Paragraph DATA REPORTING FREQUENCY

Data reports - Paragraph DATA TRANSMISSION TO THE WEB SERVICE

QA EQUIPMENT ON THE DREDGE

Clear and accurate draft marks

Ullage tape

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 35 - WATERWAY AND MARINE CONSTRUCTION

SECTION 35 20 23.23

NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM - HOPPER DREDGE

PART 1 GENERAL

- 1.1 DESCRIPTION
- 1.2 SUBMITTALS
- 1.3 PAYMENT
- 1.4 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM CERTIFICATION
 - 1.4.1 Certification
 - 1.4.2 Recertification
- 1.5 DREDGE PLANT INSTRUMENTATION PLAN (DPIP)

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

- 3.1 REQUIREMENTS FOR REPORTED DATA
 - 3.1.1 Date and Time
 - 3.1.2 Load Number
 - 3.1.3 Horizontal Positioning
 - 3.1.3.1 Vessel Horizontal Positioning
 - 3.1.3.2 Draghead Horizontal Positioning
 - 3.1.4 Hull Status
 - 3.1.5 Dredge Course
 - 3.1.6 Dredge Speed
 - 3.1.7 Dredge Heading
 - 3.1.8 Tide
 - 3.1.9 Draft
 - 3.1.10 Hopper Ullage Sounding
 - 3.1.11 Hopper Volume
 - 3.1.12 Displacement
 - 3.1.13 Empty Displacement
 - 3.1.14 Draghead Depths
 - 3.1.15 Slurry Densities
 - 3.1.16 Slurry Velocities
 - 3.1.17 Pump RPM
 - 3.1.18 Sea Suction Valve for Dragarm
 - 3.1.19 Pumpout
- 3.2 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM SYSTEM REQUIREMENTS
 - 3.2.1 Computer Requirements
 - 3.2.2 Software
 - 3.2.3 UPS
 - 3.2.4 Internet Access
 - 3.2.5 Data Routing Requirements
 - 3.2.6 Data Reporting Frequency
 - 3.2.7 Data Format
 - 3.2.8 Data Reporting

U.S. Naval Station Mayport
Duval County, Florida

- 3.2.9 Contractor Data Backup
- 3.3 PERFORMANCE REQUIREMENTS
- 3.4 LIST OF ITEMS TO BE PROVIDED BY THE CONTRACTOR

-- End of Section Table of Contents --

SECTION 35 20 23.23

NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM - HOPPER DREDGE

PART 1 GENERAL

1.1 DESCRIPTION

The work under this contract requires use of the National Dredging Quality Management Program (DQM) to monitor the dredge's status at all times during the contract and to manage data history.

This performance-based specification section identifies the minimum required output and the precision and instrumentation requirements. The requirements may be satisfied using equipment and technical procedures selected by the Contractor.

1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-07 Certificates

National Dredging Quality Management Program Certification,
Jacksonville District

1.3 PAYMENT

Separate payment for installation, operation, and maintenance of the DQM-certified system as specified herein for the duration of the dredging operations is not allowed; all costs in connection therewith are considered a subsidiary obligation of the Contractor and are covered under the contract unit price for dredging in the bidding schedule.

1.4 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM CERTIFICATION

1.4.1 Certification

The Contractor is required to have a current certification from DQM for the hopper dredge instrumentation system to be used under this contract. Criteria for certification is based on the most recent specification posted on the DQM website: <https://go.mil/4gicirhsvc>. Verify compliance with these criteria by annual onsite quality assurance (QA) checks conducted by the DQM Support Center Data Acquisition and Analysis Team and by periodic review of the transmitted data. DQM Certification is valid for one year from the date of the annual QA checks. Certification is contingent upon the system's ability to continuously meet the performance requirements as outlined in paragraph PERFORMANCE REQUIREMENTS. If issues with data quality are not corrected within 48 hours, the system certification will be revoked and additional QA checks by the Data Acquisition and Analysis Team may be necessary.

Annual DQM Certification must be based on the following:

- a. A series of quality assurance checks as outlined on the DQM website: <https://go.mil/4gicirhsvc>.
- b. Verification of data acquisition and transfer as described in paragraph Performance Requirements.
- c. Review of the Dredge Plant Instrumentation Plan (DPIP) as described in paragraph DREDGE PLANT INSTRUMENTATION PLAN (DPIP).

1.4.2 Recertification

The owner or operator of the dredge must contact DQM at <https://go.mil/4gicirhsvc> on an annual basis, or at least three weeks prior to certification expiration, to schedule QA checks for renewal. This notification is meant to make the Data Acquisition and Analysis Team aware of a target date for the annual QA checks for the dredge. At least one week prior to the target date, the Contractor must contact the Data Acquisition and Analysis Team and verbally coordinate a specific date and location. The Contractor must then follow up this conversation with a written email confirmation. The owner/operator must coordinate the QA checks with all local authorities, including but not limited to, the local USACE Contracting Officer's Representative (COR).

Recertification is required for any yard work which produces modification to displacement (change in dredge lines, or repositioning or repainting hull marks), modification to bin volume (change in bin dimensions, or addition or subtraction of structure), or changes in sensor type or location; report these changes in the sensor log section of the DPIP. A system does not have to be transmitting data between jobs; however, in order to retain its certification during this period, the system sensors or hardware should not be disconnected or removed from the dredge. If the system is powered down, retain calibration coefficients.

1.5 DREDGE PLANT INSTRUMENTATION PLAN (DPIP)

The Contractor must have a digital copy of the DPIP on file with the DQM Support Center. While working on site, the Contractor must also maintain on the dredge a copy of the DPIP which is easily accessible to Government personnel at all times. This document must describe the sensors used, configuration of the system, how sensor data will be collected, how quality control on the data will be performed, and how sensors/data reporting equipment will be calibrated and repaired if they fail. A description of the computed dredge-specific data and how the sensor data will be transmitted to the DQM database will also be included. A complete list of the required DPIP contents is provided on the DQM website <https://go.mil/4gicirhsvc>

Submit to the DQM Support Center any addendum or modifications made to the plan, subsequent to its original submission, prior to the start of work. Any changes to the computation methods must be approved by the DQM Support Center prior to their implementation.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.1 REQUIREMENTS FOR REPORTED DATA

Provide, operate, and maintain all hardware and software to meet these specifications. The Contractor is responsible for replacement, repair, and calibration of sensors and other necessary data acquisition equipment needed to supply the required data. Complete repairs within 48 hours of any sensor failure. Notify the COR upon completion of a repair, replacement, installation, modification, or calibration. The COR may request recalibration of sensors or other hardware components at any time during the contract as deemed necessary.

Keep a log of sensor repair, replacement, installation, modification, and calibration in the dredge's onboard copy of the DPIIP. The log must contain a three-year history of sensor maintenance, including the time of the sensor failures (and subsequent repairs), the time and results of sensor calibrations, the time of sensor replacements, and the time that backup sensor systems were initiated to provide required data. It must also contain the name of the person responsible for the sensor work. Install sensors that are capable of collecting parameters within specified accuracies and resolutions indicated in the following subparagraphs. Reported sensor values for ullage, draft, and draghead depth should represent a weighted average with the highest and lowest values not included in the calculated average for the given interval. This information should be documented in the DPIIP sections that say "Calculations done external to the instrumentation."

3.1.1 Date and Time

Report the date and time to the nearest second and referenced to UTC time based on a 24-hour format: mm/dd/yyyy hh:mm:ss. The reported time must be the time reported by the GPS in the NMEA string.

3.1.2 Load Number

A load number must document the end of a disposal event. Load numbering will begin at number 1 at the start of the contract and will be incremented by 1 at the completion of each disposal event or emptying of the hopper. Whenever possible, calculate the load number off of the sensors aboard the dredge, which must be a mathematically repeatable routine. Make efforts to include logic that avoids false load number increments. Do not allow the routine to miss any disposal event. If manual incrementing of the load number is in place, pay extra attention to this value in the quality control process.

3.1.3 Horizontal Positioning

Obtain all locations using a positioning system operating with a minimum accuracy level of 3 to 10 feet horizontal Circular Error Probable (CEP). Report positions as Latitude/Longitude WGS 84 in decimal degrees. West Longitude and South Latitude values are reported as negative.

3.1.3.1 Vessel Horizontal Positioning

Record vessel horizontal positioning as geographic coordinates of the vessel as indicated by the location of the GPS antenna.

3.1.3.2 Draghead Horizontal Positioning

Record draghead horizontal positioning as geographic coordinates of the heel on the centerline of the draghead(s). Describe any offset calculations from the GPS antenna in the DPIIP.

3.1.4 Hull Status

Monitor open/closed status of the hopper dredge, corresponding to the split/non-split condition of a split-hull hopper dredge. For dredges with hopper doors, the status of a single door that is the first opened during normal disposal operations may be monitored. An "open" value indicates that the hopper door is open or, in the case of split-hull dredges, that the hull is split. A "closed" value indicates that the hopper doors are closed or, in the case of split-hull dredges, that the hull is not split.

For this contract, hull status must register closed prior to leaving the disposal area.

3.1.5 Dredge Course

Provide dredge course-over-ground (COG) using industry-standard equipment. Provide dredge course-over-ground to the nearest whole degree with values from 000 (true north) to 359 degrees referenced to a clockwise positive direction convention.

3.1.6 Dredge Speed

Provide dredge speed-over-ground in knots using industry-standard equipment with a minimum accuracy of 1 knot and resolution to the nearest 0.1 knot.

3.1.7 Dredge Heading

Provide dredge heading using industry-standard equipment. The dredge heading must be accurate to within 5 degrees and reported to the nearest whole degree with values from 000 (true north) to 359 degrees referenced to a clockwise positive direction convention.

3.1.8 Tide

Obtain tide data using appropriate equipment to give the water level with an accuracy of plus or minus 0.1 foot and a resolution of 0.01 foot. Enter tide values above project datum described in the dredging specification with a positive sign and those below with a negative sign.

3.1.9 Draft

Report all draft measurements in feet, tenths, and hundredths with an accuracy of plus or minus 0.1 foot relative to observed physical draft readings. Report the measurements at a resolution of two decimal places (hundredths of a foot). The reported forward draft value is equal to the sum of the visual forward port and starboard draft mark readings divided by two. The reported aft draft value is equal to the sum of the visual aft port and starboard draft mark readings divided by two. Forward draft, aft draft, and average draft will be reported. Place sensors at an optimum location on the vessel to be reflective of observed physical draft mark readings at any trim or list. Minimum accuracies are conditional to

relatively calm water. The sensor value reported must be an average of at least ten samples per event, with at least one maximum value and one minimum value removed, and the minimum eight remaining values averaged. When the average draft is calculated for the purpose of determining displacement, maintain significant digits for average draft such that if forward draft was 0.15 and aft draft was 0.1, the average draft would be 0.125.

3.1.10 Hopper Ullage Sounding

Report all reported ullage soundings in feet, tenths, and hundredths with an accuracy of plus or minus 0.1 foot with respect to the combing and be representative of the forward and aft extents of the hopper as close to the centerline as is possible. Report the measurements at a resolution of two decimal places (hundredths of a foot). Report forward ullage and aft ullage soundings. Mount sensors to avoid discharge flume turbulence, foam, and any structure that could produce sidelobe errors. If sensors must be offset from the centerline of the hopper, they should be offset to opposite sides of the vessel. If more than one fore or one aft sensor are used, place the sensors near the corners of the hopper, and report the average value of the fore sensors and the average value of the aft sensors. The sensor value reported must be an average of at least ten samples per event, with at least one maximum value and one minimum value removed, and the minimum eight remaining values averaged. When the average ullage is calculated for the purpose of determining hopper volume, maintain significant digits for average ullage such that if forward ullage was 0.15 and aft ullage was 0.1, then the average ullage would be 0.125.

3.1.11 Hopper Volume

Report hopper volume in cubic yards, based on the most accurate method available for the dredge. The minimum standard of accuracy for hopper volume is interpolation from the certified hopper volume table, based on the average fore and aft ullage soundings.

3.1.12 Displacement

Report dredge displacement in long tons, based on the most accurate method available for the dredge. The minimum standard of accuracy for displacement is interpolation from the displacement table, based on the average draft. For this contract the density of water used to calculate displacement is 64.2* lbs/cubic foot, and it will be used for an additional interpolation between the fresh and salt water tables.

*The water density used is project-/location-specific. Enter the appropriate water density in the blank:

Fresh Water: 62.43 lbs/cubic foot

Salt Water: 64.11 - 64.30 lbs/cubic foot

3.1.13 Empty Displacement

Report empty displacement in long tons, which is the lightship value of the dredge, or the weight of the dredge with no material in the hopper, adjusted for fuel and water consumption.

3.1.14 Draghead Depths

Report draghead depths with an accuracy of plus or minus 0.5 foot and a

resolution to the nearest 0.1 foot as measured from the surface of the water with no tidal adjustments. Minimum accuracies are conditional to relatively calm water. The sensor value reported must be an average of at least ten samples per event, with at least one maximum value and one minimum value removed, and the minimum eight remaining values averaged.

3.1.15 Slurry Densities

Use a density metering device, calibrated according to the manufacturer's specifications, to record the slurry density of each dragarm to the nearest 0.001 g/cc with an accuracy of plus or minus 0.01 g/cc. If the manufacturer does not specify a frequency of recalibration, conduct calibration prior to commencement of work.

3.1.16 Slurry Velocities

Use a flow metering device, calibrated according to the manufacturer's specifications, to record the slurry velocity of each dragarm to the nearest 0.01 fps with an accuracy of plus or minus 0.5 fps. If the manufacturer does not specify a frequency of recalibration, conduct calibration prior to commencement of work. Measure the slurry velocity in the same pipeline inside diameter as that used for the slurry density measurement.

3.1.17 Pump RPM

Measure the RPM of any pump being used to move material with the highest level of accuracy that is standard on the vessel operational displays, either at the bridge, at the drag tender's controls, or in the engine room. Dredges with multiple pumps per side must report RPM for the pump that best describes the dredging process (typically the outboard pump).

3.1.18 Sea Suction Valve for Dragarm

If sea suction can be taken to bypass suction through the draghead, the sea suction location and valve status will be reported. The status of the valve will change from "closed" to "open" when the valve starts to open and will register "closed" when the valve is fully closed. When applicable, the state of the latch will be reported as "true" or "false." Report the sea suction location in a standard non-changing name string of no more than 20 characters. These field values will always occur in the XML string as a set. The DQM system can accommodate only up to four unique sea suction locations. Suggested options for the naming convention can be found in the example dataset in paragraph DATA FORMAT.

3.1.19 Pumpout

When the hopper dredge is being pumped out, report a "true" value; when it is not, report a "false" value. The only permissible values are "true" and "false."

3.2 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM SYSTEM REQUIREMENTS

The Contractor's DQM system must be capable of collecting, displaying, and transmitting information to the DQM database. Record the applicable parameters from paragraph REQUIREMENTS FOR REPORTED DATA as events locally and continually transmitted to the DQM database anytime an Internet connection is available. Equip the Dredge with a DQM computer system, consisting of a computer, monitor, keyboard, mouse, data modem, UPS, and

network hub. Provide a standalone computer system, exclusive to the DQM monitoring system, and with USACE DQM software installed on it. If a hardware problem occurs, or if a part of the system is physically damaged, the Contractor is responsible for repairing it within 48 hours of determination of the condition.

3.2.1 Computer Requirements

Provide a dedicated onboard computer for use by the DQM system. This computer must run USACE software and receive data from the Contractor's data-reporting interface. This computer must meet or exceed the following performance specifications:

CPU:	Intel or AMD processor with a (non-overclocked) clock speed of at least 1.6 gigahertz (GHz)
Hard drive:	250 gigabytes (GB); internal
RAM:	4 gigabytes (GB)
Ethernet adapter:	Internal network card with an RJ-45 connector
Ports:	1 free serial port with standard 9-pin connectors; 1 free USB port
Other hardware:	Keyboard, mouse, monitor

Install a fully licensed copy of Windows 10 Professional Operating System or higher on the computer specified above. Also install any necessary manufacturer-provided drivers for the installed hardware. Locate and orient the computer to allow data entry and data viewing, as well as to provide access to data ports for the connection of external hardware.

3.2.2 Software

The DQM computer's primary function is to transmit data to the DQM shoreside database. Do not install software which conflicts with this function on this computer. The DQM computer must have the USACE-provided Dredging Quality Management Onboard Software (DQMOBS) installed on it by DQM personnel along with USACE-selected software for remote support and management.

3.2.3 UPS

Supply an Uninterruptible Power Supply (UPS) for the computer and networking equipment. The UPS must provide backup power at 1 kVA for a minimum of ten minutes. The UPS must interface with the DQM computer to communicate UPS status. Ensure that sufficient power outlets are available to run all specified equipment.

3.2.4 Internet Access

Maintain an Internet connection capable of transmitting real-time data to the DQM server and supporting remote access, as well as enough additional bandwidth to clear historically queued data when a connection is re-obtained. If connectivity is lost, queue and transmit unsent data upon restoration of connectivity. Delays in pushing real-time data to the DQM database should not exceed four hours. Exceptions to these requirements may be granted by the DQM Support Center on a case-by-case basis with

consideration for contract-specific requirements, site-specific conditions, and extreme weather events.

Acquire and install all necessary hardware and software to make the Internet connection available for data transmission to the DQM web service. Configure the hardware and software to allow the DQM Support Center remote access to this computer. Coordination between the dredging company's IT and the DQM Support Center may be required in order to configure remote access through any security, firewall, router, and telemetry systems. Telemetry systems must be capable of meeting these minimum reporting requirements in all operating conditions.

3.2.5 Data Routing Requirements

Onboard sensors must continually monitor dredge conditions, operations and efficiency and route this information into the shipboard dredge-specific system (DSS) computer to assist in guiding dredge operations. Portions of this Contractor-collected information must be routed to the DQM computer on a real-time basis. Use an RS-232 9600- or 19200-baud serial interface to send standard sensor data to the DQM computer. Configure the serial interface as 8 bits, no parity, and no flow control.

3.2.6 Data Reporting Frequency

Log data as a series of events. Each event will consist of a dataset containing dredge information in accordance with paragraph REQUIREMENTS FOR REPORTED DATA. Each set of measurements (time, position, etc.) will be considered an event. Collect any required information in paragraph REQUIREMENTS FOR REPORTED DATA that is not an averaged variable (draft and ullage) within 1 second of the reported time. Send a data string for an event to the DQM computer every 6 to 12 seconds. This interval must remain constant throughout the contract; do not transmit data strings more than once per every 5 seconds. Collect and compute any averaged variable within this sampling interval.

3.2.7 Data Format

Report data as an extensible Markup Language (W3C standard XML 1.0) document as indicated below. Line breaks and spaces are added for readability, but the carriage return, line feed character combination is added only to delineate records (HOPPER_DREDGING_DATA tag) for actual data transmission.

```
{?xml version="1.0"?}
{HOPPER_DREDGING_DATA version = "2.0"}
  <PLANT_IDENTIFIER>integer string</PLANT_IDENTIFIER>
  {DREDGE_NAME} string32 {/DREDGE_NAME}
  {HOPPER_DATA_RECORD}
    {DATE_TIME} time date string {/DATE_TIME}
    {CONTRACT_NUMBER} string32 {/CONTRACT_NUMBER}
    {LOAD_NUMBER} integer string {/LOAD_NUMBER}
    {VESSEL_X coord_type="LL"} floating point string
  {/VESSEL_X}
    {VESSEL_Y coord_type="LL"} floating point string
  {/VESSEL_Y}
    {PORT_DRAG_X coord_type="LL"} floating point string
  {/PORT_DRAG_X}
    {PORT_DRAG_Y coord_type="LL"} floating point string
  {/PORT_DRAG_Y}
```

```
        {STBD_DRAG_X coord_type="LL"} floating point string
{/STBD_DRAG_X}
        {STBD_DRAG_Y coord_type="LL"} floating point string
{/STBD_DRAG_Y}
        {HULL_STATUS} OPEN/CLOSED string {/HULL_STATUS}
        {VESSEL_COURSE} floating point string {VESSEL_COURSE}
        {VESSEL_SPEED} floating point string {/VESSEL_SPEED}
        {VESSEL_HEADING} floating point string {/VESSEL_HEADING}
        {TIDE} floating point string {/TIDE}
        {DRAFT_FORE} floating point string {/DRAFT_FORE}
        {DRAFT_AFT} floating point string {/DRAFT_AFT}
        {ULLAGE_FORE} floating point string {/ULLAGE_FORE}
        {ULLAGE_AFT} floating point string {/ULLAGE_AFT}
        {HOPPER_VOLUME} floating point string {/HOPPER_VOLUME}
        {DISPLACEMENT} floating point string {/DISPLACEMENT}
        {EMPTY_DISPLACEMENT} floating point string
{/EMPTY_DISPLACEMENT}
        {DRAGHEAD_DEPTH_PORT} floating point string
{/DRAGHEAD_DEPTH_PORT}
        {DRAGHEAD_DEPTH_STBD} floating point string
{/DRAGHEAD_DEPTH_STBD}
        {PORT_DENSITY} floating point string {/PORT_DENSITY}
        {STBD_DENSITY} floating point string {/STBD_DENSITY}
        {PORT_VELOCITY} floating point string {/PORT_VELOCITY}
        {STBD_VELOCITY} floating point string {/STBD_VELOCITY}
        {PUMP_RPM_PORT} floating point string {/PUMP_RPM_PORT}
        {PUMP_RPM_STBD} floating point string {/PUMP_RPM_STBD}
        {VALVE_1_LOCATION} string32 {VALVE_1_LOCATION}
        {VALVE_1_STATUS} open/closed {/VALVE_1_STATUS}
        {VALVE_1_LATCHED} true/false {/VALVE_1_LATCHED}
        {VALVE_2_LOCATION} string32 {/VALVE_2_LOCATION}
        {VALVE_2_STATUS} open/closed {/VALVE_2_STATUS}
        {VALVE_2_LATCHED} true/false {/VALVE_2_LATCHED}
        {VALVE_3_LOCATION} string32 {/VALVE_3_LOCATION}
        {VALVE_3_STATUS} open/closed {/VALVE_3_STATUS}
        {VALVE_3_LATCHED} true/false {/VALVE_3_LATCHED}
        {VALVE_4_LOCATION} string32 {/VALVE_4_LOCATION}
        {VALVE_4_STATUS} open/closed {/VALVE_4_STATUS}
        {VALVE_4_LATCHED} true/false {/VALVE_4_LATCHED}
        {PUMP_OUT_ON} true/false/unknown string {/PUMP_OUT_ON}
{/HOPPER_DATA_RECORD}

{/HOPPER_DREDGING_DATA}
Carriage Return - ASCII value 13
Line Feed - ASCII value 10
```

Example

```
{?xml version="1.0"?}
{HOPPER_DREDGING_DATA version = "2.0"}
  <PLANT_IDENTIFIER>9999</PLANT_IDENTIFIER>
  {DREDGE_NAME}Essayons{/DREDGE_NAME}
  {HOPPER_DATA_RECORD}
    {DATE_TIME}04/11/2002 13:12:05{/DATE_TIME}
    {CONTRACT_NUMBER}GDSNWP-11-G-0001{/CONTRACT_NUMBER}
    {LOAD_NUMBER}102{/LOAD_NUMBER}
    {VESSEL_X coord_type="LL"}-80.123333{/VESSEL_X}
    {VESSEL_Y coord_type="LL"}10.123345{/VESSEL_Y}
    {PORT_DRAG_X coord_type="LL"}-80.1233371{/PORT_DRAG_X}
```

```
{PORT_DRAG_Y coord_type="LL"}10.12335{/PORT_DRAG_Y}
{STBD_DRAG_X coord_type="LL"}-80.123339{/STBD_DRAG_X}
{STBD_DRAG_Y coord_type="LL"}10.123347{/STBD_DRAG_Y}
{HULL_STATUS}CLOSED{/HULL_STATUS}
{VESSEL_COURSE}258{/VESSEL_COURSE}
{VESSEL_SPEED}3.4{/VESSEL_SPEED}
{VESSEL_HEADING}302{/VESSEL_HEADING}
{TIDE}-0.1{/TIDE}
{DRAFT_FORE}10.05{/DRAFT_FORE}
{DRAFT_AFT}15.13{/DRAFT_AFT}
{ULLAGE_FORE}10.11{/ULLAGE_FORE}
{ULLAGE_AFT}10.22{/ULLAGE_AFT}
{HOPPER_VOLUME}2555.2{/HOPPER_VOLUME}
{DISPLACEMENT}4444.1{/DISPLACEMENT}
{EMPTY_DISPLACEMENT}2345.0{/EMPTY_DISPLACEMENT}
{DRAGHEAD_DEPTH_PORT}55.10{/DRAGHEAD_DEPTH_PORT}
{DRAGHEAD_DEPTH_STBD}53.21{/DRAGHEAD_DEPTH_STBD}
{PORT_DENSITY}1.02{/PORT_DENSITY}
{STBD_DENSITY}1.03{/STBD_DENSITY}
{PORT_VELOCITY}22.1{/PORT_VELOCITY}
{STBD_VELOCITY}23.3{/STBD_VELOCITY}
{PUMP_RPM_PORT}55{/PUMP_RPM_PORT}
{PUMP_RPM_STBD}54{/PUMP_RPM_STBD}
{VALVE_1_LOCATION}Starboard Dragarm{VALVE_1_LOCATION}
{VALVE_1_STATUS}open{/VALVE_1_STATUS}
{VALVE_1_LATCHED>true{/VALVE_1_LATCHED}
{VALVE_2_LOCATION}Port Dragarm{/VALVE_2_LOCATION}
{VALVE_2_STATUS}closed{/VALVE_2_STATUS}
{VALVE_2_LATCHED>false{/VALVE_2_LATCHED}
{VALVE_3_LOCATION}Port Sea Chest{/VALVE_3_LOCATION}
{VALVE_3_STATUS}closed{/VALVE_3_STATUS}
{VALVE_3_LATCHED>false{/VALVE_3_LATCHED}
{VALVE_4_LOCATION}Starboard Sea Chest{/VALVE_4_LOCATION}
{VALVE_4_STATUS}open{/VALVE_4_STATUS}
{VALVE_4_LATCHED>false{/VALVE_4_LATCHED}
{PUMP_OUT_ON>false{/PUMP_OUT_ON}
{/HOPPER_DATA_RECORD}
{/HOPPER_DREDGING_DATA}
{cr}
{lf}
<?xml version="1.0"?>
  <HOPPER_DREDGING_DATA version = "2.0">
    <PLANT_IDENTIFIER>9999</PLANT_IDENTIFIER>
    {DREDGE_NAME}Essayons{/DREDGE_NAME}
    {HOPPER_DATA_RECORD}
      {DATE_TIME}04/11/2002 13:12:10{/DATE_TIME}
      {CONTRACT_NUMBER}GDSNWP-11-G-0001{/CONTRACT_NUMBER}
      {LOAD_NUMBER}102{/LOAD_NUMBER}
      {VESSEL_X coord_type="LL"}-80.123334{/VESSEL_X}
      {VESSEL_Y coord_type="LL"}10.123346{/VESSEL_Y}
      {PORT_DRAG_X coord_type="LL"}-80.1233372{/PORT_DRAG_X}
      {PORT_DRAG_Y coord_type="LL"}10.12336{/PORT_DRAG_Y}
      {STBD_DRAG_X coord_type="LL"}-80.123340{/STBD_DRAG_X}
      {STBD_DRAG_Y coord_type="LL"}10.123348{/STBD_DRAG_Y}
      {HULL_STATUS}CLOSED{/HULL_STATUS}
      {VESSEL_COURSE}259{/VESSEL_COURSE}
      {VESSEL_SPEED}3.5{/VESSEL_SPEED}
      {VESSEL_HEADING}300{/VESSEL_HEADING}
      {TIDE}-0.1{/TIDE}
```

```
{DRAFT_FORE}10.00{/DRAFT_FORE}  
{DRAFT_AFT}15.15{/DRAFT_AFT}  
{ULLAGE_FORE}10.15{/ULLAGE_FORE}  
{ULLAGE_AFT}10.20{/ULLAGE_AFT}  
{HOPPER_VOLUME}2555.5{/HOPPER_VOLUME}  
{DISPLACEMENT}4444.0{/DISPLACEMENT}  
{EMPTY_DISPLACEMENT}2345.0{/EMPTY_DISPLACEMENT}  
{DRAGHEAD_DEPTH_PORT}55.15{/DRAGHEAD_DEPTH_PORT}  
{DRAGHEAD_DEPTH_STBD}53.19{/DRAGHEAD_DEPTH_STBD}  
{PORT_DENSITY}1.00{/PORT_DENSITY}  
{STBD_DENSITY}1.01{/STBD_DENSITY}  
{PORT_VELOCITY}22.5{/PORT_VELOCITY}  
{STBD_VELOCITY}23.3{/STBD_VELOCITY}  
{PUMP_RPM_PORT}55{/PUMP_RPM_PORT}  
{PUMP_RPM_STBD}54{/PUMP_RPM_STBD}  
{VALVE_1_LOCATION}Starboard Dragarm{/VALVE_1_LOCATION}  
{VALVE_1_STATUS}open{/VALVE_1_STATUS}  
{VALVE_1_LATCHED}true{/VALVE_1_LATCHED}  
{VALVE_2_LOCATION}Port Dragarm{/VALVE_2_LOCATION}  
{VALVE_2_STATUS}closed{/VALVE_2_STATUS}  
{VALVE_2_LATCHED}false{/VALVE_2_LATCHED}  
{VALVE_3_LOCATION}Port Sea Chest{/VALVE_3_LOCATION}  
{VALVE_3_STATUS}closed{/VALVE_3_STATUS}  
{VALVE_3_LATCHED}false{/VALVE_3_LATCHED}  
{VALVE_4_LOCATION}Starboard Sea Chest{/VALVE_4_LOCATION}  
{VALVE_4_STATUS}open{/VALVE_4_STATUS}  
{VALVE_4_LATCHED}false{/VALVE_4_LATCHED}  
{PUMP_OUT_ON}false{/PUMP_OUT_ON}  
{/HOPPER_DATA_RECORD}  
{/HOPPER_DREDGING_DATA}  
{cr}  
{lf}
```

3.2.8 Data Reporting

The system must transmit correctly formatted event data XML strings to the DQM database continuously from mobilization until the last USACE post-dredging survey has been accepted. If the Internet connection (paragraph INTERNET ACCESS) is non-operable, perform manual backups from the dredge computer of the XML data string which would have been transmitted to the DQM computer over the serial connection for each day the device is inoperable and submit to the DQM Support Center within 48 hours. This submission does not replace the requirement of correcting the issue affecting the automatic transmission of data. In the event of data transfer, transmission, or hardware failure, maintain a manually recorded disposal log consisting of a series of events. These events are start of dredging, end of dredging, pre-disposal, and post-disposal. Include time stamp (GMT), position (Latitude and Longitude WGS84), draft, ullage, volume, and displacement for each event. Submit disposal logs on a daily basis to the COR during the time when the system is not operational.

3.2.9 Contractor Data Backup

Maintain an archive of all data sent to the DQM computer during the dredging contract. The COR may require, at no increase in the contract price, that the Contractor provide a copy of these data covering specified time periods. Provide the data, transmitted to the DQM computer, in the XML format with no line breaks between the parameters and each record string on a separate line. The naming convention for the files is

U.S. Naval Station Mayport
Duval County, Florida

{dredgename}_{StartYYYYMMddhhmmss}_{EndYYYYMMddhhmmss}.txt. Submit data via storage medium acceptable to the COR.

At the end of the dredging contract, contact the DQM Support Center prior to discarding the data. The DQM Support Center will verify that all data has been received and appropriately archived before giving the Contractor discard permission. Record the following information in a separate section at the end of the dredge's onboard copy of the DPIP:

Person who made the call
Date of the call
DQM representative who gave permission to discard

3.3 PERFORMANCE REQUIREMENTS

The Contractor's DQM system must be fully operational at the start of dredging operations and fully certified prior to moving dredge material on the contract (see paragraph NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM CERTIFICATION). To meet contract requirements for operability, in addition to certification, the Contractor's system must provide a data string with all values for all parameters while operating, as described in the specifications. Additionally, all hardware must be compliant with hardware requirements (paragraph COMPUTER REQUIREMENTS). Quality data strings are considered to be those providing values for all parameters reported when operating according to the specification. Make repairs necessary to restore data return compliance within 48 hours. Failure by the Contractor to report the required data within the specified time window for dredge measurements (see paragraph DATA REPORTING FREQUENCY and paragraph DATA REPORTING) will result in withholding of up to 10 percent of the contract progress payment per FAR 52.232-5 Payments under Fixed-Price Construction Contracts.

3.4 LIST OF ITEMS TO BE PROVIDED BY THE CONTRACTOR

DPIP

<https://go.mil/4gicirhsvc>

DQM System

Sensor instrumentation: paragraph REQUIREMENTS FOR REPORTED DATA
DQM computer: paragraph NATIONAL DREDGING QUALITY MANAGEMENT
PROGRAM SYSTEM REQUIREMENTS

Dredge Data

Event documentation: paragraph DATA REPORTING
Dredge data backups: paragraph CONTRACTOR DATA BACKUP

QA Equipment on the Dredge

Dragarm depth chain
Ullage tape
Refractometer
Water sampling device

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 35 - WATERWAY AND MARINE CONSTRUCTION

SECTION 35 20 23.33

NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM - PIPELINE HYDRAULIC DREDGE

PART 1 GENERAL

- 1.1 DESCRIPTION
- 1.2 SUBMITTALS
- 1.3 PAYMENT
- 1.4 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM CERTIFICATION
- 1.5 DREDGE PLANT INSTRUMENTATION PLAN (DPIP)

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

- 3.1 REQUIREMENTS FOR REPORTED DATA
 - 3.1.1 Message Bundle Data
 - 3.1.1.1 Messages
 - 3.1.1.1.1 Message Time
 - 3.1.1.1.2 Comment
 - 3.1.1.2 Dredge Events - Work Event
 - 3.1.1.2.1 Vertical Correction
 - 3.1.1.2.2 Cutter/Suction Head Location and Movement
 - 3.1.1.2.2.1 Cutter/Suction Head Horizontal Position
 - 3.1.1.2.2.2 Cutter/Suction Invert Depth
 - 3.1.1.2.2.3 Cutter/Suction Head Heading
 - 3.1.1.2.3 Dredge Activity
 - 3.1.1.2.3.1 Slurry Velocity
 - 3.1.1.2.3.2 Slurry Density
 - 3.1.1.2.3.3 Pump RPM
 - 3.1.1.2.3.4 Pump Vacuum
 - 3.1.1.2.3.5 Pump Outlet Pressure
 - 3.1.1.2.4 Outfall Information (Open Water/Spill Barge Disposal)
 - 3.1.1.2.4.1 Discharge Horizontal Position
 - 3.1.1.3 Dredge Events - State Event
 - 3.1.1.3.1 Message Time
 - 3.1.1.3.2 Contract Event
 - 3.1.1.3.2.1 Contract Number
 - 3.1.1.3.2.2 Contract Start and End
 - 3.1.1.3.3 Tide Station/River Stage Gage Event
 - 3.1.1.3.3.1 Station Name
 - 3.1.1.3.4 Length of Pipe Event
 - 3.1.1.3.4.1 Floating Pipe
 - 3.1.1.3.4.2 Submerged Pipe
 - 3.1.1.3.4.3 Shore Pipe
 - 3.1.1.3.5 Booster Pump
 - 3.1.1.3.5.1 Number of Booster Pumps
 - 3.1.1.3.6 Dredge Advance

U.S. Naval Station Mayport
Duval County, Florida

- 3.1.1.3.7 Outfall Information
 - 3.1.1.3.7.1 Discharge Location
 - 3.1.1.3.7.2 Discharge Horizontal Position
 - 3.1.1.3.7.3 Discharge Outfall Heading
 - 3.1.1.3.7.4 Discharge Pipe Elevation
- 3.1.1.3.8 Non-effective Work Event
 - 3.1.1.3.8.1 Non-effective Work Interval
 - 3.1.1.3.8.2 Dredge Function Code
 - 3.1.1.3.8.3 Additional Comments
- 3.2 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM SYSTEM REQUIREMENTS
 - 3.2.1 Computer Requirements
 - 3.2.2 Software
 - 3.2.3 UPS
 - 3.2.4 Direct Data Transmissions
 - 3.2.5 Internet Access
 - 3.2.6 Data Routing Requirements
- 3.3 DREDGE MONITORING DATA
 - 3.3.1 General
 - 3.3.2 Data Measurement Frequency
 - 3.3.2.1 Work Event Messages
 - 3.3.2.2 State Event Messages
 - 3.3.3 Parameter Transmission to the Web Service
 - 3.3.4 Contractor Data Backup
- 3.4 PERFORMANCE REQUIREMENTS
- 3.5 LIST OF ITEMS TO BE PROVIDED BY THE CONTRACTOR

-- End of Section Table of Contents --

SECTION 35 20 23.33

NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM - PIPELINE HYDRAULIC DREDGE

PART 1 GENERAL

1.1 DESCRIPTION

The work under this contract requires use of the National Dredging Quality Management Program (DQM) to monitor the dredge's status at all times during the contract and to manage data history.

This performance-based specification section identifies the minimum required output and the precision and instrumentation requirements. The requirements may be satisfied using equipment and technical procedures selected by the Contractor.

1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-07 Certificates

**National Dredging Quality Management Program Certification,
Jacksonville District**

1.3 PAYMENT

Separate payment for installation, operation, and maintenance of the DQM-certified system as specified herein for the duration of the dredging operations is not allowed; all costs in connection therewith are considered a subsidiary obligation of the Contractor and are covered under the contract unit price for dredging in the bidding schedule.

1.4 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM CERTIFICATION

The Contractor is required to have a current certification from the DQM Program for the cutter/suction head hydraulic dredge instrumentation system to be used under this contract. Standard Operating Procedures (SOP) and criteria for certification are presented on the DQM website at <https://go.mil/4gicirhsvc>.

1.5 DREDGE PLANT INSTRUMENTATION PLAN (DPIP)

The Contractor must have a digital copy of the Dredge Plant Instrumentation Plan (DPIP) on file with the DQM Support Center. While working on site, the Contractor must also maintain on the dredge a copy of the DPIP, which is easily accessible to Government personnel at all times. This document must accurately describe the sensors used, the configuration of the system, how sensor data will be collected, how quality control on the data will be performed, and how the sensors/data-reporting equipment will be calibrated and repaired if it

fails. A description of the computed dredge-specific data and how the sensor data will be transmitted to the DQM database will also be included. Prior to the start of work, the Contractor must submit to the DQM Support Center any addendum or modifications made to the plan subsequent to its original submission. Requirements and a template for the DPIP are available on the DQM website at <https://go.mil/4gicirhsvc>.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.1 REQUIREMENTS FOR REPORTED DATA

Provide, operate, and maintain all hardware and software to meet these specifications. The Contractor is also responsible for the replacement, repair, and calibration of the sensors and other necessary data acquisition equipment needed to supply the required data. Document and complete the procedure to complete a repair as soon as practical. If repair is not possible within two business days of any sensor failure, submit a plan and timeline to complete the repair. Upon completion of a repair, replacement, installation, modification, or calibration, notify the Contracting Officer's Representative (COR). The COR may request recalibration of the sensors or other hardware components at any time during the contract as deemed necessary.

Keep a log of sensor repair, replacement, installation, modification, and calibration in the dredge's onboard copy of the DPIP. The log must contain a three-year history of sensor maintenance, including the time of the sensor failures (and subsequent repairs), the time and results of sensor calibrations, the time of sensor replacements, and the time that backup sensor systems were initiated to provide the required data. It must also contain the name of the person responsible for the sensor work. Install sensors that are capable of collecting parameters within the specified accuracies and resolutions indicated in the following subparagraphs and transmit these parameters to the DQM database. Transmit all data in JSON message bundles. Each bundle can contain multiple message types. Transmit sensor data as work event messages, and transmit data which relates to the operational state of the dredge or its sensors as state event messages. (See paragraph PARAMETER TRANSMISSION TO THE WEB SERVICE.)

3.1.1 Message Bundle Data

Every message bundle must contain descriptive data that relates the message to a given dredge plant and date/time. Identify the start of a message bundle by the tag "DQM_data".

3.1.1.1 Messages

Messages contain operational data that populates the DQM database for a dredge plant. A message must consist of an event type and its associated data (as defined in paragraph DREDGE EVENTS - WORK EVENT and paragraph DREDGE EVENTS - STATE EVENT), a date/time stamp indicating when the event occurred or started, and a comment providing clarification or metadata about the situation. There are multiple event types, but they all fall into one of two categories - work events and state events.

3.1.1.1.1 Message Time

In a work event message, message time is the date and time that the data is collected from the sensors; in a state event message, message time is the date and time that the state event begins. Report and reference the message time to the nearest second and to Coordinated Universal Time (UTC) time based on a 24-hour format (YYYY-MM-DD HH:MM:SS). In order to ensure accuracy and reliability, synchronize the time stamp to UTC format from an accurate, unchangeable source (for example, a GPS National Marine Electronics Association (NMEA) datastring). Identify message time by the tag "msg_time".

3.1.1.1.2 Comment

Comments concerning the work event or state event messages being transmitted provide descriptive information that relates to the data. An example of a comment for work event data is information about a sensor issue; an example of a comment for state event data is a description of operations. Identify a comment by the introductory tag "comment", and the comment must not consist of more than 250 characters.ext

3.1.1.2 Dredge Events - Work Event

There are two types of dredge event messages - work event messages and state event messages. Work event messages contain data that are instantaneously collected or calculated from sensors and are logged as a series of events. Work events are triggered by a time interval change (as described in paragraph WORK EVENT MESSAGES). Initiate all work event messages by the header tag "work_event".

3.1.1.2.1 Vertical Correction

Obtain the variation of the water level from the vertical datum for the river stage or tidal gage described in the state events using appropriate equipment to give the water level with an accuracy of plus or minus 0.1 ft. Enter vertical correction values above project datum described in the dredging specification with a positive sign and those below with a negative sign. The tag for vertical correction is "vert_correction".

3.1.1.2.2 Cutter/Suction Head Location and Movement

Monitor the X, Y, and Z components of the cutter/suction head location. Additional calculations made from the observed values determine the rates of movement to track the progress of the dredge.

3.1.1.2.2.1 Cutter/Suction Head Horizontal Position

Obtain the forwardmost point of the cutter/suction head using a positioning system operating with a minimum accuracy level of 3-10 feet horizontal Circular Error Probable (CEP). Report it as Latitude/Longitude WGS 84 in decimal degrees with West Longitude and South Latitude values reported as negative. Identify position values by the tags "ch_latitude" and "ch_longitude".

3.1.1.2.2.2 Cutter/Suction Invert Depth

Cutter/suction invert depth is the depth of the invert of the suction mouth relative to the surface of the water. Instrumentation must be capable of reporting to an accuracy of plus or minus 0.5 foot and a

resolution to the nearest 0.1 foot with no tidal adjustments. Minimum accuracies are conditional to relatively calm water. Use the tag "ch_depth" to identify the cutter/suction head depth.

3.1.1.2.2.3 Cutter/Suction Head Heading

The cutter/suction head heading is the angle of the centerline of the cutter/suction head and dredge ladder measured relative to true north. Provide all headings using industry-standard equipment. The heading must be accurate to within 5 degrees and reported to the nearest whole degree with values from 000 (true north) to 359 degrees referenced to a clockwise positive direction convention. Use the tag "ch_heading" to identify the cutter/suction head heading.

3.1.1.2.3 Dredge Activity

Monitor dredge activity using a combination of the following parameters.

3.1.1.2.3.1 Slurry Velocity

Use a flow-metering device, calibrated according to the manufacturer's specifications, to record the slurry velocity to the nearest 0.01 fps with an accuracy of plus 0.5 fps. If the manufacturer does not specify a frequency of recalibration, conduct calibration prior to the commencement of work. Measure the slurry velocity for the same pipeline inside diameter as that used for the slurry density measurement. Associate the tag "slurry_velocity" with this value.

3.1.1.2.3.2 Slurry Density

Use a density-metering device, calibrated according to the manufacturer's specifications, to record the slurry density to the nearest 0.01 g/cc. It is understood that the accuracy of this sensor can vary based on several factors, including the type of material, the magnitude of the cut, and the length of time since calibration. If the manufacturer does not specify a frequency of recalibration, conduct calibration prior to the commencement of work. Continuous monitoring of this sensor ensures that drift and other factors inherent in the dredging process can be accounted for in monitoring dredge activity. Associate the tag "slurry_density" with this value.

3.1.1.2.3.3 Pump RPM

The pump rpm is the number of revolutions per minute measured for the slurry pump shaft. Measure the shaft revolution rate (rev/min) with the highest level of accuracy that is standard on the vessel's operational displays either at the bridge or in the engine room. Identify this value by the tag "rpm".

3.1.1.2.3.4 Pump Vacuum

Measure the vacuum pressure of the dredge pump(s) (inches of mercury) as near to the eye as practicable in the pump's suction pipe with the highest level of accuracy that is standard on the vessel's operational displays either at the leverman's controls or in the engine room. Identify vacuum pressure by the tag "vacuum".

3.1.1.2.3.5 Pump Outlet Pressure

Measure the pump outlet pressure in the discharge line on the pump side of the flap valve in terms of pounds per square inch (psi) on a gauge. Identify pump outlet pressure by the tag "outlet_psi".

3.1.1.2.4 Outfall Information (Open Water/Spill Barge Disposal)

Monitor the X and Y position of the terminal end of the outfall pipe continuously and report the position as part of the work event string.

3.1.1.2.4.1 Discharge Horizontal Position

Obtain the horizontal position of the outfall end of the discharge pipe using a positioning system operating with a minimum accuracy level of 3-10 feet horizontal Circular Error Probable (CEP). Report it as Latitude/Longitude WGS 84 in decimal degrees with West Longitude and South Latitude values being reported as negative. Identify position values by the tags "outfall_latitude" and "outfall_longitude".

3.1.1.3 Dredge Events - State Event

There are two types of dredge event messages - work event messages and state event messages. State event messages provide information about the current state of the dredge equipment or operations. They are created and sent only when a state changes. Since state events often cannot be collected in real time, state events are tagged with a date time stamp (referenced to Coordinated Universal Time (UTC)) that indicates when the state change happened relative to the work event message tag. This data is considered to be "true" until another state event tag of the same type is received. Indicate each type of state event message by a specific header tag as enumerated in the following subparagraphs. State events can be transmitted along with work event message bundles directly by the contractor using the indicated format, or they can be entered on the "State" tab in the DQM-provided software. However, they should be sent only if the state value changes.

3.1.1.3.1 Message Time

The state event time is the date and time that the event starts. Enter the leverman's time to the nearest second as local time and automatically convert to and report in UTC based on a 24-hour format (YYYY-MM-DD HH:MM:SS). Identify message time by the tag "msg_time".

3.1.1.3.2 Contract Event

Report information concerning the contract under which dredging is being performed at the start and completion of each contract using the header tag "contract_event".

3.1.1.3.2.1 Contract Number

Report the USACE-assigned contract number for the project using the tag "contract_number".

3.1.1.3.2.2 Contract Start and End

Report the start and end of a contract using the tag "event_type" with the appropriate value of "start" or "end".

3.1.1.3.3 Tide Station/River Stage Gage Event

Group together properties associated with the vertical correction (see paragraph VERTICAL CORRECTION) for the tide station/river stage gage under the header tag "station_event". This information must be sent at the start of the contract and each time the dredge has moved enough to change the station being used.

3.1.1.3.3.1 Station Name

The station name is a concise name defining the tide station/river stage gage begin referred to. It must be introduced by the tag "station_name", and it must not consist of a descriptor of more than 25 characters.

3.1.1.3.4 Length of Pipe Event

Report the leverman's estimate of the length of pipe downflow from the dredge pump, measured to the nearest whole foot, under the header tag "pipe_length_event". This information must be sent at the start of the contract and at the completion of each 24-hour period ending at midnight local time.

3.1.1.3.4.1 Floating Pipe

Report the total length of floating pipe with the tag "length_floating".

3.1.1.3.4.2 Submerged Pipe

Report the total length of floating pipe with the tag "length_submerged".

3.1.1.3.4.3 Shore Pipe

Report the total length of shore pipe with the tag "length_land".

3.1.1.3.5 Booster Pump

Include information concerning the booster pumps being used under the header tag "booster_pump_event". A message must be sent to indicate any change in the status of the booster pumps being used.

3.1.1.3.5.1 Number of Booster Pumps

Upon the addition or removal of a booster pump, report the total number of booster pumps being used with the tag "booster_total".

3.1.1.3.6 Dredge Advance

Measure the dredge advance, the total forward progress of the dredge relative to the centerline of the cut, to the nearest whole foot and cumulatively calculate over a 24-hour period from midnight to midnight local time. Identify it by the tag "advance_daily". Report in Greenwich Mean Time (GMT) the msg_time associated with this tag as the first timestamp of the following 24-hour period (based on the local time) rather than as midnight of the day for which the value was calculated.

3.1.1.3.7 Outfall Information

Monitor and send the X and Y position of the terminal end of the outfall

pipe at the start of the contract and thereafter according to the following table. Discharge Heading and Pipe Elevation may be omitted if the dredge is not discharging into an upland disposal site. For beach nourishment, the horizontal X and Y position of the outfall must be sent at the start of the contract and at the completion of each 24-hour period ending at midnight local time.

Discharge Location	Horizontal Position	Discharge Pipe Elevation	Discharge Outfall Heading
Open Water	Continuous Work Event	N/A	N/A
Scow	Upon Change	N/A	N/A
Beach	Every 24 Hours	N/A	N/A
Upland	Upon Change	Upon Change	Upon Change

3.1.1.3.7.1 Discharge Location

Report information on where the slurry is being discharged with the tag "outfall_location". Acceptable values include "upland", "open water", "beach", and "scow".

3.1.1.3.7.2 Discharge Horizontal Position

Obtain the horizontal position of the outfall end of the discharge pipe using a positioning system operating with a minimum accuracy level of 3-10 feet horizontal Circular Error Probable (CEP). Report it as Latitude/Longitude WGS 84 in decimal degrees with West Longitude and South Latitude values being reported as negative. Identify position values by the tags "outfall_latitude" and "outfall_longitude".

3.1.1.3.7.3 Discharge Outfall Heading

The discharge outfall heading is the angle relative to true north measured from the centerline of the pipe in the direction of discharge. Provide all headings using industry-standard equipment. They must be accurate to within 5 degrees and report to the nearest whole degree with values from 000 (true north) to 359 degrees referenced to a clockwise positive direction convention. Identify the discharge heading by the tag "outfall_heading".

3.1.1.3.7.4 Discharge Pipe Elevation

The discharge pipe elevation is the height of the outfall measured in feet and tenths of a foot relative to the project datum. The required accuracy is contingent upon contract requirements. Use the tag "outfall_elevation" to identify this elevation.

3.1.1.3.8 Non-effective Work Event

Report delays and dredge downtime at the conclusion of the event. Submit the reason for the non-effective work time under the header tag "non_eff_event" within 24 hours of the event.

3.1.1.3.8.1 Non-effective Work Interval

Report the start and end times for the non-effective work event using the

tags "msg_start_time" and "msg_end_time".

3.1.1.3.8.2 Dredge Function Code

Transmit the dredge operator indication of production delays, as listed on Form 4267, at the end of the non-effective interval. Identify dredge function event messages by the tag "function_code" consisting of one of the following standardized entries to indicate the operation:

AGV	Assisting Grounded Vessels
CCH	Change Cutterhead
CCSH	Clear Cutter Suction
CLPJ	Change Location Bar
COLL	Collision
CPPL	Clear Pump Pipeline
CPR	Change Impeller
DR	Dike Repair
FBD	Fire Boat Drills
HPL	Handling Pipe Line
HSL	Handling Swing Line
HSP	Handling Shore Pipe
LDNE	Loss Due to Natural Elements
LDPV	Loss Due to Passing Vessel
LNL	Transfer to New Location
MISC	Miscellaneous
MOB	Mobilization & Demobilization
MSC	Miscellaneous/Non-pay
OC	Out of Commission
OR	Operating Repairs
P	Preparation
PREP	Preparation & Making Up Tow
RPL	Repair Pipeline
SB	Sounding & Buoying
SBT	Stand-By Time as Directed
SH	Sundays-Holidays
TFS	Taking on Fuel & Supplies
TOW	Time on Tow
WAP	Waiting Attendant Plant

3.1.1.3.8.3 Additional Comments

Use the "comment" tag to provide additional explanation for the noted delays or downtimes. For example, when the code "LDPV" (Loss Due to Passing Vessel) is indicated, list the name of the vessel and the number of tows with the "comment" tag.

3.2 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM SYSTEM REQUIREMENTS

The Contractor's DQM system must be capable of collecting and transmitting information to the DQM system. Record the applicable parameters from paragraph REQUIREMENTS FOR REPORTED DATA as local events and transmit continuously to the DQM database anytime an Internet connection is available. To accomplish this transmission, there are two options. The first option is to equip the vessel with a DQM computer system consisting of a computer, monitor, keyboard, mouse, data modem, Universal Power Supply (UPS), and network hub. Provide a standalone computer system, exclusive to the DQM monitoring system, with USACE DQM software installed on it. If a hardware problem occurs, or if a part of the system is physically damaged, the Contractor is responsible for repairing it within

two business days of the determination of the condition or submitting a plan and timeline for repair if the repair will take more than two business days.

The second option is to send data directly to DQM's web endpoint as outlined in paragraph DIRECT DATA TRANSMISSION.

3.2.1 Computer Requirements

Provide a dedicated onboard computer for use by the Dredging Quality Management system. This computer must run the USACE DQM software and receive data from the Contractor's data-reporting interface. This computer must meet or exceed the following performance specifications:

CPU:	Intel or AMD processor with a (non-overclocked) clock speed of at least 1.6 gigahertz (GHz)
Hard drive:	250 gigabytes (GB); internal
RAM:	4 gigabytes (GB)
Ethernet adapter:	Internal network card with an RJ-45 connector
Ports:	1 free serial port with standard 9-pin connectors; 1 free USB port
Other hardware:	Keyboard, mouse, monitor

Install a fully licensed copy of Windows 10 Professional Operating System or newer on the computer specified above. Also install any necessary manufacturer-provided drivers for the installed hardware.

Locate and orient this computer to allow data entry and data viewing as well as to provide access to data ports for connection of external hardware.

3.2.2 Software

The DQM computer's primary function is to transmit data to the DQM shoreside database. Do not install software which conflicts with this function on this computer. The DQM computer must have the USACE-provided Dredging Quality Management Onboard Software (DQMOBS) installed on it by DQM personnel.

3.2.3 UPS

Supply an Uninterruptible Power Supply (UPS) for the computer and networking equipment. It must interface with the DQM computer to communicate UPS status, and it must provide backup power at 1 kVA for a minimum of 10 minutes. Ensure that sufficient power outlets are available to run all specified equipment.

3.2.4 Direct Data Transmissions

Option 2: Direct Data Transmissions to the DQM Web Service - Use the web service to report sensor data to the DQM database. Data must be transmitted as it is collected in real time and pushed to the DQM web service. If the web service is not available or returns an error message, the data must be stored in a queue and transmitted upon re-establishment

of the connection, starting with the oldest data in the queue and continuing until real-time transmission is restored.

Contact <https://go.mil/4gicirhsvc> to obtain the web service URL and the appropriate key credentials and communication protocol. If this option is chosen, a display of the raw data being transmitted to DQM must be easily accessible and visible onboard the dredge

3.2.5 Internet Access

Maintain an Internet connection capable of transmitting real-time data to the DQM server as well as enough additional bandwidth to clear historically queued data when a connection is re-established. If connectivity is lost, queue and transmit unsent data upon restoration of connectivity. Delays in pushing real-time data to the DQM database should not exceed four hours. Exceptions to these requirements may be granted by the DQM Support Center on a case-by-case basis with consideration for contract-specific requirements, site-specific conditions, and extreme weather events.

Acquire and install all necessary hardware and software to make the Internet connection available for data transmission to the DQM web service. Configure the hardware and software to allow the DQM Support Center remote access to this computer, and the telemetry system must be capable of meeting these minimum reporting requirements in all operating conditions.

In areas with poor cellular service and at the local District's discretion, it may be required to manually download the data on a daily basis using the protocol for retrieving and submitting backup files provided by the DQM Support Center. This method of data transmission should be used only if Internet connectivity is unavailable at the dredging site, and it should be considered a temporary measure.

3.2.6 Data Routing Requirements

Onboard sensors continually monitor dredge conditions, operations, and efficiency and route this information to the shipboard dredge-specific system (DSS) computer to assist in guiding dredge operations. Portions of this Contractor-collected information, as described in this specification, must be routed to the DQM computer on a real-time basis utilizing one of the two options outlined in this section. Use an RS-232 serial interface with a baud rate of 9600 or 19200 bps to send standard sensor data to the DQM computer. Configure the serial interface as 8 bits, no parity, and no flow control.

Digitally log and transmit information regarding changes in the state of the dredge as close to the time of the occurrence as possible. These events can either be included in a separate message bundle going to the DQM onboard computer, or they can be entered on the "State" tab in the DQM Pipeline Software, or entered online in the DQM State Data Tool.

3.3 DREDGE MONITORING DATA

3.3.1 General

Onboard sensors continuously collect dredging data in support of the dredge Contractor's operations. Portions of this Contractor-collected information, as described in this specification, and store and transmit

calculations based on them to the DQM database on a near real-time basis. Additionally, digitally log and transmit information regarding the state of the dredge.

3.3.2 Data Measurement Frequency

The frequency of data transmission is dependent on the type of message being sent. Work Event messages contain data that are instantaneously collected or calculated from sensors and are logged as a series of events. State event messages are activated by a change in the dredge state.

3.3.2.1 Work Event Messages

Log data as a series of events. Each event must consist of a dataset containing dredge information (as defined in paragraph REQUIREMENTS FOR REPORTED DATA). Consider each set of measurements (for example, time and position) an event with a 6-12 second interval between work events. This interval must remain consistent across event types for the dredge plant.

Record a standard data string within one second of an event trigger with the time stamp and all parameters reflecting when the event happened.

3.3.2.2 State Event Messages

Consider a set of descriptive information (event name, time, description, comment) a state event. Record these events within 24 hours of a change in state with the time stamp reflecting when the event happened.

3.3.3 Parameter Transmission to the Web Service

Format the data as JSON (JavaScript Object Notation, as defined at <http://www.json.org>) strings of arbitrary length. These JSON strings represent a hierarchical data structure consisting of a message bundle which may contain 0-3 automatic data messages and any number of manual data messages.

A tag/parameter is reported only when it contains a value. Do not include "Null" value strings in a message bundle.

Message bundle

```
{
  "DQM_Data": {
    "plant_identifier": <integer value 0000-9999>,
    "transmission_time": <24-hour UTC time YYYY-MM-DD HH:MM:SS>,
    "messages": {
      {
        "work_event": {
          "msg_time": <24-hour UTC time YYYY-MM-DD HH:MM:SS>,
          "vert_correction": <floating point 100th decimal place>,
          "ch_latitude": <decimal to 6 decimal places>,
          "ch_longitude": <decimal to 6 decimal places>,
          "ch_depth": <floating point 100th decimal
place>,
          "ch_heading": <integer value 000-359>,
          "slurry_velocity": <floating point 100th decimal place>,

```

U.S. Naval Station Mayport
Duval County, Florida

```

        "slurry_density":    <floating point 100th decimal place>,
        "pump_rpm":          <integer>,
        "vacuum":            <floating point 100th decimal place>,
        "outlet_psi":        <floating point 100th decimal place>,
        "comment":           <string>},
    },
    {
        "contract_event": {
            "msg_time":      <24-hour UTC time YYYY-MM-DD HH:MM:SS>,
            "contract_number": <string>,
            "event_type":    <string - "start" or "end">,
            "comment":       <string>
        }
    },
    {
        "station_event": {
            "msg_time":      <24-hour UTC time YYYY-MM-DD HH:MM:SS>,
            "station_name":  <string>,
            "comment":       <string>
        }
    },
    {
        "pipe_length_event": {
            "msg_time":      <24-hour UTC time YYYY-MM-DD HH:MM:SS>,
            "length_floating": <integer>,
            "length_submerged": <integer>,
            "length_land":    <integer>,
            "comment":       <string>
        }
    },
    {
        "booster_pump_event": {
            "msg_time":      <24-hour UTC time YYYY-MM-DDHH:MM:SS>,
            "booster_total": <integer>,
            "comment":       <string>
        }
    },
    {
        "advance_Event": {
            "msg_time":      <24-hour UTC time YYYY-MM-DD HH:MM:SS>,
            "advance_daily": <integer>,
            "comment":       <string>
        }
    },
    {
        "outfall_position": {
            "msg_time":      <24-hour UTC time YYYY-MM-DD HH:MM:SS>,
            "outfall_location": <string-"upland", "beach", "scow",
"open water">
            "outfall_latitude": <decimal to 6 decimal places>,
            "outfall_longitude": <decimal to 6 decimal places>,
            "outfall_heading": <integer value 000-359>,
            "outfall_elevation": <floating point 10th decimal place>,
            "comment":       <string>
        }
    },
    {
        "non_eff_event": {

```

U.S. Naval Station Mayport
Duval County, Florida

```
"msg_start_time":    <24-hour UTC time YYYY-MM-DD HH:MM:SS>,  
"msg_end_time":      <24-hour UTC time YYYY-MM-DD HH:MM:SS>,  
"function_code":     <string - 1 to 4 characters>,  
"comment":           <string>
```

```
    }  
  }  
}
```

3.3.4 Contractor Data Backup

Maintain an archive of all data sent to the DQM computer during the dredging contract. The COR may require, at no increase in the contract price, that the Contractor provide a copy of these data covering specified time periods. Provide the data in the same JSON format as would have been transmitted to the DQM computer. There must be no line breaks between the parameters, and each record string must be on separate line. The naming convention for the files must be `<dredgename>_<StartYYYYMMddhhmmss>_<EndYYYYMMddhhmmss>.txt`.

Data submission must be via a storage medium acceptable to the COR.

At the end of the dredging contact, call the National DQM Support Center prior to discarding the data. The DQM Support Center will verify that all data has been received and appropriately archived before giving the Contractor discard permission. Record the following information in a separate section at the end of the dredge's onboard copy of the DPIP:

Person who called the National DQM Support Center
Date of the call
DQM representative who gave permission to discard the data

3.4 PERFORMANCE REQUIREMENTS

The Contractor's National Dredging Quality Management Program's data transmission must be fully operational at the start of dredging operations. To meet contract requirements for operability, the Contractor's system must provide an accurate data string return and be compliant with hardware requirements. Data string return is defined as the number of quality records within an event or state tag sent by the contractor's system to the DQM database. Quality data strings are considered to be those providing accurate values for all parameters reported when operating according to the specification. Make repairs necessary to restore data return compliance within two business days, or submit a plan and timeline for repair if the repair will take more than two business days. Failure by the Contractor to report quality data within the specified time window for dredge measurements as stated in the specifications (see paragraph INTERNET ACCESS, paragraph DATA MEASUREMENT FREQUENCY, and paragraph PARAMETER TRANSMISSION TO THE WEB SERVICE, will result in withholding of up to 10 percent of the contract progress payment.

3.5 LIST OF ITEMS TO BE PROVIDED BY THE CONTRACTOR

DPIP

<https://go.mil/4gicirhsvc>

DQM System

Paragraph 3.2 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM SYSTEM

U.S. Naval Station Mayport
Duval County, Florida

REQUIREMENTS, including all subparagraphs

Dredge Data

Paragraph DREDGING MONITORING DATA

-- End of Section --